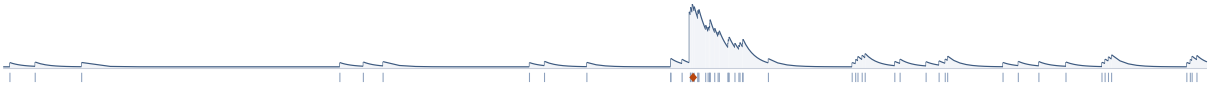
Shock and Contagion*

Self-Exciting Point Processes for Surges in Antisemitic Incidents

Kornstein Lab for Antisemitism Research

November 2026



Abstract

After salient external events, how much of a surge in antisemitic incidents is direct reaction, how much is contagion, and does online hostility spill offline? We model incident streams as a multivariate Hawkes process with exogenous shocks. Within a stream, the autocorrelation of counts cannot separate contagion from unrecorded shocks, although dominant cross-excitation is identifiable at second order; a rigid background manufactures contagion; and under-reporting attenuates it through an invertible map. We decompose post-event surges into background, direct and contagion parts. The alert rule is Bayes-optimal given the model's state, not with a plug-in state under reporting delays. Synthetic-data simulations evaluate EM, attribution, the boundary test, the thinning correction and a plug-in alert; Bayesian shrinkage, time-rescaling diagnostics, an imputation-based alert, sup-tests and nonparametric kernels were not evaluated. Without contagion, a constant background yields branching ratios of up to 0.63; for rare large shocks maximum likelihood settles 0.06–0.07 below the second-order value even with $T = 10{,}000$. Known event times largely removed the artefact where tested; a single spline background did not. The dark-figure correction restores coverage unless excitation is weak and recording poor, and a 30–50% error in the recording probability sets a floor under its precision that more data cannot lower. Forecasts that ignore future external events under-forecast; forecasts that integrate over the event process are calibrated when windows are pooled. No forecaster can time new events, which limits recall rather than pooled calibration. We pre-register a real-data analysis and report no empirical estimates.

Keywords Hawkes processes; self-exciting point processes; contagion; identification; under-reporting; surge attribution; early warning; hate crime; antisemitism.

1 Introduction

Antisemitic incidents do not arrive at a steady rate. They come in surges, and surges often follow salient external events. In the United Kingdom, the Community Security Trust (CST) recorded an average of five antisemitic incidents a day between 1 January and 6 October 2023, and 31 a day from 7 October, the day of

*Status. Methods paper; simulation study on synthetic data; pre-registered analysis plan; no empirical estimates. Every number in Section 8 is computed from synthetic data with known ground truth; the real-data analysis of Section 9 is pre-registered and has not been carried out. The companion paper (Kornstein Lab for Antisemitism Research, 2026a) estimates how many incidents go unrecorded; this paper uses its output.

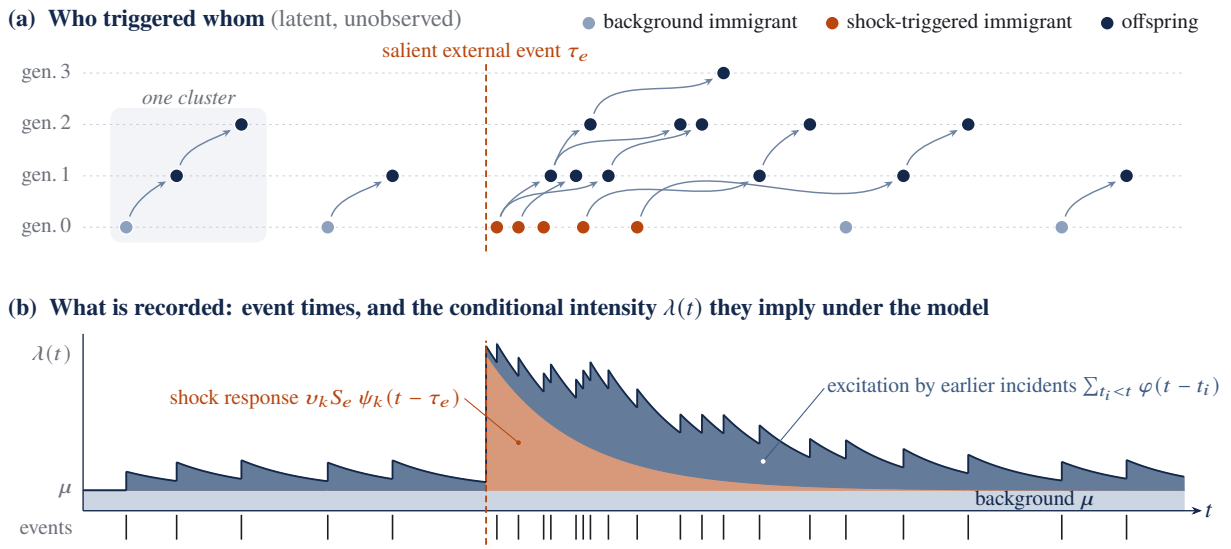


Figure 1. **The cluster representation of a Hawkes process with exogenous shocks.** *(a)* Every incident is either an *immigrant* or the *offspring* of an earlier incident. Background immigrants (light navy) arrive at rate $\mu_k(t)$; shock-triggered immigrants (orange) are the direct response to a salient external event at τ_e; each incident, of any kind, independently produces Poisson(α_{kj}) offspring at delays drawn from g_{kj} (navy), which produce offspring in turn (Hawkes and Oakes, 1974). An immigrant and all its descendants form one cluster. *(b)* The data are the event times alone (ticks). Under the model, the conditional intensity is the sum of the background, the shock response and the excitation by past incidents (Equation (3.2)). The share of each layer at an event time is the probability that the event is a background immigrant, a shock-triggered immigrant or the offspring of an earlier incident; this is how the EM algorithm reconstructs the unobserved family trees (Algorithm 1). The surge decomposition of Section 5 classifies incidents instead by the root of their cluster (Figure 3). Schematic: event times are hand-placed, and the intensity is computed exactly from them for illustrative parameters ($\mu = 0.6$, $\alpha = 0.55$, exponential kernel with $\beta = 1$, and $\upsilon_k S_e = 5$ expected direct incidents with response $\psi_k(t) = \eta e^{-\eta t}$, $\eta = 0.8$).

the Hamas attack on Israel, to the end of the year. Its daily count peaked at 80 on 11 October, the highest number ever reported to CST on a single day (Community Security Trust, 2024). On 2 October 2025, the day of a terrorist attack on the Heaton Park synagogue in Manchester, and on the following day, CST logged 40 incidents each, its highest daily totals of the year. Of those 80 incidents, 42 were antisemitic reactions to the attack itself, most of them online (Community Security Trust, 2026). In New York City, an analysis of police records found the rate of anti-Jewish hate crimes in the first year of the Israel–Hamas war to be about twice that of the preceding years of the study period (Rosenbaum, 2025). Such patterns are not peculiar to antisemitism. In the United States, contentious trial verdicts and lethal terrorist attacks have preceded spikes in racially or religiously motivated hate crimes (King and Sutton, 2013).

A surge raises three questions for the communities affected and for those who plan their security. *How much of it is a direct reaction to the event? How much is incidents begetting incidents*, through imitation, emboldenment or repeated targeting? And *does online hostility spill over into offline incidents?* The answers call for different responses. A direct reaction is concentrated in the days after the event and can be anticipated from a calendar of events. Contagion means that each incident prevented also prevents some of the incidents it would have triggered, and that a surge can outlast its trigger. A spillover from online to offline streams would make online monitoring a source of early warning. This paper develops the statistical framework needed to ask these questions of incident data, to answer them where the data allow, and to say plainly where they do not.

Self-exciting point processes. The natural model is the self-exciting, or Hawkes, point process (Hawkes, 1971), in which every incident raises the short-run rate of further incidents. Its cluster representation (Hawkes and Oakes, 1974) makes the three questions concrete (Figure 1). Each incident is either an *immigrant*, arriving from a background rate or as a direct response to a salient external event, or the *offspring* of an earlier incident. The expected numbers of offspring form a branching matrix $\boldsymbol{A} = [\alpha_{kj}]$ across streams (regions, offence types, online and offline channels), and the multiplier $\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{A})^{-1}$ counts the whole cascade that one incident sets off. Self-exciting models are established in seismology (Ogata, 1988), crime (Mohler et al., 2011; Reinhart, 2018), conflict (Lewis et al., 2012), online popularity (Rizoiu et al., 2017; Zhao et al., 2015) and gun violence (Loeffler and Flaxman, 2018), and online content has been found to predict offline hate crime (Müller and Schwarz, 2021; Williams et al., 2020). As far as we know, no study has estimated a multivariate branching matrix for antisemitic incidents, and this paper does not either. It builds and tests the tools that such an analysis requires.

Three traps. Applied naively, the model is easy to fit and easy to over-read. We formalise three traps.

(i) *Contagion and clustered shocks look alike, within a stream.* A stationary Hawkes process with an exponential kernel has exactly the same mean and Bartlett spectrum as a one-parameter family of shot-noise Cox processes, in which unrecorded shocks trigger incidents that do not trigger one another (Theorem 4.2). For a single stream, the autocorrelation of counts, the variance–time curve and the periodogram therefore cannot tell contagion from clustered exogenous shocks, at any bin width. This is a dynamic counterpart of the old distinction between true and apparent contagion (Feller, 1943). Between streams the picture is different: mutual excitation that dominates self-excitation leaves a second-order trace that no Cox process can reproduce (Example 4.6).

(ii) *A rigid background manufactures contagion.* If salient events are missing from the calendar, or the background is too stiff to follow slow variation, a Hawkes model with a constant background has a positive pseudo-true branching ratio even when there is no contagion at all (Theorem 4.7). For second-order estimators the spurious branching ratio is explicit, $n^* = 1-(1+\kappa)^{-1/2}$, where κ measures the burstiness of the unrecorded driver. Maximum likelihood need not reproduce n^* exactly. In our simulations a single spline background removed the artefact of slow rate changes but not that of fast unrecorded shocks, and known event times largely removed it where tested (Section 8.3). For rare large shocks, maximum likelihood settled 0.06–0.07 below n^* even with $T = 10{,}000$.

(iii) *Under-reporting masks contagion.* Most incidents are never recorded. That is the subject of the companion paper (Kornstein Lab for Antisemitism Research, 2026a), which estimates the size of this dark figure. If each incident is recorded independently with probability p, the recorded stream behaves, to second order, like a Hawkes process with an attenuated branching ratio n', with $pn < n' < n$ and a closed-form map between the two (Theorem 4.10). A cascade passes through unrecorded incidents too, so ignoring the dark figure understates contagion. When the most productive incidents are also the most likely to be recorded, the standard correction overshoots instead (Proposition A.6).

Two further cautions are quieter. Aggregating incidents into daily counts also attenuates estimated contagion, by an amount that can be computed (Proposition C.2). And although online-to-offline Granger non-causality is equivalent to a zero cross-excitation coefficient (Proposition 4.12, after Eichler et al., 2017), unrecorded common shocks, classifier error, under-recording and report-time stamps can each distort that coefficient (Remark 4.13).

Table 1. Notation. Symbols used throughout. Bold capitals are $M \times M$ matrices indexed by (child stream k, parent stream j).

Streams and incidents	
$k, j \in [M]$	stream: offline incidents by region or offence type; online volume
$t_i^k,\ x_i^k$	time and mark of the ith stream-k incident on $[0, T]$
$N_k(t),\ \mathcal{F}_t$	counting process; history of all streams and recorded events
$\lambda_k(t),\ \Lambda_k(t)$	conditional intensity; compensator
Self-excitation	
$\varphi_{kj} = \alpha_{kj} g_{kj}$	excitation kernel; g_{kj} a probability density
α_{kj}	expected *direct* stream-k offspring of one stream-j incident
$\boldsymbol{A} = [\alpha_{kj}],\ \varrho(\boldsymbol{A})$	branching matrix; spectral radius, < 1 for stationarity
$\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{A})^{-1}$	contagion multiplier: $\boldsymbol{M}_{kj}$ is the expected stream-k size of a cascade rooted in stream j
$\boldsymbol{\Phi}(t),\ \boldsymbol{R} = \sum_{m \geq 1} \boldsymbol{\Phi}^{*m}$	kernel matrix; resolvent
$\beta e^{-\beta t}$	exponential kernel with decay rate β (or β_{kj})
$n,\ \bar{\Lambda} = \mu/(1 - n)$	univariate branching ratio; stationary mean rate
Background and salient external events	
$\mu_k(t) = e^{B(t)^\top \theta_k}$	background; basis B of K_b functions (spline knot spacing h_b) that carries calendar effects
$e;\ \tau_e,\ S_e$	salient external event; its time and size
$\psi_k,\ \upsilon_k$	response kernel ($\int \psi_k = 1$, e.g. $\eta_k e^{-\eta_k t}$); shock loading
ϑ	full parameter vector, including the background coefficients θ_k
Recording	
$p_k = 1/\mathcal{D}_k$	recording probability; $\mathcal{D}_k$ is the companion paper's dark-figure ratio
$n' = h_p(n)$	apparent branching ratio of the p-thinned process
$d,\ F_k$	reporting delay; its distribution
$e_0,\ e_1$	classifier flag rates for non-antisemitic and antisemitic posts
Attribution and alerting	
$B_k,\ D_k,\ C_k$	background, direct and contagion parts of $\mathbb{E}\, N_k(\tau_e, \tau_e + H]$
$\boldsymbol{C}^{\text{on}},\ \sigma_k(H)$	online-channel contagion; its share
$\gamma = \beta(1 - n)$	decay rate of a cascade (univariate, exponential kernel)
$\Delta_{\text{a}},\ c$	alert window; count threshold (Δ without subscript is a bin width for counts)
$\boldsymbol{\chi}(t)$	excitation state $(\boldsymbol{x}(t), \boldsymbol{u}(t))$ of the alert rule
$\pi_t,\ \pi^*$	surge probability; alert threshold
$c_{\text{FA}},\ c_{\text{miss}}$	costs of a false alarm and of a missed surge

What this paper does, and does not, do. This is a methods paper. We set out the model, prove what can and cannot be identified, derive the surge decomposition and an alert rule, give estimation and testing procedures, evaluate some of them on synthetic data with known ground truth, and pre-register an analysis plan for real data. We report *no* empirical estimates, and nothing here should be read as a finding that antisemitic incidents are, or are not, contagious; the figures quoted above serve only to motivate the questions. Figure 2 shows the multivariate model; notation is collected in Table 1.

Contributions. Our contributions are a synthesis, generalisation and application of existing ideas to a new and sensitive domain. Classical results are attributed where they are used; we flag the results that are, to our knowledge, new in the form given.

C1 Framework. A multivariate Hawkes process with exogenous shocks for incident streams across regions, offence types and online and offline channels (Section 3), and what "contagion" does and does not mean when it is applied to antisemitic incidents (Section 3.5).

C2 Identification. For a single stream, or for streams that do not excite one another, the observational equivalence of contagion and clustered shocks is exact, for exponential (Theorem 4.2) and completely monotone kernels (Proposition 4.3). With cross-excitation, equivalence requires the excess spectral matrix $\boldsymbol{E}_H(\omega)$ to be positive semidefinite at every frequency; this fails for excite-only streams and when cross-excitation dominates self-excitation, and contagion between streams is then identified at second order (Proposition 4.5 and Example 4.6). Identification is restored by a recorded calendar, higher-order information or provenance links (Corollary 4.9), and a misspecified background yields spurious contagion (Theorem 4.7).

C3 Under-reporting. A closed-form map from n to the apparent $n' = h_p(n)$ (Theorem 4.10), whose inversion with the companion paper's estimate of p gives a dark-figure correction with a variance floor (Corollary 4.11), and the bias of that correction under heterogeneous recording (Proposition A.6).

C4 Surge attribution. An exact split of the expected count after an event into background, direct and contagion components (Theorem 5.1), hypoexponential closed forms for exponential kernels (Proposition 5.2), and a channel decomposition that isolates the online-to-offline spillover as a sum over paths (Proposition 5.4); the last two are, to our knowledge, new in this form.

C5 Estimation and testing. EM with an exact treatment of the observation window (Algorithm 1 and Proposition 6.1); Bayesian shrinkage with an entrywise description of the stationarity region (Lemma 6.2); an exact formula for the attenuation of contagion by binning (Proposition C.2); boundary-corrected and sup-type likelihood-ratio tests (Section 6.5); time-rescaling goodness of fit (Section 6.6); and placebo calendars (Section 6.7). The placebo rank test is valid whatever the misspecification of the base model, but only under the randomisation hypothesis that the real event dates are exchangeable with dates drawn from the placebo design (Lemma 6.3); without it, the placebo distribution is a calibration, not a test.

C6 Forecasting and alerting. When the model's state is known, the Bayes-optimal alert rule thresholds the predictive probability of a surge at $\pi^* = c_{\mathrm{FA}}/(c_{\mathrm{FA}} + c_{\mathrm{miss}})$, and its alert region is an upper set in the excitation state (Theorem 7.2). Reporting delays bias the state downward; a delay-weighted nowcast corrects the bias in the mean (Proposition 7.3). Plugging the nowcast into the rule is not the Bayes rule for delayed data. An imputation algorithm that targets it (Algorithm 2) is given but was not run.

C7 A simulation study. Simulations on synthetic data with known ground truth evaluate EM, surge attribution, the boundary test, the thinning correction and a plug-in alert with a nowcast state (Section 8). Under the model these tools perform as the theory says. The study also shows their limits: a single spline background did not remove spurious contagion from fast unrecorded shocks, and intervals fail when excitation is weak and recording is poor; a 30–50% error in the recording probability sets a floor under the precision of the corrected estimate that more data cannot lower. Forecasts that ignore future external events under-forecast; forecasts that integrate over the event process are calibrated when windows are pooled. What no forecaster can do is time new events, which limits recall rather than pooled calibration. On Hawkes-generated data the plug-in rule beats two simple baselines, which are not the plan's. Not evaluated are Bayesian shrinkage, time-rescaling

goodness of fit, the imputation-based alert of Algorithm 2, the sup-statistic for a decay identified only under the alternative (with its Davies approximation and bootstrap), nonparametric kernels, the INAR approximation, placebo calibration and the CUSUM monitor.

C8 A pre-registered analysis plan for New York City police hate-crime records (New York City Police Department, 2026), with an event list completed blind to incident data, registered block tests, sensitivity analyses, placebo shocks and decision rules (Section 9), and an ethics protocol (Section 10).

Related work. The likelihood theory of self-exciting processes goes back to Ogata (1978) and Ogata (1988), and their stability theory to Brémaud and Massoulié (1996). EM over the latent branching structure was developed by Veen and Schoenberg (2008) and, nonparametrically, by Marsan and Lengliné (2008); book-length treatments are Daley and Vere-Jones (2003) and Laub et al. (2021). The equivalence of Hawkes processes and INAR(∞) sequences (Kirchner, 2016) underlies our treatment of binned counts, and the Granger-causal graphs of Eichler et al. (2017) our treatment of online-to-offline dependence. Including news events in the background lowers estimated endogeneity in financial markets (Rambaldi et al., 2015), the empirical face of Theorem 4.7. Our nowcasting builds on Bayesian nowcasting in epidemiology (Höhle and an der Heiden, 2014; McGough et al., 2020).

Roadmap. Section 2 sets out what the data must provide, Section 3 defines the model, and Section 4 develops the identification results. Section 5 derives the surge decomposition, Section 6 estimation and testing, and Section 7 forecasting and alerting. Section 8 reports the simulation study, Sections 9 and 10 summarise the analysis plan and the ethics protocol, and Section 11 discusses limitations and extensions. The appendices contain the proofs (Appendices A, B and C.8), supplementary material for Sections 6 and 7 (Appendices C and D), computational and simulation details (Appendices E and F), the candidate data sources (Appendix G), and the full plan and ethics protocol (Appendices H and I).

2 Data: incident streams, external events and online signals

The framework of Section 3 needs three kinds of input: timestamped streams of recorded incidents, a calendar of salient external events (the shock times τ_e and sizes S_e), and, where available, binned series of online activity. This section sets out what each must provide; Appendix G reviews candidate sources against the providers' own documentation. No real data have been analysed for this paper, and nothing in this section is an empirical finding about antisemitism. Coverage, definitional harmonisation and under-recording are treated in the companion paper (Kornstein Lab for Antisemitism Research, 2026a); here the focus is on *time*.

2.1 What a point-process analysis needs from a record

Three clocks are attached to each incident i in stream k: the *occurrence* time t_i at which the incident happened; the *report* time $r_i = t_i + d_i$ at which it reached the recording body; and the *availability* time at which the record enters a file that an analyst can obtain. The model of Section 3 is defined on occurrence times. Two departures from that ideal are common, and they bias contagion estimates in different ways.

Report dates in place of occurrence dates. If the analysis uses r_i instead of t_i, every point is displaced by an independent delay $d_i \sim F_k$. A parent–offspring pair at lag u then appears at lag $u + d_j - d_i$, which is negative with positive probability whenever delays are dispersed, and the displaced process need not be a Hawkes process. Contagion estimated from report dates therefore mixes the excitation kernel with the delay distribution, most of all when typical delays are comparable with the kernel time scale $1/\beta$.

Coarse time stamps. Many sources record dates without times, and some record only the month of occurrence. Tied time stamps have probability zero under a continuous-time model. They can be handled by randomising times within the recorded interval and averaging over randomisations, or by working with the discrete-time INAR representation of Equation (6.8) (Kirchner, 2016), which is the natural choice for daily counts. A stream whose occurrence time is known only to the month cannot identify kernels on the scale of days from occurrence times alone.

2.2 Incident streams

Appendix G reviews four candidate incident streams, and Table 12 summarises them. Three points matter for what follows. First, the hate-crime file of the New York City Police Department (New York City Police Department, 2026), the candidate pilot, gives the month of occurrence and the day of record creation. Its occurrence times are thus interval-censored to a month, while report dates are known to the day. Second, FBI NIBRS records carry the incident date and hour (Federal Bureau of Investigation, 2025a), but national releases lag occurrence by roughly 7 to 19 months, which suits retrospective estimation of $\boldsymbol{A}$ but not alerting. Third, recording systems, scopes and referral feeds change: New York City's transition to NIBRS interrupted the state's publication of its hate-crime data (Office of the New York State Comptroller, 2024), ADL widened its recording scope from 7 October 2023 (Anti-Defamation League, 2025), and police referrals to CST have been disrupted since March 2024 (Community Security Trust, 2026). A change of scope at a candidate shock time is not identified separately from the shock unless it is modelled, and an interrupted referral feed is a step change in the recording probability $p_k(t)$ of Section 2.6. Such breaks enter the background as indicators (Appendix H.5). Monthly analyses of NYPD records (Rosenbaum, 2025) suffice for before–after contrasts, but not for kernels on the scale of days.

2.3 Reporting delays and nowcasting

Three lags separate an incident from an analyst's file: occurrence to report (d_i, driven by victims' and witnesses' decisions); report to classification or confirmation as a bias incident; and confirmation to publication, which is set by each provider's release schedule. Suppose that recording and delays are independent of the history and across incidents, with recording probability p_k (Section 2.6) and total-delay distribution F_k. Then the occurrence times of the stream-k incidents that are available at analysis time T_a form an independent thinning of the true process, with time-varying retention probability $p_k F_k(T_a - t)$. Hence

$$\mathbb{E}\big[N_k^{\mathrm{avail}}(\mathrm{d}t)\big] = p_k\, F_k(T_a - t)\, \mathbb{E}\big[\lambda_k(t)\big]\, \mathrm{d}t. \tag{2.1}$$

Near T_a the factor $F_k(T_a - t)$ is small. Without correction, a surge that is still under way looks as if it is already subsiding, which is exactly the failure an alert system must avoid. The correction is a nowcast, as developed for infectious-disease surveillance (Höhle and an der Heiden, 2014; McGough et al., 2020) and adapted to the Hawkes setting in Proposition 7.3. Estimating F_k needs records that carry both an occurrence and a report time. In the NYPD file the month of occurrence and the record-creation date bound each delay. Open-data files may also be revised after first release, so every analysis will be tied to an archived, checksummed vintage of each file.

2.4 External-event calendars

The shock term $\sum_e \upsilon_k S_e \psi_k(t - \tau_e)$ requires event times and sizes that are fixed independently of the incident data. Two established event datasets, ACLED (Raleigh et al., 2010) and GDELT (Leetaru and Schrodt, 2013), can supply candidate events; their licence terms will be recorded and respected at registration.

Neither dataset settles which events are *salient* for the outcome. The risk is that the analyst picks events after seeing surges, which would bias the shock loadings υ_k upwards and the contagion terms downwards. We therefore fix the event list before outcome data are examined, by a content-neutral procedure (Appendix H.3): event classes defined in advance by type, symmetrically and without reference to the parties involved; a pre-specified, outcome-independent salience index, such as news coverage relative to its own trailing baseline, with a fixed threshold; coders without access to incident data; and a list of times τ_e, classes and sizes S_e that is frozen and time-stamped before incident data for the study period are loaded. Calendar effects such as religious holidays and school terms are recurring and predictable, so they enter the background $\mu_k(t)$, not the shock list.

2.5 Online signals

Access to platform data, for example through the vetted-researcher procedure of Article 40 of the EU Digital Services Act (European Parliament and Council of the European Union, 2022), is not confirmed for the platforms and periods of interest (Appendix G.2). The plan therefore treats online streams as optional, so that the offline analysis does not depend on them.

Classifier-based volume series and their measurement error. An online stream enters the model as a binned count series, which is handled by the INAR approximation of Equation (6.8) or a binned Hawkes likelihood. It is constructed by applying a classifier to a defined corpus (Kornstein Lab for Antisemitism Research, 2026b,c), so it measures the sampled corpus, not platform-wide prevalence, and its error has two rates. Let U_t be the number of posts in bin t, V_t the number that are antisemitic under the reference annotation, and $\widehat{V}_t$ the number flagged. If each post is flagged independently with probability e_1 when antisemitic and e_0 otherwise, then

$$\mathbb{E}\big[\widehat{V}_t \mid U_t, V_t\big] = e_1 V_t + e_0\,(U_t - V_t), \qquad \widetilde{V}_t = \frac{\widehat{V}_t - e_0 U_t}{e_1 - e_0} \quad (e_1 > e_0), \tag{2.2}$$

where $\widetilde{V}_t$ is conditionally unbiased for V_t when e_0 and e_1 are known. The uncorrected series loads on total volume U_t with weight e_0. If total posting volume rises after a salient external event, the uncorrected series rises even when V_t does not, and a common shock can then masquerade as online-to-offline excitation. The error rates are not fixed constants. The detection proposal singles out quotation and criticism of antisemitic expression as hard cases, and if the share of such posts changes after events, e_0 moves with the shocks. The plan therefore estimates (e_0, e_1) on validation samples stratified by period, propagates their uncertainty, and reports corrected and uncorrected analyses (Appendix H.5). Any estimated online-to-offline coefficient $\alpha_{\text{off,on}}$ is interpreted as Granger-type predictive dependence in the sense of Eichler et al. (2017) (Proposition 4.12), not as a structural causal effect.

2.6 How the dark-figure estimates enter: recorded streams are thinned

Recorded incidents are a subset of true incidents. Under the working assumption that each true incident in stream k is recorded independently with probability p_k, the recorded stream is a p_k-thinning of the true one. By Theorem 4.10, contagion estimated from recorded data then targets an attenuated apparent branching ratio, not the true one. The companion paper supplies p_k. Its dark-figure ratio relative to list k, written here $\mathcal{D}_k = N/n_{k+}$, is the ratio of all incidents to those on the list, so p_k is estimated by $1/\widehat{\mathcal{D}}_k$. Its funnel decomposition shows that p_k depends on the source: a city police stream such as the NYPD file is reached through disclosure, reporting to police and bias classification ($p \approx \rho_{\text{vic}}\rho_{\text{pol}}\rho_{\text{cls}}$); the FBI stream adds agency submission ($p \approx \rho_{\text{vic}}\rho_{\text{pol}}\rho_{\text{cls}}\rho_{\text{sub}}$); and an NGO audit has a funnel of its own (Kornstein Lab

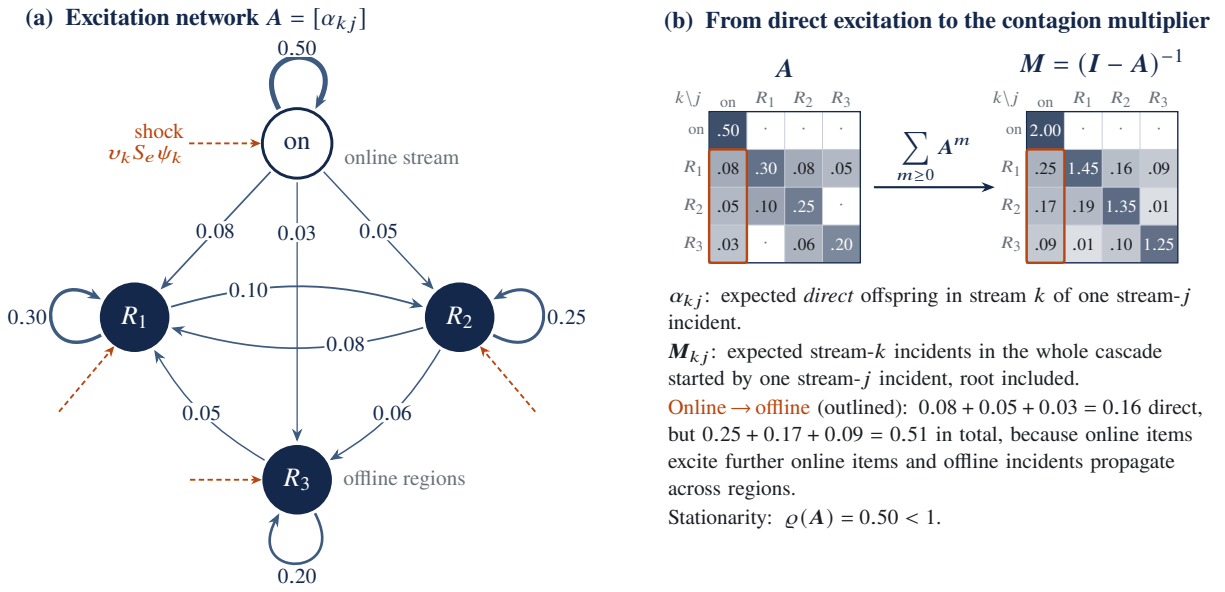


Figure 2. A multivariate excitation network and its contagion multiplier. *(a)* Four streams: an online stream and three offline regions. A directed edge $j \to k$ carries the branching coefficient α_{kj}, the expected number of stream-k incidents directly triggered by one stream-j incident; edge width grows with α_{kj}. Loops are self-excitation. Every stream also responds directly to salient external events, with loading υ_k (dashed, orange). *(b)* The branching matrix $\boldsymbol{A}$ (rows: child stream k; columns: parent stream j) and the multiplier $\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{A})^{-1} = \sum_{m\geq 0} \boldsymbol{A}^m$, which counts incidents along paths of every length (Section 3.2). Shading grows with the entry. The orange outline marks the online-to-offline spillover: direct coefficients in $\boldsymbol{A}$, total cascade effects in $\boldsymbol{M}$. The values are illustrative, chosen for legibility; they are not estimates.

for Antisemitism Research, 2026a). Two cautions follow. First, the companion paper's estimates refer to coarse domains, such as state-by-year cells, while contagion operates on the scale of days within a city. Using them assumes that p_k is constant within a domain, and this assumption is varied in the sensitivity analysis. Second, recording may itself respond to salient events, for instance if heightened attention raises the propensity to report. Recording would then no longer be independent of the history, the recorded surge would overstate the true surge, and Theorem 4.10 would not apply as stated. The plan treats time-varying p_k as a pre-specified sensitivity analysis (Appendix H.5).

3 Self-exciting point processes for incident streams

This section fixes the model class: the likelihood of multivariate point processes, a Hawkes process augmented by exogenous shocks, and its cluster representation and first- and second-order structure. The results are classical; we state them in the form needed later, with proofs in Appendix A.1. Section 3.5 explains what "contagion" means, and does not mean, when the model is applied to antisemitic incidents.

3.1 Marked multivariate point processes

We observe M *streams* of incidents on a window $[0, T]$, indexed by $k \in [M]$. A stream is any class of events that we want to model jointly. For example, offline incidents may be split by region and by offence type (assault, vandalism, harassment, threats). An online channel can also be a stream, with one point for each item that a classifier flags. The points of stream k are $0 < t_1^k < t_2^k < \cdots \leq T$, its counting process is

$N_k(t) = \#\{i : t_i^k \le t\}$, and each point may carry a mark $x_i^k \in \mathbb{X}$ (for example severity, target type or source). We assume that every N_k is simple and that no two streams share a point, almost surely. The history $\mathcal{F}_t$ is the σ-field generated by all streams and marks on $[0,t]$, together with the exogenous events (τ_e, S_e) that have occurred by time t (defined below).

The *conditional intensity* of stream k is the nonnegative, $\mathcal{F}$-predictable process λ_k for which

$$N_k(t) - \Lambda_k(t), \qquad \Lambda_k(t) = \int_0^t \lambda_k(s)\,\mathrm{d}s,$$

is a local martingale. Heuristically, $\lambda_k(t)\,\mathrm{d}t = \mathbb{P}\{\mathrm{d}N_k(t) = 1 \mid \mathcal{F}_{t-}\}$. The process Λ_k is the *compensator*. For a parametric family of intensities $\lambda_k(\cdot;\vartheta)$, the log-likelihood of the observed streams on $[0,T]$ is

$$\ell_T(\vartheta) = \sum_{k=1}^{M} \Big[\sum_{i:\, t_i^k \le T} \log \lambda_k(t_i^k;\vartheta) - \int_0^T \lambda_k(t;\vartheta)\,\mathrm{d}t \Big] \tag{3.1}$$

(Daley and Vere-Jones, 2003, Ch. 7); see Ogata (1978) for its asymptotic theory and Ogata (1988) for its use in self-exciting models. If marks are *unpredictable*, meaning that the mark of a new point is drawn from a law that does not depend on the past, the mark density adds a separate term to (3.1). The compensator also drives the time-rescaling diagnostics of Section 6.

3.2 Hawkes processes with exogenous shocks

Salient external events, such as geopolitical escalations, terror attacks or high-profile public statements, are indexed by e. Event e occurs at time τ_e and has size $S_e \ge 0$. Our working model is the linear multivariate Hawkes process (Hawkes, 1971) augmented by these exogenous shocks:

$$\lambda_k(t) = \mu_k(t) + \sum_{e:\, \tau_e < t} \upsilon_k S_e\, \psi_k(t - \tau_e) + \sum_{j=1}^{M} \int_0^{t-} \varphi_{kj}(t-s)\,\mathrm{d}N_j(s), \qquad \varphi_{kj} = \alpha_{kj}\, g_{kj}. \tag{3.2}$$

Assumption A1 (Model).
(i) The background $\mu_k(t) = \exp(B(t)^\top \theta_k)$ is built from a spline or seasonal basis B and is bounded between two positive constants on $[0,T]$. Calendar effects such as holiday periods belong to μ_k, not to the shocks.

(ii) The shock responses satisfy $\psi_k \ge 0$, $\int \psi_k = 1$ and $\psi_k(u) = 0$ for $u < 0$. The loadings satisfy $\upsilon_k \ge 0$.

(iii) Each g_{kj} is a probability density on $(0,\infty)$, and $\alpha_{kj} \ge 0$.

The constant α_{kj} is the expected number of stream-k incidents directly triggered by one stream-j incident. These constants form the *branching matrix* $\boldsymbol{A} = [\alpha_{kj}]$, and the kernels form the matrix $\boldsymbol{\Phi}(t) = [\varphi_{kj}(t)]$. The *resolvent* is $\boldsymbol{R} = \sum_{m \ge 1} \boldsymbol{\Phi}^{*m}$, where $*$ denotes convolution. When the spectral radius $\varrho(\boldsymbol{A})$ is below one, $\int \boldsymbol{R} = \sum_{m \ge 1} \boldsymbol{A}^m$ is finite and the *contagion multiplier* is

$$\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{A})^{-1} = \boldsymbol{I} + \int_0^\infty \boldsymbol{R}(t)\,\mathrm{d}t = \sum_{m \ge 0} \boldsymbol{A}^m.$$

The workhorse special case uses exponential kernels, $g_{kj}(t) = \beta_{kj} e^{-\beta_{kj} t}$. In the univariate case, $\varphi(t) = n\beta e^{-\beta t}$, the branching ratio is $n = \alpha$ and, when $n < 1$, the stationary mean rate is $\bar{\Lambda} = \mu/(1-n)$. Throughout, β (or β_{kj}) is a decay rate and υ_k a shock loading.

3.3 The cluster representation

Hawkes and Oakes (1974) showed that a self-exciting process is a Poisson cluster process. For (3.2) the construction is as follows.

(a) *Immigrants.* Stream k receives *background immigrants*, a Poisson process with intensity $\mu_k(t)$. Independently, it receives *shock immigrants*, a Poisson process with intensity $\sum_e \upsilon_k S_e \psi_k(t-\tau_e)$.

(b) *Offspring.* Each point of stream j at time s, whether an immigrant or not, independently generates offspring in each stream k. These offspring form a Poisson process with intensity $\varphi_{kj}(t-s)$, $t>s$. Equivalently, there are Poisson(α_{kj}) offspring, placed at independent delays with density g_{kj}.

(c) *Recursion.* Offspring generate offspring in the same way.

The superposition of all generations has intensity (3.2) with respect to its own history and the shock calendar (Hawkes and Oakes, 1974; Laub et al., 2021; Reinhart, 2018). Each incident belongs to exactly one cluster, the family tree of its immigrant ancestor. The expected number of generation-g descendants in stream k of a stream-j incident is $(\boldsymbol{A}^g)_{kj}$. Hence $\boldsymbol{M}_{kj}$ is the expected number of stream-k incidents in a cluster rooted at a stream-j incident, the root included. The latent parent labels drive the EM algorithm of Section 6. The split of incidents into background immigrants, shock immigrants and offspring underlies the surge decomposition of Section 5.

Proposition 3.1 (Stationarity; Brémaud and Massoulié, 1996; Hawkes and Oakes, 1974). *Suppose there are no shocks, $\mu_k(t) \equiv \mu_k > 0$ for every k, and Assumption A1(iii) holds.*

(a) If $\varrho(\boldsymbol{A}) < 1$, there is a stationary point process on $\mathbb{R}$ whose intensity with respect to its own history is $\lambda_k(t) = \mu_k + \sum_j \int_{(-\infty,t)} \varphi_{kj}(t-s)\,\mathrm{d}N_j(s)$. Its law is unique among stationary solutions with finite mean intensity. It is the Poisson cluster process of (a)–(c), and its clusters are almost surely finite. Its mean intensity vector is

$$\bar{\boldsymbol{\Lambda}} = (\boldsymbol{I}-\boldsymbol{A})^{-1}\boldsymbol{\mu} = \boldsymbol{M}\boldsymbol{\mu}.$$

The expected size of a cluster rooted at a stream-j immigrant is $\mathbf{1}^\top \boldsymbol{M}\boldsymbol{e}_j$. The long-run fraction of stream-k incidents that are offspring is $1-\mu_k/\bar{\Lambda}_k = (\boldsymbol{A}\bar{\boldsymbol{\Lambda}})_k/\bar{\Lambda}_k$.

(b) Conversely, if a stationary solution with finite mean intensity exists, then $\varrho(\boldsymbol{A}) < 1$.

(c) In the univariate case, the cluster size $\mathcal{S}$ (immigrant included) has the Borel distribution $\mathbb{P}(\mathcal{S}=k) = e^{-nk}(nk)^{k-1}/k!$, $k \geq 1$, with $\mathbb{E}\mathcal{S} = (1-n)^{-1}$, $\mathbb{E}\mathcal{S}^2 = (1-n)^{-3}$, $\mathbb{E}\mathcal{S}^3 = (1+2n)(1-n)^{-5}$ and $\mathbb{E}\mathcal{S}^4 = (1+8n+6n^2)(1-n)^{-7}$. Moreover $\mathbb{E}e^{u\mathcal{S}} < \infty$ for $u \leq n-1-\log n$, and $n-1-\log n > 0$ for $n \in (0,1)$.

The proof is in Appendix A.1.

Part (b) uses $\mu_k > 0$. Without immigrants, critical processes with finite mean intensity can be stationary under heavy-tailed kernels (Brémaud and Massoulié, 2001). In our setting $\mu_k > 0$ excludes this case.

3.4 Second-order structure

Let $\boldsymbol{N}$ be a stationary M-variate point process with mean vector $\bar{\boldsymbol{\Lambda}}$ and finite second moments. Its covariance structure has two parts. The first is an atom $\mathrm{diag}(\bar{\boldsymbol{\Lambda}})\,\delta_0$ at lag zero, which comes from each point paired with itself. The second is, we assume, a covariance density $\boldsymbol{c}(u) = [c_{kj}(u)]$:

$$\mathrm{Cov}\{\mathrm{d}N_k(t+u), \mathrm{d}N_j(t)\} = \left[\delta_{kj}\bar{\Lambda}_k\,\delta_0(u) + c_{kj}(u)\right]\mathrm{d}u\,\mathrm{d}t.$$

The *Bartlett spectrum* (Bartlett, 1963) is the Fourier transform of this covariance measure:

$$\boldsymbol{f}(\omega) = \frac{1}{2\pi}\Big[\operatorname{diag}(\bar{\boldsymbol{\Lambda}}) + \hat{\boldsymbol{c}}(\omega)\Big], \qquad \hat{\boldsymbol{c}}(\omega) = \int_{\mathbb{R}} e^{-i\omega u}\boldsymbol{c}(u)\,\mathrm{d}u \tag{3.3}$$

(Daley and Vere-Jones, 2003, Ch. 8). A Poisson process has the flat spectrum $\bar{\Lambda}/(2\pi)$. Every second-order statistic of counts is a functional of the pair $(\bar{\boldsymbol{\Lambda}}, \boldsymbol{f})$. This includes the autocovariances of the binned counts $X_t = N((t-1)\Delta, t\Delta]$ for any bin width Δ, the variance-time curve $w \mapsto \operatorname{Var} N(0,w]$, the expected periodogram, and the asymptotic index of dispersion

$$F_\infty = \lim_{w\to\infty} \frac{\operatorname{Var} N(0,w]}{\mathbb{E}N(0,w]} = \frac{2\pi f(0)}{\bar{\Lambda}}.$$

Proposition 3.2 (Bartlett spectrum of a Hawkes process; Hawkes, 1971). *Under the conditions of Proposition 3.1(a), the stationary Hawkes process has finite moments of all orders, and its Bartlett spectrum is*

$$\boldsymbol{f}(\omega) = \frac{1}{2\pi}\big(\boldsymbol{I} - \hat{\boldsymbol{\Phi}}(\omega)\big)^{-1}\operatorname{diag}(\bar{\boldsymbol{\Lambda}})\,\big(\boldsymbol{I} - \hat{\boldsymbol{\Phi}}(\omega)\big)^{-*}, \qquad \hat{\boldsymbol{\Phi}}(\omega) = \int_0^\infty e^{-i\omega t}\boldsymbol{\Phi}(t)\,\mathrm{d}t, \tag{3.4}$$

where $\boldsymbol{B}^{-*} = (\boldsymbol{B}^{-1})^*$ denotes the inverse of the conjugate transpose. In the univariate case, the covariance density is $c(u) = \bar{\Lambda}\big[R(|u|) + \int_0^\infty R(v)R(v+|u|)\,\mathrm{d}v\big]$ and $F_\infty = (1-n)^{-2}$, whatever the kernel. For the exponential kernel,

$$f_H(\omega) = \frac{\bar{\Lambda}}{2\pi}\Big[1 + \frac{\beta^2 n(2-n)}{\beta^2(1-n)^2 + \omega^2}\Big], \qquad c(u) = \frac{\bar{\Lambda}\,\beta\,n(2-n)}{2(1-n)}\,e^{-\beta(1-n)|u|}. \tag{3.5}$$

Here $n(2-n) = 1-(1-n)^2$.

A proof sketch is in Appendix A.1.

Two features of (3.5) matter later. First, the correlation time of the counts is $1/\{\beta(1-n)\}$, which is longer than the kernel time scale $1/\beta$: cascades lengthen the memory. Second, the excess dispersion is $F_\infty - 1 = n(2-n)/(1-n)^2$, which grows without bound as $n \uparrow 1$.

3.5 Interpretation for antisemitic incidents

Offline streams may be incidents recorded by police or monitoring organisations, split by region and by offence type. An online stream may be the volume of items that a classifier labels antisemitic on one platform, a *measured* quantity whose errors matter for inference (Remark 4.13). A positive α_{kj} says that, under the model, a stream-j incident raises the short-run probability of further stream-k incidents. Imitation, emboldening (visible incidents shifting perceived norms about what is tolerated), diffusion through online communities, coordinated campaigns or repeat offenders, and repeat or near-repeat targeting of the same location (Mohler et al., 2011) could all produce this; each is a *hypothesis* that the model does not by itself verify. Cross-excitation from an online to an offline stream is the formal counterpart of the hypothesis that online hostility precedes offline incidents. Three limits apply.

- The parameters describe predictive dependence in a probability model. "Contagion" is used in its technical sense of self-excitation; compare the classical distinction between true and apparent contagion (Feller, 1943). It attributes no moral responsibility to victims, to Jewish communities, or to anyone connected with an earlier incident. Responsibility for an act rests with its perpetrator.

- A positive α_{kj} is not proof of a causal mechanism. Within a single stream, unobserved shocks can mimic contagion exactly at second order; under-recording attenuates contagion; and common drivers can induce online-to-offline Granger causality without any direct link (Section 4).
- Shocks are *salient external events* whose timing is taken from an external calendar. Including an event, and estimating its loading, is a statement about incident dynamics, not a judgement about the event, its actors or its merits.

4 Identification: contagion, common shocks and under-reporting

Incidents cluster in time. The substantive question is *why*. Clustering may reflect a salient external event whose effect lasts for days (a shock with an extended response). It may reflect incidents triggering further incidents (contagion). Or it may reflect both. Inference about contagion must also cope with two features of incident data: many shocks are never recorded in any event calendar, and many incidents are never recorded at all (the dark figure of Kornstein Lab for Antisemitism Research, 2026a). This section gives four sets of results.

(1) Without a shock calendar, for a single stream or for streams that do not excite one another, and for exponential or completely monotone kernels, the autocorrelation of counts cannot distinguish contagion from clustered exogenous shocks (Theorem 4.2 and Proposition 4.3). Delayed kernels leave a second-order trace (Remark 4.4). So does cross-excitation whenever the excess spectral matrix fails to be positive semidefinite at some frequency, as it does for excite-only streams and when cross-excitation dominates self-excitation (Proposition 4.5 and Example 4.6); contagion *between* streams is then identified at second order. Omitting shocks creates apparent contagion (Theorem 4.7).

(2) Identification is restored by a recorded shock calendar, by higher-order information, or by provenance links (Corollary 4.9).

(3) Independent under-recording attenuates apparent contagion in closed form, which permits a dark-figure correction (Theorem 4.10, Corollary 4.11, and Proposition A.6).

(4) Online-to-offline Granger non-causality is equivalent to a zero cross-excitation kernel, with caveats (Proposition 4.12).

The tension between "true" and "apparent" contagion is old. Feller (1943) showed that the negative binomial law arises both from a contagion urn and from a Poisson law with a heterogeneous rate. Theorem 4.2 is a dynamic, second-order analogue for self-exciting processes. Full proofs are in Appendix A.

Conventions. $H(\mu, n, \beta)$ denotes the stationary univariate Hawkes process with kernel $\varphi(t) = n\beta e^{-\beta t}$, $n \in (0, 1)$, and mean $\bar{\Lambda} = \mu/(1 - n)$ (Proposition 3.1). Two stationary point processes are *second-order equivalent* if they have the same mean and the same Bartlett spectrum. $\mathbb{P}_X$ denotes the law of a process X, and $F_\infty = 2\pi f(0)/\bar{\Lambda}$ is the asymptotic index of dispersion (Section 3.4).

4.1 Second-order observational equivalence

Unrecorded shocks are modelled as random.

Definition 4.1 (Shot-noise Cox process). Let $\nu \geq 0$ and $r, a, \gamma > 0$. Let $\{c_\ell\}$ be a homogeneous Poisson process on $\mathbb{R}$ with rate r; these are the times of unrecorded shocks. Define

$$\xi(t) = \nu + \sum_{\ell:\, c_\ell \leq t} a\, e^{-\gamma(t-c_\ell)}.$$

Conditionally on ξ, let N be a Poisson process with intensity ξ. We write $N \sim C(\nu, r, a, \gamma)$ (Cox, 1955; Møller, 2003).

The process C has no contagion: given the shocks, incidents occur independently. Each shock produces Poisson(m) incidents, $m = a/\gamma$, at independent Exp(γ) delays, so C is also a Neyman–Scott cluster process with mean $\nu + rm$. Any stationary Cox process directed by ξ has Bartlett spectrum $(2\pi)^{-1}[\bar{\Lambda} + \hat{c}_\xi(\omega)]$, where c_ξ is the covariance function of ξ and $\hat{c}_\xi \geq 0$ (Lemma A.1). By Campbell's theorem, $c_\xi(u) = \{ra^2/(2\gamma)\}e^{-\gamma|u|}$ for C, so

$$f_C(\omega) = \frac{1}{2\pi}\left[\nu + \frac{ra}{\gamma} + \frac{ra^2}{\gamma^2 + \omega^2}\right]. \tag{4.1}$$

Theorem 4.2 (Second-order observational equivalence). *Let $H = H(\mu, n, \beta)$ and $C = C(\nu, r, a, \gamma)$.*

(a) H and C are second-order equivalent if and only if

$$\gamma = \beta(1-n), \qquad \nu + \frac{ra}{\gamma} = \bar{\Lambda}, \qquad ra^2 = \bar{\Lambda}\beta^2 n(2-n). \tag{4.2}$$

(b) For given H, the solutions of (4.2) form a one-parameter family, indexed by the shock amplitude $a \in [a_{\min}, \infty)$ with $a_{\min} = \beta n(2-n)/(1-n)$:

$$\gamma = \beta(1-n), \qquad r = \frac{\bar{\Lambda}\beta^2 n(2-n)}{a^2}, \qquad \nu = \bar{\Lambda}\left(1 - \frac{a_{\min}}{a}\right).$$

Equivalently, the mean number of incidents per shock, $m = a/\gamma$, satisfies $m \geq m_{\min} = n(2-n)/(1-n)^2 = F_\infty - 1$, and $rm^2 = \bar{\Lambda}(F_\infty - 1)$. The lower bound $a = a_{\min}$ gives $\nu = 0$: every incident is then shock-driven.

(c) For any matched pair and any bin width $\Delta > 0$, the binned counts of H and of C have the same mean and autocovariance function. Hence they also share their variance-time curves, index of dispersion and expected periodograms. As $T \to \infty$, every continuous function of finitely many empirical means and autocovariances of binned counts has the same probability limit under H and under C.

The extra degree of freedom a is invisible at second order. It trades off frequent, small shocks ($a \downarrow a_{\min}$) against rare, large ones ($a \to \infty$, with $\nu \to \bar{\Lambda}$). The shock response decays at rate $\beta(1-n)$, the inverse of the *cluster* time scale of H, rather than at the kernel rate β. Thus a Hawkes fit with branching ratio n is matched, in its autocorrelations, by a world without contagion in which unrecorded shocks each trigger on average at least $(1-n)^{-2} - 1$ incidents. The full laws of H and C do differ (Corollary 4.9(ii)).

General kernels. The exponential kernel is not special. Completely monotone kernels have the form $g(t) = \int_{(0,\infty)} e^{-bt}\sigma(\mathrm{d}b)$ for a positive measure σ. They include finite mixtures of exponentials and the Lomax (Omori-type) kernels $g(t) = (\zeta/\varsigma)(1 + t/\varsigma)^{-\zeta-1}$ with $\zeta, \varsigma > 0$.

Proposition 4.3 (Completely monotone kernels). *Let H be the stationary Hawkes process with background μ and kernel $\varphi = ng$, where $n \in (0,1)$ and g is a completely monotone probability density on $(0,\infty)$ with $g(0+) < \infty$.*

(a) The resolvent $R = \sum_{j\geq 1} \varphi^{*j}$ is completely monotone. The covariance density of H has the form $c_H(u) = \int e^{-b|u|}\,\mathcal{K}(\mathrm{d}b)$ for a positive measure $\mathcal{K}$ with $Z := \int \mathcal{K}(\mathrm{d}b)/b = \bar{\Lambda}(F_\infty - 1)/2$. If g is a mixture of K exponentials, then $\mathcal{K}$ has at most K atoms.

(b) Fix $m \geq F_\infty - 1 = n(2-n)/(1-n)^2$. Let $\{c_\ell\}$ be Poisson with rate $r = 2Z/m^2$, and give the shocks independent decay rates Γ_ℓ with law $\pi(\mathrm{d}b) = \mathcal{K}(\mathrm{d}b)/(bZ)$. The Cox process directed by

$$\xi(t) = \nu + \sum_{\ell:\, c_\ell \leq t} m\,\Gamma_\ell\, e^{-\Gamma_\ell (t - c_\ell)}, \qquad \nu = \bar{\Lambda} - rm \ \ (\geq 0),$$

is second-order equivalent to H. Each shock produces Poisson(m) *incidents with an exponential response at a random rate, and there is no contagion.*

(c) In particular, $f_H(\omega) \geq \bar{\Lambda}/(2\pi)$ for every ω.

Remark 4.4 (Delayed excitation leaves a second-order trace). By Lemma A.1, every Cox process satisfies $f \geq \bar{\Lambda}/(2\pi)$. Since $2\pi f_H(\omega)/\bar{\Lambda} = |1 - \hat{\varphi}(\omega)|^{-2}$, a Hawkes process with $2\,\mathrm{Re}\,\hat{\varphi}(\omega) < |\hat{\varphi}(\omega)|^2$ at some ω is second-order distinguishable from *every* Cox process. This happens whenever $\mathrm{Re}\,\hat{g}$ becomes negative, as it does for delayed or hump-shaped kernels. An example is the gamma kernel $g(t) \propto t^{k-1}e^{-\beta t}$ with $k > 1$, for which $\hat{g}(\omega) = (1 + i\omega/\beta)^{-k}$. For $k = 2$ a direct calculation gives

$$\min_{\omega} 2\pi f_H(\omega)/\bar{\Lambda} = 1 - n/4,$$

attained at $\omega = \beta\sqrt{3-n}$. The dip lies at frequencies of the order of the inverse delay, so it is easily lost when counts are binned coarsely.

Several streams. The equivalence results so far are proved for a single stream. If $\boldsymbol{\Phi}$ is diagonal, the streams are independent Hawkes processes, and Theorem 4.2 and Proposition 4.3 apply to each stream with its own shock process. These are the only multivariate cases in which we prove equivalence. Cross-excitation is different, because a Cox process can reproduce only an excess spectrum that is positive semidefinite at every frequency. Write $\boldsymbol{E}(\omega) = 2\pi \boldsymbol{f}(\omega) - \mathrm{diag}(\bar{\boldsymbol{\Lambda}})$ for the *excess spectral matrix*.

Proposition 4.5 (Multivariate streams). *(a) For every stationary M-variate Cox process with square-integrable driver and integrable covariance, $\boldsymbol{E}(\omega) = \hat{\boldsymbol{c}}_{\xi}(\omega)$ is Hermitian and positive semidefinite for every ω.*

(b) For the Hawkes process of Proposition 3.1, $\boldsymbol{E}_H(\omega) = \boldsymbol{G}(\omega)\,\mathrm{diag}(\bar{\boldsymbol{\Lambda}})\boldsymbol{G}(\omega)^* - \mathrm{diag}(\bar{\boldsymbol{\Lambda}})$ with $\boldsymbol{G} = (\boldsymbol{I} - \hat{\boldsymbol{\Phi}})^{-1}$. Suppose some stream k receives no excitation ($\varphi_{kl} \equiv 0$ for all l) but excites another stream j ($\alpha_{jk} > 0$). Then $\boldsymbol{E}_H(\omega)$ is not positive semidefinite for ω near 0, and no M-variate Cox process, with any common or idiosyncratic shocks, is second-order equivalent to it.

(c) Let $\{c_\ell\}$ be a homogeneous Poisson process of rate r (common unrecorded shocks), with i.i.d. random loadings $\boldsymbol{a}_\ell \in [0,\infty)^M$, independent of $\{c_\ell\}$ and with $\mathbb{E}|\boldsymbol{a}_\ell|^2 < \infty$. Let the response shapes $s_k \geq 0$ be square-integrable and stream-specific, and let $\boldsymbol{a}$ denote a generic loading:

$$\xi_k(t) = \nu_k + \sum_{\ell} a_{\ell k}\, s_k(t - c_\ell).$$

The cross-covariance is $c_{kj}(u) = r\,\mathbb{E}[a_k a_j]\int s_k(t+u)s_j(t)\,\mathrm{d}t$, where $\boldsymbol{a} = (a_k)$. It is symmetric in u when $s_k = s_j$, but asymmetric in general. Lead–lag cross-correlation between two streams is therefore not, by itself, evidence of directional excitation.

Part (a) gives a necessary condition for second-order equivalence with any Cox process: $\boldsymbol{E}_H(\omega) \succeq 0$ for every ω. Whether it is also sufficient when $\boldsymbol{\Phi}$ is not diagonal is open. Part (b) shows that it fails for excite-only streams. The next example shows that it also fails, at every frequency, when cross-excitation dominates self-excitation.

Example 4.6 (Cross-excitation that dominates self-excitation). Let $M = 2$, $\mu_1 = \mu_2$, exponential kernels with a common decay β, and $\boldsymbol{A} = \left(\begin{smallmatrix} s & c \\ c & s \end{smallmatrix}\right)$ with $s, c \geq 0$ and $s + c < 1$. Then $\bar{\Lambda}_1 = \bar{\Lambda}_2 = \bar{\Lambda}$ and $\hat{\boldsymbol{\Phi}}(\omega) = h(\omega)\boldsymbol{A}$, with $h(\omega) = \beta/(\beta + i\omega)$. The matrix $\boldsymbol{E}_H(\omega)$ has the eigenvectors $(1, 1)$ and $(1, -1)$ of $\boldsymbol{A}$, with eigenvalues $\bar{\Lambda}\{|1 - \epsilon h(\omega)|^{-2} - 1\}$ for $\epsilon = s + c$ and $\epsilon = s - c$. Since $\mathrm{Re}\, h = |h|^2$, we have $|1 - \epsilon h|^2 = 1 - |h|^2\epsilon(2 - \epsilon)$, so the eigenvalue is nonnegative for every ω if and only if $0 \leq \epsilon \leq 2$. Hence $\boldsymbol{E}_H(\omega) \succeq 0$ for all ω if and only if $c \leq s$. If $c > s$, the eigenvalue for $(1, -1)$ is negative at every ω. At $\omega = 0$ it equals $\bar{\Lambda}\{(1 + c - s)^{-2} - 1\}$, which for $\boldsymbol{A} = \left(\begin{smallmatrix} 0 & a \\ a & 0 \end{smallmatrix}\right)$ is $\bar{\Lambda}\{(1 + a)^{-2} - 1\} < 0$. Over long windows,

$$\frac{\mathrm{Var}\{N_1(0,T] - N_2(0,T]\}}{\mathbb{E}\{N_1(0,T] + N_2(0,T]\}} \longrightarrow (1 + c - s)^{-2} < 1.$$

Every Cox process has $\mathrm{Var}(N_1 - N_2) \geq \mathbb{E}(N_1 + N_2)$, because given its driver the two counts are independent Poisson variables. A direct simulation with $a = 0.5$ and $\beta = \mu_k = 1$, over 3,996 windows of length 50, gives a ratio of 0.450, against the limit $(1.5)^{-2} = 0.444$.

The example has a positive reading. Mutual excitation between two streams makes their counts move together more tightly than independent Poisson noise allows, and no configuration of common or idiosyncratic unrecorded shocks can reproduce that. Contagion between streams is therefore often identifiable at second order, even where contagion within a stream is not.

4.2 Spurious contagion from unrecorded shocks

Now reverse the direction. Suppose that the truth has no contagion and that the analyst fits a Hawkes process with a constant background.

Theorem 4.7 (Spurious contagion). *Let N be a stationary Cox process directed by a stationary $\xi \geq 0$ with mean $\bar{\Lambda} > 0$, $\mathbb{E}\xi(0)^2 < \infty$ and integrable covariance $c_\xi \not\equiv 0$. Define the* burstiness index

$$\kappa = \frac{1}{\bar{\Lambda}}\int_{\mathbb{R}} c_\xi(u)\,\mathrm{d}u = \frac{\hat{c}_\xi(0)}{\bar{\Lambda}} \geq 0, \qquad \text{so that } F_\infty = 1 + \kappa.$$

(a) Exact projection. *If $c_\xi(u) = \sigma^2 e^{-\gamma|u|}$, then exactly one exponential Hawkes process is second-order equivalent to N. Its parameters are*

$$n^* = 1 - (1 + \kappa)^{-1/2}, \qquad \beta^* = \gamma(1 + \kappa)^{1/2}, \qquad \mu^* = \bar{\Lambda}(1 + \kappa)^{-1/2}, \tag{4.3}$$

where $\kappa = 2\sigma^2/(\bar{\Lambda}\gamma)$. For $C(\nu, r, a, \gamma)$, $\kappa = ra^2/(\bar{\Lambda}\gamma^2)$. The second-order structure of a two-state regime-switching rate is also exponential: if ξ switches between λ_0 and λ_1 at rates q_{01} and q_{10}, with $\pi_1 = q_{01}/(q_{01} + q_{10})$, then $\kappa = 2(\lambda_1 - \lambda_0)^2\pi_0\pi_1/\{\bar{\Lambda}(q_{01} + q_{10})\}$.

(b) Moment-based estimators. *In the setting of (a), let N be ergodic. Let $\hat{\vartheta}_T = G(\hat{\boldsymbol{m}}_T)$, where $\hat{\boldsymbol{m}}_T$ collects the empirical mean and finitely many empirical autocovariances of the binned counts on*

$[0,T]$, *and* G *is continuous. If* $\hat{\vartheta}_T$ *is consistent for* (μ, n, β) *under every stationary exponential Hawkes process, then* $\hat{\vartheta}_T \to (\mu^*, n^*, \beta^*)$ *in probability under* N.

(c) Any covariance shape. *The dispersion functional* $n_F = 1 - F_\infty^{-1/2}$ *equals* n *for every stationary Hawkes process, whatever its kernel. Applied to* N, *it equals* $n^* = 1 - (1+\kappa)^{-1/2}$, *which is positive whenever* $\kappa > 0$.

(d) Likelihood. *Now let* N *be any stationary point process with mean* $\bar{\Lambda}$, $\mathbb{E}N(0,1]^2 < \infty$ *and covariance density* c; *for a Cox process,* $c = c_\xi$. *Fit the family* $\lambda_\vartheta(t) = \mu + n\int_{(-\infty,t)} g_\beta(t-s)\,\mathrm{d}N(s)$ *with* $\vartheta = (\mu, n, \beta) \in (0,\infty) \times [0,1) \times \mathcal{B}$, *where each* g_β *is a probability density on* $(0,\infty)$ *with* $\int_0^\infty g_\beta |c| < \infty$. *Let*

$$L(\vartheta) = \mathbb{E}\int_0^1 \log \lambda_\vartheta(t)\,\mathrm{d}N(t) - \mathbb{E}\lambda_\vartheta(0)$$

be the expected log-likelihood per unit time. If $\int_0^\infty g_{\beta_0}(u)c(u)\,\mathrm{d}u > 0$ *for some* $\beta_0 \in \mathcal{B}$, *then*

$$\sup_{\vartheta:\, n>0} L(\vartheta) > \sup_{\vartheta:\, n=0} L(\vartheta) = \bar{\Lambda}\log\bar{\Lambda} - \bar{\Lambda}.$$

Hence every maximiser of L, *that is, every pseudo-true value of the misspecified maximum likelihood estimator, has* $n > 0$. *For* $C(\nu, r, a, \gamma)$ *with exponential* g_β, $\int g_\beta c = ra^2\beta/\{2\gamma(\beta+\gamma)\} > 0$ *for every* β.

Remark 4.8 (Reading Theorem 4.7). (i) *Interpreting* κ. For C, $\kappa = s\,m$, where $s = rm/\bar{\Lambda}$ is the share of incidents triggered by unrecorded shocks and m is the mean number of incidents per shock, so $n^* = 1 - (1 + sm)^{-1/2}$; $s = 0.3$ and $m = 10$ give $n^* = 1/2$. Rare, large shocks and slow regime switches give large κ, and so apparent near-criticality (Filimonov and Sornette, 2015).

(ii) *What is and is not claimed.* Parts (a)–(c) concern second-order estimators, such as variance-time and autocovariance fits. Spectral (Whittle-type) estimators (Cheysson and Lang, 2022) are not covered by (b) as stated, but their population contrast is minimised only at ϑ^*, so under the usual M-estimation conditions, which we do not verify, they also converge to it. Part (d) proves only that the likelihood pseudo-true value has $n > 0$. That value depends on the law of N beyond second order, so it need not equal n^*. Nothing forces the members of the matched family of Theorem 4.2, which all share n^*, to share it. The simulation study compares maximum likelihood fits with n^*: they are close to n^* for frequent small shocks and below it for rare large shocks and slow regime switches (Section 8.3). For rare large shocks the gap persists at $T = 10{,}000$, so it is not finite-sample bias. Under the usual conditions for misspecified likelihoods (Ogata, 1978; White, 1982), the estimator converges to the maximisers of L; we do not verify those conditions here.

(iii) *Consequence.* Salient events that are missing from the calendar, or whose response is mis-specified, are absorbed into $\boldsymbol{A}$. Contagion estimates should therefore be reported together with the shock calendar, and with their sensitivity to omitting events from it.

4.3 Restoring identification

Three kinds of information outside the second-order structure break the equivalence: a recorded calendar of shocks, higher-order moments, and links between incidents.

Corollary 4.9 (Identification restored). *(i)* Recorded shocks. *Consider model (3.2) on $[0,T]$, started from the empty history, under Assumption A1, with the calendar (τ_e, S_e) observed. Let $m_k(t) = \mu_k(t) + \sum_e \upsilon_k S_e \psi_k(t-\tau_e)$ be the immigrant intensity. The law of $\boldsymbol{N}$ on $[0,T]$ determines m_k almost everywhere on $[0,T]$, and each φ_{kj} almost everywhere on $(0,T)$. Two further conditions complete identification of all parameters:*

- *each kernel family is determined by its restriction to $(0,T)$, as exponential, gamma and power-law families are;*
- *the map $(\theta_k, \upsilon_k, \psi_k) \mapsto m_k$ is injective. The data on $[\tau_1, \tau_2)$ identify only the product $\upsilon_k S_1 \psi_k$ there. Injectivity therefore holds, for instance, when $\tau_1 > 0$, $S_1 > 0$, μ_k is constant (or analytic in t with linearly independent basis functions), and the* scaled *family $\{c\,\psi : c > 0, \psi \in \mathfrak{S}_k\}$ is determined by its restriction to an interval. An example is a family $\mathfrak{S}_k$ of functions that are analytic on $(0,\infty)$; then $\int \psi_k = 1$ pins down υ_k. A family such as $\psi_s = s^{-1}\mathbf{1}_{[0,s]}$ fails this condition.*

Spline backgrounds are not analytic, so for the working model of Assumption A1(i) separation rests, informally, on the shock responses not being representable in the spline space, for example through a jump at τ_e. In particular, when the g_{kj} do not vanish on $(0,T)$, $\boldsymbol{A} = \boldsymbol{0}$ is distinguishable from every $\boldsymbol{A} \neq \boldsymbol{0}$: clustering explained by recorded shocks is not confounded with contagion.

(ii) Higher-order information. *Let $C = C(\nu, r, a, \gamma)$ be arbitrary. Then $\mathbb{P}_H \neq \mathbb{P}_C$. Counts over long windows already reveal this. The limits $\chi_j = \lim_{T\to\infty} T^{-1} \operatorname{cum}_j N(0,T]$ are $\mu\mathbb{E}\mathcal{S}^j$ ($\mathcal{S} \sim$ Borel(n)) for H and $\nu + r\mathbb{E}[\text{Poisson}(m)^j]$ for C. Matching χ_1, χ_2, χ_3 determines (ν, r, m) uniquely, with γ free. With $\gamma = \beta(1-n)$, this choice is a member of the family of Theorem 4.2. For every such C,*

$$\chi_4^H - \chi_4^C = \frac{\mu\, n^3\, Q(1-n)}{(1-n)^7(2-n)} > 0, \qquad Q(u) = 2u^4 + 8u^3 + 15u^2 + 16u + 6,$$

so no shot-noise Cox process matches the first four cumulant rates of H.

(iii) Provenance. *Suppose each true parent–offspring pair is recorded as a link independently with probability $q \in (0,1]$, and there are no false links. Links might come, for example, from perpetrator-confirmed imitation, a shared perpetrator or a shared campaign. In the stationary multivariate model, the expected number of links per unit time from stream j to stream k is $q\,\alpha_{kj}\bar{\Lambda}_j$. Hence the support of $\boldsymbol{A}$ (the contagion graph) is identified even when q is unknown, and $\boldsymbol{A}$ is identified when q is known.*

Part (i) is the basis of the shock-augmented likelihood in Section 6. Once shocks are recorded, the exogenous explanation becomes the nested sub-model $\boldsymbol{A} = \boldsymbol{0}$, and the boundary likelihood-ratio test of $\alpha_{kj} = 0$ applies. The result is relative to the calendar: unrecorded shocks remain subject to Theorem 4.7. Hawkes models with exogenous news events in the intensity are used in finance for the same reason (Rambaldi et al., 2015). Part (ii) is a statement of principle. Fourth-order cumulants are expensive to estimate, and the practical vehicle for higher-order information is the full likelihood.

Intensity jumps do not settle the question either. With respect to the observed history, the intensity of a Cox process also jumps at each event, because an event is evidence of an unobserved shock. What separates the models is the *form* of the observable intensity: among all laws second-order equivalent to a Hawkes process, only the Hawkes law has an observable intensity that is affine in the past (Remark A.4 in Appendix A.2).

4.4 Under-reporting masks contagion

Kornstein Lab for Antisemitism Research (2026a) model recorded incidents as an independent thinning of all incidents, with recording probability p equal to the reciprocal of the dark-figure ratio.

Theorem 4.10 (Under-reporting masks contagion). *Let $H = H(\mu, n, \beta)$, and let N^p be its independent p-thinning: each incident is recorded independently, with probability $p \in (0, 1]$, independently of H.*

(a) N^p is stationary with mean $p\bar{\Lambda}$ and Bartlett spectrum

$$f_p(\omega) = \frac{p\bar{\Lambda}}{2\pi}\left[1 + \frac{p\,\beta^2 n(2-n)}{\beta^2(1-n)^2 + \omega^2}\right].$$

(b) N^p is second-order equivalent to exactly one exponential Hawkes process, $H(\mu', n', \beta')$, with

$$\beta'(1-n') = \beta(1-n), \qquad \beta'^2 = \beta^2(1-n)^2 + p\,\beta^2 n(2-n), \qquad \mu' = p\bar{\Lambda}(1-n'),$$

$$n' = h_p(n) = 1 - \frac{1-n}{\sqrt{(1-n)^2 + p\,n(2-n)}}. \tag{4.4}$$

Equivalently, $F'_\infty - 1 = p\,(F_\infty - 1)$: thinning scales the excess dispersion by p. We write n' for the observed (apparent) branching ratio.

(c) For $p \in (0, 1)$, h_p is a strictly increasing bijection of $[0, 1)$, and $pn < h_p(n) < n$ for $n \in (0, 1)$. Moreover $h_p(n) = pn + O(n^2)$ as $n \downarrow 0$, $h_p(n) \to 1$ as $n \uparrow 1$, $h_p(n) \to 0$ as $p \downarrow 0$, $h_1(n) = n$, and $\beta(1-n) \le \beta' \le \beta$. The inverse is

$$h_p^{-1}(n') = 1 - \left[1 + \frac{(1-n')^{-2} - 1}{p}\right]^{-1/2}. \tag{4.5}$$

(d) For $p < 1$, N^p is not a Hawkes process, for any kernel and any constant background. Thus n' is an apparent *branching ratio in a precise sense. It is the branching ratio of the unique exponential Hawkes process that is second-order equivalent to N^p. It is therefore the probability limit under N^p of every moment-based estimator as in Theorem 4.7(b). We do not identify the pseudo-true value of the Hawkes maximum likelihood estimator under N^p, which need not equal n'.*

Attenuation is never worse than the naive factor p, and it is weakest near criticality. Recording reduces each individual offspring count by the factor p, but the dispersion of whole clusters, which is what second-order statistics see, is reduced less.

Corollary 4.11 (Dark-figure correction). *Let $\hat{n}'$ estimate n' from recorded incidents. Let $\hat{p}$ estimate the recording probability from the companion paper (Kornstein Lab for Antisemitism Research, 2026a) ($p = 1/\mathcal{D}$, where $\mathcal{D}$ is the dark-figure ratio), and let $\hat{p}$ be independent of $\hat{n}'$. Set $\hat{n} = h_{\hat{p}}^{-1}(\hat{n}')$ using (4.5). If $\hat{n}' \to n'$ and $\hat{p} \to p$, then $\hat{n} \to n$. By the delta method,*

$$\mathrm{Var}(\hat{n}) \approx \left[\frac{(1-n)^3}{p\,(1-n')^3}\right]^2 \mathrm{Var}(\hat{n}') + \left[\frac{n(1-n)(2-n)}{2p}\right]^2 \mathrm{Var}(\hat{p}), \qquad \mathrm{Var}(\hat{p}) \approx \frac{\mathrm{Var}(\hat{\mathcal{D}})}{\mathcal{D}^4}. \tag{4.6}$$

The second term in (4.6) does not shrink as T grows: uncertainty about coverage puts a floor under the precision of any dark-figure-corrected contagion estimate. A Bayesian alternative passes posterior draws of p from the companion paper's model through h_p^{-1}. If $\hat{n}'$ is the Hawkes maximum likelihood estimator, its limit is the pseudo-true value under N^p, so the correction is exact only to the extent that this value coincides

with n' (Theorem 4.10(d)). In the simulation study the mean maximum likelihood estimate is within 0.0104 of n' for every combination of n and p that was studied (Section 8.4).

Heterogeneous recording. Severe incidents may be both more likely to be recorded and more likely to trigger further incidents. Suppose each incident carries an *unpredictable* mark X with law π, expected direct offspring n_X and recording probability p_X, with $\bar{n} = \mathbb{E}n_X < 1$, $\bar{p} = \mathbb{E}p_X$ and an exponential kernel. The recorded process is then second-order equivalent to an exponential Hawkes process, and the homogeneous correction of Corollary 4.11 with $p = \bar{p}$ returns n_{naive} with

$$(1 - n_{\text{naive}})^{-2} = (1 - \bar{n})^{-2}\Big[1 + \frac{2(1 - \bar{n})\,\mathrm{Cov}(p_X, n_X)}{\bar{p}} + \mathrm{Var}(n_X)\Big]$$

(Proposition A.6, stated and proved in Appendix A.9). The correction therefore overstates $\bar{n}$ whenever the bracket exceeds one, in particular when more productive incidents are recorded more often; heterogeneous productivity alone already inflates the apparent branching ratio. The remedy is to model marks explicitly, with mark-specific productivities, and to use category-specific coverage when it is available. With stream-specific recording probabilities p_k, the excess spectral matrix becomes $\mathrm{diag}(\boldsymbol{p})\,\boldsymbol{E}_H\,\mathrm{diag}(\boldsymbol{p})$ (Lemma A.1(b)), which need not be of multivariate exponential-Hawkes form; because cascades pass through unrecorded incidents, the univariate correction cannot simply be applied stream by stream, and the multivariate case calls for an explicit thinning model.

4.5 Granger causality between online and offline streams

Let $V = [M]$. Stream j does *not Granger-cause* stream k, relative to V, if the intensity λ_k of N_k with respect to $\mathcal{F}^V$ has a version that is predictable with respect to $\mathcal{F}^{V\setminus\{j\}}$, the history generated by the shock calendar and the other streams. This is local independence in the sense of Didelez (2008), the continuous-time analogue of Granger (1969).

> **Proposition 4.12** (Granger non-causality; Eichler et al., 2017). *Under* (3.2) *and Assumption A1, with the process started from the empty history on* $[0, T]$*, stream* j *does not Granger-cause stream* k *if and only if* $\varphi_{kj} = 0$ *almost everywhere on* $(0, T)$*. For exponential kernels, this holds if and only if* $\alpha_{kj} = 0$*.*

Eichler et al. (2017) prove the corresponding equivalence for stationary Hawkes processes and build graphical models on it. Appendix A gives a short proof for the shock-augmented model. With $k = \text{off}$ and $j = \text{on}$, the hypothesis $\alpha_{\text{off,on}} = 0$ states that online volume carries no predictive information about offline incidents beyond that in the other modelled streams and the calendar. This is the hypothesis tested by the boundary-corrected likelihood-ratio test of Section 6.

Remark 4.13 (Caveats for online-to-offline inference).

(i) *Common shocks.* Granger causality is relative to the information set. A salient event that is missing from the calendar, and that affects online volume faster than offline incidents, induces online-to-offline Granger causality in the modelled system even if $\alpha_{\text{off,on}} = 0$ (Proposition 4.5(c)).

(ii) *Classifier error.* If, under the null, the offline intensity depends only on the offline history and the calendar, classifier errors that are independent of the offline stream cannot *create* online-to-offline causality. Under the alternative they dilute the signal, attenuating the estimated cross-excitation and the power of the test.

(iii) *Under-recording with reverse excitation.* If offline incidents are recorded with $p < 1$, self-excite and trigger online volume, the intensity of recorded offline incidents given the observed history $\mathcal{G}$,

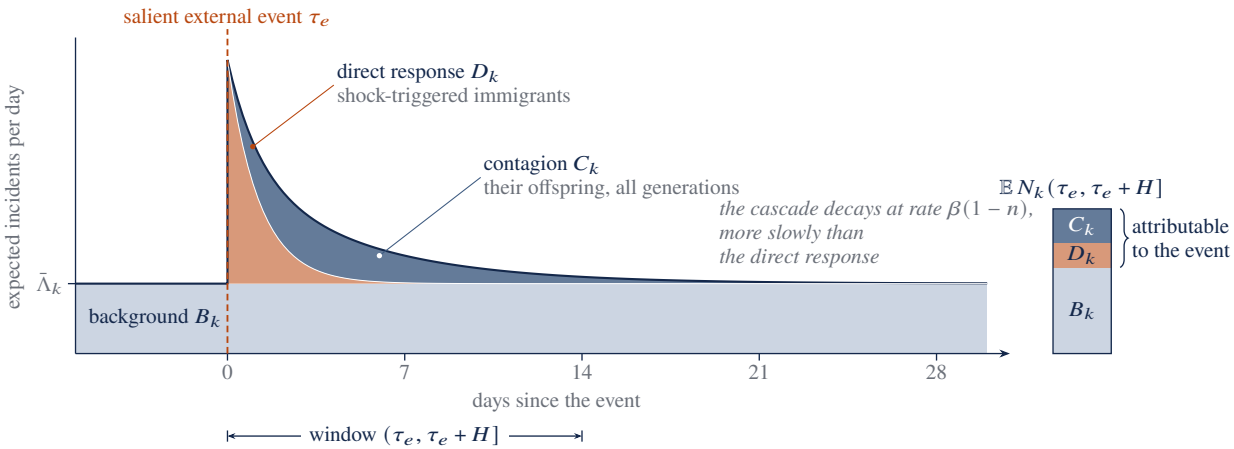


Figure 3. Surge attribution (schematic, not data). Expected incidents per day in stream k before and after a salient external event at τ_e, split by the root of each incident's cluster (Figure 1): the *background* (background immigrants and all their descendants, at the stationary level $\bar{\Lambda}_k$), the *direct* response (immigrants triggered by the event) and *contagion* (all later generations descending from those immigrants). Summing each layer over the window $(\tau_e, \tau_e + H]$ gives $\mathbb{E}\,N_k(\tau_e, \tau_e + H] = B_k + D_k + C_k$ (bar, right), which Theorem 5.1 computes in closed form through the resolvent and the multiplier $\boldsymbol{M}$. The direct response decays at the rate η of the response kernel, but the cascade it starts decays at the slower cluster rate $\beta(1-n)$ (Proposition 5.2), so a surge can outlast its trigger and be mistaken for a shift in the background. In a univariate model the contagion share $C_k/(D_k + C_k)$ rises with H towards the branching ratio n. Curves are exact expected intensities for illustrative parameters: $\mu = 1$, $n = 0.6$, kernel rate $\beta = 0.5$ per day, $\upsilon_k S_e = 10$ expected direct incidents, $\psi_k(t) = \eta e^{-\eta t}$ with $\eta = 0.8$ per day, and the plan's primary horizon $H = 14$ days (Appendix H.2).

$p\,\mathbb{E}[\lambda_{\text{off}}(t) \mid \mathcal{G}_{t-}]$, depends on the online history whenever it is informative about *unrecorded* offline incidents. Online volume may then predict recorded offline incidents with $\alpha_{\text{off,on}} = 0$.

(iv) *Timestamps.* If offline incidents are dated by report time rather than occurrence time, online reactions to an incident can precede its record and appear as online-to-offline excitation.

5 Surge attribution: background, shock and contagion

A salient external event is often followed by a rise in recorded incidents. Three questions then arise. How many incidents in the following days would have occurred without the event? How many respond directly to it? How many were triggered by other incidents in the cascade that the event set off? Under model (3.2) these questions have exact answers in terms of the parameters, because the cluster representation (Section 3.3) assigns every incident to exactly one immigrant ancestor.

Theorem 5.1 splits the expected count in a post-event window into background, direct and contagion components, Proposition 5.2 gives closed forms for exponential kernels, and Proposition 5.4 isolates the part of the contagion that passes from online to offline streams. All of these quantities are functionals of the model parameters. They describe what the fitted model implies, and they inherit the identification limits of Section 4: a component labelled "contagion" is contagion only if the model is correct, including its background and its list of events (Remark 5.5). The ingredients are classical: the resolvent and the first-moment formula go back to Hawkes (1971) (see also Bacry et al., 2015; Laub et al., 2021), the split by immigrant ancestor is the cluster representation of Hawkes and Oakes (1974), and resolvents of Volterra equations are treated in Gripenberg et al. (1990). We have not found the hypoexponential closed forms of

Proposition 5.2(d) or the channel decomposition stated in this form, although both are short consequences of classical identities. Appendix B gives the proofs.

5.1 Set-up

Throughout this section Assumption A1 holds and $\varrho(\boldsymbol{A}) < 1$. The process starts from the empty history at time 0, as in (3.2). Fix an event e with time $\tau_e \geq 0$ and size S_e, a horizon $H \in (0, \infty]$ and the post-event window $W_e = (\tau_e, \tau_e + H]$. Inequalities between vectors or matrices hold entrywise. We collect the loadings into the *direct-response vector*

$$\boldsymbol{d}_e = S_e \, (\upsilon_1, \ldots, \upsilon_M)^\top.$$

Its kth entry is the expected number of stream-k incidents that respond directly to event e. The immigrant intensities of Section 3.3 split by source into

$$\boldsymbol{\mu}(t), \qquad \boldsymbol{\nu}_e(t) = S_e\big(\upsilon_k \psi_k(t - \tau_e)\big)_{k \in [M]}, \qquad \boldsymbol{\nu}_{-e}(t) = \sum_{e' \neq e} \boldsymbol{\nu}_{e'}(t).$$

For a matrix function $\boldsymbol{F}$ and a vector function $\boldsymbol{f}$ on $[0, \infty)$, let $(\boldsymbol{F} * \boldsymbol{f})(t) = \int_0^t \boldsymbol{F}(t - s)\boldsymbol{f}(s)\,\mathrm{d}s$. Matrix convolution is defined in the same way, with $\boldsymbol{\Phi}^{*1} = \boldsymbol{\Phi}$ and $\boldsymbol{\Phi}^{*(m+1)} = \boldsymbol{\Phi} * \boldsymbol{\Phi}^{*m}$. Write $\bar{\boldsymbol{R}}(u) = \int_0^u \boldsymbol{R}$. For $\tau \geq 0$, the *cascade operator* maps an immigrant intensity to the mean intensity it generates:

$$(\mathcal{K}_\tau \boldsymbol{f})(t) = \boldsymbol{f}(t) + \int_\tau^t \boldsymbol{R}(t - s)\, \boldsymbol{f}(s)\,\mathrm{d}s, \qquad t > \tau, \qquad \mathcal{K} := \mathcal{K}_0. \tag{5.1}$$

Since $\varrho(\boldsymbol{A}) < 1$, the series $\boldsymbol{R} = \sum_{m \geq 1} \boldsymbol{\Phi}^{*m}$ converges in $L^1[0, \infty)$ with $\int_0^\infty \boldsymbol{R} = \boldsymbol{M} - \boldsymbol{I}$. In the cluster representation the incidents of generation m have mean intensity $\boldsymbol{\Phi}^{*m} * \boldsymbol{\nu}$, so an immigrant intensity $\boldsymbol{\nu}$ generates the mean intensity $\mathcal{K}\boldsymbol{\nu}$, the unique locally integrable solution of the renewal equation $\boldsymbol{m} = \boldsymbol{\nu} + \boldsymbol{\Phi} * \boldsymbol{m}$ (Lemma B.5 in Appendix B.2).

5.2 The decomposition

Theorem 5.1 (Surge attribution). *Under the standing assumptions of Section 5.1, let* $\Psi_k(H) = \int_0^H \psi_k$.

(a) Decomposition. *For every stream k,*

$$\mathbb{E}\, N_k(W_e) = B_k + D_k + C_k, \tag{5.2}$$

where

$$D_k = \upsilon_k S_e\, \Psi_k(H), \tag{5.3}$$

$$\boldsymbol{C} = \int_{W_e} (\boldsymbol{R} * \boldsymbol{\nu}_e)(t)\,\mathrm{d}t = S_e \int_0^H \bar{\boldsymbol{R}}(H - v)\, \big(\upsilon_k \psi_k(v)\big)_{k \in [M]}\,\mathrm{d}v, \tag{5.4}$$

$$\boldsymbol{B} = \int_{W_e} \mathcal{K}(\boldsymbol{\mu} + \boldsymbol{\nu}_{-e})(t)\,\mathrm{d}t = \boldsymbol{B}^{\mu} + \boldsymbol{O}, \qquad \boldsymbol{B}^{\mu} = \int_{W_e} \mathcal{K}\boldsymbol{\mu}, \quad \boldsymbol{O} = \int_{W_e} \mathcal{K}\boldsymbol{\nu}_{-e}. \tag{5.5}$$

In the cluster representation, D_k, C_k, B_k^{μ} and O_k are the expected numbers of stream-k incidents in W_e of four kinds: immigrants of event e; descendants, of any generation, of those immigrants; members of clusters rooted in background immigrants; and members of clusters rooted in other events. The baseline B_k is the expected count in W_e under the same model with $S_e = 0$, so $D_k + C_k$ is the expected excess attributable to event e.

(b) Totals. *As $H \to \infty$, $\boldsymbol{D} \uparrow \boldsymbol{d}_e$, $\boldsymbol{D} + \boldsymbol{C} \uparrow \boldsymbol{M}\boldsymbol{d}_e$ and $\boldsymbol{C} \uparrow (\boldsymbol{M} - \boldsymbol{I})\boldsymbol{d}_e = \boldsymbol{A}\boldsymbol{M}\boldsymbol{d}_e$. In the univariate case, $\boldsymbol{A} = n$ and $d_e > 0$. The contagion share of the incidents attributable to the event then equals the branching ratio:*

$$\frac{C(\infty)}{D(\infty) + C(\infty)} = n. \tag{5.6}$$

(c) Finite horizons. *For every H, $\boldsymbol{C}(H) \le \bar{\boldsymbol{R}}(H)\,\boldsymbol{D}(H) \le (\boldsymbol{M} - \boldsymbol{I})\,\boldsymbol{D}(H)$. In the univariate case, for every H with $D(H) > 0$,*

$$\frac{C(H)}{D(H) + C(H)} \le \frac{\bar{\boldsymbol{R}}(H)}{1 + \bar{\boldsymbol{R}}(H)} \le n,$$

and the ratio tends to n as $H \to \infty$. If all streams share one response kernel, $\psi_k = \psi$, then for each stream k the share $C_k(H)/\{D_k(H) + C_k(H)\}$ is at most its limit $C_k(\infty)/\{D_k(\infty) + C_k(\infty)\}$.

(d) Conditioning on the history before the event. *Let $\boldsymbol{h}_{\tau_e}(t) = \int_{[0,\tau_e]} \boldsymbol{\Phi}(t - s)\,\mathrm{d}N(s)$, $t > \tau_e$, be the excitation carried over from incidents up to τ_e. Then*

$$\mathbb{E}\big[N_k(W_e) \mid \mathcal{F}_{\tau_e}\big] = B_k^{|e} + D_k + C_k, \qquad \boldsymbol{B}^{|e} = \int_{W_e} \mathcal{K}_{\tau_e}\big(\boldsymbol{\mu} + \boldsymbol{\nu}_{-e} + \boldsymbol{h}_{\tau_e}\big)(t)\,\mathrm{d}t, \tag{5.7}$$

with D_k and C_k as in (a). Conditioning on the past changes only the baseline.

Part (b) gives the branching ratio a direct reading. In a univariate model, the fraction of event-attributable incidents that are contagion is n, whatever the excitation kernel, the response kernel ψ and the size of the event: it is a property of the cascade, not of the event. In a multivariate model the stream-k share is $1 - d_{e,k}/(\boldsymbol{M}\boldsymbol{d}_e)_k$, which depends on how the event loads on the other streams and can exceed α_{kk} (Appendix B.8). Part (c) says that contagion takes time, so shares reported at short horizons understate the eventual contagion. The scalar ordering can fail when streams respond on different time scales. With $\boldsymbol{A} = \left(\begin{smallmatrix} 0.5 & 0 \\ 0.4 & 0.2 \end{smallmatrix}\right)$, all $\beta_{kj} = 3$, $\boldsymbol{\eta} = (5, 0.2)$ and $\boldsymbol{d}_e = (10, 5)^\top$, the share of stream 2 is 0.879 at $H = 0.83$ but 0.692 in the limit; the vector bound in (c) holds in every case. This is one reason why the analysis plan reports several horizons (Appendix H.2). Part (d) assigns the excitation left by incidents before the event to the baseline, so a surge that begins during an already elevated period is not credited to the event. It is the version to use when decomposing the window after a specific event, given the observed history.

Two supplements are in Appendix B.4. A time-varying background is passed through the filter $\delta_0 \boldsymbol{I} + \boldsymbol{R}$, which amplifies slow variation by up to $\boldsymbol{M}$; a calendar effect that coincides with τ_e but is omitted from $\boldsymbol{\mu}$ is attributed to $\boldsymbol{D}$ and $\boldsymbol{C}$, a time-local analogue of Theorem 4.7 (Remark B.6). And the latent parent labels used by EM split a *realised* surge into direct, contagion and baseline parts whose expectations are the components of (5.2) (Remark B.7).

5.3 Impulse responses for exponential kernels

For the exponential kernels of Section 3.2, the resolvent, the mean intensity and the finite-horizon totals reduce to matrix exponentials.

Proposition 5.2 (Impulse responses). *(a)* Univariate. *If $M = 1$ and $\varphi(t) = n\beta e^{-\beta t}$ with $n < 1$, then*

$$\boldsymbol{R}(t) = n\beta\, e^{-\beta(1-n)t}, \qquad \bar{\boldsymbol{R}}(t) = \frac{n}{1 - n}\big(1 - e^{-\beta(1-n)t}\big). \tag{5.8}$$

(b) Decay rates set by the receiving stream. *Suppose* $\varphi_{kj}(t) = \alpha_{kj}\beta_{k\bullet}e^{-\beta_{k\bullet}t}$, *and let* $\boldsymbol{\Omega} = \operatorname{diag}(\beta_{1\bullet}, \dots, \beta_{M\bullet})$. *Then*

$$\boldsymbol{R}(t) = e^{-\boldsymbol{\Omega}(\boldsymbol{I}-\boldsymbol{A})t}\,\boldsymbol{\Omega}\boldsymbol{A}, \tag{5.9}$$

and $-\boldsymbol{\Omega}(\boldsymbol{I}-\boldsymbol{A})$ *is Hurwitz, that is, all its eigenvalues have negative real part. The excitation part* $\boldsymbol{y} = \boldsymbol{R} * \boldsymbol{\nu}$ *of the mean intensity* $\mathcal{K}\boldsymbol{\nu} = \boldsymbol{\nu} + \boldsymbol{y}$ *solves the linear system* $\boldsymbol{y}' = -\boldsymbol{\Omega}(\boldsymbol{I}-\boldsymbol{A})\boldsymbol{y} + \boldsymbol{\Omega}\boldsymbol{A}\boldsymbol{\nu}$ *with* $\boldsymbol{y}(0) = \boldsymbol{0}$. *With a common rate,* $\boldsymbol{\Omega} = \beta\boldsymbol{I}$, *this gives* $\boldsymbol{R}(t) = \beta\boldsymbol{A}e^{-\beta(\boldsymbol{I}-\boldsymbol{A})t}$.

(c) General exponential kernels. *Suppose* $\varphi_{kj}(t) = \alpha_{kj}\beta_{kj}e^{-\beta_{kj}t}$. *Index* $\mathbb{R}^{M^2}$ *by pairs* (k, j), *and define* $\boldsymbol{P} : \mathbb{R}^M \to \mathbb{R}^{M^2}$, $\boldsymbol{S} : \mathbb{R}^{M^2} \to \mathbb{R}^M$ *and* $\boldsymbol{Q} : \mathbb{R}^{M^2} \to \mathbb{R}^{M^2}$ *by*

$$(\boldsymbol{P}\boldsymbol{v})_{kj} = \alpha_{kj}\beta_{kj}v_j, \qquad (\boldsymbol{S}\boldsymbol{z})_k = \sum_j z_{kj}, \qquad (\boldsymbol{Q}\boldsymbol{z})_{kj} = -\beta_{kj}z_{kj} + \alpha_{kj}\beta_{kj}\sum_l z_{jl}.$$

Then $\boldsymbol{R}(t) = \boldsymbol{S}\,e^{\boldsymbol{Q}t}\boldsymbol{P}$, *and* $\boldsymbol{Q}$ *is Hurwitz. The coordinate* z_{kj} *of* $\boldsymbol{z}(t) = \int_0^t e^{\boldsymbol{Q}(t-s)}\boldsymbol{P}\boldsymbol{\nu}(s)\,\mathrm{d}s$ *is the mean excitation of stream* k *by stream-*j *incidents.*

(d) Finite-horizon totals for exponential shock responses. *Let* $\psi_k(t) = \eta_k e^{-\eta_k t}$. *For rates* $\eta, \gamma > 0$, *let* $\Pi_{\eta,\gamma}(H) = \mathbb{P}(X_\eta + X_\gamma \le H)$ *be the hypoexponential distribution function. Here* X_η *and* X_γ *are independent exponential variables with rates* η *and* γ, *and* $\dot{\Pi}_{\eta,\gamma}$ *denotes the density. Explicitly,*

$$\Pi_{\eta,\gamma}(H) = \begin{cases} 1 - \dfrac{\gamma e^{-\eta H} - \eta e^{-\gamma H}}{\gamma - \eta}, & \eta \neq \gamma, \\ 1 - (1 + \gamma H)\,e^{-\gamma H}, & \eta = \gamma \quad \textit{(resonance)}. \end{cases} \tag{5.10}$$

In the univariate case, let $d = \upsilon_1 S_e$ *and* $\gamma = \beta(1-n)$. *Then*

$$D(H) = d\left(1 - e^{-\eta H}\right), \qquad C(H) = \frac{n}{1-n}\,d\,\Pi_{\eta,\gamma}(H). \tag{5.11}$$

The intensity attributable to the event at lag t *is* $d\,\{\eta e^{-\eta t} + \frac{n}{1-n}\dot{\Pi}_{\eta,\gamma}(t)\}$. *Its contagion part peaks at* $t^* = \log(\eta/\gamma)/(\eta - \gamma)$, *which equals* $1/\gamma$ *at resonance.*

If all kernels share one decay rate β *and all responses share one rate* η, *then*

$$\boldsymbol{D}(H) = \left(1 - e^{-\eta H}\right)\boldsymbol{d}_e, \qquad \boldsymbol{C}(H) = (\boldsymbol{M} - \boldsymbol{I})\,\Pi_{\eta,\boldsymbol{G}}(H)\,\boldsymbol{d}_e, \qquad \boldsymbol{G} = \beta(\boldsymbol{I} - \boldsymbol{A}). \tag{5.12}$$

Here $\Pi_{\eta,\boldsymbol{G}}(H) = \int_0^H \eta e^{-\eta v}\left(\boldsymbol{I} - e^{-\boldsymbol{G}(H-v)}\right)\mathrm{d}v$ *is the matrix function induced by the entire map* $x \mapsto \Pi_{\eta,x}(H)$.

In general, with pair-specific β_{kj} *and stream-specific* η_k, *consider the state* $\boldsymbol{w} = (\boldsymbol{\nu}_e, \boldsymbol{z}, \boldsymbol{D}, \boldsymbol{C})$: *the event's immigrant intensity, the excitation state of (c), and the two running totals. Then* $(\boldsymbol{D}(H), \boldsymbol{C}(H))$ *are the last two blocks of* $\exp(H\boldsymbol{Z})\,\boldsymbol{w}_0$, *where*

$$\boldsymbol{Z} = \begin{pmatrix} -\operatorname{diag}(\boldsymbol{\eta}) & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{P} & \boldsymbol{Q} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{I} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{S} & \boldsymbol{0} & \boldsymbol{0} \end{pmatrix}, \qquad \boldsymbol{w}_0 = \begin{pmatrix} \operatorname{diag}(\boldsymbol{\eta})\,\boldsymbol{d}_e \\ \boldsymbol{0} \\ \boldsymbol{0} \\ \boldsymbol{0} \end{pmatrix}. \tag{5.13}$$

Three features of Proposition 5.2 matter for interpretation. First, the cascade decays at rate $\gamma = \beta(1-n)$, more slowly than a single kernel, which matches the correlation time in Section 3.4. As $n \uparrow 1$, $C(H)$ approaches its total only slowly, so the horizon should exceed a few multiples of $\max\{1/\eta, 1/\gamma\}$ before shares are compared across events. Second, in the univariate model the contagion count up to H is a fixed multiple of the distribution function of a sum of two delays: from the event to a direct response (rate η), and from that response to a randomly chosen descendant (rate γ, since $\boldsymbol{R}/\int \boldsymbol{R}$ is the $\mathrm{Exp}(\gamma)$ density). Third, at resonance, $\eta = \gamma$, the singularity in (5.10) is removable; the matrix exponential (5.13) needs no special case (Van Loan, 1978), and the simulation code evaluates it directly. The background part $\boldsymbol{B}^{\mu}$ is computed in the same way, with forcing $\boldsymbol{\mu}$ in place of $\boldsymbol{\nu}_e$.

5.4 Online-to-offline spillover: a channel decomposition

Partition the streams into offline streams $\mathcal{S}_{\text{off}}$ and online streams $\mathcal{S}_{\text{on}}$. Let $\boldsymbol{E}$ keep the online-to-offline entries of $\boldsymbol{A}$, $E_{kj} = \alpha_{kj}$ for $k \in \mathcal{S}_{\text{off}}$ and $j \in \mathcal{S}_{\text{on}}$, with all other entries zero. Set $\boldsymbol{A}^{\circ} = \boldsymbol{A} - \boldsymbol{E}$ and $\boldsymbol{M}^{\circ} = (\boldsymbol{I} - \boldsymbol{A}^{\circ})^{-1}$. Write $\boldsymbol{C}(H; \boldsymbol{A})$ for the contagion component of Theorem 5.1 computed with branching matrix $\boldsymbol{A}$, holding fixed every other parameter: the kernel shapes g_{kj}, the background, the loadings and the response kernels. Only the online-to-offline entries are set to zero. The online-to-online entries are kept; zeroing the whole column of an online stream would also remove its excitation of other online streams, and would answer a different question.

Definition 5.3 (Online channel). The *online-channel contagion* of event e over horizon H is

$$\boldsymbol{C}^{\text{on}}(H) = \boldsymbol{C}(H; \boldsymbol{A}) - \boldsymbol{C}(H; \boldsymbol{A}^{\circ}).$$

For an offline stream k with $C_k(H) > 0$, its *online-channel share* is $\sigma_k(H) = C_k^{\text{on}}(H)/C_k(H)$.

The direct component does not depend on $\boldsymbol{A}$. Hence $\boldsymbol{C}^{\text{on}}$ is also the reduction in all event-attributable incidents when the online-to-offline entries are set to zero.

Proposition 5.4 (Path sums and bounds). *Under the standing assumptions:*

(a) Ancestral reading. *$C_k^{\text{on}}(H)$ is the expected number of stream-k incidents in W_e that descend from an immigrant of event e along an ancestral line that contains at least one online-to-offline transmission.*

(b) Path sum. *$\varrho(\boldsymbol{M}^{\circ}\boldsymbol{E}) < 1$ and*

$$\boldsymbol{M} = \sum_{r \geq 0} (\boldsymbol{M}^{\circ}\boldsymbol{E})^r \, \boldsymbol{M}^{\circ}. \tag{5.14}$$

Entry (k, j) of the rth term is the total weight $\prod \alpha$ of the walks from j to k in the excitation graph with exactly r online-to-offline steps. Consequently

$$\boldsymbol{C}^{\text{on}}(\infty) = (\boldsymbol{M} - \boldsymbol{M}^{\circ})\,\boldsymbol{d}_e = \boldsymbol{M}\boldsymbol{E}\boldsymbol{M}^{\circ}\boldsymbol{d}_e = \boldsymbol{M}^{\circ}\boldsymbol{E}\boldsymbol{M}\boldsymbol{d}_e. \tag{5.15}$$

(c) Bounds. *For every $H \in (0, \infty]$, $\mathbf{0} \leq \boldsymbol{C}^{\text{on}}(H) \leq \boldsymbol{C}(H)$, so $\sigma_k(H) \in [0, 1]$. Moreover:*

- *$C_k^{\text{on}}(\infty) > 0$ if and only if some walk of positive weight runs from a stream j with $d_{e,j} > 0$ to k and contains an online-to-offline step.*
- *$\boldsymbol{C}^{\text{on}}(\infty) \geq \boldsymbol{M}^{\circ}\boldsymbol{E}\boldsymbol{M}^{\circ}\boldsymbol{d}_e$, which counts the walks with a single crossing.*

- *Equality holds when there is no offline-to-online excitation, that is, $\alpha_{kj} = 0$ for all $k \in \mathcal{S}_{\text{on}}$ and $j \in \mathcal{S}_{\text{off}}$. Then, on the offline streams,*

$$\boldsymbol{C}^{\text{on}}_{\text{off}}(\infty) = (\boldsymbol{I} - \boldsymbol{A}_{\text{off,off}})^{-1}\boldsymbol{A}_{\text{off,on}}(\boldsymbol{I} - \boldsymbol{A}_{\text{on,on}})^{-1}\boldsymbol{d}_{e,\text{on}}, \tag{5.16}$$

which vanishes when the event has no direct online response.

(d) Non-additivity. *Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be disjoint sets of entries of $\boldsymbol{A}$, for instance the channels from two platforms. Let $\boldsymbol{C}^{\mathcal{E}}$ be defined as in Definition 5.3, with the entries in $\mathcal{E}$ set to zero. Then, for every H,*

$$\max\left(\boldsymbol{C}^{\mathcal{E}_1}, \boldsymbol{C}^{\mathcal{E}_2}\right) \le \boldsymbol{C}^{\mathcal{E}_1 \cup \mathcal{E}_2} \le \boldsymbol{C}^{\mathcal{E}_1} + \boldsymbol{C}^{\mathcal{E}_2}.$$

The gap $\boldsymbol{C}^{\mathcal{E}_1} + \boldsymbol{C}^{\mathcal{E}_2} - \boldsymbol{C}^{\mathcal{E}_1 \cup \mathcal{E}_2}$ is the expected number of descendants whose ancestral line uses both channels.

Part (d) means that channel shares are not additive. The shares of two platforms can sum to more than the share of both together. Each share answers its own counterfactual question. Shares should not be presented as slices of one pie.

Remark 5.5 (What the channel decomposition does and does not show). $\boldsymbol{C}^{\text{on}}$ is a *model-based counterfactual*: the expected reduction in event-attributable offline incidents if online-to-offline excitation were switched off and nothing else changed. It becomes a causal statement, for instance about the effect of moderation, only under further assumptions.

(i) *Excitation is influence.* By Proposition 4.12, $\alpha_{\text{off,on}} > 0$ is equivalent to Granger causality, which is predictive, not causal. Cross-excitation sometimes leaves a second-order trace that no common-shock model reproduces (Example 4.6), but an omitted common driver can still create apparent excitation (Theorem 4.7, Remark 4.13(i)).

(ii) *Modularity.* An intervention must remove exactly the entries in $\boldsymbol{E}$ and leave everything else unchanged; interventions on platforms may displace activity rather than remove it.

(iii) *Measurement and timestamps.* The online stream is a classifier output and offline streams are thinned, so the decomposition should be computed on the true scale (Remark 5.6); and report-time stamps can make online reactions precede the record of an incident (Remark 4.13(iv)).

In line with the analysis plan (Appendix H.7), $\boldsymbol{C}^{\text{on}}$ and σ_k are reported as model-based quantities and are not described as causal effects.

5.5 Uncertainty

Every component above is a smooth function $g(\vartheta)$ of the parameter vector ϑ on the region $\varrho(\boldsymbol{A}) < 1$. Given posterior or parametric-bootstrap draws $\vartheta^{(1)}, \dots, \vartheta^{(R)}$, we compute $(\boldsymbol{B}, \boldsymbol{D}, \boldsymbol{C}, \boldsymbol{C}^{\text{on}})^{(r)}$ exactly for each draw and report quantiles. This propagates the dependence between components and respects the constraints $\sigma_k \in [0, 1]$ and, in univariate models, $C/(D + C) \le n$; infinite-horizon totals are reported only for draws with $\varrho(\boldsymbol{A}^{(r)}) < 1$. The analysis plan uses this route. The delta method, with the closed-form gradients of Appendix B.6, is a cheaper alternative, but it is unreliable near the boundary $\alpha_{kj} = 0$ (Section 6.5) and near criticality, where $\boldsymbol{M}$ has a pole.

Remark 5.6 (Under-recording). Suppose each stream-k incident is recorded independently with probability p_k. Then the expected number of recorded incidents in W_e is $p_k(B_k + D_k + C_k)$. Recording scales all three

components of a stream by the same factor. The within-stream shares $C_k/(D_k + C_k)$ and σ_k are therefore invariant to p_k, provided they are computed from true-scale parameters.

Fitting the recorded streams as if they were complete does not have this invariance. By Theorem 4.10, moment-based estimators of the branching ratio from a recorded univariate stream converge to $h_p(n) < n$. The contagion share implied by such a fit, which equals the fitted branching ratio by (5.6), understates contagion. The dark-figure correction (Corollary 4.11) should therefore precede attribution.

Appendix B.8 works through a two-stream example with invented parameters, in which the offline contagion share is about twice $\alpha_{\text{off,off}}$ and most of the offline contagion passes through the online stream.

6 Estimation, testing and model checking

This section turns the model of Section 3 into inference. Everything rests on the log-likelihood (3.1). Section 6.1 evaluates it exactly for exponential kernels and discusses the background. Section 6.2 maximises it by EM over the latent branching structure, Section 6.3 gives a Bayesian version with shrinkage on $\boldsymbol{A}$, and Section 6.4 treats binned counts. We then test whether one stream excites another (Section 6.5), check goodness of fit (Section 6.6), and calibrate shock effects against placebo calendars (Section 6.7). Most ingredients are known, and we cite them as we go. To our knowledge, two results are new in this form:

- with the observation window kept exactly in the complete-data likelihood, the EM update of an exponential decay rate is a strictly concave one-dimensional problem with a moment-matching reading (Proposition 6.1);
- an exact formula for the attenuation of contagion estimates computed from binned counts (Proposition C.2).

We also make explicit an elementary fact about M-matrices, which turns the stationarity constraint into a truncated-gamma Gibbs update (Lemma 6.2). Proofs, the Gibbs sampler, nonparametric kernels, the conditions and proofs of the classical boundary and time-rescaling results, and the Davies problem are in Appendix C; computational details are in Appendix E.

Throughout, ϑ denotes the full parameter vector: the background coefficients θ_k, the shock loadings υ_k, the response parameters, $\boldsymbol{A}$ and the kernel parameters. We reserve θ_k for the background, as in Assumption A1. We write $n_k = N_k(T)$ and $n_{\text{ev}} = \sum_k n_k$ for the total number of incidents; n without a subscript is the branching ratio, as in Section 3.

6.1 Maximum likelihood

Separation by target stream. The log-likelihood (3.1) is a sum $\ell_T = \sum_k \ell_{T,k}$ of stream terms. The term $\ell_{T,k}$ depends only on the parameters of λ_k: on θ_k, on υ_k and ψ_k, and on row k of $\boldsymbol{A}$ with its kernels. Unless a parameter is shared across rows, such as a common decay rate, the likelihood therefore separates. Each row of the model is fitted to its own stream, and the other streams enter only as history. We use this separation for testing (Section 6.5) and for parallel computation (Appendix E).

Exponential kernels. Let $g_{kj}(u) = \beta_{kj} e^{-\beta_{kj} u}$ and $\psi_k(u) = \eta_k e^{-\eta_k u}$ for $u > 0$, and let $\Psi_k(u) = \int_0^u \psi_k$, with $\Psi_k(u) = 0$ for $u \le 0$. With the decay sums $R_{kj}(t) = \sum_{l:\, t_l^j < t} e^{-\beta_{kj}(t - t_l^j)}$, the intensity and the compensator on $[0, T]$ are

$$\lambda_k(t) = \mu_k(t) + \upsilon_k \sum_{e:\, \tau_e < t} S_e \, \eta_k e^{-\eta_k (t - \tau_e)} + \sum_{j=1}^{M} \alpha_{kj} \beta_{kj} R_{kj}(t), \tag{6.1}$$

$$\Lambda_k(T) = \int_0^T \mu_k(t) \, \mathrm{d}t + \upsilon_k \sum_e S_e \Psi_k^T(e) + \sum_{j=1}^{M} \alpha_{kj} G_{kj}, \tag{6.2}$$

where $\Psi_k^T(e) = \Psi_k(T - \tau_e) - \Psi_k(-\tau_e)$ is the share of the response to event e that falls inside the window, and $G_{kj} = \sum_{l=1}^{n_j} \{1 - e^{-\beta_{kj}(T - t_l^j)}\}$ is the expected number of in-window offspring per unit of α_{kj}. For a pair (k, j), let $s_1 < s_2 < \cdots$ be the merged event times of streams j and k, with d_m the stream of s_m. The exponential kernel is memoryless, so

$$R_{kj}(s_m) = e^{-\beta_{kj}(s_m - s_{m-1})} \big\{ R_{kj}(s_{m-1}) + \mathbf{1}\{d_{m-1} = j\} \big\}, \qquad R_{kj}(s_1) = 0. \tag{6.3}$$

For each pair, (6.3) costs $O(n_j + n_k)$, so all intensities at the events cost $O(M n_{\mathrm{ev}})$ operations, against $O(n_{\mathrm{ev}}^2)$ for direct summation. The recursion is due to Ozaki (1979); Appendix E gives the analogous recursions for the gradient and the EM statistics.

Asymptotic theory. Ogata (1978) proved consistency, asymptotic normality and efficiency of the maximum likelihood estimator for stationary point processes. His conditions are stationarity and ergodicity, identifiability, a compact parameter space with ϑ_0 in its interior, and smoothness and moment conditions on the intensity and its first three derivatives. Under them, $\sqrt{T}(\hat{\vartheta} - \vartheta_0) \Rightarrow N(0, \boldsymbol{I}^{-1})$, where $\boldsymbol{I} = \sum_k \mathbb{E}\{\lambda_k(0)^{-1} \nabla \lambda_k(0) \nabla \lambda_k(0)^\top\}$ is the Fisher information per unit time under the stationary law. The shock-augmented model with a spline background is not stationary. A fixed-dimensional theory applies if the number K_b of basis functions is held fixed as T grows. Our pre-registered design instead fixes the knot spacing h_b (Appendix H.4), so K_b grows in proportion to T and the background is a semiparametric nuisance, whose theory we do not develop. We use the normal approximation with the observed information (Appendix E) away from the boundary, and parametric bootstrap or posterior draws where it is in doubt: near $\alpha_{kj} = 0$ (Section 6.5), near criticality, and for nonlinear functionals such as the surge decomposition (Section 5.5).

Constraints. The parameters satisfy $\alpha_{kj} \ge 0$, $\upsilon_k \ge 0$ and $\beta_{kj}, \eta_k > 0$, so $\lambda_k \ge \mu_k > 0$ and the likelihood is defined for every $\boldsymbol{A} \ge \mathbf{0}$. The condition $\varrho(\boldsymbol{A}) < 1$ is needed only for the stationary interpretation, for $\boldsymbol{M}$ and for the surge decomposition. We fit without it and report $\varrho(\hat{\boldsymbol{A}})$ with its uncertainty, reading an estimate near or above one as a symptom of misspecification (Theorem 4.7) rather than as evidence of an explosive process. If the constraint is imposed, the M-step for $\boldsymbol{A}$ with the decays fixed stays convex, because $\log \varrho(\boldsymbol{A})$ is convex in the log-entries (Kingman, 1961); in the Bayesian fit we impose it by truncation (Lemma 6.2).

The background, and why its flexibility matters. The background is $\mu_k(t) = \exp\{B(t)^\top \theta_k\}$. Here B consists of cubic B-splines in calendar time with K_b basis functions, plus day-of-week and holiday indicators (Appendix H.4). We maximise the penalised log-likelihood

$$\ell_T^{\mathrm{pen}}(\vartheta) = \ell_T(\vartheta) - \tfrac{1}{2} \sum_k \lambda_k^{\mathrm{pen}} \, \theta_k^\top P \, \theta_k. \tag{6.4}$$

Two choices of P are natural: the second-difference penalty on adjacent spline coefficients (Eilers and Marx, 1996), which leaves linear trends unpenalised, and $P = \boldsymbol{I} - \mathbf{1}\mathbf{1}^{\top}/K_b$, which shrinks towards a constant background. Our implementation offers the second, and by default uses no penalty, with the flexibility set by the knot spacing alone. The background is not a nuisance to be fitted as flexibly as possible. It is part of the identification strategy, and it can fail in two directions.

- *Too rigid.* Rate variation that the background cannot represent is absorbed by the excitation terms. Theorem 4.7 makes this exact for a constant background. A process with no contagion but a fluctuating rate, such as one driven by unrecorded shocks or a regime-switching level, has second-order pseudo-true branching ratio $n^{*} = 1 - (1+\kappa)^{-1/2} > 0$, and its likelihood pseudo-true value also has $n > 0$.
- *Too flexible.* A background that can vary on the time scale of a cascade can fit the realised clusters directly. In the limit the problem is degenerate. If μ_k ranges over all nonnegative functions, then $\sup_{\mu} \ell_T(\mu, \boldsymbol{A} = \mathbf{0}) = +\infty$: take μ equal to $(2\epsilon)^{-1}$ on ϵ-neighbourhoods of the events and let $\epsilon \downarrow 0$. The likelihood alone therefore cannot separate contagion from background. The separation comes only from restricting the roughness of μ_k.

Identification thus rests on a separation of time scales: the background should be smooth on the cascade time scale $1/\{\beta(1-n)\}$ of Proposition 3.2, while still following seasonal and secular movement. For the primary analysis we fix the knot spacing h_b in advance at a multiple of the longest plausible cascade time scale, and report the estimates for a pre-registered ladder of spacings (Appendices H.4 and H.5). An excitation estimate that vanishes as the background becomes more flexible, while the fit stays adequate, is not robustly distinguishable from background variation. As a check, we compare the pre-registered spacing with the one that maximises a blocked cross-validated log score,

$$\mathrm{CV} = \sum_{I \in \mathcal{I}} \sum_{k=1}^{M} \Big[\sum_{t_i^k \in I} \log \hat{\lambda}_k^{(-I)}(t_i^k) - \int_I \hat{\lambda}_k^{(-I)}(t)\, \mathrm{d}t \Big], \tag{6.5}$$

over contiguous blocks I much longer than the cascade time scale, where $\hat{\lambda}^{(-I)}$ is fitted without block I but conditions on the full history. In a held-out block, background bumps fitted to clusters elsewhere do not help, whereas genuine excitation still predicts clustering.

6.2 EM via the latent branching structure

By the cluster representation (Section 3.3), every incident is a background immigrant, a shock immigrant, or an offspring of an earlier incident. Let Z_i^k be the parent label of t_i^k. It takes the value 0 (background), e (shock), or (j,l) with $t_l^j < t_i^k$ (offspring of t_l^j). Given the labels, the immigrants and the offspring of each incident form independent Poisson processes on $[0,T]$. The complete-data log-likelihood is therefore

$$\begin{aligned} \ell_T^c(\vartheta) = {} & \sum_k \Big[\sum_{i:\, Z_i^k = 0} \log \mu_k(t_i^k) - \int_0^T \mu_k \Big] + \sum_{k,e} \Big[\sum_{i:\, Z_i^k = e} \log\{\upsilon_k S_e \psi_k(t_i^k - \tau_e)\} - \upsilon_k S_e \Psi_k^T(e) \Big] \\ & + \sum_{k,j} \Big[\sum_{(i,l):\, Z_i^k = (j,l)} \log\{\alpha_{kj} g_{kj}(t_i^k - t_l^j)\} - \alpha_{kj}\, \mathcal{G}_{kj} \Big], \qquad \mathcal{G}_{kj} = \sum_{l=1}^{n_j} \int_0^{T - t_l^j} g_{kj}(u)\, \mathrm{d}u. \end{aligned} \tag{6.6}$$

The window terms $\Psi_k^T(e)$ and $\mathcal{G}_{kj}$ are the parts of the response and offspring kernels that fall inside $[0,T]$. For exponential kernels, $\mathcal{G}_{kj} = G_{kj}$ of (6.2). Summing $\exp(\ell_T^c)$ over all labellings recovers the likelihood,

because $\lambda_k(t_i^k)$ is the sum of the parent contributions. This is the EM of Veen and Schoenberg (2008) for the ETAS model, the stochastic declustering of Marsan and Lengliné (2008) and the crime model of Mohler et al. (2011), extended here to exogenous shocks.

E-step. The posterior label probabilities are

$$p_i^{k,0} = \frac{\mu_k(t_i^k)}{\lambda_k(t_i^k)}, \qquad p_i^{k,e} = \frac{\upsilon_k S_e \psi_k(t_i^k - \tau_e)}{\lambda_k(t_i^k)}, \qquad p_i^{k,jl} = \frac{\alpha_{kj} g_{kj}(t_i^k - t_l^j)}{\lambda_k(t_i^k)}.$$

The M-step needs only the aggregates

$$\begin{aligned} N_k^0 = \sum_i p_i^{k,0}, \qquad N_{ke} = \sum_i p_i^{k,e}, \qquad W_{ke} = \sum_i p_i^{k,e}(t_i^k - \tau_e), \\ N_{kj} = \sum_{i,l} p_i^{k,jl}, \qquad W_{kj} = \sum_{i,l} p_i^{k,jl}(t_i^k - t_l^j). \end{aligned} \tag{6.7}$$

For exponential kernels, $\sum_l p_i^{k,jl} = \alpha_{kj}\beta_{kj} R_{kj}(t_i^k)/\lambda_k(t_i^k)$ and $\sum_l p_i^{k,jl}(t_i^k - t_l^j) = \alpha_{kj}\beta_{kj} R_{kj}^{(1)}(t_i^k)/\lambda_k(t_i^k)$, where $R_{kj}^{(1)}(t) = \sum_{t_l^j < t}(t - t_l^j)e^{-\beta_{kj}(t - t_l^j)}$ obeys a recursion like (6.3) (Appendix E). The E-step therefore costs $O(M n_{\mathrm{ev}})$ operations, and the $n_{\mathrm{ev}} \times n_{\mathrm{ev}}$ matrix of parent probabilities is never formed.

Proposition 6.1 (M-step). *Fix the aggregates* (6.7)*, with $N_{kj} > 0$ and $\sum_e N_{ke} > 0$. Let $Q(\vartheta) = \mathbb{E}\{\ell_T^c(\vartheta) \mid \text{data}, \vartheta^{(r)}\}$, which is the expression obtained from* (6.6) *by replacing label indicators with the probabilities p.*

(a) Background. *For constant μ_k, Q is maximised at $\mu_k = N_k^0/T$. For $\mu_k = \exp(B^\top\theta_k)$, θ_k maximises the strictly concave function $\sum_i p_i^{k,0} B(t_i^k)^\top\theta_k - \int_0^T e^{B(t)^\top\theta_k}\mathrm{d}t - \frac{1}{2}\lambda_k^{\mathrm{pen}}\theta_k^\top P\theta_k$, provided the basis functions are linearly independent on $[0,T]$.*

(b) Branching ratios and loadings. *For fixed kernel shapes,*

$$\alpha_{kj} = \frac{N_{kj}}{\mathcal{G}_{kj}}, \qquad \upsilon_k = \frac{\sum_e N_{ke}}{\sum_e S_e \Psi_k^T(e)}.$$

(c) Decay rates. *For exponential g_{kj}, write $r_l = T - t_l^j$ (for the window with a burn-in segment, see Remark C.1) and*

$$m_j(\beta) = \frac{\sum_l \int_0^{r_l} u\, e^{-\beta u}\,\mathrm{d}u}{\sum_l \int_0^{r_l} e^{-\beta u}\,\mathrm{d}u} = \frac{\sum_l\{1 - (1+\beta r_l)e^{-\beta r_l}\}}{\beta \sum_l\{1 - e^{-\beta r_l}\}}.$$

With α_{kj} profiled out, Q is strictly concave in β_{kj}. The function m_j decreases strictly from $m_j(0+) = \sum_l r_l^2/(2\sum_l r_l)$ to 0. If $W_{kj}/N_{kj} < m_j(0+)$, the maximiser is the unique root of

$$m_j(\beta_{kj}) = W_{kj}/N_{kj}.$$

Otherwise Q increases as $\beta_{kj} \downarrow 0$, and the update stops at a pre-set lower bound. In words, the mean lag of the in-window kernel equals the responsibility-weighted mean lag. A decay shared by a set $\mathcal{P}$ of pairs solves $\sum_{(k,j)\in\mathcal{P}} N_{kj} m_j(\beta) = \sum_{(k,j)\in\mathcal{P}} W_{kj}$. The response rate η_k of an exponential ψ_k solves the same equation, with N_{kj}, W_{kj} replaced by $\sum_e N_{ke}, \sum_e W_{ke}$, and the parent windows replaced by the shock windows with weights S_e.

(d) Monotonicity. *If $\vartheta^{(r+1)}$ maximises Q, or merely satisfies $Q(\vartheta^{(r+1)}) \geq Q(\vartheta^{(r)})$, then $\ell_T(\vartheta^{(r+1)}) \geq \ell_T(\vartheta^{(r)})$. With the penalty* (6.4)*, the same holds for ℓ_T^{pen}.*

The proof is in Appendix C.8. The window terms $\mathcal{G}_{kj}$ and $\Psi_k^T(e)$ matter. Dropping them gives the familiar edge-free updates $\alpha_{kj} = N_{kj}/n_j$ and $\beta_{kj} = N_{kj}/W_{kj}$, an EM algorithm for a pseudo-likelihood rather than for ℓ_T. It underestimates α_{kj} and the loadings, by a relative amount of about $1/(\beta_{kj}T)$, and overestimates decay rates. This is negligible for a short kernel in a long window, but not for a salient event shortly before T, which is exactly the case of real-time attribution. Remark C.1 in Appendix C.1 gives the details and the window terms for a burn-in segment.

Algorithm 1 in Appendix C summarises the iteration. Three practical points about Algorithm 1 matter. First, the likelihood can be multimodal in the decay rates; when excitation is weak, a spurious mode with tiny α and very large β can appear, so our implementation starts EM from each local maximum of a grid profile in the decays. When excitation is absent, the likelihood can also have a ridge on which $\beta \to 0$ while α grows and the expected number of in-window offspring per incident stays small; on that ridge $\hat{\alpha}$ alone is not a usable estimate of the branching ratio (Section 8.3). Second, EM converges linearly and is slow when some α_{kj} is near 0; SQUAREM extrapolation (Varadhan and Roland, 2008), with steps kept only when they raise the likelihood, accelerates it, and null fits set $\alpha_{kj} = 0$ exactly. Third, the probabilities $p_i^{k,\cdot}$ give a stochastic declustering of the realised incidents (Marsan and Lengliné, 2008), used in Remark B.7. The partial M-steps of Veen and Schoenberg (2008) and the nonparametric kernels of Lewis and Mohler (2011) (Appendix C.3) fit the same template.

6.3 Bayesian estimation with shrinkage on A

With many streams, $\boldsymbol{A}$ has M^2 entries, most of which we expect to be small. We use priors that shrink small entries strongly and leave large ones nearly untouched: a local–global gamma prior, whose marginal is a horseshoe-type prior placed directly on the nonnegative α_{kj} (Armagan et al., 2011; Carvalho et al., 2010), or a spike-and-slab prior with inclusion indicators ι_{kj} (Linderman and Adams, 2014); Appendix C.2 gives both. Stationarity is imposed by truncating the prior to $\{\varrho(\boldsymbol{A}) < 1\}$. The next lemma, a standard rank-one fact about nonsingular M-matrices, makes the truncation explicit one entry at a time, in the form the sampler needs.

Lemma 6.2 (The stationarity region, entry by entry). *Let $\boldsymbol{A} \geq \boldsymbol{0}$, fix (k, j), and let $\boldsymbol{A}^{(-kj)}$ be $\boldsymbol{A}$ with α_{kj} set to 0. Assume $\varrho(\boldsymbol{A}^{(-kj)}) < 1$; this holds whenever $\varrho(\boldsymbol{A}) < 1$, by Perron–Frobenius monotonicity. Let $\boldsymbol{M}^{(-kj)} = (\boldsymbol{I} - \boldsymbol{A}^{(-kj)})^{-1}$. For $x \geq 0$,*

$$\varrho\big(\boldsymbol{A}^{(-kj)} + x\, \boldsymbol{e}_k \boldsymbol{e}_j^{\top}\big) < 1 \quad \Longleftrightarrow \quad x\, \boldsymbol{M}_{jk}^{(-kj)} < 1.$$

If $\boldsymbol{M}_{jk}^{(-kj)} = 0$, every $x \geq 0$ is admissible.

The proof is in Appendix C.8.

The threshold has a direct reading: adding the edge $j \to k$ with weight x closes a feedback loop with gain $x\, \boldsymbol{M}_{jk}^{(-kj)}$, and the cascade stays subcritical exactly when the gain is below one. An edge that closes no loop, such as an online-to-offline entry in a model without offline-to-online feedback, is unconstrained.

Gibbs sampling by data augmentation. Given the parent labels Z, the complete-data likelihood (6.6) is a product of Poisson-process likelihoods, so each parameter has a simple full conditional. In particular the window term enters the branching ratios exactly, and $\alpha_{kj} \mid \cdot \sim \text{Gamma}(a_1 + N_{kj},\, \omega_{kj} + G_{kj})$ truncated to $[0, 1/\boldsymbol{M}_{jk}^{(-kj)})$, which is sampled by inversion (Lemma 6.2). Appendix C.2 gives the full sweep, including a block update of the spike-and-slab indicators and a slice-sampling update of the decay rates. Each posterior

draw of ϑ gives draws of $\varrho(\boldsymbol{A})$, $\boldsymbol{M}$ and the surge decomposition of Theorem 5.1, so parameter uncertainty propagates to every reported quantity (Section 5.5).

6.4 Binned counts: a discrete-time approximation

Online volumes, and some offline sources, are available only as counts per hour or per day. Let $X_{k,t} = N_k((t-1)\Delta, t\Delta]$ be the count of stream k in bin t, where Δ is the bin width. The integer-valued autoregression INAR(p) with Poisson offspring and Poisson immigration is

$$\begin{aligned} X_{k,t} \mid \mathcal{F}_{(t-1)\Delta} &\sim \text{Poisson}\left(\nu_{k,t} + \sum_{j=1}^{M}\sum_{\ell=1}^{p} \varphi_{kj,\ell}\, X_{j,t-\ell}\right), \\ \nu_{k,t} &= \int_{(t-1)\Delta}^{t\Delta} \left\{\mu_k(u) + \upsilon_k \sum_{e} S_e \psi_k(u - \tau_e)\right\} \mathrm{d}u, \end{aligned} \tag{6.8}$$

independently over k. In the INAR representation, each of the $X_{j,t-\ell}$ incidents has Poisson($\varphi_{kj,\ell}$) offspring in bin t. Sums of independent Poisson variables are Poisson, so (6.8) is the identity-link Poisson autoregression (Fokianos et al., 2009). Its conditional likelihood is exact for the binned model. It is concave in the lag coefficients, and in any linear parametrisation of ν.

Under the parameter mapping $\nu = \Delta\mu$, $\varphi_{kj,\ell} = \Delta\varphi_{kj}(\ell\Delta)$ of Kirchner (2016, 2017), the INAR(∞) bin-count process converges to the Hawkes process as $\Delta \to 0$, and conditional least squares (CLS) for the INAR(p) representation gives $\hat{\alpha}_{kj} = \sum_{\ell \le p} \hat{\varphi}_{kj,\ell}$ (Appendix C.4). The mapping is a first-order statement. What a fitted autoregression converges to at a fixed bin width is answered exactly by Proposition C.2 in Appendix C.4. For a stationary univariate Hawkes process with branching ratio n, the coefficients of the best linear predictor of a bin count from its past, which are the limit of CLS, sum to $n_\Delta = 1 - (1-n)\,\varpi_\Delta^{1/2} \le n$, where $\varpi_\Delta \ge 1$ is the innovation dispersion; for an exponential kernel, $n_\Delta = n - \frac{1}{2}n(1-n)\beta\Delta + O(\Delta^2)$. The bias of autoregressive contagion estimates is therefore of order $\beta\Delta$, and it does not vanish for a fixed bin width, however long the series: offspring born in their parent's bin are invisible to lagged regressors. The same identity holds for a Cox process without contagion, so binned autoregressions are exposed to spurious contagion in the same way as continuous-time fits, and within-bin cross-excitation, such as same-day online-to-offline spillover, never enters the lag coefficients (Appendix C.4).

Nonparametric kernels and sparse excitation. Histogram and spline kernels (Lewis and Mohler, 2011; Marsan and Lengliné, 2008) keep every M-step of Algorithm 1 in closed form, and ℓ_1 or nuclear-norm penalties give sparse or low-rank estimates of $\boldsymbol{A}$ when there are many streams (Hansen et al., 2015; Zhou et al., 2013a). Appendix C.3 gives the updates and their costs. Neither is evaluated in this paper.

6.5 Testing whether one stream excites another

The hypothesis $H_0 : \alpha_{kj} = 0$ says that stream j does not excite stream k. For $k \neq j$ it is Granger non-causality from j to k (Proposition 4.12). With k = off and j = on, it is the hypothesis of no online-to-offline spillover. Under H_0 the parameter lies on the boundary of $\{\alpha_{kj} \ge 0\}$, and the usual χ^2_1 reference is wrong. Write $\vartheta = (\alpha, \vartheta_2)$, where $\alpha = \alpha_{kj}$ and $\vartheta_2 \in \Theta_2 \subset \mathbb{R}^d$ collects the remaining parameters, and let $\mathrm{LR}_T = 2\{\ell_T(\hat{\vartheta}) - \ell_T(\tilde{\vartheta})\}$, where $\hat{\vartheta}$ and $\tilde{\vartheta}$ maximise ℓ_T over $[0,\infty) \times \Theta_2$ and over $\{0\} \times \Theta_2$.

We use the classical result for a parameter on the boundary (Chernoff, 1954; Self and Liang, 1987), which Appendix C.5 restates with its conditions and a proof (Proposition C.3). The conditions are that

both maximisers are consistent, that the normalised score is asymptotically normal with positive definite information $\boldsymbol{I}$, and that the observed information converges to $\boldsymbol{I}$ locally uniformly. Under them,

$$\mathrm{LR}_T \Rightarrow \tfrac{1}{2}\chi_0^2 + \tfrac{1}{2}\chi_1^2. \tag{6.9}$$

The asymptotic p-value of an observed value $\mathrm{LR}_T = x > 0$ is $\frac{1}{2}\mathbb{P}(\chi_1^2 \geq x)$, and it is 1 if $x = 0$. The 5% critical value is 2.71, not 3.84. Moreover $\sqrt{T}\,\hat{\alpha}$ converges to the positive part of a centred normal variable, so Wald intervals for α_{kj} are invalid near the boundary. The conditions require the decay of φ_{kj} to be identified under H_0: it must be fixed in advance, or shared with kernels whose branching ratios are positive under H_0, as with a common decay within a block (Appendix H.4). Remark C.4 in Appendix C.5 discusses this and the other conditions.

Decays identified only under the alternative. If φ_{kj} has its own decay rate, or a whole block of entries that share one decay is tested at once, the decay is not identified under H_0 and the chi-bar-square reference is invalid. We then use the sup-statistic over a grid $\beta_1 < \cdots < \beta_G$ in a pre-registered interval $\mathcal{B} = [\beta_L, \beta_U]$,

$$\mathrm{LR}_{\sup} = \max_{g \leq G} \mathrm{LR}_T(\beta_g), \tag{6.10}$$

where $\mathrm{LR}_T(\beta)$ fixes the decay at β. Its null limit (Davies, 1977, 1987) is stochastically larger than $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$, so our reference is the parametric bootstrap: simulate the full model at the null fit $\tilde{\vartheta}$ (Appendix E), recompute the sup-statistic with refits, and report

$$p_{\mathrm{B}} = \frac{1 + \#\{b \leq B : \mathrm{LR}_{\sup}^{(b)} \geq \mathrm{LR}_{\sup}\}}{B + 1}. \tag{6.11}$$

Appendix C.6 gives an upcrossing approximation and a cheaper score-based bootstrap. A common decay within each block (Appendix H.4) keeps tests of single entries within the scope of (6.9) when another entry of the block is positive, but a test that a whole block vanishes needs the sup-statistic. The analysis plan registers such tests for its model hierarchy (Appendix H.4); none is evaluated in the simulation study.

Multiple testing. A full screen of $\boldsymbol{A}$ involves up to M^2 tests. Statistics for different target streams are asymptotically independent when no parameter is shared across rows, because their scores are stochastic integrals against martingales without common jumps; statistics for the same target are correlated. The pre-registered plan controls the family-wise error rate with Holm's procedure (Holm, 1979) (Appendix H.4). For exploratory screens of large matrices we control the false discovery rate with the step-up procedure of Benjamini and Hochberg (1995), in the variant of Benjamini and Yekutieli (2001) that is valid under arbitrary dependence, because scores within a row can be negatively correlated.

6.6 Goodness of fit by time rescaling

By the time-rescaling theorem (Brown et al., 2002; Papangelou, 1972), restated with a proof as Proposition C.5 in Appendix C.7, if the model is correct then the rescaled times $\Lambda_k(t_1^k) < \Lambda_k(t_2^k) < \cdots$ of each stream form a unit-rate Poisson process. The increments $z_i^k = \Lambda_k(t_i^k) - \Lambda_k(t_{i-1}^k)$ are i.i.d. Exp(1), and the $u_i^k = 1 - e^{-z_i^k}$ are i.i.d. uniform on $(0, 1)$. The positivity condition holds because $\lambda_k \geq \mu_k > 0$. The points of different streams, rescaled by their own compensators, are moreover independent (Brown and Nair, 1988; Meyer, 1971), so increments can be pooled across streams.

Diagnostics. For each stream we use a Kolmogorov–Smirnov test of the u_i^k against the uniform law, displayed as a uniform quantile plot with the asymptotic 95% band $\pm 1.36/\sqrt{n_k}$ (Brown et al., 2002); a lag plot and the autocorrelations of the u_i^k, which check independence; and the residual process $N_k(t) - \Lambda_k(t)$ against t (Ogata, 1988), which shows *where* in time a model fails. A misspecified background shows up as long excursions, a misspecified shock response as excursions that start at the τ_e.

Estimated parameters. With $\hat{\vartheta}$ in place of ϑ_0, the u_i^k are no longer exactly i.i.d. uniform, and the plug-in Kolmogorov–Smirnov p-values are not calibrated. In the i.i.d. setting with maximum likelihood estimates the plug-in test is conservative (Durbin, 1973; Lilliefors, 1967); Appendix C.7 explains why the same mechanism plausibly operates here and when it can fail. Either way, we calibrate by parametric bootstrap, as pre-registered in Appendix H.6. Simulate from $\hat{\vartheta}$ with the observed calendar, refit, recompute the Kolmogorov–Smirnov statistic $\mathrm{KS}^{(b)}$, and report $(1 + \#\{\mathrm{KS}^{(b)} \geq \mathrm{KS}\})/(B+1)$. The time-rescaling check is not evaluated in the simulation study.

What the check can and cannot show. Time rescaling uses the full conditional intensity, so in principle it sees beyond second order: a Hawkes model fitted to a shot-noise Cox process misplaces where the intensity is high. In moderate samples, however, its power against the matched Cox processes of Theorem 4.2 may be low. A passed check supports the fitted model as a description of the data; it does not establish contagion.

6.7 Placebo calibration of shock effects

The loadings υ_k and the direct components D_k of Theorem 5.1 are estimated in a model whose background and kernels are approximations. Suppose the background is too rigid around the dates of salient events, or that the calendar overlaps with other periods of elevated activity. Then an extra shock term would pick up a response at almost any date, and the asymptotic test of $\upsilon_k = 0$, itself a boundary test, would be miscalibrated. Placebo calendars provide a reference distribution that does not rely on the model being correct. The calibration is done one class c of real events at a time. The *base model* $\mathcal{M}_{-c}$ is the selected model with class c removed. A *design* $\mathcal{R}_c$ draws calendars with the same number of events as class c: dates uniformly from eligible days, with the sizes S_e of class c randomly permuted over them. The eligible days and any spacing rule may depend on the window and on the calendars of the other classes, but not on the real dates of class c, and they must include every day on which a real event could fall; excluding holidays, for instance, rules out the randomisation hypothesis below whenever a real event falls on a holiday. The design should also reproduce rules that constrain the real calendar, such as merging events less than 48 hours apart (Appendix H.3). Let C_0 be the real class-c calendar and $C_1, \ldots, C_B$ independent draws from $\mathcal{R}_c$. For $b = 0, \ldots, B$, fit $\mathcal{M}_{-c}$ with C_b added as a shock class and record the same *statistic* $\Gamma^{(b)} = \Gamma(C_b; \text{data})$: the loading, its likelihood-ratio statistic, or the direct component $\hat{D}_k$. *Report* the rank p-value $p = (1 + \#\{b \geq 1 : \Gamma^{(b)} \geq \Gamma^{(0)}\})/(B+1)$, and the rejection frequency of the asymptotic shock test across the placebo fits, which is the empirical false-positive rate at the nominal level.

> **Lemma 6.3** (Validity of the placebo rank test). *Let $\Gamma(\cdot\,; data)$ be computed by the same procedure for every calendar. Suppose that, conditionally on the incident data and on the calendars of the other classes, $C_0, C_1, \ldots, C_B$ are exchangeable. This holds, for example, when $C_1, \ldots, C_B$ are drawn independently from $\mathcal{R}_c$ and, under the* randomisation hypothesis*, C_0 is a further independent draw from $\mathcal{R}_c$ that is conditionally independent of the incidents. Then $\mathbb{P}(p \leq a) \leq a$ for every $a \in [0,1]$, whatever the misspecification of $\mathcal{M}_{-c}$.*

The proof is in Appendix C.8.

Two tempting shortcuts break exchangeability and make the rank test liberal: excluding placebo dates near the real class-c dates, which makes the design depend on C_0; and fitting the placebo classes with the real class still in the model, which makes the statistics different functions of the data. Under the alternative, placebo dates can fall inside real response windows. This reduces power and makes the false-positive estimate conservative, but does not affect validity under the null.

The randomisation hypothesis is strong. Salient events cluster in periods of tension, which may differ systematically from typical eligible days. Without it, the rank p-value is best read as a calibration, not a test: does the base model's response at the real dates exceed what the same model attributes to comparable dates drawn without reference to them? The placebo distribution of $\hat{D}_k$ then estimates the false-attribution floor, the number of incidents the model would credit to an arbitrary date. A real-event direct component is interpreted only if it ranks high among its placebo counterparts, and if the empirical false-positive rate clearly exceeds the nominal level, inference on shocks uses the placebo distribution instead of the asymptotic reference (Appendices H.6 and H.7). Placebo calibration is not evaluated in the simulation study.

7 Forecasting and alerting

Community-security organisations and public agencies need to know, on a given day, whether the coming days are likely to bring an unusual number of incidents. This section develops four pieces:

- predictive distributions from the fitted model;
- an alert rule with an explicit trade-off between the costs of false alarms and missed surges;
- a correction of the model's state for reporting delays;
- methods to evaluate forecasts and alerts.

Self-exciting models have been used for short-term crime forecasting (Mohler et al., 2011; Reinhart, 2018) and for forecasting the popularity of online content (Rizoiu et al., 2017; Zhao et al., 2015). The decision rule (Theorem 7.2(a)) and the scoring rules are classical. The monotone coupling behind Theorem 7.2(b)–(c) is the standard cluster, or Poisson-embedding, argument (Brémaud and Massoulié, 1996; Hawkes and Oakes, 1974). What we add is its consequence for the shape of the Bayes alert region and the delay-corrected state with its bias (Proposition 7.3). Proofs are in Appendix B; exact predictive distributions, an imputation algorithm for alerting from delayed data and a monitor for changes in the parameters are in Appendix D. Throughout, parameters are treated as known. Parameter uncertainty enters through posterior or bootstrap draws.

7.1 Predictive distributions

The basic fact is that, given the past, the future is again a Poisson cluster process. Future salient events are unknown at time t. We forecast conditionally on a given calendar of future events, which is empty unless stated. If event arrivals are modelled as a marked Poisson process, the results below hold conditionally on its realisation, and therefore also after averaging over it. The choice of future calendar matters for calibration. A forecaster that sets the future calendar to empty while events in fact keep arriving is misspecified: its predictive law is not the conditional law of the count, and it under-forecasts. A forecaster that averages over a correct model for event arrivals gives the conditional law, so its probability integral transform is uniform when windows are pooled (Lemma 7.4). What no forecaster can do is *time* new events, which limits the recall of an alert rule rather than its pooled calibration (Section 8.6).

Lemma 7.1 (The future given the past). *Assume Assumption A1, and fix $t \geq 0$ and the calendar of events. For $s > t$, let $\boldsymbol{u}_t(s) = \sum_e \boldsymbol{\nu}_e(s)$ be the remaining immigrant intensity of all events in the calendar, and let $\boldsymbol{h}_t(s) = \int_{[0,t]} \boldsymbol{\Phi}(s-r)\,\mathrm{d}\boldsymbol{N}(r)$. Conditionally on $\mathcal{F}_t$, the incidents in (t,∞) form a Poisson cluster process. Its immigrants in stream k form a Poisson process with intensity*

$$\nu_{t,k}(s) = \mu_k(s) + u_{t,k}(s) + h_{t,k}(s), \qquad s > t,$$

and offspring are generated as in Section 3.3. If $\varrho(\boldsymbol{A}) < 1$, then $\mathbb{E}[\boldsymbol{N}(t, t+\Delta_{\mathrm{a}}] \mid \mathcal{F}_t] = \int_t^{t+\Delta_{\mathrm{a}}} (\mathcal{K}_t \boldsymbol{\nu}_t)(s)\,\mathrm{d}s$, with $\mathcal{K}_t$ as in (5.1).
Suppose in addition that $\varphi_{kj}(u) = \alpha_{kj}\beta_{kj}e^{-\beta_{kj}u}$ and $\psi_k(u) = \eta_k e^{-\eta_k u}$, and that there are no future events. Then, for $s > t$,

$$\begin{aligned} h_{t,k}(s) &= \sum_{j=1}^{M} x_{kj}(t)\, e^{-\beta_{kj}(s-t)}, \quad x_{kj}(t) = \sum_{i:\, t_i^j \leq t} \alpha_{kj}\beta_{kj} e^{-\beta_{kj}(t-t_i^j)}, \\ u_{t,k}(s) &= u_k(t)\, e^{-\eta_k(s-t)}, \qquad\qquad u_k(t) = \sum_{e:\, \tau_e \leq t} \upsilon_k S_e \eta_k e^{-\eta_k(t-\tau_e)}. \end{aligned} \tag{7.1}$$

The state $\boldsymbol{\chi}(t) = (\boldsymbol{x}(t), \boldsymbol{u}(t)) \in [0,\infty)^{M^2+M}$ is a piecewise-deterministic Markov process. Between incidents and events it decays exponentially. At a stream-j incident, x_{kj} jumps by $\alpha_{kj}\beta_{kj}$ for each k. At event e, u_k jumps by $\upsilon_k S_e \eta_k$. The conditional law of the future given $\mathcal{F}_t$ depends on $\mathcal{F}_t$ only through $\boldsymbol{\chi}(t)$, and the predictive mean is affine in $\boldsymbol{\chi}(t)$. The intensity just after t is $\lambda_k(t+) = \mu_k(t) + u_k(t) + \sum_j x_{kj}(t)$.

Lemma 7.1 makes exact forecasting possible, with no discretisation error. Generation 0 of the future consists of background immigrants on $(t, t+\Delta_{\mathrm{a}}]$, the remaining immigrants of past events, and the pending offspring of past incidents; those of stream j in stream k number Poisson$\{x_{kj}(t)(1-e^{-\beta_{kj}\Delta_{\mathrm{a}}})/\beta_{kj}\}$. Each generation-0 point then cascades as in Section 3.3, truncated at $t+\Delta_{\mathrm{a}}$, and J repetitions give a Monte Carlo predictive distribution with error of order $J^{-1/2}$. The predictive law can also be computed without simulation. Its probability generating function solves M^2 ordinary differential equations backwards in time, and a discrete Fourier transform then gives the predictive probability mass function (Remark D.1 in Appendix D.1).

7.2 Bayes-optimal alerting

Fix a window length Δ_{a} and a *surge set* $\mathcal{U} \subseteq \mathbb{Z}_+^M$ that is an upper set: if $\boldsymbol{n} \in \mathcal{U}$ and $\boldsymbol{n}' \geq \boldsymbol{n}$, then $\boldsymbol{n}' \in \mathcal{U}$. The leading example is $\mathcal{U} = \{\boldsymbol{n} : n_k \geq c\}$, at least c incidents in stream k. Weighted totals $\{\boldsymbol{n} : \boldsymbol{w}^\top \boldsymbol{n} \geq c\}$ with $\boldsymbol{w} \geq \boldsymbol{0}$ also qualify. The surge indicator is $Y_t = \mathbf{1}\{\boldsymbol{N}(t, t+\Delta_{\mathrm{a}}] \in \mathcal{U}\}$.

At time t the analyst holds information $\mathcal{G}_t$. This is the history $\mathcal{F}_t$ when data are complete and timely, and less under reporting delays (Section 7.3). The analyst chooses an alert $a_t \in \{0,1\}$ that is $\mathcal{G}_t$-measurable, and incurs the loss $c_{\mathrm{FA}}\, a_t(1-Y_t) + c_{\mathrm{miss}}(1-a_t)Y_t$ with $c_{\mathrm{FA}}, c_{\mathrm{miss}} > 0$. Let

$$\pi_t = \mathbb{P}(Y_t = 1 \mid \mathcal{G}_t), \qquad \pi^* = \frac{c_{\mathrm{FA}}}{c_{\mathrm{FA}} + c_{\mathrm{miss}}}.$$

Theorem 7.2 (Bayes-optimal alerting). *(a)* Bayes rule. *Among all $\mathcal{G}_t$-measurable alerts, the rule $a_t^* = \mathbf{1}\{\pi_t \geq \pi^*\}$ minimises the expected loss. Its risk is $\mathbb{E}\min\{c_{\mathrm{FA}}(1-\pi_t),\, c_{\mathrm{miss}}\pi_t\}$. Every other minimiser agrees with a_t^* almost surely outside $\{\pi_t = \pi^*\}$.*

(b) Monotonicity in the history. *Let $\mathcal{G}_t = \mathcal{F}_t$, and fix the parameters and the calendar of events. Let $\omega \subseteq \omega'$ be two histories on $[0,t]$, where ω' contains every incident of ω, with the same stream*

labels, and possibly more. Then there is a coupling of the predictive laws of $\boldsymbol{N}(t, t+\Delta_a]$ under ω and ω', $\boldsymbol{N}$ and $\boldsymbol{N}'$, with $\boldsymbol{N}' \geq \boldsymbol{N}$. In particular $\pi_t(\omega) \leq \pi_t(\omega')$ for every upper set $\mathcal{U}$.

(c) Exponential kernels: a monotone decision region. *Under the exponential specification of Lemma 7.1, $\pi_t = G_t(\chi(t))$ for a function G_t that is nondecreasing in each coordinate and Lipschitz. The Bayes alert region $\mathcal{A}_t = \{\chi : G_t(\chi) \geq \pi^*\}$ is therefore a closed upper set: if the rule alerts at χ, it alerts at every $\chi' \geq \chi$. The function G_t depends on t only through $\boldsymbol{\mu}$ on $(t, t+\Delta_a]$ and the calendar of future events. For constant $\boldsymbol{\mu}$ and no future events it does not depend on t.*

(d) Univariate case: a threshold curve. *Let $M = 1$, with $\mu > 0$ on $(t, t+\Delta_a]$ and $\mathcal{U} = \{n \geq c\}$ for some $c \geq 1$. Then $\pi_t = g_t(x(t), u(t))$, where g_t is continuous, nondecreasing in u and strictly increasing in x. The Bayes rule alerts if and only if*

$$x(t) \;\geq\; x_t^*(u(t)), \qquad x_t^*(u) = \inf\{x \geq 0 : g_t(x, u) \geq \pi^*\} < \infty.$$

The threshold curve $x_t^*(u)$ is nonincreasing in u, nondecreasing in c and in π^*, and nonincreasing in Δ_a. If $\eta = \beta$, or if no event has occurred so that $u \equiv 0$, then g_t depends on $x + u$ only. The rule is then a threshold on the intensity: alert if and only if $\lambda(t+) \geq \lambda_t^* := \mu(t) + x_t^*(0)$. In general, replacing $u(t)$ by 0 changes π_t by at most $L_t u(t)$, where L_t is a Lipschitz constant of G_t in (c), for instance $L_t = \Delta_a/(1-n)$.

Part (a) is standard Bayes decision theory. Part (b) says that the predictive law of $\boldsymbol{N}(t, t+\Delta_a]$ increases with the history in the usual multivariate stochastic order (Shaked and Shanthikumar, 2007). The decision depends on the two costs only through π^*, which is agreed with partners. The analysis plan's grid of cost ratios $c_{\text{miss}}/c_{\text{FA}} \in \{2, 5, 10\}$ corresponds to $\pi^* \in \{1/3, 1/6, 1/11\}$ (Appendix H.2).

The structural parts have three operational consequences. Under the model, an additional incident never turns an alert off. In the univariate case, the curve $x_t^*(u)$ can be tabulated once, and checking for an alert means comparing the current excitation with it; since an event's contribution $u(t)$ never vanishes, the single intensity threshold λ_t^* is exact only when $\eta = \beta$ or before any event. In the multivariate case, $\mathcal{A}_t$ is an upper set but in general not a half-space. The predictive mean is affine in $\chi(t)$, so the rule "alert if the predictive mean exceeds a level" has a half-space as its region. That rule is not optimal in general, because the tail of the future count also depends on its dispersion, which is larger when the excitation sits in a strongly self-exciting stream. With unknown parameters, part (a) applies to the posterior predictive probability. Parts (b)–(d) hold for each parameter value, but they need not hold for the posterior mixture, because the posterior itself depends on the history.

7.3 Nowcasting under reporting delays

Incidents reach the analyst late (Section 2.3). Near the forecast origin the record is therefore incomplete. The model's state is then too low, exactly when an alert matters most.

Assumption A2 (Recording and reporting). Each stream-k incident is recorded with probability $p_k \in (0, 1]$. A recorded incident that occurs at s becomes available to the analyst at $s + d$, where the delay d has distribution function F_k on $[0, \infty)$. Recording indicators and delays are independent across incidents, and independent of the incident process and the calendar of events.

Fix the analysis time t, and let $O_k^{(t)}$ count the occurrence times of the stream-k incidents available by t. Let $\mathcal{G}_s^\circ$ be generated by $\mathcal{F}_s$ together with the recording indicators and delays of the incidents in $[0, s]$. We call an incident *available* if it is recorded and has been reported by t.

Proposition 7.3 (Nowcasting the state). *Assume Assumptions A1 and A2.*

(a) Thinning. *On* $[0,t]$, $O_k^{(t)}$ *has* $\mathcal{G}^\circ$*-intensity* $p_k F_k(t-s)\,\lambda_k(s)$, *and the incidents not yet available have* $\mathcal{G}^\circ$*-intensity* $\{1-p_kF_k(t-s)\}\lambda_k(s)$. *Hence* $\mathbb{E}\,O_k^{(t)}(\mathrm{d}s) = p_kF_k(t-s)\,\mathbb{E}\lambda_k(s)\,\mathrm{d}s$.

(b) Unbiased nowcast. *Let* $\delta \geq 0$ *satisfy* $F_j(\delta) > 0$ *for every* j. *For* $s \geq t-\delta$, *define the weighted carried-over excitation*

$$\hat{\boldsymbol{h}}(s) = \sum_{j=1}^{M} \sum_{\substack{i:\, s_i^j \leq t-\delta,\\ \textit{available}}} \frac{\boldsymbol{\Phi}_{\cdot j}(s-s_i^j)}{p_j F_j(t-s_i^j)}, \tag{7.2}$$

where $\boldsymbol{\Phi}_{\cdot j}$ *is the* j*th column of* $\boldsymbol{\Phi}$. *Then* $\mathbb{E}[\hat{\boldsymbol{h}}(s) \mid \mathcal{F}_{t-\delta}] = \boldsymbol{h}_{t-\delta}(s)$ *for every* $s \geq t-\delta$. *If* $\varrho(\boldsymbol{A}) < 1$, *the nowcast* $\hat{\boldsymbol{\lambda}}(t) = \mathcal{K}_{t-\delta}(\boldsymbol{\mu} + \boldsymbol{u}_{t-\delta} + \hat{\boldsymbol{h}})(t)$ *satisfies* $\mathbb{E}[\hat{\boldsymbol{\lambda}}(t) \mid \mathcal{F}_{t-\delta}] = \mathbb{E}[\boldsymbol{\lambda}(t) \mid \mathcal{F}_{t-\delta}]$, *and hence* $\mathbb{E}\hat{\boldsymbol{\lambda}}(t) = \mathbb{E}\boldsymbol{\lambda}(t)$.

For exponential kernels the same holds for the state. Let $\hat{\boldsymbol{x}}(t)$ *be obtained by computing the state at* $t-\delta$ *with the weights of* (7.2) *and propagating it to* t *with the mean dynamics of Proposition 5.2(c). Then* $\hat{\boldsymbol{x}}(t)$ *is conditionally unbiased for* $\mathbb{E}[\boldsymbol{x}(t) \mid \mathcal{F}_{t-\delta}]$. *If every* $F_j(0) > 0$, *one can take* $\delta = 0$. *In that case* $\hat{\boldsymbol{h}}$ *is the Horvitz–Thompson estimator of* $\boldsymbol{h}_t$, *and it is conditionally unbiased given the complete history* $\mathcal{F}_t$.

(c) The naive state is biased downward. *Let* $\tilde{x}_{kj}(t) = \sum_{i:\, s_i^j \leq t,\ \textit{available}} \varphi_{kj}(t-s_i^j)/p_j$ *be the state computed from the available incidents, with recording corrected but delays ignored. Then*

$$\mathbb{E}\big[\tilde{x}_{kj}(t) \mid \mathcal{F}_t\big] = \sum_{i:\, s_i^j \leq t} F_j(t-s_i^j)\,\varphi_{kj}(t-s_i^j) \;\leq\; x_{kj}(t) = \sum_{i:\, s_i^j \leq t} \varphi_{kj}(t-s_i^j). \tag{7.3}$$

The inequality is strict whenever some incident with $\varphi_{kj}(t-s_i^j) > 0$ *has* $F_j(t-s_i^j) < 1$. *Now suppose* $\boldsymbol{\mu}$ *is constant, there are no events,* $\varrho(\boldsymbol{A}) < 1$ *and the process is stationary. Then* $\mathbb{E}\tilde{x}_{kj}(t) = \mathbb{P}(d_j \leq U_{kj})\,\mathbb{E}x_{kj}(t)$, *where* $d_j \sim F_j$ *and* $U_{kj} \sim g_{kj}$ *are independent. For exponential kernels,* $\mathbb{P}(d_j \leq U_{kj}) = \mathbb{E}e^{-\beta_{kj}d_j}$.

(d) Consequence for alerts. *If* $p_k \equiv 1$, *then* $\tilde{\boldsymbol{x}}(t) \leq \boldsymbol{x}(t)$ *pathwise. By Theorem 7.2(c), the alert probability computed from the naive state is then at most the one computed from the true state. A rule based on the naive state alerts only when the rule based on the true state alerts. Ignoring delays can postpone an alert but never brings one forward.*

Part (a) restates (2.1) at the level of intensities: any likelihood for the data available at t must carry the factor $p_kF_k(t-s)$ on $\lambda_k(s)$, and integrate over the incidents not yet available. Data augmentation, by Markov chain Monte Carlo or stochastic EM, imputes them, the self-exciting analogue of Bayesian nowcasting (Höhle and an der Heiden, 2014; McGough et al., 2020). A simpler route, adequate for estimating ϑ, restricts the likelihood to the settled part of the record, $s \leq t-L$ with $F_k(L) \approx 1$, and uses (b) only for the state near t.

The size of the bias in (c) can be large. With exponentially distributed delays of mean two days and a kernel rate $\beta = 1.5$ per day, $\mathbb{E}e^{-\beta d} = 1/(1+2\beta) = 0.25$. On average, the naive state then carries a quarter of the true excitation.

The unbiased nowcast has a cost in variance. The weights $1/\{p_jF_j(t-s)\}$ are large for recent incidents, and without a lag floor they are unbounded: if $F_j(0) = 0$, $F_j(u) = O(u)$ as $u \downarrow 0$ and $\varphi(0) > 0$, the

conditional variance of the weighted sum over all incidents up to t has infinite expectation. This is why (b) requires $F_j(\delta) > 0$; a floor $\delta > 0$ discards the available incidents in $(t - \delta, t]$.

An unbiased state does not give a calibrated tail probability. G_t is nonlinear, and a noisy state that happens to be high produces a high plug-in probability more often than a surge. Plugging $\hat{\boldsymbol{x}}(t)$ into G_t, or simulating forward from it, therefore does not give $\mathbb{P}(Y_t = 1 \mid \mathcal{G}_t)$ for the available data $\mathcal{G}_t$. The Bayes optimality of Theorem 7.2(a) then holds only for the oracle rule that knows the true state. Plugging a nowcast into the threshold rule is a shortcut, not the Bayes rule for delayed data. Algorithm 2 in Appendix D.2 approximates the predictive probability given the available data instead. It imputes the incidents that are not yet available, by sequential importance sampling with the intensities of (a) (Lemma D.2), and so targets the Bayes rule under delays. We have not run it: the simulation study evaluates only the plug-in shortcut (Section 8.6). Whichever rule is used, the probability integral transform (Lemma 7.4) and a reliability diagram of the alert probabilities should be checked on held-out data before operational use. If they show miscalibration, the probabilities should be recalibrated, for instance by isotonic regression of Y_t on $\hat{\pi}_t$ over the training period, before the threshold π^* is applied.

7.4 Evaluation

Forecasts are evaluated against *settled* counts. These are the counts in each window once reporting has essentially completed, not the counts available at the forecast origin. Let P_t be the predictive distribution of a count $Y_t = N_k(t, t + \Delta_a]$, with distribution function $P_t(y) = P_t(Y \le y)$.

Proper scoring rules. A scoring rule is strictly proper if the true predictive distribution uniquely optimises its expected value (Gneiting and Raftery, 2007). We use two scores, both negatively oriented:

$$\mathrm{LogS}(P, y) = -\log P(Y = y), \qquad \mathrm{CRPS}(P, y) = \sum_{z=0}^{\infty} \big\{P(z) - \mathbf{1}\{y \le z\}\big\}^2 = \mathbb{E}_P|X - y| - \tfrac{1}{2}\mathbb{E}_P|X - X'|, \tag{7.4}$$

where X and X' are independent with law P. The log score is local and requires $P(y) > 0$, so Monte Carlo probability mass functions need a floor, or the exact law of Remark D.1 should be used. The CRPS for counts is strictly proper relative to distributions with a finite mean, is insensitive to small probabilities, and is estimated from predictive draws by the sample version of the second expression (Czado et al., 2009; Gneiting and Raftery, 2007). For the surge event itself, the Brier score $(\pi_t - Y_t)^2$ is strictly proper.

Calibration. For discrete counts the probability integral transform (PIT) is randomised:

$$U_t = P_t(Y_t - 1) + V_t\big\{P_t(Y_t) - P_t(Y_t - 1)\big\},$$

with V_t uniform on $(0, 1)$ and independent of everything else, and $P_t(-1) = 0$. Czado et al. (2009) also give a non-randomised version.

Lemma 7.4 (PIT for counts). *(a) If P_t is the conditional law of Y_t given the forecaster's information $\mathcal{G}_t$, then U_t is uniform on $(0, 1)$ and independent of $\mathcal{G}_t$.*

(b) Suppose forecasts are issued at $t_m = t_0 + m\Delta_a$, so that the windows do not overlap. Suppose also that each past count Y_{t_l}, $l < m$, is $\mathcal{G}_{t_m}$-measurable, that is, past windows have settled by the next forecast origin. Then $U_{t_1}, U_{t_2}, \ldots$ are independent and uniform on $(0, 1)$.

This is the count-data form of the prequential principle of Dawid (1984); see also Gneiting et al. (2007) and Czado et al. (2009). A slope in the PIT histogram indicates bias, a U shape forecasts that are too narrow,

and a hump forecasts that are too wide. With overlapping windows, or with counts that have not settled, the U_t are uniform but dependent, so tests of uniformity need block resampling.

Alerts. For a threshold π, the hit rate (recall) is $\mathrm{HR}(\pi) = \mathbb{P}(\pi_t \geq \pi \mid Y_t = 1)$, the false-alarm rate $\mathrm{FAR}(\pi) = \mathbb{P}(\pi_t \geq \pi \mid Y_t = 0)$ and the precision $\mathrm{PPV}(\pi) = \mathbb{P}(Y_t = 1 \mid \pi_t \geq \pi)$. The ROC curve, (FAR, HR), does not depend on how common surges are. When surges are rare, the precision–recall curve, (HR, PPV), is more informative about the burden of false alarms. The realised cost over a test period, $\sum_t \{c_{\mathrm{FA}} a_t (1 - Y_t) + c_{\mathrm{miss}} (1 - a_t) Y_t\}$, is the criterion of Theorem 7.2. The comparison with baselines in the analysis plan (Appendix H.7) is made in these units. Alerts issued daily for overlapping windows have dependent outcomes, so intervals use a block bootstrap.

The Brier score is twice the average realised cost when the normalised cost π^* is drawn uniformly from $(0, 1)$ (Remark D.3 in Appendix D.3). It summarises performance when partners cannot agree on one cost ratio; when they can, the realised cost at π^* is the relevant measure.

Regime shifts. Alerts concern the next Δ_{a} days with the parameters held fixed. Whether the parameters themselves have changed, through a sustained rise either in the background or in contagion, is a different question. Appendix D.4 sketches a CUSUM monitor for it (Remark D.4). It inherits the identification limits of Section 4 and must be run on settled data or with the nowcast of Proposition 7.3.

8 Simulation study

All data in this section are synthetic. Every dataset is simulated from a model whose parameters we chose, so the truth is known. No real incident data are used, and nothing in this section is a finding about antisemitic incidents. Every number is taken from `results/sim_results.json`, which `code/run_experiments.py` writes; the design table, configurations, seeds, software versions, run times and code hashes are in Appendix F. Time is measured in arbitrary units, which we call days.

8.1 Design

Six experiments test the claims of Sections 4 to 7 (Table 8): recovery by EM (E1), surge attribution (E5), spurious contagion (E2), thinning (E3), the online-to-offline test (E4) and alerting (E6). Data are drawn by exact cluster simulation (Appendix E.3) from the empty history at time 0, with exponential kernels and shock responses. Fits use Algorithm 1 with the exact window terms, one decay rate β for all excitation kernels, one response rate η for all streams unless stated otherwise, the default multistart over decay rates and a relative tolerance of 10^{-8} (10^{-9} in E4). Intervals are nominal 95% Wald intervals from the observed information (Appendix E.4), delta-method for derived quantities. Monte Carlo standard errors (MC SE) are in parentheses (Appendix F.5); for a coverage of 0.95 it is 0.022 with $R = 100$ replicates and 0.034 with $R = 40$.

8.2 E1 and E5: recovery and attribution

E1 checks EM on the shock-augmented model, whose calendar of events breaks the stationarity assumed in Section 6.1; E5 checks the closed forms of Theorem 5.1 and Proposition 5.2 and the ancestral reading of the online channel (Proposition 5.4(a)) against branching labels. Appendix F.4 gives the design and full results. In E1 ($M = 3$, $\varrho(\boldsymbol{A}) = 0.485$, $R = 100$ per T) all 200 fits converged with positive definite information. At $T = 2000$ the nine α_{kj} have mean |bias| 0.002, mean RMSE 0.026 and coverage 0.92–0.99, as do the entries of the multiplier $\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{A})^{-1}$; doubling T reduces the RMSE by about $2^{-1/2}$. The online loading

$\upsilon_3 = 6$ is overestimated by 0.137 (0.079), with the same sign at $T = 1000$. The design is benign: $\varrho(\boldsymbol{A})$ is far from one and every α_{kj} far from zero. In E5 the exact decomposition agrees with the branching labels of 20,000 paths: over 11 quantities per horizon the largest $|z|$ is 1.50, and the online-channel contagion of offline region A at $H = 14$ is 5.779 exactly and 5.800 (0.026) by simulation. Plug-in estimates from the E1 fits cover in 0.91–0.96 of replicates over 45 attribution quantities at $T = 2000$ (Table 11(b) in Appendix F.4). The direct component is less precise than the contagion component (RMSE 1.598 against 1.153 for region A): the loadings rest on 17 events, the branching matrix on thousands of incidents.

8.3 E2: spurious contagion

Claim. Without contagion but with a fluctuating rate, a constant-background Hawkes fit finds contagion. For second-order estimators the pseudo-true branching ratio is $n^* = 1 - (1 + \kappa)^{-1/2}$ (Theorem 4.7(a)–(c)); for maximum likelihood, Theorem 4.7(d) proves only that it is positive, and Remark 4.8(ii) leaves its relation to n^* open. Known shock times restore identification (Corollary 4.9(i)); we also ask whether a flexible background does.

Design. Every dataset has mean rate $\Lambda = 2$ and no contagion. Two families are shot-noise Cox processes (Definition 4.1) with shock decay $\gamma = 0.5$ and m incidents per shock: *rare* ($m = 16$, $\kappa \in \{0.5, 1, 2, 4, 8\}$) and *frequent* ($m = 4$, $\kappa \in \{0.5, 1, 2, 3\}$). The third family switches the rate between $\Lambda \pm \delta$ at rate 0.01 in each direction, with $\kappa \in \{0.5, 2, 8\}$. With $T = 1000$ and $R = 20$, rare-family paths carry 1–77 shocks (including a 40-day lead-in) and frequent-family paths 54–424 (Table 2(a)), so a difference between the families could be finite-sample bias. To separate the two, the rare family at $\kappa \in \{2, 4\}$ was also run with $T = 10{,}000$ (135–341 shocks per path; constant background only). We fit (i) a constant background; (ii) a log-linear cubic B-spline background with 20 basis functions, knots about 59 days apart; and (iii) a constant background with the shock times as a calendar ($S_e = 1$; loading and response rate estimated), for all rare configurations and the frequent ones with $\kappa \le 1$ (Appendix F.2 explains the cost of the others). For the shot-noise families we also compute the dispersion functional $n_F = 1 - F^{-1/2}$ of Theorem 4.7(c) from counts in 25-day bins; its finite-window target is n_w^* (Appendix F.2).

What is reported. Without contagion the likelihood has a ridge on which the decay tends to zero while $\hat{\alpha}$ grows and the in-window offspring stay few. Flagging a fit as on the ridge when $\hat{\beta}T < 1$ (its kernel has not decayed within the window), nine of the 620 fits at $T = 1000$ are on it: six of the 140 shocks-known fits, two constant-background fits and one spline fit for slow switching (Table 2(a)). Eight have $\hat{\beta}T < 0.1$ and $\hat{\alpha}$ between 1.7 and 29.1; no long-window fit is on the ridge. We therefore report the in-window branching ratio $\hat{n}_{\text{eff}} = \hat{\alpha}\, N(T)^{-1} \sum_l \{1 - e^{-\hat{\beta}(T - t_l)}\}$, the expected number of in-window offspring per incident and the share of incidents that EM attributes to self-excitation. For the constant-background fits to the shot-noise families it equals $\hat{\alpha}$ to within 0.003. Its distribution is skewed near zero, so Table 2(a) gives medians with the means: the mean spline estimate for slow switching at $\kappa = 8$, for example, is 0.028, but 0.018 without its ridge fit, and the median is 0.003.

Constant background. Spurious contagion appears in every configuration and grows with κ (Figure 4(a)). For the rare family the gaps $\hat{n}_{\text{eff}} - n^*$ are −0.059 (0.021), −0.018 (0.017), −0.058 (0.014), −0.066 (0.0085) and −0.039 (0.008) for $\kappa = 0.5, 1, 2, 4, 8$; for the frequent family they lie within ±0.009, each within one MC SE. The rare-family gaps are not finite-sample bias. With $T = 10{,}000$ they are −0.062 (0.004) at $\kappa = 2$ and −0.068 (0.002) at $\kappa = 4$, as large as at $T = 1000$, although the long paths carry as many shocks as the frequent-family paths. Over the same tenfold increase in T the dispersion estimator settles on its target (0.408 and 0.541, MC SE 0.006 and 0.004, against $n_w^* = 0.407$ and 0.538). The maximum likelihood

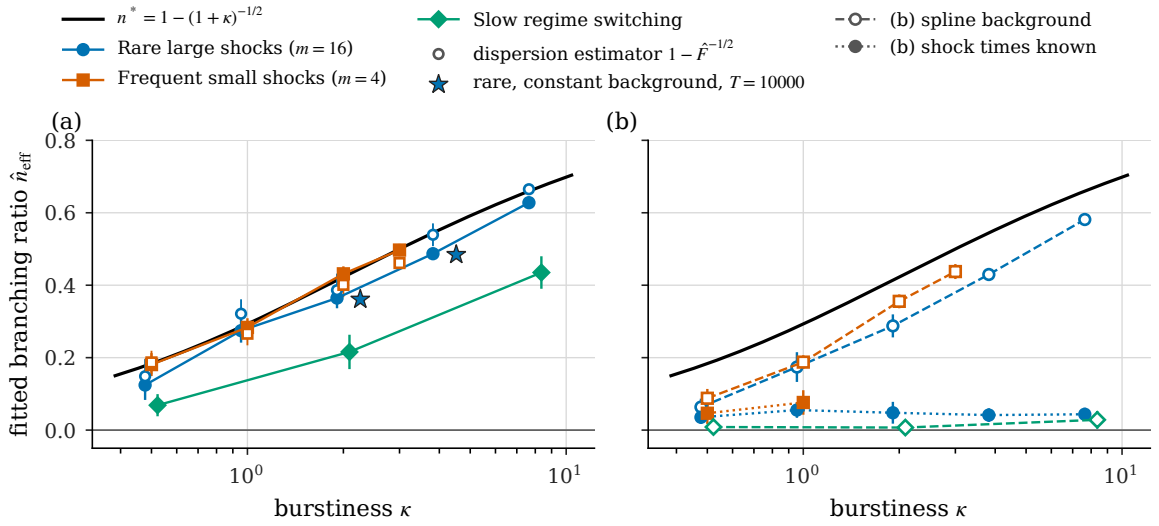


Figure 4. E2, spurious contagion in synthetic data with *no* contagion ($R = 20$ per point; bars ±1.96 MC SE). (a) $\hat{n}_{\text{eff}}$ of constant-background fits against the burstiness κ ($T = 1000$), with n^* (line), the dispersion estimator (open markers) and the rare family with $T = 10{,}000$ (stars, shifted right). (b) Spline background (open markers, dashed) and shock times known (filled markers, dotted). Means; medians are in Table 2(a).

estimator thus converges to a pseudo-true value below n^* when shocks are rare and large, as Remark 4.8(ii) allows: the likelihood responds to structure beyond second order, where the members of the matched family of Theorem 4.2 differ. For slow switching the fits fall far short of n^* (0.069, 0.216 and 0.435 against 0.184, 0.423 and 0.667); with about ten regime switches in 1000 days, a finite-window effect remains a second explanation there, which we did not test.

Remedies. A spline background removes the artefact of slow switching (means 0.008, 0.007 and 0.028; medians 0.005, 0.002 and 0.003) but not that of fast shocks (Figure 4(b)): it lowers $\hat{n}_{\text{eff}}$ by only 0.05–0.10, for example from 0.628 to 0.581 for the rare family at $\kappa = 8$. Responses that decay within days cannot be represented by knots two months apart (Section 6.1); a background flexible enough to follow them would also absorb genuine contagion, which E2 does not measure. Known shock times largely remove the artefact, where tested: for the rare family at every κ and the frequent family at $\kappa \leq 1$, $\hat{n}_{\text{eff}}$ has means 0.035–0.076 (medians 0.012–0.055) against 0.12–0.63 without the calendar, and the loadings are recovered (15.2–16.4 against $m = 16$; 3.94 and 3.87 against $m = 4$). The small positive remainder is expected of an estimator constrained to $\alpha \geq 0$ at a true value on the boundary (Proposition C.3(b)). Where tested, the conclusion is that of Corollary 4.9(i): spurious contagion from unrecorded shocks is removed by recording the shocks, not by smoothing.

8.4 E3: thinning and the dark-figure correction

Claim. Independent recording with probability p maps n to $n' = h_p(n)$ (Theorem 4.10), and h_p^{-1} undoes the attenuation (Corollary 4.11), with a floor on its precision set by the uncertainty about p. Theorem 4.10(d) leaves open whether the maximum likelihood estimator converges to n'. Under heterogeneous recording the homogeneous correction errs as Proposition A.6 predicts.

Design. A univariate exponential Hawkes process with $\beta = 1$, $\Lambda = 2$, $T = 1500$ and $n \in \{0.3, 0.5, 0.7\}$ ($R = 40$) is thinned at $p \in \{1, 0.8, 0.6, 0.4, 0.2\}$, with nested recorded sets, and refitted. The *naive* estimate

Table 2. Results of E2 and E3 (synthetic data). (a) E2, data *without* contagion: in-window branching ratio $\hat{n}_{\text{eff}}$, mean (median) over $R = 20$ replicates; shocks per path, median [minimum, maximum], including the 40-day lead-in; Ridge: number of fits with $\hat{\beta}T < 1$ among the constant, spline and shocks-known fits (a hyphen: fit not run). (b) E3 at $p = 0.4$: naive and corrected estimates of n ($R = 40$). Floor: the p-term $r\,n(1-n)(2-n)/2$ of the standard deviation in (4.6), for relative error r of $\hat{p}$. Cov.: coverage of nominal 95% intervals.

(a) E2: spurious contagion

					$\hat{n}_{\text{eff}}$, mean (median)			
Family	κ	T	Shocks per path	n^*	Constant	Spline	Shocks known	Ridge
Rare ($m = 16$)	0.5	1000	3.5 [1, 8]	0.184	0.124 (0.119)	0.064 (0.053)	0.035 (0.023)	0/0/0
	1	1000	9 [4, 20]	0.293	0.275 (0.294)	0.174 (0.192)	0.055 (0.055)	0/0/1
	2	1000	15.5 [9, 30]	0.423	0.365 (0.392)	0.288 (0.299)	0.048 (0.017)	0/0/2
	4	1000	34.5 [25, 43]	0.553	0.487 (0.483)	0.430 (0.436)	0.041 (0.033)	0/0/0
	8	1000	65.5 [52, 77]	0.667	0.628 (0.628)	0.581 (0.582)	0.044 (0.023)	0/0/0
Frequent ($m = 4$)	0.5	1000	69.5 [54, 86]	0.184	0.181 (0.174)	0.088 (0.085)	0.047 (0.012)	0/0/1
	1	1000	133.5 [104, 149]	0.293	0.284 (0.292)	0.188 (0.181)	0.076 (0.048)	0/0/2
	2	1000	258 [240, 290]	0.423	0.431 (0.426)	0.355 (0.347)	—	0/0/-
	3	1000	390 [349, 424]	0.500	0.497 (0.494)	0.438 (0.434)	—	0/0/-
Slow switching	0.5	1000	—	0.184	0.069 (0.066)	0.008 (0.005)	—	0/0/-
	2	1000	—	0.423	0.216 (0.221)	0.007 (0.002)	—	2/0/-
	8	1000	—	0.667	0.435 (0.425)	0.028 (0.003)	—	0/1/-
Rare ($m = 16$), long	2	10000	157.5 [135, 183]	0.423	0.361 (0.359)	—	—	0/-/-
	4	10000	322 [293, 341]	0.553	0.484 (0.483)	—	—	0/-/-

(b) E3: thinning and the dark-figure correction

Setting	Estimator	Target	Mean	Bias	RMSE	Floor	Cov. (%)
$n = 0.5$, $p = 0.4$	$\hat{n}'$ (naive)	0.500	0.315	−0.185	0.189	—	2.5
	$h_p^{-1}(\hat{n}')$, p known	0.500	0.488	−0.012	0.042	—	95.0
	$h_{\hat{p}}^{-1}(\hat{n}')$, 10% error	0.500	0.491	−0.009	0.046	0.019	97.5
	$h_{\hat{p}}^{-1}(\hat{n}')$, 30% error	0.500	0.503	+0.003	0.073	0.056	87.5
	$h_{\hat{p}}^{-1}(\hat{n}')$, 50% error	0.500	0.525	+0.025	0.128	0.094	85.0
$n = 0.7$, $p = 0.4$	$\hat{n}'$ (naive)	0.700	0.553	−0.147	0.152	—	5.0
	$h_p^{-1}(\hat{n}')$, p known	0.700	0.698	−0.002	0.031	—	95.0
	$h_{\hat{p}}^{-1}(\hat{n}')$, 10% error	0.700	0.704	+0.004	0.035	0.014	92.5
	$h_{\hat{p}}^{-1}(\hat{n}')$, 30% error	0.700	0.720	+0.020	0.061	0.041	80.0
	$h_{\hat{p}}^{-1}(\hat{n}')$, 50% error	0.700	0.742	+0.042	0.101	0.068	80.0
heterog., $\mathrm{Cov}(p_X, n_X) > 0$	$h_{\bar{p}}^{-1}(\hat{n}'')$	0.500	0.567	+0.067	0.076	—	—

is $\hat{n}'$; the *corrected* estimate is $h_p^{-1}(\hat{n}')$ with the delta-method variance (4.6), with p known or estimated by $\hat{p} = p(1+rZ)$, $Z \sim N(0,1)$, clipped to $[0.02, 1]$, with standard error $r\hat{p}$ and relative error $r \in \{10, 30, 50\}\%$ (one Z for all three). In the heterogeneous design two equally likely marks have $n_x = (0.1, 0.9)$, so $\bar{n} = 0.5$, and are recorded with $(p_1, p_2) \in \{(0.2, 0.8), (0.5, 0.5), (0.8, 0.2), (1, 1)\}$ ($R = 30$).

Results. The maximum likelihood estimate tracks $n'(p)$: over the 15 combinations of n and p, |mean of $\hat{n}' - n'(p)| \le 0.0104$ (MC SE 0.0062). Ignoring thinning is not harmless: at $n = 0.5$ the naive interval covers the true n in 0.975, 0.825, 0.175, 0.025 and 0 of replicates for $p = 1, \dots, 0.2$ (Figure 5(a)). The correction with p known has bias between −0.0004 and −0.0155 and coverage 0.925–1 at $n = 0.5$ and 0.90–0.95 at $n = 0.7$, and 10% error in $\hat{p}$ changes little (Table 2(b)). The exception is weak excitation with heavy thinning: at $n = 0.3$ and $p \le 0.4$, n' is within a few standard errors of the boundary (Proposition C.3(b)), the fitted decay is erratic (means 23.4 and 110.8 against 0.83 and 0.77) and coverage falls to 0.85 and 0.70; profile-likelihood or bootstrap intervals are needed there.

Larger errors in $\hat{p}$ expose the floor of Corollary 4.11. With $\mathrm{SD}(\hat{p}) = rp$, the second term of (4.6) contributes the standard deviation $r\,n(1-n)(2-n)/2$, which depends on neither p nor T; at $n = 0.5$ it is 0.019, 0.056 and 0.094 for $r = 10$, 30 and 50%. Across the five values of p the RMSE of the corrected estimate is 0.030–0.076 with 10% error, as with p known, but 0.051–0.096 with 30% and 0.099–0.167 with 50% (Figure 5(c)), never much below the floor (the lowest values are at $p = 1$, where $\hat{p}$ is clipped). Uncertainty about coverage, not the number of incidents, then limits precision, and the delta method understates it: the correction grows faster as $\hat{p}$ falls than it shrinks as $\hat{p}$ rises, so the estimates are skewed upwards (at $n = 0.5$ and $r = 50\%$, means 0.503–0.569, medians 0.497–0.531), and coverage falls to 0.875–0.975 for $r = 30\%$ and 0.775–0.925 for $r = 50\%$ ($n = 0.5$). With coverage estimates this uncertain, p is better propagated by draws.

Under heterogeneous recording the homogeneous correction errs in the direction Proposition A.6 predicts (Figure 5(d)): its bias relative to $\bar{n}$ is +0.0665, +0.0273, +0.0217 and −0.0301 for positive, zero (equal or complete recording) and negative $\mathrm{Cov}(p_X, n_X)$, against theoretical +0.077, +0.036, +0.036 and −0.021; the fitted $\hat{n}''$ lies 0.008–0.014 below its second-order value.

8.5 E4: the online-to-offline test and common shocks

Under $\alpha_{\mathrm{off,on}} = 0$ the likelihood-ratio statistic should have the chi-bar-square limit $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ (Proposition C.3); an omitted common shock that reaches online volume first should produce online-to-offline Granger causality (Remark 4.13(i)), and recording it should remove it (Corollary 4.9(i)). Two streams have $\boldsymbol{A} = \left(\begin{smallmatrix} 0.3 & \alpha_{\mathrm{off,on}} \\ 0.2 & 0.4 \end{smallmatrix}\right)$ and $T = 600$; the shared decay stays identified under H_0 (Remark C.4(i)). Common shocks arrive at rate 0.05 and reach the online stream within about half a day, the offline stream over about four days. The test holds its size (Table 11(a) and Figure 9, both in Appendix F.4). Under H_0 ($R = 160$) it rejects in 0.0875 (0.022), 0.0625 (0.019) and 0.01875 (0.011) of replicates at the 10, 5 and 1% levels, and the statistic is zero in 0.5625 (0.039) of them; the naive χ_1^2 reference rejects in 0.025 at 5%. The power is 0.617, 0.983 and 1.0 for $\alpha_{\mathrm{off,on}} = 0.05$, 0.1 and 0.2 ($R = 60$). With the common shocks omitted the test rejects in all 60 replicates (mean estimate 0.072), with no online-to-offline excitation; with the calendar in the model it rejects in 0.083 (0.036), with mean estimate 0.010. The Davies problem (Section 6.5) was not studied (its bootstrap needs about 600 EM fits per dataset); the block tests of the plan need it, so it must pass the synthetic dry run (Appendix H.6, V4).

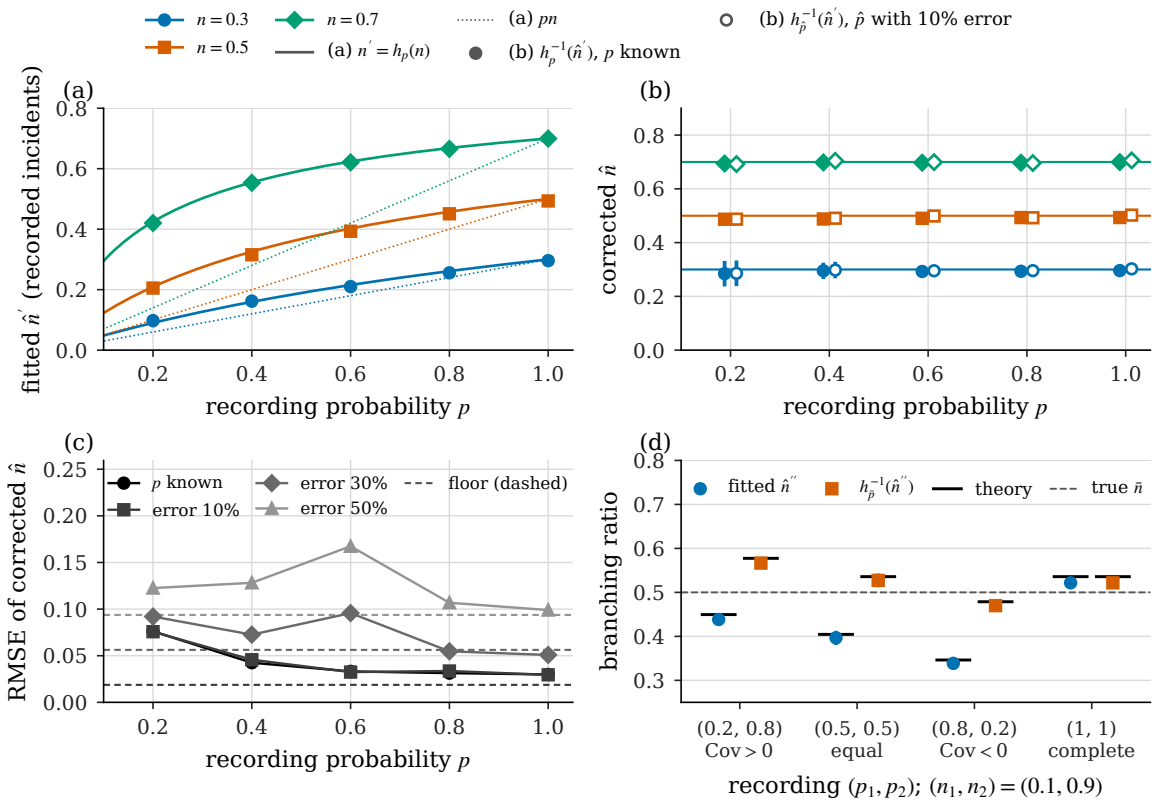


Figure 5. E3, thinning (synthetic data; bars ±1.96 MC SE). (a) Fitted branching ratio of the recorded incidents against p, with $n' = h_p(n)$ (solid) and pn (dotted); $R = 40$. (b) Corrected estimates with p known (filled) or estimated with 10% error (open); lines mark the true n. (c) The variance floor at $n = 0.5$: RMSE of the corrected estimate with p known and with 10, 30 and 50% error in $\hat{p}$, and the floor $r\,n(1-n)(2-n)/2$ (dashed); at $p = 1$, $\hat{p}$ is clipped at 1. (d) Heterogeneous recording of two marks with $(n_1, n_2) = (0.1, 0.9)$: fitted $\hat{n}''$ and the homogeneous correction against Proposition A.6 (solid) and $\bar{n} = 0.5$ (dashed); $R = 30$.

8.6 E6: alerting and nowcasting

Claim. Thresholding the predictive probability of a surge at π^* minimises the expected cost (Theorem 7.2(a)). If the predictive distribution is the conditional law given the forecaster's information, the randomised PIT is uniform (Lemma 7.4(a)). Reporting delays bias the naive state downward, by the factor $\mathbb{E}e^{-\beta d}$ under stationarity, and the delay-weighted state is unbiased (Proposition 7.3).

Design. Two streams, offline (the target) and online, have $\boldsymbol{A} = \left(\begin{smallmatrix} 0.35 & 0.10 \\ 0.25 & 0.45 \end{smallmatrix}\right)$, and external events arrive as a Poisson process at rate $1/30$ per day with log-normal sizes. Each of 12 replicates has a training period $[0, 1000]$ and 150 weekly forecast windows. A surge is a window with at least $c = 21$ offline incidents, the 90th percentile in a separate pilot; the surge rate in the 1800 test windows is 0.0967 (0.0070). At each origin t the predictive distribution of the offline count is estimated from 2000 cluster continuations of the state at t (Lemma 7.1), with the true parameters or with EM estimates from the training period, in two versions. One assumes *no future events* after t. The other *integrates* over them, adding to each continuation the clusters of events drawn from the event process for $(t, t + 7]$ (true rate and size law, or those of the training calendar with fitted loadings); this is the conditional law given the past, to which Lemma 7.4(a) applies (Section 7.1). The baselines are a moving-average Poisson forecast (mean of the last four weekly counts) and climatology. Alerts use $c_{\mathrm{FA}} = 1$ and $c_{\mathrm{miss}} \in \{2, 5, 10\}$, so $\pi^* \in \{1/3, 1/6, 1/11\}$. Under delays (offline only: zero with probability 0.3, else exponential with mean three days), the fitted forecast plugs the naive or the Horvitz–Thompson state (Proposition 7.3(b)) into the continuation. This plug-in is the shortcut that Section 7.3 warns is not the Bayes rule for delayed data; Algorithm 2 itself was not run. Surges are defined by the 90th rather than the plan's 95th percentile, and the baselines are not the plan's background-only model and persistence rule (Appendix H.2).

Decisions. The Hawkes rules have the lowest realised cost at every cost ratio (Table 3, Figure 6(a)), for example 0.285 per window at $c_{\mathrm{miss}} = 5$ with true parameters against 0.451 for the moving average and 0.483 for climatology, which is essentially the best constant rule. The comparison favours the Hawkes model: the data are generated by it, and the baselines are simple ones, not the plan's. Fitting the parameters costs almost nothing. Integrating over future events changes the costs little (paired differences -0.001, -0.002 and -0.014, MC SEs 0.002, 0.007 and 0.016) but improves the Brier score from 0.0655 to 0.0628 (paired MC SE 0.0006). It raises the surge probability in every window, trading precision for recall at $\pi^* = 1/6$ (0.503 and 0.511 against 0.614 and 0.466).

Calibration. Pooled over all windows, the forecast that integrates over future events is calibrated (Figure 6(d)). With true parameters the PIT gives $\chi^2 = 3.3$ on 9 degrees of freedom ($p = 0.95$), and the mean forecast exceeds the outcome by 0.07 (0.18) incidents per window. With fitted parameters $\chi^2 = 17.1$ ($p = 0.048$) and the bias is +0.38 (0.22), because the event rate estimated from the training calendar, 0.0353 (0.0018) on average, exceeds $1/30$. The forecast that assumes no future events is not calibrated: $\chi^2 = 175.3$, and it is too low by 1.91 (0.18) incidents per window with true parameters. That miscalibration comes from a misspecified forecaster that leaves out the event process, not from the method. What no forecaster can do is *time* the events. In 108 of the 174 surge windows (62%), an external event arrives after the forecast origin. Conditionally on whether one arrives, even the integrating forecast is not calibrated, and cannot be: its PIT values are too low in the 1424 windows without a new event and too high in the 376 with one, 36.4% of them in the top decile (Figure 6(e)). The limitation shows in recall: at $\pi^* = 1/6$ the integrating rule with true parameters detects 89% of the surges in windows without a new event but 28% of those in windows with one. Alerts should be reissued as soon as a salient event occurs.

Table 3. E6, alerting (synthetic data; 12 replicates × 150 weekly windows). Realised cost per window at $c_{\text{miss}}/c_{\text{FA}} = 2$, 5 and 10 (MC SE 0.008–0.06); precision and recall at $\pi^* = 1/6$; Brier score of the surge probability; pooled PIT statistic χ^2_9 (5% critical value 16.9); mean forecast minus outcome (Bias). "No future events": empty calendar after the forecast origin; "integrating": future events drawn from the event process. Delayed rows use fitted parameters; climatology never alerts at $\pi^* = 1/6$.

	Cost per window							
Forecaster	2	5	10	Prec.	Recall	Brier	PIT χ^2_9	Bias
Complete data								
Hawkes, true par., no future events	0.137	0.287	0.526	0.614	0.466	0.0655	175.3	−1.91
Hawkes, true par., integrating	0.136	0.285	0.512	0.503	0.511	0.0628	3.3	+0.07
Hawkes, fitted, no future events	0.137	0.287	0.534	0.594	0.471	0.0654	169.8	−1.75
Hawkes, fitted, integrating	0.134	0.289	0.514	0.484	0.511	0.0629	17.1	+0.38
Moving-average Poisson	0.221	0.451	0.792	0.216	0.247	0.1141	1994.3	+0.06
Climatology	0.193	0.483	0.898	—	0.000	0.0880	26.5	+0.94
Reporting delays								
Fitted, naive state, no future events	0.138	0.304	0.518	0.629	0.420	0.0668	247.8	−2.18
Fitted, HT nowcast, no future events	0.135	0.297	0.523	0.572	0.454	0.0654	190.5	−1.75
Fitted, naive state, integrating	0.137	0.279	0.502	0.550	0.506	0.0633	6.8	−0.05
Fitted, HT nowcast, integrating	0.135	0.287	0.510	0.481	0.517	0.0630	13.3	+0.37
Moving-average Poisson, available counts	0.226	0.478	0.848	0.177	0.161	0.1130	1879.8	−0.65

Delays. The naive state is biased downward as Proposition 7.3(c) predicts: with true parameters it averages 0.4752 of the complete-data state, against $\mathbb{E}e^{-\beta d} = 0.475$, and never exceeds it; the Horvitz–Thompson state averages 1.0033 of it. In windows without a new event the no-future-event forecast from the naive state is miscalibrated ($p = 0.033$) and that from the nowcast is not ($p = 0.435$). Neither decisions nor pooled calibration show a benefit, however (Table 3, Figure 6(b),(f)). With integration the three costs differ by at most 0.012 at each ratio, and at $c_{\text{miss}} = 5$ the naive state is even slightly cheaper than the nowcast (paired difference −0.008, MC SE 0.003): its forecasts are lower by 0.42 incidents per window, which offsets the fitted forecaster's over-forecast. Here the delays are short and the undelayed online stream carries much of the excitation. The study confirms the correction of the state, not that the nowcast improves alert decisions.

8.7 What the study does and does not show

Under the model the tools work as the theory says: EM recovers the branching matrix and the surge decomposition with close to nominal coverage, the boundary test holds its size, and the dark-figure correction restores coverage except when excitation is weak and recording poor, with a floor of about 0.04–0.09 set by a 30–50% error in p. The traps are real: a constant background finds contagion where there is none (for rare large shocks 0.06–0.07 below n^* even with $T = 10{,}000$); smoothing does not remove the artefact of fast shocks, while the shock times largely do, where tested; and an omitted common shock creates an online-to-offline effect. On Hawkes-generated data a plug-in Hawkes alert rule beats two simple baselines that are not the plan's. Forecasts that integrate over future events are calibrated when pooled; no forecaster can time the events, and that limits recall.

The study has clear limits. Each experiment uses a few parameter points, at most three streams, exponential kernels and, apart from E2, constant backgrounds; there are no near-critical systems and no misspecified kernels. The replicate numbers resolve differences of a few hundredths in branching ratios

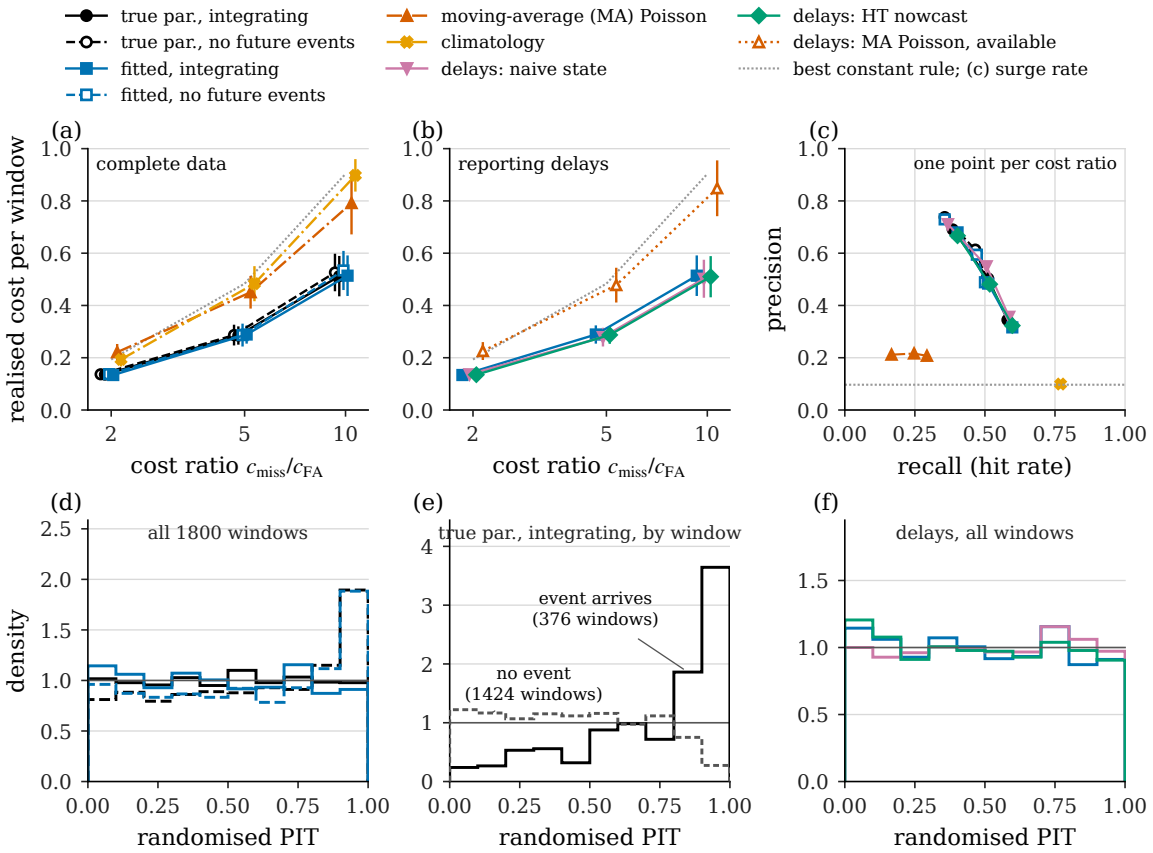


Figure 6. E6, alerting and nowcasting (synthetic data; 12 replicates × 150 weekly windows; bars ±1.96 MC SE). Hawkes forecasts with true or fitted parameters, assuming no future events (dashed, open markers) or integrating over them (solid, filled). (a) Realised cost per window against the cost ratio; dotted: best constant rule. (b) The same under reporting delays (fitted, integrating). (c) Precision against recall at the three thresholds; dotted: surge rate. (d) PIT histograms, all windows. (e) PIT of the integrating forecast with true parameters, for windows in which an external event arrives after the origin and for the others. (f) PIT under delays, all windows.

and about 0.01–0.04 in costs, not smaller ones. Not evaluated were the Davies sup-test and its bootstraps, Bayesian estimation and shrinkage, the INAR approximation, nonparametric kernels, the time-rescaling check, placebo calibration and the imputation algorithm (Algorithm 2); the delay correction was evaluated only as a plug-in state. Nor did we test whether a flexible background absorbs genuine contagion, long windows for slow switching, or the selection rules of the analysis plan. These checks belong to the synthetic dry run of the plan (Appendix H.6, V4).

9 A pre-registered analysis plan

This section summarises the analysis plan that we intend to register before any outcome data are examined. Appendix H gives it in full, with the replication package and the timeline. The numerical settings are proposals until registration; later changes will be reported as deviations. No part of the plan has been carried out on real data. The simulation study (Section 8) evaluates EM estimation, surge attribution, the boundary test, the thinning correction and a plug-in alert, but not the block tests, the placebo calibration or the selection and decision rules of the plan. These must first pass a synthetic dry run (V4).

Data and design. The candidate pilot is the New York City Police Department's hate-crime file (New York City Police Department, 2026), restricted to anti-Jewish records from 2019 onwards, with the five boroughs as offline streams and the final 12 months as a hold-out test period (Appendix H.1). The design is chosen on data availability alone. If day-level occurrence dates can be obtained under a data-use agreement, the model is fitted to them (Design A). Otherwise it is fitted to record-creation dates as a report-time process, with an explicit delay distribution and the occurrence month as a constraint (Design B). Online, ADL and CST streams are conditional on access being confirmed.

Estimands. (E1) The branching matrix $\boldsymbol{A}$, with $\varrho(\boldsymbol{A})$ and $\boldsymbol{M}$, on the recorded scale first and thinning-corrected alongside. (E2) The surge decomposition $B_k + D_k + C_k$ of Theorem 5.1 for every pre-registered event at $H = 14$ days, with 7 and 28 days as secondary horizons, and the online-channel share where online streams are included. (E3) Alert performance with window $\Delta_a = 7$ days against a background-only model and a persistence rule, at cost ratios agreed with community-security partners (Appendix H.2).

Events. The event list is fixed blind to incident data by the procedure of Section 2.4: classes defined by type, a salience index computed from event data alone with a threshold that yields 10 to 30 events, independent coding by two coders without access to incident data, and a hashed freeze before incident data are loaded (Appendix H.3).

Models and tests. The models are nested: background only (M0), plus shocks (M1), self-excitation (M2), cross-borough excitation (M3) and online↔offline blocks (M4). Each block of $\boldsymbol{A}$ has one common exponential decay, and the background is a cubic B-spline with knot spacing $h_b = 60$ days plus day-of-week and holiday effects. A step is taken only if its block test rejects at level 0.05 and the out-of-sample log score improves. A block's decay is not identified when all its entries vanish, so each block test uses the sup-LR statistic (6.10) over nine decays with half-lives from 0.5 to 21 days, referred to a parametric bootstrap with $B = 499$ replicates from the null fit. The block tests are applied in a fixed sequence. Entry-level tests within the accepted blocks form one Holm family at level 0.05. If the dry run shows a block test to be too slow or oversized, registered fallbacks apply (Appendix H.4).

Sensitivity, validation and decisions. Eight sensitivity analyses vary the background flexibility ($h_b \in \{30, 60, 120, 365\}$ days), the recording model, the kernel family, the delay model, the event list, the spatial resolution, the correction for online measurement error and the treatment of ties (Appendix H.5). Validation uses time-rescaling goodness of fit with a bootstrap reference (V1), out-of-sample scores and PIT histograms (V2), placebo calendars whose dates are drawn from *all* days of the training period, so that the randomisation hypothesis of Lemma 6.3 can hold (V3), and the synthetic dry run (V4) (Appendix H.6). Contagion in a stream is claimed only if its Holm-adjusted test rejects, its interval excludes zero under the most flexible background, goodness of fit is not rejected and the placebo false-positive rate is at most 0.10. Cross-stream effects are described as Granger-type predictive dependence, never as causation. Alerts are recommended for operational use only if they beat both baselines in realised cost (Appendix H.7).

10 Ethics and responsible use

Models that forecast surges in hate incidents can protect people. They can also cause harm through how outputs are worded, where they are shared and what they are used for. Appendix I gives in full the protocol that accompanies the plan; it complements the companion paper's protocol for victim privacy and record linkage (Kornstein Lab for Antisemitism Research, 2026a). Its main commitments are the following.

- *No stigmatising outputs.* The spatial units are recording areas, not communities, and the model contains no information about perpetrators. Alerts and maps are at the borough level or coarser, never street- or venue-level, and no output ranks neighbourhoods or communities (Appendix I.1).
- *Controlled disclosure.* During prospective evaluation, alerts go only to vetted community-security partners whose agreements limit use to protective purposes, after review by a named member of the research team. Alerts are not published, and aggregate evaluations are released only after a 90-day embargo (Appendix I.2).
- *Timing, not blame.* Events are described by class and date. No community, nationality or religious group is described as a source of incidents, and online-to-offline coefficients are reported as Granger-type predictive dependence (Appendix I.3).
- *Procedural neutrality.* Event classes apply to all parties alike, the list is frozen before outcome data are seen, every event is reported whatever the result, and the coding protocol is reviewed externally before registration (Appendix I.4).
- *Data protection.* Online data enter only as aggregate counts, without account-level profiles or linkage to offline incidents (Appendix I.5). The protocol is submitted for ethics review before any data are accessed, and outputs follow a minimum-cell rule (Appendix I.6).

11 Discussion

What the framework makes possible. After a salient external event, the public argument is usually about whether the event "caused" the surge that followed. The framework replaces that single question with several that have precise answers under a stated model. How many incidents in the window would have occurred anyway, how many responded directly to the event, and how many were set off by other incidents (Theorem 5.1)? How much of that cascade passed through online streams (Proposition 5.4)? How large is the cascade that one incident starts in each stream, as measured by the multiplier $\boldsymbol{M}$? Each answer comes with an interval and with the assumptions it rests on.

The framework is equally explicit about what the data cannot settle. For a single stream, or for streams that do not excite one another, the autocorrelation of counts cannot tell contagion from clustered shocks (Theorem 4.2). Between streams the picture is more favourable: when cross-excitation dominates self-excitation, or a stream only excites, the cross-spectrum carries a trace that no common-shock model reproduces (Proposition 4.5 and Example 4.6). An incomplete event calendar or a rigid background manufactures contagion (Theorem 4.7). Identification therefore has to come largely from outside the counts: from a calendar of events fixed blind to the outcome, from higher-order information that only the full likelihood exploits, or from provenance links between incidents (Corollary 4.9). In our simulations a single spline background did not remove the artefact of fast unrecorded shocks, and known event times largely removed it where tested (Section 8.3). The analysis plan is built around that requirement (Appendix H.3). The companion paper enters at a precise point: thinning attenuates apparent contagion in a known way (Theorem 4.10), so a contagion estimate is only as good as the coverage estimate behind it, and the uncertainty about coverage sets a floor under its precision (Corollary 4.11). For those who must act on a forecast, the alert rule makes the trade-off between false alarms and missed surges an explicit choice, agreed with the people who bear its costs (Theorem 7.2). Reporting delays bias the model's state downward, and the nowcast removes that bias in the mean (Proposition 7.3). In our simulations the nowcast removed the

bias of the state but did not lower alert costs, and plugging it into the rule is not the Bayes rule for delayed data.

Limitations. We stress six.

Linearity and the absence of inhibition. In a linear Hawkes process an incident can only raise the rate of later incidents. Deterrence, protective deployment after an alert, arrests and saturation all lower it, and the model cannot represent them. Where they operate, the fitted excitation is a net effect, and an alert that prompts successful protection makes itself look like a false alarm (Appendix I.1). Nonlinear Hawkes processes (Brémaud and Massoulié, 1996) remove this restriction at the cost of the cluster representation on which the surge decomposition rests.

Stationarity and constant parameters. The theory assumes $\varrho(\boldsymbol{A}) < 1$ and parameters that do not change over the study period. Contagion may itself change, and recording probabilities can move with attention and with breaks in reporting systems (Section 2.2). The change-point monitor of Remark D.4 can detect such shifts but does not model them. Estimates close to criticality deserve particular caution, since a misspecified background pushes the apparent branching ratio towards one (Remark 4.8).

Pre-specification of events. Everything rests on the event calendar. Events missing from it are absorbed into $\boldsymbol{A}$ (Theorem 4.7). The salience index and its threshold are choices that the sensitivity analysis can vary but not remove (Appendix H.5). The shock term also treats event times as exogenous, while some candidate events, such as public statements, may themselves respond to a wave of incidents; blind coding does not prevent such reverse dependence. Forecasts depend on the calendar in a further way. A forecaster that ignores future events under-forecasts, a misspecification that integrating over the event process removes when windows are pooled. What no model can do is time a new event, and that limits recall rather than pooled calibration (Sections 7.1 and 8.6). In the simulations 62% of surges fell in forecast windows in which a new event arrived after the forecast origin; the integrating alert rule with true parameters detected 28% of these and 89% of the others.

Classifier error in online streams. An online stream measures a defined corpus, not platform-wide prevalence, and its error rates may change after salient events precisely when they matter most (Equation (2.2)). Independent errors cannot create online-to-offline Granger causality under the null, but they dilute it under the alternative, and errors that move with common shocks can mimic it (Remark 4.13). Every online-to-offline coefficient is a statement about predictive dependence between measured series, not about the influence of online speech on offline behaviour.

Ecological interpretation. The streams are aggregates over recording areas and offence categories. A coefficient α_{kj} summarises how the counts of two streams move together in time; it does not say who influenced whom, and it says nothing about the people who live in a region. The mechanisms that could produce self-excitation are hypotheses that the model does not separate (Section 3.5), and family trees reconstructed by EM are probabilistic summaries under the model, not findings about individual incidents.

Independent recording and untested tools. The thinning results assume that each incident is recorded independently of the history. If salient events raise the propensity to report, or if severe incidents are both more productive and more often recorded, the correction can err in either direction (Proposition A.6). Several tools, including Bayesian shrinkage, time-rescaling diagnostics, the imputation-based alert, the sup-tests for the model hierarchy and placebo calibration, were not evaluated in our simulations and must first pass the synthetic dry run of the plan (Appendix H.6).

Extensions. Four extensions seem most valuable: *spatio-temporal kernels* of the ETAS kind (Mohler et al., 2011; Ogata, 1988; Reinhart, 2018), with outputs that respect the ethics protocol's ban on street- or venue-level maps; *marked and typed incidents*, whose mark-specific productivities address the bias of

Proposition A.6, with sparse or low-rank estimates of $\boldsymbol{A}$ for many streams (Hansen et al., 2015; Zhou et al., 2013a); *nonlinear Hawkes processes* with inhibitory kernels, whose Granger-causal graphs remain available (Eichler et al., 2017), so that deterrence and protection enter the model rather than its residuals; and, closest to this series, *joint estimation with the dark-figure model*, treating each list as a delayed, time-varying thinning of one latent self-exciting process. The companion paper names continuous-time, open-population estimation as its own leading extension (Kornstein Lab for Antisemitism Research, 2026a); the framework developed here supplies the point process it needs.

Why model the dynamics. A count says how many incidents were recorded. It does not say where they came from. A surge that is mostly a direct reaction to an event calls for vigilance in the days after similar events. A surge sustained by contagion calls for breaking the chain early, because every incident prevented also prevents the incidents it would have triggered. A surge that travels from online to offline calls for watching the online streams. These are different claims about how hostility spreads, and they carry different responsibilities. The data can support some of these claims, and on some questions they cannot decide between rival explanations. Stating which is which, with assumptions in the open and a plan fixed before the data are seen, is how a measurement programme can take seriously both the people who live through a surge and the evidence about it.

Declarations

Funding. None.

Competing interests. None declared.

AI assistance. This paper was drafted with substantial assistance from AI language models (Anthropic's Claude), directed and reviewed by the Kornstein Lab; all derivations were checked numerically and by independent automated review, and all simulation numbers are produced by the released code.

Code and data availability. The code for the simulation study (the `hawkes` Python package, the experiment driver `run_experiments.py` with its deterministic seeds, and the script that produces every figure and table of Section 8) is in the paper's repository, together with the simulation output (Appendix F.6). All data in this paper are synthetic and are regenerated by that code. No real incident data were used.

References

T. W. Anderson. The integral of a symmetric unimodal function over a symmetric convex set and some probability inequalities. *Proceedings of the American Mathematical Society*, 6(2):170–176, 1955. doi: 10.2307/2032333.

Anti-Defamation League. Audit of antisemitic incidents 2024. Technical report, ADL Center on Extremism, New York, April 2025. URL https://www.adl.org/resources/report/audit-antisemitic-incidents-2024. Accessed November 2026.

Anti-Defamation League. H.E.A.T. Map: Hate, extremism, antisemitism, terrorism. Interactive map and downloadable incident data, ADL Center on Extremism, n.d. URL https://www.adl.org/resources/tools-to-track-hate/heat-map. Accessed November 2026.

Artin Armagan, Merlise Clyde, and David B. Dunson. Generalized beta mixtures of Gaussians. In *Advances in Neural Information Processing Systems 24 (NIPS 2011)*, pages 523–531, 2011.

Emmanuel Bacry, Iacopo Mastromatteo, and Jean-François Muzy. Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1):1550005, 2015. doi: 10.1142/S2382626615500057.

M. S. Bartlett. The spectral analysis of point processes. *Journal of the Royal Statistical Society: Series B (Methodological)*, 25(2):264–281, 1963. doi: 10.1111/j.2517-6161.1963.tb00508.x.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

Yoav Benjamini and Daniel Yekutieli. The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4):1165–1188, 2001. doi: 10.1214/aos/1013699998.

Pierre Brémaud. *Point Processes and Queues: Martingale Dynamics*. Springer Series in Statistics. Springer, New York, 1981. doi: 10.1007/978-1-4684-9477-8.

Pierre Brémaud and Laurent Massoulié. Stability of nonlinear Hawkes processes. *The Annals of Probability*, 24(3): 1563–1588, 1996. doi: 10.1214/aop/1065725193.

Pierre Brémaud and Laurent Massoulié. Hawkes branching point processes without ancestors. *Journal of Applied Probability*, 38(1):122–135, 2001. doi: 10.1239/jap/996986648.

Peter J. Brockwell and Richard A. Davis. *Time Series: Theory and Methods*. Springer Series in Statistics. Springer, New York, second edition, 1991. doi: 10.1007/978-1-4419-0320-4.

Emery N. Brown, Riccardo Barbieri, Valérie Ventura, Robert E. Kass, and Loren M. Frank. The time-rescaling theorem and its application to neural spike train data analysis. *Neural Computation*, 14(2):325–346, 2002. doi: 10.1162/08997660252741149.

Timothy C. Brown and M. Gopalan Nair. A simple proof of the multivariate random time change theorem for point processes. *Journal of Applied Probability*, 25(1):210–214, 1988. doi: 10.2307/3214247.

Carlos M. Carvalho, Nicholas G. Polson, and James G. Scott. The horseshoe estimator for sparse signals. *Biometrika*, 97(2):465–480, 2010. doi: 10.1093/biomet/asq017.

Herman Chernoff. On the distribution of the likelihood ratio. *The Annals of Mathematical Statistics*, 25(3):573–578, 1954. doi: 10.1214/aoms/1177728725.

Felix Cheysson and Gabriel Lang. Spectral estimation of Hawkes processes from count data. *The Annals of Statistics*, 50(3):1722–1746, 2022. doi: 10.1214/22-AOS2173.

Community Security Trust. Antisemitic incidents report 2023. Technical report, Community Security Trust, London, February 2024. URL https://cst.org.uk/public/data/file/9/f/Antisemitic_Incidents_Report_2023.pdf. Accessed November 2026.

Community Security Trust. Antisemitic incidents report 2025. Technical report, Community Security Trust, London, February 2026. URL https://cst.org.uk/research/cst-publications/antisemitic-incidents-report-2025. Accessed November 2026.

Congressional Research Service. Federal data on hate crimes in the United States. CRS Report R46318, Congressional Research Service, 2025. URL https://www.congress.gov/crs-product/R46318. Updated December 16, 2025. Accessed November 2026.

D. R. Cox. Some statistical methods connected with series of events. *Journal of the Royal Statistical Society: Series B (Methodological)*, 17(2):129–157, 1955. doi: 10.1111/j.2517-6161.1955.tb00188.x.

Claudia Czado, Tilmann Gneiting, and Leonhard Held. Predictive model assessment for count data. *Biometrics*, 65 (4):1254–1261, 2009. doi: 10.1111/j.1541-0420.2009.01191.x.

Daryl J. Daley and David Vere-Jones. *An Introduction to the Theory of Point Processes. Volume I: Elementary Theory and Methods*. Probability and Its Applications. Springer, New York, second edition, 2003. doi: 10.1007/b97277.

Daryl J. Daley and David Vere-Jones. *An Introduction to the Theory of Point Processes. Volume II: General Theory and Structure*. Probability and Its Applications. Springer, New York, second edition, 2008. doi: 10.1007/978-0-387-49835-5.

Robert B. Davies. Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*, 64 (2):247–254, 1977. doi: 10.1093/biomet/64.2.247.

Robert B. Davies. Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*, 74 (1):33–43, 1987. doi: 10.1093/biomet/74.1.33.

A. P. Dawid. Present position and potential developments: Some personal views: Statistical theory: The prequential approach. *Journal of the Royal Statistical Society: Series A (General)*, 147(2):278–292, 1984. doi: 10.2307/2981683.

A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(1):1–38, 1977. doi: 10.1111/j.2517-6161.1977.tb01600.x.

Vanessa Didelez. Graphical models for marked point processes based on local independence. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 70(1):245–264, 2008. doi: 10.1111/j.1467-9868.2007.00634.x.

J. Durbin. Weak convergence of the sample distribution function when parameters are estimated. *The Annals of Statistics*, 1(2):279–290, 1973. doi: 10.1214/aos/1176342365.

Meyer Dwass. The total progeny in a branching process and a related random walk. *Journal of Applied Probability*, 6:682–686, 1969. doi: 10.2307/3212112.

Michael Eichler, Rainer Dahlhaus, and Johannes Dueck. Graphical modeling for multivariate Hawkes processes with nonparametric link functions. *Journal of Time Series Analysis*, 38(2):225–242, 2017. doi: 10.1111/jtsa.12213.

Paul H. C. Eilers and Brian D. Marx. Flexible smoothing with B-splines and penalties. *Statistical Science*, 11(2):89–121, 1996. doi: 10.1214/ss/1038425655.

European Parliament and Council of the European Union. Regulation (EU) 2022/2065 of 19 October 2022 on a single market for digital services and amending Directive 2000/31/EC (Digital Services Act). Official Journal of the European Union, L 277, pp. 1–102, 27 October 2022. URL https://eur-lex.europa.eu/eli/reg/2022/2065/oj. Accessed November 2026.

Federal Bureau of Investigation. Supplemental hate crime statistics, 2021. Technical report, Uniform Crime Reporting Program, Criminal Justice Information Services Division, March 2023. URL https://le.fbi.gov/file-repository/supplemental-hate-crime-statistics-2021.pdf. Accessed November 2026.

Federal Bureau of Investigation. *National Incident-Based Reporting System User Manual, Version 2025.0*. U.S. Department of Justice, Federal Bureau of Investigation, Criminal Justice Information Services Division, 2025a. URL https://le.fbi.gov/file-repository/nibrs-user-manual-2025-0-062625.pdf. Accessed November 2026.

Federal Bureau of Investigation. *National Incident-Based Reporting System Technical Specification, Version 2025.0*. U.S. Department of Justice, Federal Bureau of Investigation, Criminal Justice Information Services Division, 2025b. URL https://le.fbi.gov/file-repository/2025-0-nibrs-technical-specification-062625.pdf. Accessed November 2026.

Federal Bureau of Investigation. FBI releases 2024 reported crimes in the nation statistics. Press release, 5 August 2025c. URL https://www.fbi.gov/news/press-releases/fbi-releases-2024-reported-crimes-in-the-nation-statistics. Accessed November 2026.

Federal Bureau of Investigation. National Incident-Based Reporting System (NIBRS). FBI Uniform Crime Reporting Program web page, 2026. URL https://www.fbi.gov/how-we-can-help-you/more-fbi-services-and-information/ucr/nibrs. Accessed November 2026.

William Feller. On a general class of "contagious" distributions. *The Annals of Mathematical Statistics*, 14(4):389–400, 1943. doi: 10.1214/aoms/1177731359.

Vladimir Filimonov and Didier Sornette. Apparent criticality and calibration issues in the Hawkes self-excited point process model: application to high-frequency financial data. *Quantitative Finance*, 15(8):1293–1314, 2015. doi: 10.1080/14697688.2015.1032544.

Konstantinos Fokianos, Anders Rahbek, and Dag Tjøstheim. Poisson autoregression. *Journal of the American Statistical Association*, 104(488):1430–1439, 2009. doi: 10.1198/jasa.2009.tm08270.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(2):243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x.

C. W. J. Granger. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3):424–438, 1969. doi: 10.2307/1912791.

Peter J. Green. On use of the EM algorithm for penalized likelihood estimation. *Journal of the Royal Statistical Society: Series B (Methodological)*, 52(3):443–452, 1990. doi: 10.1111/j.2517-6161.1990.tb01798.x.

Gustaf Gripenberg, Stig-Olof Londen, and Olof Staffans. *Volterra Integral and Functional Equations*, volume 34 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 1990. doi: 10.1017/CBO9780511662805.

Niels Richard Hansen, Patricia Reynaud-Bouret, and Vincent Rivoirard. Lasso and probabilistic inequalities for multivariate point processes. *Bernoulli*, 21(1):83–143, 2015. doi: 10.3150/13-BEJ562.

Alan G. Hawkes. Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1):83–90, 1971. doi: 10.1093/biomet/58.1.83.

Alan G. Hawkes and David Oakes. A cluster process representation of a self-exciting process. *Journal of Applied Probability*, 11(3):493–503, 1974. doi: 10.2307/3212693.

Michael Höhle and Matthias an der Heiden. Bayesian nowcasting during the STEC O104:H4 outbreak in Germany, 2011. *Biometrics*, 70(4):993–1002, 2014. doi: 10.1111/biom.12194.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

Ryan D. King and Gretchen M. Sutton. High times for hate crimes: Explaining the temporal clustering of hate-motivated offending. *Criminology*, 51(4):871–894, 2013. doi: 10.1111/1745-9125.12022.

J. F. C. Kingman. A convexity property of positive matrices. *The Quarterly Journal of Mathematics*, 12(1):283–284, 1961. doi: 10.1093/qmath/12.1.283.

Matthias Kirchner. Hawkes and INAR(∞) processes. *Stochastic Processes and their Applications*, 126(8):2494–2525, 2016. doi: 10.1016/j.spa.2016.02.008.

Matthias Kirchner. An estimation procedure for the Hawkes process. *Quantitative Finance*, 17(4):571–595, 2017. doi: 10.1080/14697688.2016.1211312.

Kornstein Lab for Antisemitism Research. The dark figure: Estimating unreported antisemitic hate crime with Bayesian record linkage and multiple-systems estimation. Companion methods paper, 2026a.

Kornstein Lab for Antisemitism Research. Evaluating algorithms for antisemitism detection. Design-stage research proposal, 2026b. Reports no empirical results.

Kornstein Lab for Antisemitism Research. Measuring antisemitism in online discourse. Design-stage research proposal, 2026c. Reports no empirical results.

Patrick J. Laub, Young Lee, and Thomas Taimre. *The Elements of Hawkes Processes*. Springer, Cham, 2021. doi: 10.1007/978-3-030-84639-8.

Kalev Leetaru and Philip A. Schrodt. GDELT: Global data on events, location and tone, 1979–2012. Paper presented at the International Studies Association Annual Convention, San Francisco, CA, April 2013.

Erik Lewis and George Mohler. A nonparametric EM algorithm for multiscale Hawkes processes. Unpublished manuscript, 2011.

Erik Lewis, George Mohler, P. Jeffrey Brantingham, and Andrea L. Bertozzi. Self-exciting point process models of civilian deaths in Iraq. *Security Journal*, 25(3):244–264, 2012. doi: 10.1057/sj.2011.21.

P. A. W. Lewis and G. S. Shedler. Simulation of nonhomogeneous Poisson processes by thinning. *Naval Research Logistics Quarterly*, 26(3):403–413, 1979. doi: 10.1002/nav.3800260304.

Hubert W. Lilliefors. On the Kolmogorov–Smirnov test for normality with mean and variance unknown. *Journal of the American Statistical Association*, 62(318):399–402, 1967. doi: 10.1080/01621459.1967.10482916.

Scott W. Linderman and Ryan P. Adams. Discovering latent network structure in point process data. In *Proceedings of the 31st International Conference on Machine Learning (ICML)*, volume 32 of *Proceedings of Machine Learning Research*, pages 1413–1421, 2014.

Charles Loeffler and Seth Flaxman. Is gun violence contagious? A spatiotemporal test. *Journal of Quantitative Criminology*, 34(4):999–1017, 2018. doi: 10.1007/s10940-017-9363-8.

Thomas A. Louis. Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 44(2):226–233, 1982. doi: 10.1111/j.2517-6161.1982.tb01203.x.

David Marsan and Olivier Lengliné. Extending earthquakes' reach through cascading. *Science*, 319(5866):1076–1079, 2008. doi: 10.1126/science.1148783.

Sarah F. McGough, Michael A. Johansson, Marc Lipsitch, and Nicolas A. Menzies. Nowcasting by Bayesian Smoothing: A flexible, generalizable model for real-time epidemic tracking. *PLoS Computational Biology*, 16(4): e1007735, 2020. doi: 10.1371/journal.pcbi.1007735.

Paul-André Meyer. Démonstration simplifiée d'un théorème de Knight. In *Séminaire de Probabilités V*, volume 191 of *Lecture Notes in Mathematics*, pages 191–195. Springer, Berlin, 1971. doi: 10.1007/BFb0058859.

G. O. Mohler, M. B. Short, P. J. Brantingham, F. P. Schoenberg, and G. E. Tita. Self-exciting point process modeling of crime. *Journal of the American Statistical Association*, 106(493):100–108, 2011. doi: 10.1198/jasa.2011.ap09546.

Jesper Møller. Shot noise Cox processes. *Advances in Applied Probability*, 35(3):614–640, 2003. doi: 10.1239/aap/1059486821.

Jesper Møller and Jakob G. Rasmussen. Perfect simulation of Hawkes processes. *Advances in Applied Probability*, 37(3):629–646, 2005. doi: 10.1239/aap/1127483739.

Karsten Müller and Carlo Schwarz. Fanning the flames of hate: Social media and hate crime. *Journal of the European Economic Association*, 19(4):2131–2167, 2021. doi: 10.1093/jeea/jvaa045.

Radford M. Neal. Slice sampling. *The Annals of Statistics*, 31(3):705–767, 2003. doi: 10.1214/aos/1056562461.

New York City Police Department. NYPD hate crimes. NYC Open Data, dataset bqiq-cu78, 2026. URL https://data.cityofnewyork.us/Public-Safety/NYPD-Hate-Crimes/bqiq-cu78. Accessed November 2026.

Office of the New York State Comptroller. The concerning growth of hate crime in New York State. Technical report, Office of Budget Policy and Analysis, Office of the New York State Comptroller, Albany, NY, August 2024. URL https://www.osc.ny.gov/reports/concerning-growth-hate-crime-new-york-state. Accessed November 2026.

Yosihiko Ogata. The asymptotic behaviour of maximum likelihood estimators for stationary point processes. *Annals of the Institute of Statistical Mathematics*, 30(2):243–261, 1978. doi: 10.1007/BF02480216.

Yosihiko Ogata. On Lewis' simulation method for point processes. *IEEE Transactions on Information Theory*, 27(1): 23–31, 1981. doi: 10.1109/TIT.1981.1056305.

Yosihiko Ogata. Statistical models for earthquake occurrences and residual analysis for point processes. *Journal of the American Statistical Association*, 83(401):9–27, 1988. doi: 10.1080/01621459.1988.10478560.

Tohru Ozaki. Maximum likelihood estimation of Hawkes' self-exciting point processes. *Annals of the Institute of Statistical Mathematics*, 31(1):145–155, 1979. doi: 10.1007/BF02480272.

E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954. doi: 10.1093/biomet/41.1-2.100.

F. Papangelou. Integrability of expected increments of point processes and a related random change of scale. *Transactions of the American Mathematical Society*, 165:483–506, 1972. doi: 10.1090/S0002-9947-1972-0314102-9.

Clionadh Raleigh, Andrew Linke, Håvard Hegre, and Joakim Karlsen. Introducing ACLED: An armed conflict location and event dataset. *Journal of Peace Research*, 47(5):651–660, 2010. doi: 10.1177/0022343310378914.

Marcello Rambaldi, Paris Pennesi, and Fabrizio Lillo. Modeling foreign exchange market activity around macroeconomic news: Hawkes-process approach. *Physical Review E*, 91(1):012819, 2015. doi: 10.1103/PhysRevE.91.012819.

Alex Reinhart. A review of self-exciting spatio-temporal point processes and their applications. *Statistical Science*, 33(3):299–318, 2018. doi: 10.1214/17-STS629.

Marian-Andrei Rizoiu, Young Lee, Swapnil Mishra, and Lexing Xie. A tutorial on Hawkes processes for events in social media. arXiv preprint arXiv:1708.06401, 2017.

Janet E. Rosenbaum. Antisemitic hate crimes in New York City: an analysis of administrative data, 2019–2024. *Journal of Public Health Policy*, 46(4):816–829, 2025. doi: 10.1057/s41271-025-00596-4.

Mark J. Schervish. A general method for comparing probability assessors. *The Annals of Statistics*, 17(4):1856–1879, 1989. doi: 10.1214/aos/1176347398.

René L. Schilling, Renming Song, and Zoran Vondraček. *Bernstein Functions: Theory and Applications*, volume 37 of *De Gruyter Studies in Mathematics*. De Gruyter, Berlin, second edition, 2012. doi: 10.1515/9783110269338.

Frederic Paik Schoenberg. Facilitated estimation of ETAS. *Bulletin of the Seismological Society of America*, 103(1):601–605, 2013. doi: 10.1785/0120120146.

Steven G. Self and Kung-Yee Liang. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610, 1987. doi: 10.1080/01621459.1987.10478472.

Moshe Shaked and J. George Shanthikumar. *Stochastic Orders*. Springer Series in Statistics. Springer, New York, 2007. doi: 10.1007/978-0-387-34675-5.

Aad W. van der Vaart and Jon A. Wellner. *Weak Convergence and Empirical Processes: With Applications to Statistics*. Springer Series in Statistics. Springer, New York, 1996. doi: 10.1007/978-1-4757-2545-2.

Charles F. Van Loan. Computing integrals involving the matrix exponential. *IEEE Transactions on Automatic Control*, 23(3):395–404, 1978. doi: 10.1109/TAC.1978.1101743.

Ravi Varadhan and Christophe Roland. Simple and globally convergent methods for accelerating the convergence of any EM algorithm. *Scandinavian Journal of Statistics*, 35(2):335–353, 2008. doi: 10.1111/j.1467-9469.2007.00585.x.

Alejandro Veen and Frederic P. Schoenberg. Estimation of space–time branching process models in seismology using an EM-type algorithm. *Journal of the American Statistical Association*, 103(482):614–624, 2008. doi: 10.1198/016214508000000148.

Halbert White. Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1):1–25, 1982. doi: 10.2307/1912526.

Matthew L. Williams, Pete Burnap, Amir Javed, Han Liu, and Sefa Ozalp. Hate in the machine: Anti-Black and anti-Muslim social media posts as predictors of offline racially and religiously aggravated crime. *The British Journal of Criminology*, 60(1):93–117, 2020. doi: 10.1093/bjc/azz049.

Qingyuan Zhao, Murat A. Erdogdu, Hera Y. He, Anand Rajaraman, and Jure Leskovec. SEISMIC: A self-exciting point process model for predicting tweet popularity. In *Proceedings of the 21st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1513–1522. ACM, 2015. doi: 10.1145/2783258.2783401.

Ke Zhou, Hongyuan Zha, and Le Song. Learning social infectivity in sparse low-rank networks using multi-dimensional Hawkes processes. In *Proceedings of the Sixteenth International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 31 of *Proceedings of Machine Learning Research*, pages 641–649, 2013a.

Ke Zhou, Hongyuan Zha, and Le Song. Learning triggering kernels for multi-dimensional Hawkes processes. In *Proceedings of the 30th International Conference on Machine Learning (ICML)*, volume 28 of *Proceedings of Machine Learning Research*, pages 1301–1309, 2013b.

Appendix A Proofs for Sections 3 and 4

Throughout, a stationary M-variate point process $\boldsymbol{N}$ with mean $\bar{\boldsymbol{\Lambda}}$ has a covariance measure $\operatorname{diag}(\bar{\boldsymbol{\Lambda}})\delta_0(\mathrm{d}u) + \boldsymbol{c}(u)\,\mathrm{d}u$, and Bartlett spectrum (3.3). All Bartlett spectra below have continuous densities. Equality of spectra therefore means equality of these densities at every ω.

A.1 Proofs for Section 3

Proof of Proposition 3.1. (a) The offspring structure is a multitype Galton–Watson process with mean matrix $\boldsymbol{A}$, where column j lists the expected offspring of a type-j parent. It is subcritical when $\varrho(\boldsymbol{A}) < 1$, so every cluster is almost surely finite. Its expected type-k content, given a type-j root, is $\sum_{g\ge 0}(\boldsymbol{A}^g)_{kj} = \boldsymbol{M}_{kj}$. The immigrants form a homogeneous Poisson process on $\mathbb{R}$, so the superposition of clusters is stationary, with mean $\boldsymbol{M}\boldsymbol{\mu}$. That it has the stated intensity is shown by Hawkes and Oakes (1974). Uniqueness in law is shown by Brémaud and Massoulié (1996). The offspring fraction follows from $\bar{\boldsymbol{\Lambda}} - \boldsymbol{\mu} = \boldsymbol{A}\bar{\boldsymbol{\Lambda}}$.

(b) Taking expectations in the intensity gives $\bar{\boldsymbol{\Lambda}} = \boldsymbol{\mu} + \boldsymbol{A}\bar{\boldsymbol{\Lambda}}$ with $\bar{\boldsymbol{\Lambda}}$ finite. Every entry of $\bar{\boldsymbol{\Lambda}}$ is at least $\mu_k > 0$. Let $\boldsymbol{w} \ge \boldsymbol{0}$, $\boldsymbol{w} \ne \boldsymbol{0}$, be a left Perron vector of the nonnegative matrix $\boldsymbol{A}$, so that $\boldsymbol{w}^\top \boldsymbol{A} = \varrho(\boldsymbol{A})\boldsymbol{w}^\top$. Then $(1 - \varrho(\boldsymbol{A}))\,\boldsymbol{w}^\top\bar{\boldsymbol{\Lambda}} = \boldsymbol{w}^\top\boldsymbol{\mu} > 0$. Since $\boldsymbol{w}^\top\bar{\boldsymbol{\Lambda}} > 0$, it follows that $\varrho(\boldsymbol{A}) < 1$.

(c) The cluster size is the total progeny of a Galton–Watson process with Poisson(n) offspring. Its law is the classical Borel law (see Dwass, 1969). The probability generating function G satisfies $G(z) = z\exp\{n(G(z) - 1)\}$, which gives the moments by repeated differentiation at $z = 1$. Stirling's formula gives $\mathbb{P}(S = k) \asymp k^{-3/2}(ne^{1-n})^k$, which yields the exponential-moment bound. □

Proof sketch of Proposition 3.2. Moments of all orders exist because the cluster sizes have exponential moments (Proposition 3.1(c) and its multitype analogue). Write $\mathrm{d}N = \boldsymbol{\lambda}\,\mathrm{d}t + \mathrm{d}\boldsymbol{\mathcal{M}}$, where $\boldsymbol{\mathcal{M}}$ is a martingale with orthogonal increments and $\mathbb{E}[\mathrm{d}\mathcal{M}_k(t)\,\mathrm{d}\mathcal{M}_j(t)] = \delta_{kj}\bar{\Lambda}_k\,\mathrm{d}t$. The streams have no common points. Since $\bar{\boldsymbol{\Lambda}} = \boldsymbol{\mu} + \boldsymbol{A}\bar{\boldsymbol{\Lambda}}$, the centred measure $\mathrm{d}N - \bar{\boldsymbol{\Lambda}}\,\mathrm{d}t$ satisfies the renewal-type equation $\mathrm{d}N - \bar{\boldsymbol{\Lambda}}\mathrm{d}t = \boldsymbol{\Phi} * (\mathrm{d}N - \bar{\boldsymbol{\Lambda}}\mathrm{d}t) + \mathrm{d}\boldsymbol{\mathcal{M}}$. Because $\int \boldsymbol{R} < \infty$, its stationary solution is $(\delta_0\boldsymbol{I} + \boldsymbol{R}) * \mathrm{d}\boldsymbol{\mathcal{M}}$. Since $\boldsymbol{I} + \hat{\boldsymbol{R}} = (\boldsymbol{I} - \hat{\boldsymbol{\Phi}})^{-1}$, the Fourier transform of the covariance measure is (3.4). Moreover $\boldsymbol{I} - \hat{\boldsymbol{\Phi}}(\omega)$ is invertible for every ω, because $\varrho(\hat{\boldsymbol{\Phi}}(\omega)) \le \varrho(|\hat{\boldsymbol{\Phi}}(\omega)|) \le \varrho(\boldsymbol{A}) < 1$, where $|\cdot|$ is taken entrywise. Hawkes (1971) gives a direct derivation from the integral equation for $\boldsymbol{c}$. For the univariate formulas, expand $|1 + \hat{R}|^2 = 1 + \hat{R} + \overline{\hat{R}} + |\hat{R}|^2$ and invert term by term. Then $F_\infty = |1 - \hat{\varphi}(0)|^{-2} = (1 - n)^{-2}$. For the exponential kernel, $\hat{\varphi}(\omega) = n\beta/(\beta + i\omega)$ and $|1 - \hat{\varphi}(\omega)|^{-2} = (\beta^2 + \omega^2)/(\beta^2(1 - n)^2 + \omega^2)$. □

A.2 Auxiliary lemmas

Lemma A.1 (Cox processes and independent thinning). *(a) Let $\boldsymbol{N}$ be an M-variate stationary Cox process directed by $\boldsymbol{\xi} = (\xi_k)$. Assume $\mathbb{E}\xi_k(0)^2 < \infty$, write $\bar{\boldsymbol{\Lambda}}$ for the mean and $\boldsymbol{c}_{\xi}(u) = \operatorname{Cov}\{\boldsymbol{\xi}(t + u), \boldsymbol{\xi}(t)\}$ for the covariance, and assume $\boldsymbol{c}_{\xi}$ has integrable entries. Then $\boldsymbol{N}$ has covariance density $\boldsymbol{c}_{\xi}$ and Bartlett spectrum $(2\pi)^{-1}[\operatorname{diag}(\bar{\boldsymbol{\Lambda}}) + \hat{\boldsymbol{c}}_{\xi}(\omega)]$, where $\hat{\boldsymbol{c}}_{\xi}(\omega)$ is Hermitian and positive semidefinite for every ω.*

(b) Let $\boldsymbol{N}$ be stationary with mean $\bar{\boldsymbol{\Lambda}}$ and covariance density $\boldsymbol{c}$. Record each point of stream k independently with probability p_k, and let $\boldsymbol{p} = (p_k)$. The recorded process $\boldsymbol{N}^{\boldsymbol{p}}$ has mean $\operatorname{diag}(\boldsymbol{p})\bar{\boldsymbol{\Lambda}}$ and covariance density $\operatorname{diag}(\boldsymbol{p})\,\boldsymbol{c}(u)\operatorname{diag}(\boldsymbol{p})$.

Proof. (a) Given $\boldsymbol{\xi}$, the streams are independent Poisson processes. Hence, for bounded Borel sets I, J,

$$\mathbb{E}[N_k(I)N_j(J) \mid \boldsymbol{\xi}] = \delta_{kj}\int_{I\cap J}\xi_k + \int_I \xi_k \int_J \xi_j.$$

Taking expectations gives $\operatorname{Cov}\{N_k(I), N_j(J)\} = \delta_{kj}\bar{\Lambda}_k|I \cap J| + \int_I \int_J c_{\xi,kj}(t-s)\,\mathrm{d}t\,\mathrm{d}s$. For $\boldsymbol{w} \in \mathbb{C}^M$, the function $\boldsymbol{w}^* \boldsymbol{c}_\xi(u)\boldsymbol{w}$ is the covariance function of the stationary process $\boldsymbol{w}^*\boldsymbol{\xi}$. It is integrable, so by Fejér's theorem its Fourier transform is

$$\lim_{T\to\infty} T^{-1}\,\mathbb{E}\left|\int_0^T e^{-i\omega t}\{\boldsymbol{w}^*\boldsymbol{\xi}(t) - \mathbb{E}\boldsymbol{w}^*\boldsymbol{\xi}(t)\}\,\mathrm{d}t\right|^2 \geq 0.$$

(b) Let R_i be the recording indicators. Pairs of distinct points contribute $\mathbb{E}[R_i R_{i'}] = p_k p_j$ times the second factorial moment measure, whose density off the diagonal is $\bar{\Lambda}_k\bar{\Lambda}_j + c_{kj}$. Each point paired with itself contributes $p_k\bar{\Lambda}_k|I \cap J|$ when $k = j$. Subtracting $p_k p_j \bar{\Lambda}_k\bar{\Lambda}_j|I||J|$ gives the claim. □

Lemma A.2 (Poisson cluster processes). *Let N be a stationary Poisson cluster process on $\mathbb{R}$. Cluster centres form a Poisson process of rate r. Each centre x carries an independent copy of a finite random configuration $\{y_1, \dots, y_{\mathcal{S}}\}$, possibly empty, placed at $x + y_i$. Assume $\mathbb{E}e^{u_0\mathcal{S}} < \infty$ for some $u_0 > 0$. Then:*

(a) for every $j \geq 1$, $\chi_j = \lim_{T\to\infty} T^{-1} \operatorname{cum}_j N(0,T] = r\,\mathbb{E}\mathcal{S}^j$;

(b) the law of N determines $w(k) = r\mathbb{P}(\mathcal{S} = k)$ for every $k \geq 1$.

Proof. (a) Let $Y(x)$ be the number of points of a cluster centred at x that fall in $(0,T]$. The marked Poisson process of centres has Laplace functional

$$\log \mathbb{E}e^{-sN(0,T]} = -r\int_{\mathbb{R}} \mathbb{E}[1 - e^{-sY(x)}]\,\mathrm{d}x, \qquad s \geq 0.$$

For every cluster, $Y(x)^j \leq \mathcal{S}^{j-1}Y(x)$ and $\int Y(x)\,\mathrm{d}x = \mathcal{S}T$. Hence $\int \mathbb{E}Y(x)^j\,\mathrm{d}x \leq T\,\mathbb{E}\mathcal{S}^j < \infty$. By dominated convergence, the right-hand side is j times differentiable at $s = 0+$, with jth derivative $(-1)^j r\int \mathbb{E}Y(x)^j\,\mathrm{d}x$. Therefore $\mathbb{E}N(0,T]^j < \infty$ by monotone convergence, and

$$\operatorname{cum}_j N(0,T] = r\int_{\mathbb{R}} \mathbb{E}Y(x)^j\,\mathrm{d}x.$$

Let $\operatorname{diam} = \max_i y_i - \min_i y_i$. Then $Y(x) = \mathcal{S}$ on an interval of x-values of length $(T - \operatorname{diam})_+$, so

$$\mathcal{S}^j(1 - \operatorname{diam}/T)_+ \leq T^{-1}\int Y(x)^j\,\mathrm{d}x \leq \mathcal{S}^j.$$

Dominated convergence gives $T^{-1}\operatorname{cum}_j N(0,T] \to r\mathbb{E}\mathcal{S}^j$.

(b) Suppose two such processes have the same law. By (a), $r\mathbb{E}\mathcal{S}^j = r'\mathbb{E}\mathcal{S}'^j$ for every $j \geq 1$. The finite measures $\nu(k) = k\,w(k)$ and $\nu'(k) = k\,w'(k)$ on $\{1, 2, \dots\}$ then have the same moments of every order $j \geq 0$. Both have finite exponential moments for $u < u_0$. Their generating functions $\sum_k \nu(k)z^k$ are therefore analytic for $|z| < e^{u_0}$ and agree, so $\nu = \nu'$ and hence $w = w'$. □

By Proposition 3.1, a stationary Hawkes process with any kernel satisfies Lemma A.2 with $r = \mu$ and $\mathcal{S} \sim \text{Borel}(n)$. The process $C(\nu, r, a, \gamma)$ superposes a Poisson process of rate ν (clusters of size 1) and a Neyman–Scott process with centre rate r and $\mathcal{S} \sim \text{Poisson}(m)$. It therefore satisfies the lemma with $w_C(k) = \nu\mathbf{1}\{k = 1\} + re^{-m}m^k/k!$. Independent p-thinning of a Poisson cluster process gives a Poisson cluster process with cluster size $\text{Binomial}(\mathcal{S}, p)$.

Lemma A.3 (Janossy densities identify the intensity). *Let $\boldsymbol{N}$ be an M-variate point process on $[0,T]$ with $\mathbb{E}\boldsymbol{N}[0,T] < \infty$. Its intensity, with respect to its own history and a deterministic calendar, is*

$\lambda_k(t) = h_k(t; \boldsymbol{N}|_{[0,t)})$ for measurable functionals with $h_k \geq h_{\min} > 0$. For a configuration x on $[0,t]$ with points $(s_1,k_1),\dots,(s_m,k_m)$, $s_1 < \cdots < s_m$, the Janossy density of $\boldsymbol{N}|_{[0,t]}$ is

$$j_t(x) = \prod_{i=1}^{m} h_{k_i}(s_i; x_{<s_i}) \exp\Big\{ - \sum_k \int_0^t h_k(u; x_{<u})\, \mathrm{d}u \Big\} > 0$$

(Daley and Vere-Jones, 2003, Ch. 7). For a configuration x on $[0,t)$, $h_k(t;x) = j_t(x \cup \{(t,k)\})/j_t(x)$. Consequently, if two such specifications $h^{(1)}, h^{(2)}$ give the same law on $[0,T]$, then $h_k^{(1)}(t;x) = h_k^{(2)}(t;x)$ for almost every (t,x) with respect to Lebesgue measure times the Lebesgue–Poisson measure. In particular this holds for almost every t at $x = \emptyset$, and for almost every (s,t) at $x = \{(s,j)\}$.

Proof. Adding a point at the right end t multiplies j_t by $h_k(t;x)$ and leaves the integral unchanged, which gives the ratio formula. The law determines its Janossy densities up to null sets. Since $j_t > 0$, almost-everywhere statements under the law transfer to the Lebesgue–Poisson reference measure. The empty configuration is an atom of that measure, and the one-point configurations carry the measure $\mathrm{d}s$. □

Remark A.4 (Why intensity jumps do not settle the question). It is sometimes said that a Hawkes intensity jumps at events whereas a Cox intensity does not. This is true only of the Cox intensity *given* the latent driver ξ, which is unobservable. With respect to the observed history, the Cox intensity is the filter $\lambda^N(t) = \mathbb{E}[\xi(t) \mid \mathcal{F}_{t-}^N]$. Wherever ξ is continuous, Bayes' rule shows that λ^N jumps at each event by $\mathrm{Var}(\xi(t) \mid \mathcal{F}_{t-}^N)/\mathbb{E}(\xi(t) \mid \mathcal{F}_{t-}^N) \geq 0$ (cf. Brémaud, 1981). Learning about an unobserved shock looks like self-excitation.

What does separate the models is the *form* of the filter. For a Hawkes process the observable intensity is affine in the past. The next lemma shows that, among all stationary laws second-order equivalent to H, $\mathbb{P}_H$ is the only one whose observable intensity is affine in the past. Here "affine" means $\lambda(t) = c + \int_{(-\infty,t)} h(t-s)\,\mathrm{d}N(s)$ with $c > 0$, $h \geq 0$, $h \in L^1 \cap L^2$ and $\int h < 1$. By Corollary 4.9(ii), the filter of every matched C is therefore nonlinear. The information that separates contagion from shocks lies in this nonlinearity.

Lemma A.5 (Affine observable intensity characterises the Hawkes law). *Let $H = H(\mu,\varphi)$ be a stationary Hawkes process with kernel $\varphi \in L^1 \cap L^2$, $\int \varphi = n < 1$. Let N be a stationary point process, second-order equivalent to H, whose intensity with respect to its own history is $\lambda(t) = c + \int_{(-\infty,t)} h(t-s)\,\mathrm{d}N(s)$, with $c > 0$, $h \geq 0$, $h \in L^1 \cap L^2$ and $\int h < 1$. Then $h = \varphi$ almost everywhere, $c = \mu$, and $\mathbb{P}_N = \mathbb{P}_H$.*

Proof. Let c_N be the covariance density of N. Since $\mathrm{d}N(t) - \lambda(t)\mathrm{d}t$ is orthogonal to $\mathcal{F}_{t-}$, we have $\mathrm{Cov}\{\mathrm{d}N(t) - \lambda(t)\mathrm{d}t, \mathrm{d}N(s)\} = 0$ for $s < t$. Expanding $\mathrm{Cov}\{\lambda(t), \mathrm{d}N(s)\}$ gives, for $u > 0$,

$$c_N(u) = \bar{\Lambda} h(u) + \int_0^\infty h(v)\, c_N(u-v)\,\mathrm{d}v. \tag{A.1}$$

The kernel φ solves (A.1) with the same $c_N = c_H$, since H has affine intensity with kernel φ. The difference $d = h - \varphi \in L^1 \cap L^2$ is supported on $(0,\infty)$ and satisfies $\bar{\Lambda} d(u) + \int_0^\infty d(v) c_N(u-v)\,\mathrm{d}v = 0$ for $u > 0$. Multiply by $d(u)$, integrate over $u > 0$ and apply Parseval:

$$0 = \frac{1}{2\pi} \int |\hat{d}(\omega)|^2 \{\bar{\Lambda} + \hat{c}_N(\omega)\}\,\mathrm{d}\omega = \int |\hat{d}(\omega)|^2 f_H(\omega)\,\mathrm{d}\omega.$$

Since $|1 - \hat{\varphi}| \leq 1 + n$, we have $f_H \geq \bar{\Lambda}(1+n)^{-2}/(2\pi) > 0$. Hence $\hat{d} = 0$, and so $d = 0$. Taking expectations of the intensity gives $c = \bar{\Lambda}(1-n) = \mu$. Thus N is a stationary solution of the Hawkes equation with parameters (μ,φ), and $\mathbb{P}_N = \mathbb{P}_H$ by the uniqueness result of Brémaud and Massoulié (1996). □

A.3 Proof of Theorem 4.2

(a) By (3.5) and (4.1), the means are $\bar{\Lambda}$ and $\nu + ra/\gamma$. Given equal means, the spectra agree if and only if

$$\frac{\bar{\Lambda}\beta^2 n(2-n)}{\beta^2(1-n)^2 + x} = \frac{ra^2}{\gamma^2 + x} \qquad \text{for all } x = \omega^2 \geq 0.$$

Both numerators are positive. Cross-multiplying and comparing coefficients in x gives $ra^2 = \bar{\Lambda}\beta^2 n(2-n)$ and $\gamma^2 = \beta^2(1-n)^2$. The converse is immediate.

(b) Solving (4.2) for given a gives $r = \bar{\Lambda}\beta^2 n(2-n)/a^2$ and

$$\nu = \bar{\Lambda} - \frac{ra}{\gamma} = \bar{\Lambda}\Big(1 - \frac{\beta n(2-n)}{a(1-n)}\Big),$$

which is nonnegative if and only if $a \geq a_{\min}$. With $m = a/\gamma$, $m_{\min} = a_{\min}/\{\beta(1-n)\} = n(2-n)/(1-n)^2$, which equals $F_\infty - 1$ by Proposition 3.2. Also $rm^2 = ra^2/\gamma^2 = \bar{\Lambda} n(2-n)/(1-n)^2$.

(c) For bins I, J, $\mathrm{Cov}\{N(I), N(J)\}$ integrates the covariance measure, which is the inverse Fourier transform of the Bartlett spectrum. The variance-time curve $\operatorname{Var} N(0,w] = \bar{\Lambda} w + 2\int_0^w (w-u)c(u)\,\mathrm{d}u$, the index of dispersion and the expected periodogram are functionals of the same objects. Both H and C are stationary Poisson cluster processes with almost surely finite clusters, so they are mixing (Daley and Vere-Jones, 2008, Ch. 12). Their binned count sequences are therefore stationary and ergodic, with finite second moments. By Birkhoff's theorem, empirical means and autocovariances converge almost surely to the common population values. The continuous mapping theorem concludes the proof. □

A.4 Proof of Proposition 4.3

Step 1: finite mixtures. Let $\varphi(t) = \sum_{i=1}^K c_i e^{-b_i t}$ with $0 < b_1 < \cdots < b_K$, $c_i > 0$ and $\sum_i c_i/b_i = n < 1$. Its Laplace transform $\tilde{\varphi}(z) = \sum_i c_i/(z + b_i)$ is strictly decreasing on each interval between consecutive poles. On $(-b_1, \infty)$ it falls from $+\infty$ to 0 and equals $n < 1$ at $z = 0$. Hence $\tilde{\varphi} = 1$ at exactly one point $z_1 \in (-b_1, 0)$. On each $(-b_{i+1}, -b_i)$ it falls from $+\infty$ to $-\infty$, which gives one more root z_{i+1}.

Write $1 - \tilde{\varphi} = P_1/P_0$ with $P_0(z) = \prod_i (z + b_i)$, so that P_1 is monic of degree K. These K roots are then all the roots of P_1; they are real, negative and simple. Moreover $P_0 - P_1$ has degree $K - 1$, so

$$\tilde{R} = \frac{\tilde{\varphi}}{1 - \tilde{\varphi}} = \frac{P_0 - P_1}{P_1} = \sum_{l=1}^K \frac{\rho_l}{z - z_l}, \qquad \rho_l = \frac{1}{-\tilde{\varphi}'(z_l)} > 0,$$

since $\tilde{\varphi}(z_l) = 1$ and $\tilde{\varphi}' < 0$. The function $\sum_l \rho_l e^{z_l t}$ and the resolvent $R = \sum_j \varphi^{*j} \in L^1$ have the same Laplace transform on $z \geq 0$. Hence $R(t) = \sum_l \rho_l e^{z_l t}$, a positive combination of decaying exponentials, which is completely monotone. Also $R(0+) = \lim_{z\to\infty} z\tilde{R}(z) = \varphi(0+)$.

Step 2: general completely monotone kernels. By Bernstein's theorem, $\varphi(t) = n\int_{(0,\infty)} e^{-bt}\sigma(\mathrm{d}b)$ with $\int \sigma(\mathrm{d}b)/b = 1$. Fix $L \geq 1$ and partition $(2^{-L}, 2^L]$ into cells of width 2^{-L}. Let

$$\varphi_L(t) = n\sum_I \sigma(I) e^{-b_I^+ t},$$

where b_I^+ is the right end of cell I. Each φ_L is a finite exponential mixture. Refining a cell or extending the range can only increase φ_L, so $\varphi_L \leq \varphi_{L+1} \leq \varphi$, and $\varphi_L \uparrow \varphi$ pointwise by monotone convergence. Hence $\int \varphi_L \leq n < 1$ and, by Step 1, each $R_L = \sum_k \varphi_L^{*k}$ is completely monotone.

Convolution powers are monotone in the kernel, so $R_L \uparrow R$ pointwise by monotone convergence. The limit R is nonincreasing and integrable, with $\int R = n/(1-n)$, so it is finite on $(0,\infty)$. A pointwise limit of completely monotone functions is completely monotone (Schilling et al., 2012, Cor. 1.6). Moreover

$$R(0+) = \sup_L R_L(0+) = \sup_L \varphi_L(0+) \le \varphi(0+) < \infty.$$

Step 3: covariance. By Proposition 3.2, $2\pi f_H = \bar{\Lambda}|1+\hat{R}|^2$, so for $u \neq 0$

$$c_H(u) = \bar{\Lambda}\Big[R(|u|) + \int_0^\infty R(v)R(v+|u|)\,\mathrm{d}v\Big].$$

Write $R(t) = \int e^{-bt}\varpi(\mathrm{d}b)$. By Tonelli, $\int_0^\infty R(v)R(v+u)\,\mathrm{d}v = \iint e^{-b'u}(b+b')^{-1}\varpi(\mathrm{d}b)\varpi(\mathrm{d}b')$. Hence $c_H(u) = \int e^{-b|u|}\mathcal{K}(\mathrm{d}b)$ with

$$\mathcal{K}(\mathrm{d}b) = \bar{\Lambda}\Big[1 + \int \frac{\varpi(\mathrm{d}b')}{b+b'}\Big]\varpi(\mathrm{d}b) \ge 0.$$

If ϖ has K atoms, so does $\mathcal{K}$. Also $Z = \int \mathcal{K}(\mathrm{d}b)/b = \frac{1}{2}\int_{\mathbb{R}} c_H = \frac{1}{2}\{2\pi f_H(0) - \bar{\Lambda}\} = \bar{\Lambda}(F_\infty - 1)/2$, and $\int \mathcal{K}(\mathrm{d}b) = c_H(0) < \infty$ by Step 2.

Step 4: a Cox process with the same second-order structure. Each shot integrates to m, so $\mathbb{E}\xi = \nu + rm = \bar{\Lambda}$. By Campbell's theorem for the marked Poisson process $\{(c_\ell, \Gamma_\ell)\}$, for $u \ge 0$,

$$\mathrm{Cov}\{\xi(t), \xi(t+u)\} = r\,\mathbb{E}\int_{-\infty}^{t} m^2\Gamma^2 e^{-\Gamma(t-c)}e^{-\Gamma(t+u-c)}\,\mathrm{d}c = \frac{rm^2}{2}\int b\,e^{-bu}\,\pi(\mathrm{d}b) = \frac{rm^2}{2Z}\int e^{-bu}\mathcal{K}(\mathrm{d}b),$$

which equals $c_H(u)$ for $r = 2Z/m^2$. Then $\nu = \bar{\Lambda} - 2Z/m = \bar{\Lambda}\{1 - (F_\infty - 1)/m\} \ge 0$ if and only if $m \ge F_\infty - 1$. Also $\mathbb{E}\xi^2 < \infty$ because $c_H(0) < \infty$. By Lemma A.1(a), the Cox process has mean $\bar{\Lambda}$ and covariance density c_H, so it is second-order equivalent to H.

Step 5. $2\pi f_H(\omega) = \bar{\Lambda} + \int 2b(b^2+\omega^2)^{-1}\mathcal{K}(\mathrm{d}b) \ge \bar{\Lambda}$.

Gamma kernel (Remark 4.4). For $g(t) = \beta^2 t e^{-\beta t}$, put $x = \omega^2/\beta^2$. Then

$$|1-\hat{\varphi}|^2 = \frac{(1-n-x)^2 + 4x}{(1+x)^2}.$$

Its derivative in x has the sign of $3-n-x$, so it is maximised at $x = 3-n$, where it equals $4/(4-n)$. Hence $\min_\omega 2\pi f_H/\bar{\Lambda} = 1 - n/4$. □

A.5 Proof of Proposition 4.5

(a) This is Lemma A.1(a).

(b) Row k of $\boldsymbol{I} - \hat{\boldsymbol{\Phi}}$ is $\boldsymbol{e}_k^\top$, so $\boldsymbol{e}_k^\top = \boldsymbol{e}_k^\top \boldsymbol{G}$. Hence $(\boldsymbol{G}\boldsymbol{\Delta}\boldsymbol{G}^*)_{kk} = \bar{\Lambda}_k$ and $(\boldsymbol{G}\boldsymbol{\Delta}\boldsymbol{G}^*)_{jk} = G_{jk}\bar{\Lambda}_k$, where $\boldsymbol{\Delta} = \mathrm{diag}(\bar{\boldsymbol{\Lambda}})$. Thus $E_{H,kk} \equiv 0$ and $E_{H,jk} = G_{jk}\bar{\Lambda}_k$. At $\omega = 0$, $\boldsymbol{G}(0) = \boldsymbol{M} = \sum_m \boldsymbol{A}^m \ge \boldsymbol{A}$ entrywise, so $G_{jk}(0) \ge \alpha_{jk} > 0$. Since $\hat{\boldsymbol{\Phi}}$ is continuous, $G_{jk} \neq 0$ on a neighbourhood U of 0. On U, the principal 2×2 submatrix of $\boldsymbol{E}_H$ indexed by $\{k,j\}$ has determinant $-|E_{H,jk}|^2 < 0$, so $\boldsymbol{E}_H$ is not positive semidefinite there. By (a), no Cox process has this spectrum.

(c) Campbell's theorem gives $\mathrm{Cov}\{\xi_k(t+u), \xi_j(t)\} = r\mathbb{E}[a_k a_j]\int s_k(t+u-c)s_j(t-c)\,\mathrm{d}c$. □

A.6 Proof of Theorem 4.7

(a) By Lemma A.1, $2\pi f_N(\omega) = \bar{\Lambda} + 2\sigma^2\gamma/(\gamma^2+\omega^2)$. An exponential Hawkes process has $2\pi f(\omega) = \bar{\Lambda}_H + \bar{\Lambda}_H\beta^2 n(2-n)/\{\beta^2(1-n)^2+\omega^2\}$. As in the proof of Theorem 4.2(a), second-order equivalence holds if and only if $\bar{\Lambda}_H = \bar{\Lambda}$, $\beta(1-n) = \gamma$ and $\bar{\Lambda}\beta^2 n(2-n) = 2\sigma^2\gamma$. Using $\beta^2 n(2-n) = \gamma^2\{(1-n)^{-2}-1\}$, the last condition becomes

$$(1-n)^{-2} = 1 + 2\sigma^2/(\bar{\Lambda}\gamma) = 1+\kappa,$$

since $\int c_\xi = 2\sigma^2/\gamma$. This has the unique solution $n^* \in (0,1)$, and then $\beta^* = \gamma/(1-n^*)$ and $\mu^* = \bar{\Lambda}(1-n^*)$. For the shot-noise process, $\sigma^2 = ra^2/(2\gamma)$. For the two-state chain, $\mathbb{P}(X_u = 1 \mid X_0 = 1) = \pi_1 + \pi_0 e^{-(q_{01}+q_{10})u}$, which gives $c_\xi(u) = (\lambda_1-\lambda_0)^2\pi_0\pi_1 e^{-(q_{01}+q_{10})|u|}$.

(b) Let $\boldsymbol{m}(\vartheta)$ denote the population counterpart of $\hat{\boldsymbol{m}}_T$. Since N is stationary and ergodic with finite second moments, $\hat{\boldsymbol{m}}_T \to \boldsymbol{m}(\vartheta^*)$ almost surely under N, by Birkhoff's theorem and (a). Under $H(\vartheta^*)$, which is ergodic (proof of Theorem 4.2(c)), also $\hat{\boldsymbol{m}}_T \to \boldsymbol{m}(\vartheta^*)$, and by assumption $G(\hat{\boldsymbol{m}}_T) \to \vartheta^*$. Continuity of G then forces $G(\boldsymbol{m}(\vartheta^*)) = \vartheta^*$. Hence, under N, $\hat{\vartheta}_T = G(\hat{\boldsymbol{m}}_T) \to G(\boldsymbol{m}(\vartheta^*)) = \vartheta^*$.

(c) $\hat{c}_\xi$ is continuous because $c_\xi \in L^1$, so $F_\infty(N) = 1 + \hat{c}_\xi(0)/\bar{\Lambda} = 1+\kappa$. For a Hawkes process, $F_\infty = |1-\hat{\varphi}(0)|^{-2} = (1-n)^{-2}$, so $n_F = n$.

(d) Fix β and set $X(t) = \int_{(-\infty,t)} g_\beta(t-s)\,\mathrm{d}N(s) \geq 0$, a predictable stationary process with $\mathbb{E}X = \bar{\Lambda}$. The second factorial moment measure of N has density $\bar{\Lambda}^2 + c(t-s)$ off the diagonal, so

$$\mathbb{E}\int_0^1 X(t)\,\mathrm{d}N(t) = \int_0^1\int_{-\infty}^t g_\beta(t-s)\{\bar{\Lambda}^2 + c(t-s)\}\,\mathrm{d}s\,\mathrm{d}t = \bar{\Lambda}^2 + I_\beta, \qquad I_\beta = \int_0^\infty g_\beta c. \tag{A.2}$$

Since $\log\mu \leq \log\lambda_\vartheta \leq \log\mu + nX/\mu$, the function L is finite.

For $n=0$, $L(\mu,0,\beta) = \bar{\Lambda}\log\mu - \mu$, which is uniquely maximised at $\mu = \bar{\Lambda}$ with value $\bar{\Lambda}\log\bar{\Lambda} - \bar{\Lambda}$. Now fix $\mu = \bar{\Lambda}$ and $\beta = \beta_0$. By concavity, the difference quotients $D_n = \{\log(\bar{\Lambda}+nX) - \log\bar{\Lambda}\}/n \geq 0$ increase to $X/\bar{\Lambda}$ as $n \downarrow 0$. By monotone convergence and (A.2),

$$\frac{L(\bar{\Lambda},n,\beta_0) - L(\bar{\Lambda},0,\beta_0)}{n} = \mathbb{E}\int_0^1 D_n\,\mathrm{d}N - \bar{\Lambda} \longrightarrow \frac{I_{\beta_0}}{\bar{\Lambda}} > 0.$$

Hence $L(\bar{\Lambda},n,\beta_0) > \bar{\Lambda}\log\bar{\Lambda} - \bar{\Lambda}$ for all small $n>0$. For C, $c(u) = \{ra^2/(2\gamma)\}e^{-\gamma|u|}$, and integrating against $\beta e^{-\beta u}$ gives the stated I_β. □

A.7 Proof of Corollary 4.9

(i) Under Assumption A1 the Volterra resolvent of $\boldsymbol{\Phi}$ on $[0,T]$ is integrable, so $\mathbb{E}\boldsymbol{N}[0,T] < \infty$, and the intensity has the form of Lemma A.3 with

$$h_k(t;x) = m_k(t) + \sum_j \sum_{s\in x_j} \varphi_{kj}(t-s), \qquad h_k \geq \inf\mu_k > 0.$$

Take two parameter values with the same law. Taking $x = \emptyset$ in Lemma A.3 gives $m_k^{(1)} = m_k^{(2)}$ almost everywhere on $[0,T]$. Taking $x = \{(s,j)\}$ then gives $\varphi_{kj}^{(1)}(t-s) = \varphi_{kj}^{(2)}(t-s)$ for almost every $0<s<t<T$. By Fubini, $\varphi_{kj}^{(1)} = \varphi_{kj}^{(2)}$ almost everywhere on $(0,T)$.

For the sufficient condition, $m_k = \mu_k$ on $(0,\tau_1)$. This determines μ_k when μ_k is constant, and on all of $[0,T]$ by analytic continuation when it is analytic. Then $\sum_e S_e\upsilon_k\psi_k(t-\tau_e) = m_k(t) - \mu_k(t)$ is known. On $[\tau_1,\tau_2)$ only the first shock is active, so this determines the product $\upsilon_k S_1\psi_k$ there. Since $S_1 > 0$ is observed,

it determines $\upsilon_k\psi_k$ on $[0,\tau_2-\tau_1)$. By the condition on the scaled family, $\upsilon_k\psi_k$ is then determined on $(0,\infty)$. Then $\int\psi_k=1$ gives $\upsilon_k=\int\upsilon_k\psi_k$, and hence ψ_k. Finally, if $\alpha_{kj}>0$ and g_{kj} does not vanish on $(0,T)$, then $\varphi_{kj}\neq 0$ on a set of positive measure, so the law differs from that with $\alpha_{kj}=0$.

(ii) By Lemma A.2(a) and Proposition 3.1(c), the cumulant rates of H are $\mu\mathbb{E}\mathcal{S}^j$. Normalise $\mu=1$ (all χ_j scale linearly in μ) and write $u=1-n$. Then

$$\chi_1=u^{-1},\quad \chi_2=u^{-3},\quad \chi_3=(3-2u)u^{-5},\quad \chi_4=(15-20u+6u^2)u^{-7}.$$

The cumulant rates of C are $\nu+r\mathbb{E}[\mathrm{Poisson}(m)^j]$, with $\mathbb{E}\,\mathrm{Poisson}(m)^j=m$, $m+m^2$, $m+3m^2+m^3$ and $m+7m^2+6m^3+m^4$ for $j=1,\dots,4$. Matching $j=1,2,3$ gives $rm^2=\chi_2-\chi_1$ and $r(3m^2+m^3)=\chi_3-\chi_1$, hence

$$m+3=\frac{\chi_3-\chi_1}{\chi_2-\chi_1}=\frac{3+u+u^2+u^3}{u^2(1+u)},\qquad m=\frac{(1-u)(3+4u+2u^2)}{u^2(1+u)}>0.$$

This determines (ν,r,m) uniquely, with γ free. Moreover

$$m-m_{\min}=\frac{(1-u)(2+2u+u^2)}{u^2(1+u)}>0,$$

so with $\gamma=\beta(1-n)$ this is a member of the family of Theorem 4.2. Since $7m^2+6m^3+m^4=m^2\{(m+3)^2-2\}$,

$$\chi_4^C=\chi_1+\frac{(\chi_3-\chi_1)^2}{\chi_2-\chi_1}-2(\chi_2-\chi_1),\qquad \chi_4^H-\chi_4^C=\frac{\mathcal{Y}}{\chi_2-\chi_1},$$

where

$$\mathcal{Y}=(\chi_4-\chi_1)(\chi_2-\chi_1)+2(\chi_2-\chi_1)^2-(\chi_3-\chi_1)^2.$$

Direct expansion gives

$$u^{10}\mathcal{Y}=6-8u-13u^2+20u^3+2u^4-4u^5-5u^6+2u^8=(1-u)^4\,Q(u).$$

Using $\chi_2-\chi_1=n(2-n)(1-n)^{-3}$ yields the stated difference, which is positive for $n\in(0,1)$ since $Q>0$ on $(0,1)$. If $\mathbb{P}_H=\mathbb{P}_C$, all cumulant rates would coincide, which is impossible.

A second proof uses Lemma A.2(b). For the Borel law, $\mathbb{P}(\mathcal{S}=2)\mathbb{P}(\mathcal{S}=4)/\mathbb{P}(\mathcal{S}=3)^2=32/27$. For the cluster law of C the same ratio is $3/4$, since the ν-component only affects $k=1$.

(iii) In the cluster representation, each stream-j incident has Poisson(α_{kj}) stream-k offspring, independently of everything else. By stationarity, the parent–offspring pairs of type (j,k) therefore occur at rate $\alpha_{kj}\bar{\Lambda}_j$ per unit time, and each is recorded with probability q. □

A.8 Proof of Theorem 4.10 and Corollary 4.11

(a) This follows from Lemma A.1(b) with $M=1$ and (3.5).

(b) By the argument of Theorem 4.2(a), $H(\mu',n',\beta')$ is second-order equivalent to N^p if and only if $\bar{\Lambda}'=p\bar{\Lambda}$, $\beta'(1-n')=\beta(1-n)=:b$ and $\beta'^2n'(2-n')=p\beta^2n(2-n)$. Put $F(x)=(1-x)^{-2}$. Then $\beta^2n(2-n)=b^2\{F(n)-1\}$ and similarly for n', so the last condition reads $F(n')-1=p\{F(n)-1\}$. This determines n' uniquely, and then $\beta'^2=b^2F(n')=\beta^2(1-n)^2+p\beta^2n(2-n)$ and $\mu'=\bar{\Lambda}'(1-n')$.

(c) The function F is a strictly increasing, strictly convex bijection $[0,1)\to[1,\infty)$ with $F(0)=1$, and

$$h_p=F^{-1}\circ\{1+p(F-1)\}.$$

Hence h_p is an increasing bijection. Next, $h_p(n) < n$ if and only if $(1-p)\{F(n)-1\} > 0$. Also $h_p(n) > pn$ if and only if $F(pn) < (1-p)F(0) + pF(n)$, which is strict convexity. The expansion follows from $F(n) = 1 + 2n + O(n^2)$. Finally,

$$\beta' = b\,F(n')^{1/2} = \beta\{F(n')/F(n)\}^{1/2} \in [\beta(1-n), \beta],$$

and solving $F(n) = 1 + \{F(n')-1\}/p$ gives (4.5).

(d) N^p is a Poisson cluster process with $r = \mu$ and cluster size $\mathcal{S}_p \mid \mathcal{S} \sim \text{Binomial}(\mathcal{S}, p)$, where $\mathcal{S} \sim \text{Borel}(n)$, so $\mathbb{E}e^{u\mathcal{S}_p} < \infty$ for small $u > 0$. Suppose N^p were a stationary Hawkes process with constant background $\mu'' \geq 0$ and some kernel $\varphi'' \geq 0$ of mass n''.

First suppose $\mu'' = 0$. Taking expectations of the intensity gives $p\bar{\Lambda} = n'' p\bar{\Lambda}$, so $n'' = 1$. The process N^p has finite second moments, so $\mathrm{d}N^p - \lambda''\mathrm{d}t$ is a martingale measure with spectral density $p\bar{\Lambda}/(2\pi)$. Since $\mathrm{d}N^p - \lambda''\mathrm{d}t$ is the output of the filter $\delta_0 - \varphi''$ applied to $\mathrm{d}N^p$, its spectral measure is $|1-\hat{\varphi}''(\omega)|^2 f_p(\omega)\,\mathrm{d}\omega$. Hence $|1-\hat{\varphi}''(\omega)|^2 f_p(\omega) = p\bar{\Lambda}/(2\pi)$ for almost every ω. As $\omega \to 0$, $\hat{\varphi}''(\omega) \to n'' = 1$ while f_p stays bounded by (a), since $F'_\infty < \infty$. The left side therefore tends to 0, a contradiction.

So $\mu'' > 0$. Then Proposition 3.1(b) gives $n'' < 1$, and N^p is a Poisson cluster process with Borel(n'') clusters. By Lemma A.2(b), $\mu\mathbb{P}(\mathcal{S}_p = k) = \mu''\mathbb{P}(\text{Borel}(n'') = k)$ for $k \geq 1$.

For any Borel law with $n'' > 0$,

$$\frac{\mathbb{P}(1)\mathbb{P}(3)}{\mathbb{P}(2)^2} = \frac{e^{-n''}\frac{3}{2}n''^2 e^{-3n''}}{(n''e^{-2n''})^2} = \frac{3}{2}.$$

If $n'' = 0$, then $\mathbb{P}(2) = 0 < \mathbb{P}(\mathcal{S}_p = 2)$, which is already a contradiction. For $\mathcal{S}_p$, write the Borel generating function as $G(z) = \Theta(\zeta z)/n$, where $\zeta = ne^{-n}$ and $\Theta(y) = \sum_{k\geq 1} k^{k-1}y^k/k!$ is the tree function, which satisfies $\Theta = ye^{\Theta}$. Then $\mathbb{P}(\mathcal{S}_p = k) = p^k G^{(k)}(1-p)/k! = (p\zeta)^k \Theta^{(k)}(y)/(n\,k!)$ with $y = \zeta(1-p) \in (0, e^{-1})$. Differentiating $\Theta = ye^{\Theta}$ and writing $v = \Theta(y) \in (0,1)$,

$$\Theta' = \frac{e^v}{1-v}, \qquad \Theta'' = \frac{e^{2v}(2-v)}{(1-v)^3}, \qquad \Theta''' = \frac{e^{3v}(2v^2-8v+9)}{(1-v)^5}.$$

Therefore

$$\frac{\mathbb{P}(\mathcal{S}_p=1)\mathbb{P}(\mathcal{S}_p=3)}{\mathbb{P}(\mathcal{S}_p=2)^2} = \frac{2}{3}\frac{\Theta'\Theta'''}{(\Theta'')^2} = \frac{2}{3}\frac{2v^2-8v+9}{(2-v)^2} = \frac{3}{2} + \frac{v(4-v)}{6(2-v)^2} > \frac{3}{2},$$

a contradiction. The statement about moment-based estimators follows as in Theorem 4.7(b), since N^p is a Poisson cluster process (hence ergodic) and is second-order equivalent to $H(\mu', n', \beta')$.

Proof of Corollary 4.11. Write $h^{-1}(n', p) = 1 - \mathcal{F}^{-1/2}$ with $\mathcal{F} = 1 + \{(1-n')^{-2} - 1\}/p = (1-n)^{-2}$. Then

$$\frac{\partial h^{-1}}{\partial n'} = \frac{\mathcal{F}^{-3/2}}{p\,(1-n')^3} = \frac{(1-n)^3}{p\,(1-n')^3}, \qquad \frac{\partial h^{-1}}{\partial p} = -\frac{\mathcal{F}^{-3/2}\{(1-n')^{-2}-1\}}{2p^2} = -\frac{n(1-n)(2-n)}{2p},$$

using $(1-n')^{-2} - 1 = p\{(1-n)^{-2} - 1\}$. The map h^{-1} is C^1 on $(0,1)\times(0,1]$. Consistency follows from the continuous mapping theorem, and (4.6) from the delta method and independence. □

A.9 Heterogeneous recording

Severe incidents may be both more likely to be recorded and more likely to trigger further incidents. Suppose each incident carries an *unpredictable* mark $X \in \{1, \ldots, K\}$, such as a severity class. Marks are i.i.d. with law π, independent of the past. An incident with mark x has expected direct offspring n_x and is recorded independently with probability p_x, and the kernel is $\beta e^{-\beta t}$. Write $\mathbb{E}$, Var and Cov for moments under π, and set $\bar{n} = \mathbb{E}n_X < 1$ and $\bar{p} = \mathbb{E}p_X$.

Proposition A.6 (Heterogeneous recording). *The recorded process is stationary with mean $\bar{p}\bar{\Lambda}$, where $\bar{\Lambda} = \mu/(1-\bar{n})$, and it is second-order equivalent to an exponential Hawkes process with branching ratio*

$$n'' = 1 - \frac{1-\bar{n}}{\sqrt{(1-\bar{n})^2 + 2(1-\bar{n})\mathbb{E}[p_X n_X] + \bar{p}\,\mathbb{E}[n_X^2]}}.$$

The homogeneous correction of Corollary 4.11 with $p = \bar{p}$ returns $n_{\text{naive}} = h_{\bar{p}}^{-1}(n'')$, where

$$(1-n_{\text{naive}})^{-2} = (1-\bar{n})^{-2}\Big[1 + \frac{2(1-\bar{n})\,\mathrm{Cov}(p_X, n_X)}{\bar{p}} + \mathrm{Var}(n_X)\Big].$$

It overstates $\bar{n}$ whenever $2(1-\bar{n})\,\mathrm{Cov}(p_X, n_X)/\bar{p} + \mathrm{Var}(n_X) > 0$, in particular when more productive incidents are recorded more often.

With $p_x \equiv p$ and $n_x \equiv n$, Proposition A.6 reduces to Theorem 4.10. Heterogeneous productivity alone ($p \equiv 1$) inflates the apparent branching ratio of unmarked data to $1 - (1-\bar{n})/\sqrt{1+\mathrm{Var}(n_X)}$.

Proof of Proposition A.6. Marks are unpredictable with law π, so incidents with mark x have intensity $\pi_x\{\mu + \sum_i n_{X_i} g(t-t_i)\}$. The marked process is therefore a K-variate Hawkes process with background $\mu\boldsymbol{\pi}$, branching matrix $\boldsymbol{A} = \boldsymbol{\pi}\boldsymbol{n}^\top$ and common kernel g. Since $\boldsymbol{A}$ has rank one, $\varrho(\boldsymbol{A}) = \boldsymbol{n}^\top\boldsymbol{\pi} = \bar{n} < 1$. By the Sherman–Morrison formula, $(\boldsymbol{I}-\boldsymbol{A})^{-1} = \boldsymbol{I} + \boldsymbol{\pi}\boldsymbol{n}^\top/(1-\bar{n})$, so $\bar{\boldsymbol{\Lambda}} = \mu\boldsymbol{\pi}/(1-\bar{n}) = \bar{\Lambda}\boldsymbol{\pi}$. Similarly, with $\hat{\boldsymbol{\Phi}} = \hat{g}\,\boldsymbol{\pi}\boldsymbol{n}^\top$,

$$\boldsymbol{G} = \boldsymbol{I} + \mathsf{H}\boldsymbol{\pi}\boldsymbol{n}^\top, \qquad \mathsf{H} = \hat{g}/(1-\bar{n}\hat{g}).$$

The recorded process is $\mathbf{1}^\top \boldsymbol{N}^{\boldsymbol{p}}$ with $\boldsymbol{p} = (p_x)$. By Lemma A.1(b) and Proposition 3.2, with $\boldsymbol{\Delta} = \bar{\Lambda}\,\mathrm{diag}(\boldsymbol{\pi})$,

$$2\pi f_{\text{rec}} = \boldsymbol{p}^\top\bar{\boldsymbol{\Lambda}} + \boldsymbol{p}^\top(\boldsymbol{G}\boldsymbol{\Delta}\boldsymbol{G}^* - \boldsymbol{\Delta})\boldsymbol{p}.$$

Since $\boldsymbol{G}\boldsymbol{\Delta}\boldsymbol{G}^* - \boldsymbol{\Delta} = \mathsf{H}\boldsymbol{\pi}\boldsymbol{n}^\top\boldsymbol{\Delta} + \bar{\mathsf{H}}\boldsymbol{\Delta}\boldsymbol{n}\boldsymbol{\pi}^\top + |\mathsf{H}|^2(\boldsymbol{n}^\top\boldsymbol{\Delta}\boldsymbol{n})\boldsymbol{\pi}\boldsymbol{\pi}^\top$, and $\boldsymbol{p}^\top\boldsymbol{\pi} = \bar{p}$, $\boldsymbol{n}^\top\boldsymbol{\Delta}\boldsymbol{p} = \bar{\Lambda}\mathbb{E}[p_X n_X]$, $\boldsymbol{n}^\top\boldsymbol{\Delta}\boldsymbol{n} = \bar{\Lambda}\mathbb{E}[n_X^2]$,

$$2\pi f_{\text{rec}} = \bar{p}\bar{\Lambda}\big[1 + 2\,\mathrm{Re}(\mathsf{H})\mathbb{E}[p_X n_X] + |\mathsf{H}|^2\bar{p}\,\mathbb{E}[n_X^2]\big].$$

For $g(t) = \beta e^{-\beta t}$, $\mathsf{H} = \beta/(b+i\omega)$ with $b = \beta(1-\bar{n})$, so $2\,\mathrm{Re}\,\mathsf{H} = 2\beta b/(b^2+\omega^2)$ and $|\mathsf{H}|^2 = \beta^2/(b^2+\omega^2)$. This gives the Bartlett spectrum

$$f_{\text{rec}}(\omega) = \frac{\bar{p}\bar{\Lambda}}{2\pi}\Big[1 + \frac{\beta^2\{2(1-\bar{n})\mathbb{E}[p_X n_X] + \bar{p}\,\mathbb{E}[n_X^2]\}}{\beta^2(1-\bar{n})^2 + \omega^2}\Big].$$

Matching with an exponential Hawkes spectrum, as in Theorem 4.10(b), gives $b'' = b$ and

$$(1-n'')^{-2} = 1 + \frac{2(1-\bar{n})\mathbb{E}[p_X n_X] + \bar{p}\,\mathbb{E}[n_X^2]}{(1-\bar{n})^2}.$$

Finally, $(1-n_{\text{naive}})^{-2} = 1 + \{(1-n'')^{-2} - 1\}/\bar{p}$. Substituting $\mathbb{E}[p_X n_X] = \bar{p}\bar{n} + \mathrm{Cov}(p_X, n_X)$, $\mathbb{E}[n_X^2] = \bar{n}^2 + \mathrm{Var}(n_X)$ and $1-(1-\bar{n})^2 = 2\bar{n}(1-\bar{n}) + \bar{n}^2$ gives the stated expression. □

A.10 Proof of Proposition 4.12

As in the proof of Corollary 4.9(i), the intensity has the form of Lemma A.3 with $h_k(t;x) = m_k(t) + \sum_l \sum_{s\in x_l} \varphi_{kl}(t-s)$.

If $\varphi_{kj} = 0$ almost everywhere on $(0, T)$, then $h_k(t; x)$ depends on x only through x_{-j}, the configuration with stream j removed. Hence λ_k is $\mathcal{F}^{V\setminus\{j\}}$-predictable.

Conversely, suppose λ_k has a $\mathcal{F}^{V\setminus\{j\}}$-predictable version. Predictable processes of a point-process history are measurable functionals of the history strictly before t (Daley and Vere-Jones, 2008, Ch. 14). So $h_k(t; x) = \tilde{h}_k(t, x_{-j})$ for $\mathbb{P} \otimes \mathrm{d}t$-almost every (t, x), and, because the Janossy densities are positive, for almost every (t, x) with respect to Lebesgue measure times the Lebesgue–Poisson measure. The configurations $x = \emptyset$ and $x = \{(s, j)\}$ have the same $x_{-j} = \emptyset$. Hence

$$m_k(t) = \tilde{h}_k(t, \emptyset) = m_k(t) + \varphi_{kj}(t - s) \qquad \text{for almost every } 0 < s < t < T,$$

that is, $\varphi_{kj} = 0$ almost everywhere on $(0, T)$. For exponential kernels, $g_{kj} > 0$ everywhere, so this is equivalent to $\alpha_{kj} = 0$. □

Appendix B Proofs and supplements for Sections 5 and 7

Throughout this appendix, inequalities between vectors, matrices and matrix functions hold entrywise, and $\mathbf{1}$ is the vector of ones. For nonnegative functions, integrals and sums are exchanged by Tonelli's theorem without further comment.

B.1 Auxiliary lemmas

Lemma B.1 (Convolution bounds). *Let $\boldsymbol{F}, \boldsymbol{G} \geq 0$ be locally integrable matrix functions on $[0, \infty)$ of compatible sizes. For every $T \in (0, \infty]$, $\int_0^T (\boldsymbol{F} * \boldsymbol{G}) \leq \big(\int_0^T \boldsymbol{F}\big)\big(\int_0^T \boldsymbol{G}\big)$, with equality when $T = \infty$.*

Proof. By Tonelli's theorem, $\int_0^T \int_0^t \boldsymbol{F}(t - s)\boldsymbol{G}(s)\,\mathrm{d}s\,\mathrm{d}t = \int_0^T \big(\int_0^{T-s} \boldsymbol{F}\big)\boldsymbol{G}(s)\,\mathrm{d}s$. The inner integral is at most $\int_0^T \boldsymbol{F}$, and it equals $\int_0^\infty \boldsymbol{F}$ when $T = \infty$. □

Lemma B.2 (Splitting and pruning cluster processes). *Consider the cluster construction of Section 3.3 with immigrant intensity $\boldsymbol{\nu} \geq 0$ and kernel matrix $\boldsymbol{\Phi}$.*

(a) If $\boldsymbol{\nu} = \boldsymbol{\nu}^1 + \boldsymbol{\nu}^2$ with $\boldsymbol{\nu}^i \geq 0$, the process is the superposition of two independent cluster processes with immigrant intensities $\boldsymbol{\nu}^1$ and $\boldsymbol{\nu}^2$ and kernel $\boldsymbol{\Phi}$.

(b) Let $\mathcal{E}$ be a set of index pairs (k, j). In every cluster, delete each stream-k child of a stream-j parent with $(k, j) \in \mathcal{E}$, together with all of its descendants. The remaining points form a cluster process with the same immigrants and kernel $\boldsymbol{\Phi}^{\mathcal{E}}$, where $\varphi^{\mathcal{E}}_{kj} = \varphi_{kj}\mathbf{1}\{(k, j) \notin \mathcal{E}\}$.

Proof. (a) A Poisson process with intensity $\boldsymbol{\nu}^1 + \boldsymbol{\nu}^2$ is the superposition of independent Poisson processes with intensities $\boldsymbol{\nu}^1$ and $\boldsymbol{\nu}^2$; this is the colouring theorem. Clusters are generated independently given their roots.

(b) Given a stream-j parent at s, its children in the different streams k form independent Poisson processes with intensities $\varphi_{kj}(\cdot - s)$. Deleting the processes with $(k, j) \in \mathcal{E}$ leaves independent Poisson processes with intensities $\varphi^{\mathcal{E}}_{kj}(\cdot - s)$. Every retained child generates its own offspring in the same way, so by induction over generations the retained points follow the cluster construction with kernel $\boldsymbol{\Phi}^{\mathcal{E}}$. □

Lemma B.3 (A Hurwitz criterion). *Let $\boldsymbol{\mathfrak{D}} = \mathrm{diag}(\mathfrak{d}_1, \ldots, \mathfrak{d}_m)$ with $\mathfrak{d}_i > 0$, and let $\boldsymbol{P} \in [0, \infty)^{m\times M}$ and $\boldsymbol{S} \in [0, \infty)^{M\times m}$ satisfy $\varrho(\boldsymbol{S}\boldsymbol{\mathfrak{D}}^{-1}\boldsymbol{P}) < 1$. Then every eigenvalue of $\boldsymbol{Q} = -\boldsymbol{\mathfrak{D}} + \boldsymbol{P}\boldsymbol{S}$ has negative real part.*

Proof. The matrices $\boldsymbol{T} = \mathfrak{D}^{-1}\boldsymbol{P}\boldsymbol{S}$ and $\boldsymbol{S}\mathfrak{D}^{-1}\boldsymbol{P}$ have the same nonzero eigenvalues, so $\varrho(\boldsymbol{T}) < 1$. Choose $r \in (\varrho(\boldsymbol{T}), 1)$, and let $\boldsymbol{v} = \sum_{n\geq 0}(\boldsymbol{T}/r)^n\mathbf{1} = (\boldsymbol{I} - \boldsymbol{T}/r)^{-1}\mathbf{1} \geq \mathbf{1}$. Then $\boldsymbol{T}\boldsymbol{v} = r(\boldsymbol{v} - \mathbf{1})$, and so

$$\boldsymbol{Q}\boldsymbol{v} = \mathfrak{D}(\boldsymbol{T}\boldsymbol{v} - \boldsymbol{v}) = -\mathfrak{D}\{(1-r)\boldsymbol{v} + r\mathbf{1}\} < \mathbf{0}.$$

The off-diagonal entries of $\boldsymbol{Q}$ are those of $\boldsymbol{P}\boldsymbol{S}$, which are nonnegative. Since $(\boldsymbol{Q}\boldsymbol{v})_i < 0$, it follows that $Q_{ii} < 0$ and $\sum_{l\neq i} Q_{il}v_l < -Q_{ii}v_i$.

Let $\boldsymbol{Q}\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\boldsymbol{x} \neq \mathbf{0}$, and choose i to maximise $|x_i|/v_i$. Then $|\lambda - Q_{ii}|\,|x_i| = |\sum_{l\neq i} Q_{il}x_l| \leq \sum_{l\neq i} Q_{il}v_l\,|x_i|/v_i$. Hence $|\lambda - Q_{ii}| < -Q_{ii}$. So λ lies in an open disc centred at $Q_{ii} < 0$ with radius $|Q_{ii}|$, and $\operatorname{Re}\lambda < 0$. □

Lemma B.4 (Resolvent of an exponential-sum kernel). *Let $\mathfrak{D}$, $\boldsymbol{P}$ and $\boldsymbol{S}$ be as in Lemma B.3, let $\boldsymbol{\Phi}(t) = \boldsymbol{S}e^{-\mathfrak{D}t}\boldsymbol{P}$, and assume that $\varrho(\int_0^\infty \boldsymbol{\Phi}) = \varrho(\boldsymbol{S}\mathfrak{D}^{-1}\boldsymbol{P}) < 1$. Then $\boldsymbol{R}(t) = \boldsymbol{S}e^{\boldsymbol{Q}t}\boldsymbol{P}$, where $\boldsymbol{Q} = -\mathfrak{D} + \boldsymbol{P}\boldsymbol{S}$.*

Proof. Let $\tilde{\boldsymbol{R}}(t) = \boldsymbol{S}e^{\boldsymbol{Q}t}\boldsymbol{P}$. Define $\boldsymbol{U}(t) = e^{\boldsymbol{Q}t}\boldsymbol{P} - e^{-\mathfrak{D}t}\boldsymbol{P}$ and $\boldsymbol{V}(t) = \int_0^t e^{-\mathfrak{D}(t-s)}\boldsymbol{P}\boldsymbol{S}e^{\boldsymbol{Q}s}\boldsymbol{P}\,\mathrm{d}s$. Both vanish at $t = 0$. Moreover,

$$\boldsymbol{U}' = \boldsymbol{Q}e^{\boldsymbol{Q}t}\boldsymbol{P} + \mathfrak{D}e^{-\mathfrak{D}t}\boldsymbol{P} = -\mathfrak{D}\boldsymbol{U} + \boldsymbol{P}\boldsymbol{S}e^{\boldsymbol{Q}t}\boldsymbol{P}, \qquad \boldsymbol{V}' = -\mathfrak{D}\boldsymbol{V} + \boldsymbol{P}\boldsymbol{S}e^{\boldsymbol{Q}t}\boldsymbol{P}.$$

The two functions solve the same linear equation with the same initial value, so $\boldsymbol{U} = \boldsymbol{V}$. Multiplying on the left by $\boldsymbol{S}$ gives $\tilde{\boldsymbol{R}} - \boldsymbol{\Phi} = \boldsymbol{S}\boldsymbol{V} = \boldsymbol{\Phi} * \tilde{\boldsymbol{R}}$. Since $\tilde{\boldsymbol{R}}$ is continuous, uniqueness in Lemma B.5(a) gives $\tilde{\boldsymbol{R}} = \boldsymbol{R}$. □

B.2 Resolvent and mean intensity

The following lemma, used throughout Section 5, collects classical facts about the resolvent (Gripenberg et al., 1990; Hawkes, 1971).

Lemma B.5 (Resolvent and mean intensity). *Let $\boldsymbol{\Phi} \geq 0$ be integrable on $[0,\infty)$, with $\int_0^\infty \boldsymbol{\Phi} = \boldsymbol{A}$ and $\varrho(\boldsymbol{A}) < 1$.*

(a) The series $\boldsymbol{R} = \sum_{m\geq 1} \boldsymbol{\Phi}^{*m}$ converges in $L^1[0,\infty)$, and $\int_0^\infty \boldsymbol{R} = \boldsymbol{M} - \boldsymbol{I}$. The resolvent $\boldsymbol{R}$ is the unique locally integrable solution of $\boldsymbol{R} = \boldsymbol{\Phi} + \boldsymbol{\Phi} * \boldsymbol{R}$, and it also satisfies $\boldsymbol{R} = \boldsymbol{\Phi} + \boldsymbol{R} * \boldsymbol{\Phi}$.

(b) Let N have intensity $\boldsymbol{\lambda}(t) = \boldsymbol{\nu}(t) + \int_0^{t-} \boldsymbol{\Phi}(t-s)\,\mathrm{d}N(s)$ for a locally integrable $\boldsymbol{\nu} \geq 0$. In the cluster representation, the incidents of generation m have mean intensity $\boldsymbol{\Phi}^{*m} * \boldsymbol{\nu}$, with $\boldsymbol{\Phi}^{*0} * \boldsymbol{\nu} := \boldsymbol{\nu}$. Hence $\mathbb{E}\,\mathrm{d}N(t) = (\mathcal{K}\boldsymbol{\nu})(t)\,\mathrm{d}t$, and $\mathcal{K}\boldsymbol{\nu}$ is the unique locally integrable solution of the renewal equation $\boldsymbol{m} = \boldsymbol{\nu} + \boldsymbol{\Phi} * \boldsymbol{m}$.

Proof of Lemma B.5. (a) By Lemma B.1 with $T = \infty$ and induction, $\int_0^\infty \boldsymbol{\Phi}^{*m} = \boldsymbol{A}^m$. Since $\varrho(\boldsymbol{A}) < 1$, the series $\sum_{m\geq 1} \boldsymbol{A}^m = \boldsymbol{M} - \boldsymbol{I}$ converges. By monotone convergence, the nonnegative partial sums of $\sum_m \boldsymbol{\Phi}^{*m}$ therefore converge in L^1 to $\boldsymbol{R}$, with $\int \boldsymbol{R} = \boldsymbol{M} - \boldsymbol{I}$. Monotone convergence inside the convolution integral gives $\boldsymbol{\Phi} * \boldsymbol{R} = \sum_{m\geq 1} \boldsymbol{\Phi}^{*(m+1)} = \boldsymbol{R} - \boldsymbol{\Phi}$. Convolution is associative, so $\boldsymbol{\Phi}^{*m} * \boldsymbol{\Phi} = \boldsymbol{\Phi}^{*(m+1)}$, and the same argument gives $\boldsymbol{R} * \boldsymbol{\Phi} = \boldsymbol{R} - \boldsymbol{\Phi}$.

For uniqueness, let $\tilde{\boldsymbol{R}}$ be another locally integrable solution, and let $\boldsymbol{\Delta} = \boldsymbol{R} - \tilde{\boldsymbol{R}}$. Then $\boldsymbol{\Delta} = \boldsymbol{\Phi} * \boldsymbol{\Delta}$, and by iteration $\boldsymbol{\Delta} = \boldsymbol{\Phi}^{*m} * \boldsymbol{\Delta}$, so $|\boldsymbol{\Delta}| \leq \boldsymbol{\Phi}^{*m} * |\boldsymbol{\Delta}|$ entrywise. For $T > 0$, Lemma B.1 gives

$$\int_0^T |\boldsymbol{\Delta}| \leq \Big(\int_0^T \boldsymbol{\Phi}^{*m}\Big)\Big(\int_0^T |\boldsymbol{\Delta}|\Big) \leq \boldsymbol{A}^m \int_0^T |\boldsymbol{\Delta}| \xrightarrow[m\to\infty]{} \mathbf{0}.$$

Hence $\boldsymbol{\Delta} = \mathbf{0}$ almost everywhere.

(b) Generation 0 is a Poisson process with intensity $\boldsymbol{\nu}$. Let $\boldsymbol{m}_g$ be the mean intensity of generation g. Given generations $0, \dots, g$, generation $g+1$ is a superposition of independent Poisson processes with total intensity $\sum \boldsymbol{\Phi}_{\cdot j}(t-s)$, summed over the points (s, j) of generation g. By Campbell's theorem its mean intensity is $\boldsymbol{m}_{g+1} = \boldsymbol{\Phi} * \boldsymbol{m}_g$, so $\boldsymbol{m}_g = \boldsymbol{\Phi}^{*g} * \boldsymbol{\nu}$. Summing over g by monotone convergence gives $\mathbb{E}\,\mathrm{d}N(t) = (\boldsymbol{\nu} + \boldsymbol{R} * \boldsymbol{\nu})(t)\,\mathrm{d}t = (\mathcal{K}\boldsymbol{\nu})(t)\,\mathrm{d}t$. By Lemma B.1, $\int_0^T \mathcal{K}\boldsymbol{\nu} \le (\boldsymbol{I} + \int_0^T \boldsymbol{R}) \int_0^T \boldsymbol{\nu} \le \boldsymbol{M} \int_0^T \boldsymbol{\nu} < \infty$, so the process is locally finite. That it has the stated intensity is the result of Hawkes and Oakes (1974) recalled in Section 3.3.

Using $\boldsymbol{\Phi} * \boldsymbol{R} = \boldsymbol{R} - \boldsymbol{\Phi}$, $\boldsymbol{\nu} + \boldsymbol{\Phi} * \mathcal{K}\boldsymbol{\nu} = \boldsymbol{\nu} + \boldsymbol{\Phi} * \boldsymbol{\nu} + (\boldsymbol{R} - \boldsymbol{\Phi}) * \boldsymbol{\nu} = \mathcal{K}\boldsymbol{\nu}$, so $\mathcal{K}\boldsymbol{\nu}$ solves the renewal equation. If $\tilde{\boldsymbol{m}}$ is another locally integrable solution, the difference $\boldsymbol{\delta}$ satisfies $\boldsymbol{\delta} = \boldsymbol{\Phi} * \boldsymbol{\delta}$, and the argument of (a) gives $\boldsymbol{\delta} = \boldsymbol{0}$. □

B.3 Proof of Theorem 5.1

(a) By Lemma B.2(a), $N = N^{\mu} + N^{e} + N^{-e}$, a superposition of independent cluster processes with immigrant intensities $\boldsymbol{\mu}$, $\boldsymbol{\nu}_e$ and $\boldsymbol{\nu}_{-e}$. Every incident belongs to exactly one of them. In N^e, generation 0 consists of the immigrants of event e, with intensity $\boldsymbol{\nu}_e$. By Lemma B.5(b), the later generations have total mean intensity $\boldsymbol{R} * \boldsymbol{\nu}_e$. Integrating over W_e gives $D_k = \upsilon_k S_e \int_{\tau_e}^{\tau_e + H} \psi_k(t - \tau_e)\,\mathrm{d}t = \upsilon_k S_e \Psi_k(H)$ and the first expression in (5.4).

For the second expression, $\boldsymbol{\nu}_e$ vanishes before τ_e, so $(\boldsymbol{R} * \boldsymbol{\nu}_e)(\tau_e + u) = \int_0^u \boldsymbol{R}(u - v)\boldsymbol{\nu}_e(\tau_e + v)\,\mathrm{d}v$. Hence

$$\int_0^H \int_0^u \boldsymbol{R}(u - v)\boldsymbol{\nu}_e(\tau_e + v)\,\mathrm{d}v\,\mathrm{d}u = \int_0^H \bar{\boldsymbol{R}}(H - v)\,\boldsymbol{\nu}_e(\tau_e + v)\,\mathrm{d}v, \qquad \boldsymbol{\nu}_e(\tau_e + v) = S_e\big(\upsilon_k \psi_k(v)\big)_k.$$

The same lemma applied to N^{μ} and N^{-e} gives $\boldsymbol{B}^{\mu}$ and $\boldsymbol{O}$. Setting $S_e = 0$ removes N^e and leaves the law of $N^{\mu} + N^{-e}$ unchanged, so $B_k = \mathbb{E}_{S_e = 0} N_k(W_e)$.

(b) Since $\int \psi_k = 1$, $D_k(H) \uparrow \upsilon_k S_e$. In (5.4), for each v the integrand $\mathbf{1}\{v \le H\}\,\bar{\boldsymbol{R}}(H - v)$ increases with H to $\int_0^{\infty} \boldsymbol{R} = \boldsymbol{M} - \boldsymbol{I}$. By monotone convergence, $\boldsymbol{C}(H) \uparrow (\boldsymbol{M} - \boldsymbol{I}) S_e \int_0^{\infty} (\upsilon_k \psi_k)_k = (\boldsymbol{M} - \boldsymbol{I})\boldsymbol{d}_e$. From $\boldsymbol{M}(\boldsymbol{I} - \boldsymbol{A}) = \boldsymbol{I}$ we get $\boldsymbol{M} - \boldsymbol{I} = \boldsymbol{M}\boldsymbol{A} = \boldsymbol{A}\boldsymbol{M}$, because $\boldsymbol{M}$ is a power series in $\boldsymbol{A}$. In the univariate case, $\boldsymbol{M} = (1 - n)^{-1}$ and $C(\infty)/\{D(\infty) + C(\infty)\} = (\boldsymbol{M} - 1)d_e/(\boldsymbol{M} d_e) = 1 - (1 - n) = n$.

(c) Since $\boldsymbol{R} \ge 0$, $\bar{\boldsymbol{R}}$ is nondecreasing and bounded by $\int_0^{\infty} \boldsymbol{R} = \boldsymbol{M} - \boldsymbol{I}$. In (5.4), $\bar{\boldsymbol{R}}(H - v) \le \bar{\boldsymbol{R}}(H)$ for $0 \le v \le H$, so

$$\boldsymbol{C}(H) \le \bar{\boldsymbol{R}}(H)\, S_e \int_0^H \big(\upsilon_k \psi_k(v)\big)_k\,\mathrm{d}v = \bar{\boldsymbol{R}}(H)\,\boldsymbol{D}(H) \le (\boldsymbol{M} - \boldsymbol{I})\,\boldsymbol{D}(H).$$

In the univariate case $C \le \bar{\boldsymbol{R}}(H) D$. Since $x \mapsto x/(D + x)$ is increasing, $C/(D + C) \le \bar{\boldsymbol{R}}(H)/\{1 + \bar{\boldsymbol{R}}(H)\} \le (\boldsymbol{M} - 1)/\boldsymbol{M} = n$. The limit follows from (b).

With a common response kernel, $\boldsymbol{D}(H) = \Psi(H)\boldsymbol{d}_e$, so $\boldsymbol{C}(H) \le \Psi(H)(\boldsymbol{M} - \boldsymbol{I})\boldsymbol{d}_e = \Psi(H)\boldsymbol{C}(\infty)$. Since $D_k(H) = \Psi(H) d_{e,k}$, the same monotonicity gives $C_k(H)/\{D_k(H) + C_k(H)\} \le C_k(\infty)/\{d_{e,k} + C_k(\infty)\}$.

(d) Apply Lemma 7.1 at time τ_e, proved in Appendix B.9. Conditionally on $\mathcal{F}_{\tau_e}$, the incidents after τ_e form a cluster process with immigrant intensity $\boldsymbol{\mu} + \boldsymbol{\nu}_e + \boldsymbol{\nu}_{-e} + \boldsymbol{h}_{\tau_e}$ on (τ_e, ∞). Split it by source using Lemma B.2(a), and apply Lemma B.5(b) to each part with time origin τ_e, so that $\mathcal{K}$ becomes $\mathcal{K}_{\tau_e}$. The part rooted at event e has immigrant intensity $\boldsymbol{\nu}_e$, exactly as in (a), so D_k and C_k are unchanged. The remaining part gives $\boldsymbol{B}^{|e}$. □

B.4 Time-varying backgrounds and realised attribution

The following two remarks supplement Theorem 5.1; their derivations follow.

Remark B.6 (Time-varying background). The background-rooted component $\boldsymbol{B}^{\mu} = \int_{W_e} \boldsymbol{\mu} + \int_{W_e} \boldsymbol{R} * \boldsymbol{\mu}$ counts background immigrants plus the cascades that they trigger. If $\boldsymbol{\mu}$ is constant and the process has run for long enough, or is started in the infinite past (Proposition 3.1), then $\mathcal{K}\boldsymbol{\mu} \to \boldsymbol{M}\boldsymbol{\mu}$ and $\boldsymbol{B}^{\mu} = H\,\boldsymbol{M}\boldsymbol{\mu}$.

A time-varying background is passed through a linear filter with impulse response $\delta_0 \boldsymbol{I} + \boldsymbol{R}$, where δ_0 is the Dirac delta. The filter amplifies slow variation by up to $\boldsymbol{M}$ and smooths it over the cascade time scale. Suppose each μ_j is Lipschitz with constant L. Then

$$\big|(\boldsymbol{R} * \boldsymbol{\mu})(t) - (\boldsymbol{M} - \boldsymbol{I})\boldsymbol{\mu}(t)\big| \le L \int_0^t u\,\boldsymbol{R}(u)\,\mathrm{d}u\;\mathbf{1} + \Big(\int_t^{\infty} \boldsymbol{R}\Big)\boldsymbol{\mu}(t).$$

For the univariate exponential kernel, $\int_0^{\infty} u\boldsymbol{R}(u)\,\mathrm{d}u = n/\{\beta(1-n)^2\}$. When the background varies slowly relative to $1/\{\beta(1-n)\}$, the background-driven rate therefore tracks $\boldsymbol{M}\boldsymbol{\mu}(t)$.

A calendar effect that coincides with τ_e belongs in $\boldsymbol{\mu}$ (Assumption A1(i)). If it is omitted, its incidents are attributed to $\boldsymbol{D}$ and $\boldsymbol{C}$. This is a time-local analogue of the spurious contagion of Theorem 4.7.

Remark B.7 (Attribution of the realised incidents). Theorem 5.1 decomposes an expectation. For a realised surge, with parameters fixed, the latent structure used by EM (Marsan and Lengliné, 2008; Mohler et al., 2011; Veen and Schoenberg, 2008) gives a decomposition after the fact. Given the realised event times, each incident i (in stream k_i at time t_i) independently selects one of the following:

- an earlier incident j as its parent, with probability $p_{ij} = \varphi_{k_i k_j}(t_i - t_j)/\lambda_{k_i}(t_i)$;
- event e, with probability $p_{ie} = \upsilon_{k_i} S_e \psi_{k_i}(t_i - \tau_e)/\lambda_{k_i}(t_i)$;
- the background or another event, with the remaining probability.

The probabilities ρ_{ie} that the root of i is an immigrant of event e satisfy $\rho_{ie} = p_{ie} + \sum_{j:t_j<t_i} p_{ij}\rho_{je}$. They are computed in one forward pass. The realised decomposition over W_e has three parts: the direct part $\sum_i p_{ie}$; the contagion part $\sum_i(\rho_{ie} - p_{ie})$; and the baseline $\sum_i(1-\rho_{ie})$, each summed over the stream-k incidents in W_e. These add up to the realised count $N_k(W_e)$. At the true parameters, their expectations are the components of (5.2).

Background bound. We can write $(\boldsymbol{R}*\boldsymbol{\mu})(t) - (\boldsymbol{M}-\boldsymbol{I})\boldsymbol{\mu}(t) = \int_0^t \boldsymbol{R}(u)\{\boldsymbol{\mu}(t-u) - \boldsymbol{\mu}(t)\}\,\mathrm{d}u - (\int_t^{\infty} \boldsymbol{R})\boldsymbol{\mu}(t)$. Each entry of $\boldsymbol{\mu}(t-u) - \boldsymbol{\mu}(t)$ is bounded in absolute value by Lu, and the second term is nonnegative. For the exponential kernel, $\int_0^{\infty} u\,n\beta e^{-\gamma u}\,\mathrm{d}u = n\beta/\gamma^2 = n/\{\beta(1-n)^2\}$.

Realised attribution. Label each incident i by its parent: an earlier incident, an event, or the background. The joint density of the event times and the labels is $\prod_i c_i(\ell_i)\exp\{-\sum_k \int \lambda_k\}$, where $c_i(\ell)$ is the contribution of label ℓ to $\lambda_{k_i}(t_i)$ (Veen and Schoenberg, 2008). Summing over labels recovers (3.1). Conditionally on the times, the labels are therefore independent, with $\mathbb{P}(\ell_i = \ell) = c_i(\ell)/\lambda_{k_i}(t_i)$.

The root of i is an immigrant of e if and only if either $\ell_i = e$, or $\ell_i = j$ and the root of j is an immigrant of e. The root of j is determined by labels of incidents earlier than t_i, which are independent of ℓ_i. This gives the recursion for ρ_{ie}. Taking expectations of the conditional probabilities recovers the expected counts in (5.2), by the tower property.

B.5 Proof of Proposition 5.2

Parts (a)–(c) follow from Lemmas B.3 and B.4 with three choices of $(\mathfrak{D}, \boldsymbol{P}, \boldsymbol{S})$.

- *(a)* $m = M = 1$, $\mathfrak{D} = \beta$, $P = n\beta$ and $S = 1$, so $\boldsymbol{R}(t) = n\beta e^{-\beta(1-n)t}$. Integration gives $\bar{\boldsymbol{R}}$.
- *(b)* $m = M$, $\mathfrak{D} = \boldsymbol{\Omega}$, $\boldsymbol{P} = \boldsymbol{\Omega}\boldsymbol{A}$ and $\boldsymbol{S} = \boldsymbol{I}$. Then $\boldsymbol{\Phi}(t) = e^{-\boldsymbol{\Omega}t}\boldsymbol{\Omega}\boldsymbol{A}$ has entries $\alpha_{kj}\beta_{k\bullet}e^{-\beta_{k\bullet}t}$, $\boldsymbol{Q} = -\boldsymbol{\Omega}(\boldsymbol{I} - \boldsymbol{A})$ and $\boldsymbol{S}\mathfrak{D}^{-1}\boldsymbol{P} = \boldsymbol{A}$. The function $\boldsymbol{y}(t) = \int_0^t e^{\boldsymbol{Q}(t-s)}\boldsymbol{\Omega}\boldsymbol{A}\boldsymbol{\nu}(s)\,\mathrm{d}s = (\boldsymbol{R} * \boldsymbol{\nu})(t)$ is the

variation-of-constants solution of $\boldsymbol{y}' = \boldsymbol{Q}\boldsymbol{y} + \boldsymbol{\Omega}\boldsymbol{A}\boldsymbol{\nu}$, $\boldsymbol{y}(0) = \boldsymbol{0}$. With $\boldsymbol{\Omega} = \beta\boldsymbol{I}$, the matrix $\boldsymbol{A}$ commutes with $e^{-\beta(\boldsymbol{I}-\boldsymbol{A})t}$.

- *(c)* $m = M^2$, $\mathfrak{D} = \operatorname{diag}(\beta_{kj})$, and $\boldsymbol{P}, \boldsymbol{S}$ as defined. Then $(\boldsymbol{S}e^{-\mathfrak{D}t}\boldsymbol{P}\boldsymbol{v})_k = \sum_j \alpha_{kj}\beta_{kj}e^{-\beta_{kj}t}v_j$, $(\boldsymbol{P}\boldsymbol{S}\boldsymbol{z})_{kj} = \alpha_{kj}\beta_{kj}\sum_l z_{jl}$ and $(\boldsymbol{S}\mathfrak{D}^{-1}\boldsymbol{P}\boldsymbol{v})_k = \sum_j \alpha_{kj}v_j = (\boldsymbol{A}\boldsymbol{v})_k$. The function $\boldsymbol{z}(t) = \int_0^t e^{\boldsymbol{Q}(t-s)}\boldsymbol{P}\boldsymbol{\nu}(s)\,\mathrm{d}s$ solves $z'_{kj} = -\beta_{kj}z_{kj} + \alpha_{kj}\beta_{kj}m_j$, where $\boldsymbol{m} = \boldsymbol{\nu} + \boldsymbol{S}\boldsymbol{z} = \mathcal{K}\boldsymbol{\nu}$. Hence $z_{kj}(t) = \int_0^t \varphi_{kj}(t-s)m_j(s)\,\mathrm{d}s$, the mean excitation of stream k by stream j.

(d) *Univariate.* $D(H) = d\,\Psi(H) = d(1 - e^{-\eta H})$. By (5.4) and (5.8),

$$\begin{aligned} C(H) &= d\int_0^H \eta e^{-\eta v}\,\frac{n}{1-n}\bigl(1 - e^{-\gamma(H-v)}\bigr)\,\mathrm{d}v \\ &= \frac{n}{1-n}\,d\int_0^H \eta e^{-\eta v}\mathbb{P}(X_\gamma \le H - v)\,\mathrm{d}v = \frac{n}{1-n}\,d\,\mathbb{P}(X_\eta + X_\gamma \le H). \end{aligned}$$

For $\eta \neq \gamma$, the density of $X_\eta + X_\gamma$ is $\dot{\Pi}_{\eta,\gamma}(t) = \eta\gamma(e^{-\eta t} - e^{-\gamma t})/(\gamma - \eta)$. Integration gives the distribution function

$$\frac{\gamma(1 - e^{-\eta H}) - \eta(1 - e^{-\gamma H})}{\gamma - \eta},$$

which is (5.10). For $\eta = \gamma$, the sum has the Gamma$(2, \gamma)$ law.

The intensity attributable to the event at lag t is

$$d\eta e^{-\eta t} + d\int_0^t n\beta e^{-\gamma(t-v)}\eta e^{-\eta v}\,\mathrm{d}v = d\,\{\eta e^{-\eta t} + \tfrac{n}{1-n}\dot{\Pi}_{\eta,\gamma}(t)\},$$

since $n\beta/\gamma = n/(1-n)$. The density $\dot{\Pi}_{\eta,\gamma}$ vanishes at 0 and at infinity. For $\eta \neq \gamma$ its only critical point solves $\eta e^{-\eta t} = \gamma e^{-\gamma t}$, namely $t^* = \log(\eta/\gamma)/(\eta - \gamma) > 0$. At resonance, $\gamma^2 t e^{-\gamma t}$ is maximised at $1/\gamma$.

Common rates. By (b), $\bar{\boldsymbol{R}}(u) = \beta\boldsymbol{A}\boldsymbol{G}^{-1}(\boldsymbol{I} - e^{-\boldsymbol{G}u})$ with $\boldsymbol{G} = \beta(\boldsymbol{I} - \boldsymbol{A})$, which is invertible because $\varrho(\boldsymbol{A}) < 1$. Since $\beta\boldsymbol{A}\boldsymbol{G}^{-1} = \boldsymbol{A}(\boldsymbol{I} - \boldsymbol{A})^{-1} = \boldsymbol{M} - \boldsymbol{I}$, (5.4) gives $\boldsymbol{C}(H) = (\boldsymbol{M} - \boldsymbol{I})\int_0^H \eta e^{-\eta v}(\boldsymbol{I} - e^{-\boldsymbol{G}(H-v)})\,\mathrm{d}v\,\boldsymbol{d}_e$. The scalar map $x \mapsto \int_0^H \eta e^{-\eta v}(1 - e^{-x(H-v)})\,\mathrm{d}v$ is entire, with a power series that converges absolutely and uniformly on bounded sets. Substituting $\boldsymbol{G}$ into the series therefore yields the matrix integral. For real $x > 0$ the map equals $\Pi_{\eta,x}(H)$.

General case. The immigrant intensity $\boldsymbol{\nu}_e(\tau_e + u) = \operatorname{diag}(\boldsymbol{\eta})e^{-\operatorname{diag}(\boldsymbol{\eta})u}\boldsymbol{d}_e$ solves $\boldsymbol{\nu}' = -\operatorname{diag}(\boldsymbol{\eta})\boldsymbol{\nu}$. The function $\boldsymbol{z}(u) = \int_0^u e^{\boldsymbol{Q}(u-v)}\boldsymbol{P}\boldsymbol{\nu}(v)\,\mathrm{d}v$ solves $\boldsymbol{z}' = \boldsymbol{Q}\boldsymbol{z} + \boldsymbol{P}\boldsymbol{\nu}$ with $\boldsymbol{z}(0) = \boldsymbol{0}$, and $\boldsymbol{S}\boldsymbol{z} = \boldsymbol{R} * \boldsymbol{\nu}$ by (c). Finally $\boldsymbol{D}' = \boldsymbol{\nu}$ and $\boldsymbol{C}' = \boldsymbol{S}\boldsymbol{z}$. Together these form the linear system $\boldsymbol{w}' = \boldsymbol{Z}\boldsymbol{w}$ with $\boldsymbol{w}(0) = \boldsymbol{w}_0$, so $\boldsymbol{w}(H) = e^{H\boldsymbol{Z}}\boldsymbol{w}_0$. □

B.6 The delta method for the decomposition

If $\hat{\vartheta}$ is asymptotically normal with estimated covariance $\hat{\boldsymbol{V}}$ (Section 6), then $\operatorname{Var} g(\hat{\vartheta}) \approx \nabla g^\top \hat{\boldsymbol{V}} \nabla g$, with the joint gradient for joint covariances. The infinite-horizon totals have closed-form gradients:

$$\frac{\partial \boldsymbol{M}}{\partial \alpha_{kj}} = \boldsymbol{M}\boldsymbol{e}_k\boldsymbol{e}_j^\top\boldsymbol{M}, \qquad \frac{\partial(\boldsymbol{M}\boldsymbol{d}_e)}{\partial \alpha_{kj}} = (\boldsymbol{M}\boldsymbol{d}_e)_j\,\boldsymbol{M}\boldsymbol{e}_k, \qquad \frac{\partial(\boldsymbol{M}\boldsymbol{d}_e)}{\partial \upsilon_j} = S_e\,\boldsymbol{M}\boldsymbol{e}_j. \tag{B.1}$$

For the online channel, $\partial \boldsymbol{C}^{\mathrm{on}}(\infty)/\partial\alpha_{kj}$ is $(\boldsymbol{M}\boldsymbol{d}_e)_j\boldsymbol{M}\boldsymbol{e}_k$, minus $(\boldsymbol{M}^\circ\boldsymbol{d}_e)_j\boldsymbol{M}^\circ\boldsymbol{e}_k$ if (k, j) is not an online-to-offline entry. Finite-horizon gradients are derivatives of the matrix exponential in (5.13) (Van Loan, 1978). Shares should be handled on the logit scale. The delta method is unreliable near the boundary $\alpha_{kj} = 0$ (Proposition C.3) and near criticality, where $\boldsymbol{M}$ has a pole.

B.7 Proof of Proposition 5.4

(a) Build the cluster process under $\boldsymbol{A}$, and prune it as in Lemma B.2(b) with $\mathcal{E}$ the online-to-offline pairs. The pruned process is the cluster process with branching matrix $\boldsymbol{A}^{\circ}$, the same kernel shapes and the same immigrants. Its descendants of event e are exactly those whose ancestral line contains no online-to-offline step. Hence $\boldsymbol{C}(H;\boldsymbol{A}^{\circ})$ is the expected number of such descendants in W_e. Subtracting from $\boldsymbol{C}(H;\boldsymbol{A})$ leaves the expected number of descendants whose line contains at least one such step.

(b) Since $\boldsymbol{0} \le \boldsymbol{A}^{\circ} \le \boldsymbol{A}$, we have $\varrho(\boldsymbol{A}^{\circ}) \le \varrho(\boldsymbol{A}) < 1$, so $\boldsymbol{M}^{\circ} = \sum_n (\boldsymbol{A}^{\circ})^n \ge \boldsymbol{I}$. Expand $\boldsymbol{A}^n = (\boldsymbol{A}^{\circ} + \boldsymbol{E})^n$ as a sum of ordered products of n factors, each equal to $\boldsymbol{A}^{\circ}$ or $\boldsymbol{E}$. A product with exactly r factors $\boldsymbol{E}$ can be written uniquely as $(\boldsymbol{A}^{\circ})^{a_r}\boldsymbol{E}(\boldsymbol{A}^{\circ})^{a_{r-1}}\boldsymbol{E}\cdots\boldsymbol{E}(\boldsymbol{A}^{\circ})^{a_0}$, with $a_i \ge 0$ and $\sum_i a_i = n - r$. All terms are nonnegative. Rearranging the double series over n and over products by the number r of factors $\boldsymbol{E}$ gives

$$\boldsymbol{M} = \sum_{n\ge 0} \boldsymbol{A}^n = \sum_{r\ge 0} \sum_{a_0,\dots,a_r \ge 0} (\boldsymbol{A}^{\circ})^{a_r}\boldsymbol{E}\cdots\boldsymbol{E}(\boldsymbol{A}^{\circ})^{a_0} = \sum_{r\ge 0} (\boldsymbol{M}^{\circ}\boldsymbol{E})^r \boldsymbol{M}^{\circ}.$$

Entry (k, j) of $\boldsymbol{A}^n$ is the sum over walks $j = i_0 \to i_1 \to \cdots \to i_n = k$ of $\prod_l \alpha_{i_{l+1} i_l}$. In the expansion, step l carries the factor $\boldsymbol{E}$ exactly when i_l is online and i_{l+1} is offline, which gives the walk reading of the rth term.

Because $\boldsymbol{M}^{\circ} \ge \boldsymbol{I}$, we have $\sum_r (\boldsymbol{M}^{\circ}\boldsymbol{E})^r \le \sum_r (\boldsymbol{M}^{\circ}\boldsymbol{E})^r \boldsymbol{M}^{\circ} = \boldsymbol{M} < \infty$. For a nonnegative matrix $\boldsymbol{T}$ with Perron vector $\boldsymbol{w} \ge \boldsymbol{0}$, $\boldsymbol{w} \ne \boldsymbol{0}$, the convergence of $\sum_r \boldsymbol{T}^r \boldsymbol{w} = \sum_r \varrho(\boldsymbol{T})^r \boldsymbol{w}$ forces $\varrho(\boldsymbol{T}) < 1$. Hence $\varrho(\boldsymbol{M}^{\circ}\boldsymbol{E}) < 1$.

Then $\boldsymbol{M} - \boldsymbol{M}^{\circ} = \sum_{r\ge 1} (\boldsymbol{M}^{\circ}\boldsymbol{E})^r \boldsymbol{M}^{\circ} = \boldsymbol{M}^{\circ}\boldsymbol{E}\boldsymbol{M}$. The resolvent identity $\boldsymbol{M} - \boldsymbol{M}^{\circ} = \boldsymbol{M}\{(\boldsymbol{I} - \boldsymbol{A}^{\circ}) - (\boldsymbol{I} - \boldsymbol{A})\}\boldsymbol{M}^{\circ} = \boldsymbol{M}\boldsymbol{E}\boldsymbol{M}^{\circ}$ gives the other form. By Theorem 5.1(b), applied with $\boldsymbol{A}$ and with $\boldsymbol{A}^{\circ}$, $\boldsymbol{C}^{\mathrm{on}}(\infty) = (\boldsymbol{M} - \boldsymbol{I})\boldsymbol{d}_e - (\boldsymbol{M}^{\circ} - \boldsymbol{I})\boldsymbol{d}_e = (\boldsymbol{M} - \boldsymbol{M}^{\circ})\boldsymbol{d}_e$.

(c) The bounds $\boldsymbol{0} \le \boldsymbol{C}^{\mathrm{on}}(H) \le \boldsymbol{C}(H)$ hold because (a) counts a subset of the descendants that $\boldsymbol{C}(H)$ counts. By (b), $C^{\mathrm{on}}_k(\infty) = \sum_{r\ge 1}\{(\boldsymbol{M}^{\circ}\boldsymbol{E})^r \boldsymbol{M}^{\circ}\boldsymbol{d}_e\}_k$ is a sum of products of nonnegative walk weights and the entries $d_{e,j}$. It is positive if and only if one of these products is positive, which gives the walk criterion. The lower bound is the term $r = 1$.

If $\alpha_{kj} = 0$ for all k online and j offline, no walk returns from an offline stream to an online one. A walk therefore takes at most one online-to-offline step, and the terms with $r \ge 2$ vanish. With streams ordered (offline, online), this case gives $\boldsymbol{A}^{\circ} = \mathrm{diag}(\boldsymbol{A}_{\mathrm{off,off}}, \boldsymbol{A}_{\mathrm{on,on}})$ and $\boldsymbol{E} = \left(\begin{smallmatrix} \boldsymbol{0} & \boldsymbol{A}_{\mathrm{off,on}} \\ \boldsymbol{0} & \boldsymbol{0} \end{smallmatrix}\right)$. Multiplying out $\boldsymbol{M}^{\circ}\boldsymbol{E}\boldsymbol{M}^{\circ}\boldsymbol{d}_e$ gives (5.16).

(d) Prune one realisation of the cluster process with each of $\mathcal{E}_1$, $\mathcal{E}_2$ and $\mathcal{E}_1 \cup \mathcal{E}_2$. By (a), each $\boldsymbol{C}^{\mathcal{E}}$ is the expected number $\mathbb{E}X_{\mathcal{E}}$ of descendants in W_e whose ancestral line uses an entry of $\mathcal{E}$. Pathwise, $X_{\mathcal{E}_1\cup\mathcal{E}_2} = X_{\mathcal{E}_1} + X_{\mathcal{E}_2} - X_{12}$, where X_{12} counts the lines that use both channels, and $\max(X_{\mathcal{E}_1}, X_{\mathcal{E}_2}) \le X_{\mathcal{E}_1\cup\mathcal{E}_2}$. Taking expectations proves (d). □

Gradients (B.1). The identity $\mathrm{d}(\boldsymbol{I} - \boldsymbol{A})^{-1} = (\boldsymbol{I} - \boldsymbol{A})^{-1}(\mathrm{d}\boldsymbol{A})(\boldsymbol{I} - \boldsymbol{A})^{-1}$ with $\mathrm{d}\boldsymbol{A} = \boldsymbol{e}_k \boldsymbol{e}_j^{\top}$ gives $\partial \boldsymbol{M}/\partial \alpha_{kj}$. The other formulas follow from $\boldsymbol{d}_e = S_e(\upsilon_1, \dots, \upsilon_M)^{\top}$. The online-channel formula follows because $\boldsymbol{M}^{\circ}$ depends on α_{kj} only when (k, j) is not an online-to-offline pair.

B.8 An illustrative example

The following parameter values are invented for illustration. They are not estimates, and the example describes no real data. Take two streams, online volume (stream 1) and offline incidents (stream 2), with time measured in days, and

$$\boldsymbol{A} = \begin{pmatrix} 0.5 & 0 \\ 0.1 & 0.3 \end{pmatrix}, \qquad \boldsymbol{\Omega} = \mathrm{diag}(2,\ 0.5), \qquad \boldsymbol{\mu} = (2,\ 0.5)^{\top}, \qquad \boldsymbol{d}_e = (20,\ 5)^{\top}, \qquad \eta_1 = \eta_2 = 1.$$

Thus each online item triggers on average 0.1 offline incidents. Offline incidents trigger no online volume. Online excitation decays within about half a day, and offline excitation decays over about two days. The background is constant, and we take the process to have run long before the event (Proposition 3.1), so $\boldsymbol{B} = H\,\boldsymbol{M}\boldsymbol{\mu}$ with $\boldsymbol{M}\boldsymbol{\mu} = (4,\ 1.29)^\top$ per day. For a process started from the empty history at time 0, as in the standing assumptions, $\boldsymbol{B}$ approaches this value as τ_e grows. Table 4 reports the decomposition, computed from (5.9)–(5.13).

Table 4. Illustrative surge decomposition with invented parameters (not data). B, D and C are the expected background, direct and contagion counts over H days after the event. C^{on} is the offline contagion that passes through the online stream, and $\sigma = C^{\text{on}}/C$ is its share. At $H = \infty$, the background total is infinite and is omitted.

Stream	H	B_k	D_k	C_k	$\dfrac{C_k}{D_k + C_k}$	C_k^{on}	σ_k
Online	7	28.00	19.98	19.85	0.498	—	—
	14	56.00	20.00	20.00	0.500	—	—
	∞	—	20.00	20.00	0.500	—	—
Offline	7	9.00	5.00	6.63	0.570	4.77	0.719
	14	18.00	5.00	7.75	0.608	5.63	0.727
	∞	—	5.00	7.86	0.611	5.71	0.727

The table illustrates the results above. The online stream receives no excitation from offline incidents, so it behaves as a univariate process: its contagion share is at most $n_{\text{on}} = 0.5$, and the bound binds in the limit (Theorem 5.1(b),(c)). The offline share, 0.611 in the limit, is about twice $\alpha_{\text{off,off}} = 0.3$, and most offline contagion arrives through the online stream ($\sigma = 0.727$). With no offline-to-online excitation, $C^{\text{on}}_{\text{off}}(\infty) = 0.7^{-1} \times 0.1 \times 0.5^{-1} \times 20 = 5.71$ is exactly the single-crossing term (5.16); an event with no direct online response, $\boldsymbol{d}_e = (0, 5)^\top$, would have no online channel at all, and with feedback $\alpha_{\text{on,off}} = 0.2$ it would have $\sigma_{\text{off}}(\infty) = 0.168$ (from (5.15)). Over the first week the expected offline count, $B + D + C = 20.62$, is 2.3 times the baseline. Under the model, most of this illustrative surge would be event-driven cascade, and most of that cascade would pass through the online stream. A model with an omitted common driver of both streams, and no online-to-offline excitation at all, could produce similar surges (Remark 5.5).

B.9 Proof of Lemma 7.1

For $s > t$, by (3.2), the intensity of the process on (t, ∞) with respect to $\mathcal{F}_t$ joined with its own history is

$$\boldsymbol{\lambda}(s) = \boldsymbol{\mu}(s) + \boldsymbol{u}_t(s) + \boldsymbol{h}_t(s) + \int_{(t,s)} \boldsymbol{\Phi}(s - r)\, \mathrm{d}N(r).$$

Here $\boldsymbol{\nu}_t = \boldsymbol{\mu} + \boldsymbol{u}_t + \boldsymbol{h}_t$ is $\mathcal{F}_t$-measurable. Conditionally on $\mathcal{F}_t$, this is the intensity of a Hawkes process on (t, ∞) that starts from the empty history, with immigrant intensity $\boldsymbol{\nu}_t$ and kernel $\boldsymbol{\Phi}$. The cluster process with immigrant intensity $\boldsymbol{\nu}_t$ has this intensity (Hawkes and Oakes, 1974), and the intensity determines the law of a point process (Daley and Vere-Jones, 2003, Ch. 7). The two conditional laws therefore agree. The mean formula is Lemma B.5(b) with time origin t.

For exponential kernels, $h_{t,k}(s) = \sum_j \sum_{t_i^j \le t} \alpha_{kj}\beta_{kj} e^{-\beta_{kj}(s - t_i^j)} = \sum_j e^{-\beta_{kj}(s-t)} x_{kj}(t)$, and $u_{t,k}(s) = e^{-\eta_k(s-t)} u_k(t)$ in the same way. Hence $\boldsymbol{\nu}_t$, and with it the conditional law of the future, depends on $\mathcal{F}_t$ only through $\boldsymbol{\chi}(t)$. The dynamics of $\boldsymbol{\chi}$ follow from (7.1). The predictive mean $\int_t^{t+\Delta_a} \mathcal{K}_t \boldsymbol{\nu}_t$ is linear in $\boldsymbol{\nu}_t$, which is affine in $\boldsymbol{\chi}(t)$. □

B.10 Proof of Theorem 7.2

(a) Conditionally on $\mathcal{G}_t$, alerting costs $c_{\mathrm{FA}}(1-\pi_t)$ in expectation and not alerting costs $c_{\mathrm{miss}}\pi_t$. For any $\mathcal{G}_t$-measurable a_t,

$$\mathbb{E}\,\mathrm{loss} = \mathbb{E}\big[a_t c_{\mathrm{FA}}(1-\pi_t) + (1-a_t)c_{\mathrm{miss}}\pi_t\big] \geq \mathbb{E}\min\{c_{\mathrm{FA}}(1-\pi_t), c_{\mathrm{miss}}\pi_t\}.$$

Equality holds if and only if a_t selects the minimum almost surely wherever the two costs differ. Finally, $c_{\mathrm{FA}}(1-\pi) \leq c_{\mathrm{miss}}\pi$ if and only if $\pi \geq \pi^*$.

(b) By Lemma 7.1, the future under ω is a cluster process with immigrant intensity $\boldsymbol{\nu} = \boldsymbol{\mu} + \boldsymbol{u}_t + \boldsymbol{h}_t^{\omega}$. Under ω' it is a cluster process with $\boldsymbol{\nu}' = \boldsymbol{\nu} + \sum \boldsymbol{\Phi}_{\cdot j}(\cdot - r)$, where the sum runs over the incidents (r, j) in $\omega' \setminus \omega$, so $\boldsymbol{\nu}' - \boldsymbol{\nu} \geq \mathbf{0}$. Let $\boldsymbol{N}$ be a cluster process with immigrant intensity $\boldsymbol{\nu}$, and let $\boldsymbol{N}^+$ be an independent cluster process with immigrant intensity $\boldsymbol{\nu}' - \boldsymbol{\nu}$. By Lemma B.2(a), $\boldsymbol{N}' = \boldsymbol{N} + \boldsymbol{N}^+$ has the predictive law under ω', and $\boldsymbol{N}'(t, t+\Delta_{\mathrm{a}}] \geq \boldsymbol{N}(t, t+\Delta_{\mathrm{a}}]$ pathwise. For an upper set $\mathcal{U}$, $\{\boldsymbol{N} \in \mathcal{U}\} \subseteq \{\boldsymbol{N}' \in \mathcal{U}\}$.

(c) By Lemma 7.1, $\pi_t = G_t(\boldsymbol{\chi}(t))$, where $G_t(\boldsymbol{\chi})$ is the probability that a cluster process with immigrant intensity

$$\nu_k^{\boldsymbol{\chi}}(s) = \mu_k(s) + u_k e^{-\eta_k(s-t)} + \sum_j x_{kj} e^{-\beta_{kj}(s-t)}$$

has $\boldsymbol{N}(t, t+\Delta_{\mathrm{a}}] \in \mathcal{U}$. If $\boldsymbol{\chi} \leq \boldsymbol{\chi}'$, then $\boldsymbol{\nu}^{\boldsymbol{\chi}'} - \boldsymbol{\nu}^{\boldsymbol{\chi}} \geq \mathbf{0}$, and the coupling of (b) gives $G_t(\boldsymbol{\chi}) \leq G_t(\boldsymbol{\chi}')$.

For the Lipschitz bound with $\boldsymbol{\chi} \leq \boldsymbol{\chi}'$, write $W = (t, t+\Delta_{\mathrm{a}}]$. Then

$$|G_t(\boldsymbol{\chi}') - G_t(\boldsymbol{\chi})| \leq \mathbb{P}\{\boldsymbol{N}^+(W) \neq \mathbf{0}\} \leq \mathbb{E}\,\mathbf{1}^\top \boldsymbol{N}^+(W) \leq \mathbf{1}^\top \boldsymbol{M} \int_W (\boldsymbol{\nu}^{\boldsymbol{\chi}'} - \boldsymbol{\nu}^{\boldsymbol{\chi}}) \leq \Delta_{\mathrm{a}} \max_j (\mathbf{1}^\top \boldsymbol{M})_j \, \|\boldsymbol{\chi}' - \boldsymbol{\chi}\|_1,$$

using Lemma B.5(b), Lemma B.1 and the fact that each exponential factor is at most 1. If $\varrho(\boldsymbol{A}) \geq 1$, replace $\boldsymbol{M}$ by the finite matrix $\boldsymbol{I} + \int_0^{\Delta_{\mathrm{a}}} \boldsymbol{R}$, where $\boldsymbol{R}$ is the locally integrable resolvent (Gripenberg et al., 1990, Ch. 2). For general $\boldsymbol{\chi}, \boldsymbol{\chi}'$, pass through $\boldsymbol{\chi} \wedge \boldsymbol{\chi}'$ and note that $\|\boldsymbol{\chi} - \boldsymbol{\chi} \wedge \boldsymbol{\chi}'\|_1 + \|\boldsymbol{\chi}' - \boldsymbol{\chi} \wedge \boldsymbol{\chi}'\|_1 = \|\boldsymbol{\chi} - \boldsymbol{\chi}'\|_1$.

Monotonicity makes $\mathcal{A}_t$ an upper set, and continuity makes it closed. The law of the cluster process on W depends on t only through $\boldsymbol{\nu}^{\boldsymbol{\chi}}$ on W. The exponential terms depend on $s - t$ alone, so the dependence on t is through $\boldsymbol{\mu}$ on W and the future calendar.

(d) Let $M = 1$, and write $\nu^{x,u}(s) = \mu(s) + xe^{-\beta(s-t)} + ue^{-\eta(s-t)}$ and $\pi_t = g_t(x, u)$. By (c), g_t is continuous and nondecreasing in both arguments.

For strict monotonicity in x, let $0 \leq x < x'$ and couple as in (b), with N^+ an independent cluster process with immigrant intensity $(x' - x)e^{-\beta(s-t)}$. Then

$$g_t(x', u) - g_t(x, u) = \mathbb{P}(N < c \leq N + N^+) \geq \mathbb{P}(N = c-1)\mathbb{P}(N^+ \geq 1).$$

Here $\mathbb{P}(N^+ \geq 1) \geq 1 - \exp\{-(x' - x)(1 - e^{-\beta\Delta_{\mathrm{a}}})/\beta\} > 0$. The number of immigrants in W is Poisson with mean $m = \int_W \nu^{x,u} > 0$, since $\mu > 0$, and each immigrant has no offspring at all with probability e^{-n}. Hence $\mathbb{P}(N = c-1) \geq e^{-m} m^{c-1} e^{-n(c-1)}/(c-1)! > 0$.

The immigrants contributed by x are Poisson with mean $x(1 - e^{-\beta\Delta_{\mathrm{a}}})/\beta \to \infty$, so $g_t(x, u) \to 1$ as $x \to \infty$. Since $\pi^* < 1$, the set $\{x : g_t(x, u) \geq \pi^*\}$ is a nonempty closed ray $[x_t^*(u), \infty)$, and the rule alerts if and only if $x(t) \geq x_t^*(u(t))$. Because g_t is nondecreasing in u, $x_t^*(u)$ is nonincreasing in u.

If $\eta = \beta$, then $\nu^{x,u}$ depends on (x, u) only through $x + u$, and so does g_t. The same holds trivially when $u \equiv 0$. In either case the rule alerts if and only if $x + u \geq x_t^*(0)$, that is, $\lambda(t+) \geq \mu(t) + x_t^*(0)$. The bound $|g_t(x, u) - g_t(x, 0)| \leq L_t u$ is the Lipschitz property of (c).

The comparative statics follow from three facts. First, $\mathbb{P}(N \geq c)$ is nonincreasing in c. Second, the set $\{g_t \geq \pi^*\}$ shrinks as π^* grows. Third, $N(t, t+\Delta_{\mathrm{a}}]$ is nondecreasing in Δ_{a} on every realisation of the future, whose law does not depend on Δ_{a}. □

B.11 Proof of Lemma D.2

Condition throughout on the history up to $t - L$. Mark each incident in $(t - L, t]$ as available by t or not. By Proposition 7.3(a), the marked process has $\mathcal{G}^\circ$-intensities $p_k F_k(t - s)\lambda_k(s)$ for available and $\{1 - p_k F_k(t - s)\}\lambda_k(s)$ for unavailable incidents. Its density with respect to a unit-rate Poisson reference on $(t - L, t]$, one for each mark and stream, is therefore

$$\prod_{i \in O} p_{k_i} F_{k_i}(t - s_i)\lambda_{k_i}(s_i) \prod_{j \in U} \{1 - p_{k_j} F_{k_j}(t - s_j)\}\lambda_{k_j}(s_j) \, \exp\Big\{-\sum_k \int_{t-L}^{t} \lambda_k(s)\, \mathrm{d}s\Big\}$$

up to a constant, where λ is computed from the history up to $t - L$ and from $O \cup U$ (Daley and Vere-Jones, 2003, Ch. 7). As a function of U for fixed O, this is proportional to the conditional density of U given O.

The proposal inserts the fixed points of O and generates U with the predictable intensity $\{1 - p_k F_k(t - s)\}\lambda_k(s)$. Its density is $\prod_{j \in U}\{1 - p_{k_j} F_{k_j}(t - s_j)\}\lambda_{k_j}(s_j) \exp\{-\sum_k \int \{1 - p_k F_k(t - s)\}\lambda_k(s)\, \mathrm{d}s\}$. The ratio of the two densities is

$$\prod_{i \in O} p_{k_i} F_{k_i}(t - s_i)\lambda_{k_i}(s_i) \exp\Big\{-\sum_k \int_{t-L}^{t} p_k F_k(t - s)\lambda_k(s)\, \mathrm{d}s\Big\}.$$

The factors $p_{k_i} F_{k_i}(t - s_i)$ do not depend on U, which gives (D.3).

Given the completed history, Lemma 7.1 shows that the probability of a surge is $G_t(\chi(t))$, which is bounded. The consistency of self-normalised importance sampling with an absolutely continuous proposal and bounded integrand then gives the limit, by the strong law of large numbers applied to numerator and denominator. □

B.12 Proof of Proposition 7.3

(a) Mark each incident with its recording indicator R_i and delay d_i. By Assumption A2, the marks are independent of the incident process and of the past, so $\boldsymbol{\lambda}$ remains the $\mathcal{G}^\circ$-intensity of $\boldsymbol{N}$. The marked process has $\mathcal{G}^\circ$-compensator $\lambda_k(s)\, \mathrm{d}s$ times the law of the marks (Daley and Vere-Jones, 2003, Ch. 6–7). Integrating the indicator $\mathbf{1}\{R = 1,\ d \le t - s\}$ against this compensator gives the compensator $\int p_k F_k(t - s)\lambda_k(s)\, \mathrm{d}s$ for $O_k^{(t)}$, and the complement gives the second statement. Taking expectations gives the mean.

(b) For an incident of stream j at $s_i \le t - \delta$, the availability indicator $A_i = \mathbf{1}\{R_i = 1,\ d_i \le t - s_i\}$ is independent of $\mathcal{F}_\infty$, with $\mathbb{P}(A_i = 1) = p_j F_j(t - s_i) \ge p_j F_j(\delta) > 0$. Hence $\mathbb{E}[A_i / \{p_j F_j(t - s_i)\} \mid \mathcal{F}_{t-\delta}] = 1$, and $\mathbb{E}[\hat{\boldsymbol{h}}(s) \mid \mathcal{F}_{t-\delta}] = \sum_{s_i \le t-\delta} \boldsymbol{\Phi}_{\cdot j}(s - s_i) = \boldsymbol{h}_{t-\delta}(s)$, since the sum is finite almost surely.

The map $\boldsymbol{f} \mapsto \mathcal{K}_{t-\delta}\boldsymbol{f}(t)$ is linear with a nonnegative kernel. By conditional Fubini and Lemma 7.1,

$$\mathbb{E}[\hat{\boldsymbol{\lambda}}(t) \mid \mathcal{F}_{t-\delta}] = \mathcal{K}_{t-\delta}(\boldsymbol{\mu} + \boldsymbol{u}_{t-\delta} + \boldsymbol{h}_{t-\delta})(t) = \mathbb{E}[\boldsymbol{\lambda}(t) \mid \mathcal{F}_{t-\delta}].$$

The unconditional statement follows by the tower property.

For exponential kernels, the state at $t - \delta$ with weights $A_i / \{p_j F_j(t - s_i)\}$ is conditionally unbiased for $\boldsymbol{x}(t - \delta)$ by the same argument. The mean dynamics $\boldsymbol{z}' = \boldsymbol{Q}\boldsymbol{z} + \boldsymbol{P}(\boldsymbol{\mu} + \boldsymbol{u})$ of Proposition 5.2(c) map the state at $t - \delta$ affinely to $\mathbb{E}[\boldsymbol{x}(t) \mid \mathcal{F}_{t-\delta}]$, which proves the claim. With $\delta = 0$, the same computation conditions on $\mathcal{F}_t$.

(c) By the independence in (b), $\mathbb{E}[\tilde{x}_{kj}(t) \mid \mathcal{F}_t] = \sum_i \varphi_{kj}(t - s_i)\mathbb{P}(A_i = 1)/p_j = \sum_i F_j(t - s_i)\varphi_{kj}(t - s_i)$. This is at most $x_{kj}(t)$ because $F_j \le 1$, with the stated case of strict inequality. Under stationarity, stream j has constant mean intensity $\bar{\Lambda}_j$. By Campbell's theorem, $\mathbb{E}\sum_{s_i \le t} f(t - s_i) = \bar{\Lambda}_j \int_0^\infty f$ for $f \ge 0$. Hence

$$\mathbb{E}\tilde{x}_{kj}(t) = \bar{\Lambda}_j \alpha_{kj} \int_0^\infty F_j(u) g_{kj}(u)\, \mathrm{d}u = \mathbb{P}(d_j \le U_{kj})\, \alpha_{kj}\bar{\Lambda}_j, \qquad \mathbb{E}x_{kj}(t) = \alpha_{kj}\bar{\Lambda}_j.$$

For $g_{kj}(u) = \beta_{kj} e^{-\beta_{kj} u}$, $\mathbb{P}(d_j \le U_{kj}) = \mathbb{E}\mathbb{P}(U_{kj} \ge d_j \mid d_j) = \mathbb{E}e^{-\beta_{kj} d_j}$.

(d) If $p_k \equiv 1$, the available incidents are a subset of all incidents, so $\tilde{x}_{kj}(t) \le x_{kj}(t)$ for every realisation. The event state $\boldsymbol{u}(t)$ is known exactly. By Theorem 7.2(c), $G_t(\tilde{\boldsymbol{x}}(t), \boldsymbol{u}(t)) \le G_t(\boldsymbol{x}(t), \boldsymbol{u}(t))$, so the naive rule alerts only where the rule based on the true state alerts. □

B.13 Proofs for Section 7.4 and Remark D.1

Lemma 7.4. (a) Let $P = P_t$ and $p(y) = P(y) - P(y-1)$. For $u \in (0,1)$,

$$\mathbb{P}(U_t \le u \mid \mathcal{G}_t) = \sum_y p(y) \mathbb{P}\Big(V \le \frac{u - P(y-1)}{p(y)}\Big) = \sum_y \min\big\{\max\{u - P(y-1), 0\},\, p(y)\big\} = u,$$

because the intervals $[P(y-1), P(y))$ partition $[0,1)$. The conditional law does not depend on $\mathcal{G}_t$, so U_t is uniform and independent of $\mathcal{G}_t$.

(b) Let $\mathcal{H}_m = \mathcal{G}_{t_m} \vee \sigma(V_{t_1}, \dots, V_{t_{m-1}})$. The randomisations are independent of everything else, so P_{t_m} is also the conditional law of Y_{t_m} given $\mathcal{H}_m$. By (a), U_{t_m} is therefore uniform and independent of $\mathcal{H}_m$. Each U_{t_l} with $l < m$ is a function of P_{t_l}, Y_{t_l} and V_{t_l}, all of which are $\mathcal{H}_m$-measurable by assumption. Induction over m gives independence.

Remark D.1. Conditionally on its roots, the future is a sum of independent cluster counts. For a Poisson process of roots with intensity ν_t, the probability generating functional gives $\mathbb{E}\prod_i f(s_i) = \exp\{-\int (1-f)\,\nu_t\}$ with $f(s) = G_j(z;s)$ for roots in stream j, which is (D.2). A root in stream j at s has offspring in each stream i forming independent Poisson processes with intensities $\varphi_{ij}(\cdot - s)$, and each child roots an independent subcluster. The same identity then gives (D.1), with $G_i(z;r) = 1$ for $r > t + \Delta_{\mathrm{a}}$.

In the univariate case, $I(s) = \int_s^{t+\Delta_{\mathrm{a}}} n\beta e^{-\beta(r-s)}\{G(z;r) - 1\}\,\mathrm{d}r$, and differentiation gives $I'(s) = \beta I(s) - n\beta\{G(z;s) - 1\}$.

The coordinate χ_i of the state adds the immigrant intensity $\chi_i e^{-b_i(s-t)}$ to one stream $k(i)$, with b_i the corresponding decay rate. Its factor in (D.2) is $\exp\{\chi_i c_i(\hat{G}_i(z) - 1)\}$, where $c_i = (1 - e^{-b_i \Delta_{\mathrm{a}}})/b_i$ and $\hat{G}_i(z) = c_i^{-1} \int_W e^{-b_i(s-t)} G_{k(i)}(z;s)\,\mathrm{d}s$ is a probability generating function. This factor is a compound Poisson law with rate $\chi_i c_i$.

For the aliasing bound, the discrete Fourier transform of the values at the Lth roots of unity returns $\sum_{m \ge 0} p(y + mL)$ for $y < L$. Its error is therefore at most $\mathbb{P}(N \ge L)$.

Remark D.3. If $Y = 1$, then $\int_0^1 (1-c)\mathbf{1}\{\pi < c\}\,\mathrm{d}c = (1-\pi)^2/2$. If $Y = 0$, then $\int_0^1 c\,\mathbf{1}\{\pi \ge c\}\,\mathrm{d}c = \pi^2/2$.

Appendix C Supplement to Section 6

This appendix collects the material behind Section 6 that the main text cites: the treatment of the observation window in EM, the Gibbs sampler, nonparametric kernels, the classical boundary and time-rescaling results with their conditions, the Davies problem, and the proofs of the results of Section 6.

Algorithm 1 summarises the EM iteration of Section 6.2.

C.1 Edge effects

Remark C.1 (Edge effects). The finite window enters the M-step only through the window terms $\mathcal{G}_{kj}$ and $\Psi_k^T(e)$. Ignoring them means setting $\mathcal{G}_{kj} = n_j$ and $\Psi_k^T(e) = 1$, and letting $m_j(\beta) = 1/\beta$. This gives the familiar updates $\alpha_{kj} = N_{kj}/n_j$, $\beta_{kj} = N_{kj}/W_{kj}$ and $\upsilon_k = \sum_e N_{ke} / \sum_e S_e$. The resulting iteration is an EM algorithm for an edge-free pseudo-likelihood, in which every incident's offspring and every shock's response

Algorithm 1 EM for the shock-augmented Hawkes model with exponential kernels.

Require: Events $\{t_i^k\}$ on $[0,T]$; calendar $\{(\tau_e, S_e)\}$; basis B, penalty $(\lambda^{\text{pen}}, P)$; start $\vartheta^{(0)}$ with every free $\alpha_{kj}^{(0)} > 0$; tolerance ϵ.

Ensure: $\hat{\vartheta}$ and the non-decreasing trace $\ell_T(\vartheta^{(r)})$.

for $r = 0, 1, 2, \ldots$ **do**

  E-step. For every pair (k, j), run (6.3) and its lag analogue to obtain $R_{kj}(t_i^k)$ and $R_{kj}^{(1)}(t_i^k)$; form $\lambda_k(t_i^k)$ by (6.1).

  Accumulate N_k^0, $N_{kj} = \sum_i \alpha_{kj}\beta_{kj} R_{kj}(t_i^k)/\lambda_k(t_i^k)$, $W_{kj} = \sum_i \alpha_{kj}\beta_{kj} R_{kj}^{(1)}(t_i^k)/\lambda_k(t_i^k)$, N_{ke} and W_{ke}.

  $\ell_T(\vartheta^{(r)}) \leftarrow \sum_{k,i} \log \lambda_k(t_i^k) - \sum_k \Lambda_k(T)$, with $\Lambda_k(T)$ from (6.2).

  if $r \geq 1$ and $\ell_T(\vartheta^{(r)}) - \ell_T(\vartheta^{(r-1)}) \leq \epsilon\{1 + |\ell_T(\vartheta^{(r)})|\}$ **then**

    return $\vartheta^{(r)}$

  M-step (Proposition 6.1). Background: $\mu_k \leftarrow N_k^0/T$, or Newton steps with step halving on θ_k.

  Decay rates: solve $m_j(\beta_{kj}) = W_{kj}/N_{kj}$ (or the pooled equation for a shared decay) by a one-dimensional root search; m_j is monotone.

  Branching ratios $\alpha_{kj} \leftarrow N_{kj}/G_{kj}$; response rates η_k by the same root equation; loadings $\upsilon_k \leftarrow \sum_e N_{ke} / \sum_e S_e \Psi_k^T(e)$.

are counted in full, and it is monotone for that criterion. It is not an EM algorithm for ℓ_T: its ℓ_T-trace need not increase, and its fixed point is biased. Offspring of incidents near T, and the responses to shocks near T, fall partly after T. As a result, α_{kj} and υ_k are underestimated, and decay rates are overestimated because the observed lags are right-truncated. The relative error in α_{kj} is about $n_j^{-1} \sum_l e^{-\beta_{kj}(T - t_l^j)} \approx 1/(\beta_{kj} T)$. It is negligible when the kernel is short relative to the window. It is not negligible for a salient event shortly before T, which is exactly the case of real-time attribution (Sections 5 and 7). For ETAS models, a closely related approximation replaces the integral of the triggering function over the observation region by its integral over the whole domain; Schoenberg (2013) discusses it. The exact window terms cost nothing extra here, and we use them throughout.

Incidents before time 0 excite the start of the window. We therefore condition on a burn-in segment $[0, T_0]$, whose incidents enter the intensities as history but contribute no likelihood terms. The window terms then become

$$\mathcal{G}_{kj} = \sum_l \int_{(T_0 - t_l^j)^+}^{T - t_l^j} g_{kj}(u)\, \mathrm{d}u, \qquad \Psi_k^T(e) = \Psi_k(T - \tau_e) - \Psi_k(T_0 - \tau_e),$$

with $\int_{T_0}^T \mu_k$ in place of $\int_0^T \mu_k$, and the E-step sums run over incidents in $(T_0, T]$. Proposition 6.1 holds unchanged, with $\nu(\mathrm{d}u) = \sum_l \mathbf{1}\left\{(T_0 - t_l^j)^+ \leq u \leq T - t_l^j\right\} \mathrm{d}u$ in its proof. This ν is again a positive measure that is not a point mass, so its Laplace transform is strictly log-convex. The function m_j is the mean of u under $e^{-\beta u}\nu(\mathrm{d}u)$, and $m_j(0+)$ is the mean of $\nu/\nu(\mathbb{R})$.

C.2 Gibbs sampling by data augmentation

Priors. With many streams (regions × offence types × channels), $\boldsymbol{A}$ has M^2 entries, most of which we expect to be small. We use priors that shrink small entries strongly and leave large ones nearly untouched. In a local–global gamma prior, $\alpha_{kj} \mid \omega_{kj} \sim \text{Gamma}(a_1, \omega_{kj})$ (shape, rate), $\omega_{kj} \mid \varsigma \sim \text{Gamma}(a_2, \varsigma)$ and

$\varsigma \sim \text{Gamma}(a_\varsigma, b_\varsigma)$, so that $\alpha_{kj}/\varsigma \sim \text{BetaPrime}(a_1, a_2)$; with $a_1 = a_2 = \frac{1}{2}$ this is the law of the squared half-Cauchy local scale of the horseshoe (Carvalho et al., 2010), a horseshoe-type prior placed directly on the nonnegative α_{kj} (cf. Armagan et al., 2011). In a spike-and-slab prior, $\alpha_{kj} = \iota_{kj}\tilde{\alpha}_{kj}$ with inclusion indicators $\iota_{kj} \sim \text{Bernoulli}(\pi_{kj})$ and $\tilde{\alpha}_{kj} \sim \text{Gamma}(a_s, b_s)$; the ι_{kj} may follow a random-graph model, as in the network Hawkes process of Linderman and Adams (2014). In both cases the diagonal (self-excitation) entries get their own, less aggressive prior, and the prior is truncated to $\{\varrho(\boldsymbol{A}) < 1\}$ as in Lemma 6.2.

The sweep. Given the labels Z, the complete-data likelihood (6.6) is a product of Poisson-process likelihoods. Each parameter then has a simple full conditional. One sweep consists of the following steps.

(1) *Labels.* Draw each Z_i^k from the E-step probabilities: first the category (background, shock e, or stream j), in $O(M + E)$ operations; then, given stream j, the parent l with probability proportional to $e^{-\beta_{kj}(t_i^k - t_l^j)}$. The second draw is a binary search on rescaled cumulative sums, in $O(\log n_{\text{ev}})$ operations. Record the realised counts N_{kj}, N_{ke}, N_k^0 and lag sums W_{kj}, W_{ke}.

(2) *Branching ratios.* The complete-data likelihood in α_{kj} is $\alpha_{kj}^{N_{kj}} e^{-\alpha_{kj} G_{kj}}$, and this includes the window term exactly. Hence

$$\alpha_{kj} \mid \cdot \;\sim\; \text{Gamma}\left(a_1 + N_{kj},\; \omega_{kj} + G_{kj}\right) \text{ truncated to } \left[0,\; 1/\boldsymbol{M}_{jk}^{(-kj)}\right),$$

which is sampled by inversion (Lemma 6.2). The scales follow from $\omega_{kj} \mid \cdot \sim \text{Gamma}(a_1 + a_2, \alpha_{kj} + \varsigma)$ and $\varsigma \mid \cdot \sim \text{Gamma}(a_\varsigma + a_2|\mathcal{P}|,\; b_\varsigma + \sum \omega_{kj})$, where $\mathcal{P}$ is the set of shrunk entries. (In the spike-and-slab version, the slab replaces (a_1, ω_{kj}) by (a_s, b_s).)

(3) *Spike and slab.* Update (ι_{kj}, Z^k) as a block. First draw ι_{kj} with Z^k integrated out. The odds are $\pi_{kj} \exp\{\ell_{T,k}(\tilde{\alpha}_{kj})\} / [(1 - \pi_{kj}) \exp\{\ell_{T,k}(0)\}]$, where $\ell_{T,k}(x)$ is the stream-k log-likelihood with $\alpha_{kj} = x$, computable in $O(n_{\text{ev}})$ by (6.3). Set $\iota_{kj} = 1$ only if the truncation allows it. Then draw Z^k given ι_{kj}. When $\iota_{kj} = 0$, draw $\tilde{\alpha}_{kj}$ from its prior. Updating ι_{kj} given Z^k instead would mix badly: any label pointing to stream j forces $\iota_{kj} = 1$.

(4) *Loadings and background.* With gamma priors, $\upsilon_k \mid \cdot \sim \text{Gamma}(a_\upsilon + \sum_e N_{ke},\; b_\upsilon + \sum_e S_e \Psi_k^T(e))$. A constant background has $\mu_k \mid \cdot \sim \text{Gamma}(a_\mu + N_k^0,\; b_\mu + T)$. For the spline background, a Gaussian prior $\theta_k \sim N(0, (\lambda_k^{\text{pen}} P + \varepsilon \boldsymbol{I})^{-1})$ is the Bayesian form of (6.4). Its full conditional, that of a Poisson process with log-linear intensity, is log-concave and is updated by a Langevin or Laplace-proposal Metropolis step.

(5) *Decay rates.* The full conditional density of β_{kj} is proportional to

$$\pi(\beta)\, \beta^{N_{kj}} e^{-\beta W_{kj}} \exp\left\{\alpha_{kj} \sum_l e^{-\beta(T - t_l^j)}\right\}.$$

The last factor is the window correction. We update $\log \beta_{kj}$ by slice sampling (Neal, 2003). When the window correction is negligible and π is a gamma density, the update is conjugate.

Each posterior draw of ϑ gives draws of $\varrho(\boldsymbol{A})$, $\boldsymbol{M}$ and the surge decomposition of Theorem 5.1, so parameter uncertainty propagates to every reported quantity (Section 5.5).

C.3 Nonparametric kernels and sparse excitation

Histogram and spline kernels by EM.. Let $\varphi_{kj}(u) = \sum_{m=1}^{L} c_{kj,m} b_m(u)$, where the $b_m \geq 0$ are densities on $(0, u_L]$ and $c_{kj,m} \geq 0$, so that $\alpha_{kj} = \sum_m c_{kj,m}$. Normalised indicators of the bins $[u_{m-1}, u_m)$ give

the histogram kernels of the model-independent stochastic declustering of Marsan and Lengliné (2008). B-spline densities give smooth kernels. In both cases φ_{kj} is a nonnegative mixture. Adding the mixture component m to the parent label therefore keeps every M-step in closed form:

$$c_{kj,m} = \frac{N_{kj,m}}{\sum_l \int_0^{T-t_l^j} b_m(u)\,\mathrm{d}u}, \qquad N_{kj,m} = \sum_{i,l} \frac{c_{kj,m} b_m(t_i^k - t_l^j)}{\lambda_k(t_i^k)}.$$

The denominator is the window correction of Proposition 6.1(b), applied component by component. A roughness penalty on $(c_{kj,m})_m$ can be added through the one-step-late EM (Green, 1990) or a penalised Newton step, as in the nonparametric EM of Lewis and Mohler (2011). Zhou et al. (2013b) estimate multivariate triggering kernels by a related majorise–minimise scheme. There is no recursion for these kernels. The E-step sums over pairs closer than the support u_L, at a cost of $O(n_{\mathrm{ev}} w)$, where w is the mean number of incidents in a window of length u_L. We choose u_L below the knot spacing of the background. This builds the separation of time scales of Section 6.1 into the design. Otherwise the long tail of a free kernel and a slow background compete for the same variation.

Sparse branching matrices. For many streams, sparsity can also be imposed by penalisation. Hansen et al. (2015) use the least-squares contrast $\sum_k \{-2\int_0^T \lambda_k\,\mathrm{d}N_k + \int_0^T \lambda_k^2\,\mathrm{d}t\}$. Its expectation is minimised by the true intensity, and it is quadratic in the coefficients of a linear intensity. With weighted ℓ_1 penalties whose data-driven weights come from martingale Bernstein inequalities, they obtain a Lasso with oracle inequalities. Zhou et al. (2013a) combine ℓ_1 and nuclear-norm penalties on $\boldsymbol{A}$ for networks that are sparse and of low rank. Within EM, an ℓ_1 penalty $\lambda^{\mathrm{pen}}\alpha_{kj}$ keeps the M-step in closed form, $\alpha_{kj} = N_{kj}/(G_{kj} + \lambda^{\mathrm{pen}})$. It shrinks every entry but never sets one exactly to zero. The Bayesian priors of Section 6.3 quantify uncertainty about the sparsity pattern and are our default.

C.4 Binned counts

Parameter mapping. Kirchner (2016) showed that, with $\nu = \Delta\mu$ and $\varphi_\ell = \Delta\,\varphi(\ell\Delta)$, the INAR($\infty$) bin-count process converges weakly to the Hawkes process as $\Delta \to 0$. Kirchner (2017) extended the mapping to the multivariate case, $\varphi_{kj,\ell} = \Delta\varphi_{kj}(\ell\Delta)$, and turned it into a conditional least squares (CLS) estimator for the INAR(p) representation, with $\hat\alpha_{kj} = \sum_{\ell \le p} \hat\varphi_{kj,\ell}$. For exponential kernels the lag weights are geometric, and our implementation estimates the decay as $\hat\beta_{kj} = -\log(1-r)/\Delta$ with $r = \sum_\ell \hat\varphi_{kj,\ell} / \sum_\ell \ell\,\hat\varphi_{kj,\ell}$. The mapping is a first-order statement: its lag coefficients sum to $\alpha\beta\Delta/(e^{\beta\Delta}-1) = \alpha(1 - \beta\Delta/2 + \dots)$. What a fitted autoregression converges to is answered exactly by the next result, which Section 6.4 summarises.

Proposition C.2 (Binning attenuates contagion). *Let N be a stationary univariate Hawkes process (Proposition 3.1) with branching ratio $n \in [0,1)$ and mean rate $\bar\Lambda$. Let X_t be its counts in bins of width Δ. Let π_ℓ be the coefficients of the best linear predictor of X_t from $(X_{t-\ell})_{\ell\ge1}$; this is the limit of CLS for INAR(p) as the number of bins and then p tend to infinity. Assume $\sum_\ell |\pi_\ell| < \infty$. Let σ_Δ^2 be the one-step prediction variance, and let $\varpi_\Delta = \sigma_\Delta^2/(\bar\Lambda\Delta)$ be the innovation dispersion.*

(a) General kernel. $\varpi_\Delta \ge 1$ *and*

$$n_\Delta := \sum_{\ell\ge1} \pi_\ell = 1 - (1-n)\,\varpi_\Delta^{1/2} \;\le\; n.$$

(b) Exponential kernel $\varphi(u) = n\beta e^{-\beta u}$. *Let* $\gamma = \beta(1-n)$, $\rho = e^{-\gamma\Delta}$ *and* $K = \bar{\Lambda}\beta n(2-n)/\{2(1-n)\}$. *Then* X *has the autocovariances of an ARMA*$(1,1)$ *process:*

$$V := \operatorname{Var} X_t = \bar{\Lambda}\Delta + 2K\Big(\frac{\Delta}{\gamma} - \frac{1-e^{-\gamma\Delta}}{\gamma^2}\Big), \qquad \operatorname{Cov}(X_t, X_{t+\ell}) = C\rho^{\ell} \quad (\ell \geq 1),$$

where $C = 4K\gamma^{-2}\sinh^2(\gamma\Delta/2)$. *With* $v_0 = V(1+\rho^2) - 2C\rho^2$ *and* $v_1 = \rho(C-V)$,

$$\sigma_\Delta^2 = \tfrac{1}{2}\Big\{v_0 + \big(v_0^2 - 4v_1^2\big)^{1/2}\Big\}, \qquad \pi_\ell = (\rho-\zeta)\zeta^{\ell-1}, \qquad n_\Delta = \frac{\rho-\zeta}{1-\zeta}, \qquad \zeta = -\frac{v_1}{\sigma_\Delta^2} \in (-1,1).$$

As $\Delta \to 0$,

$$\varpi_\Delta = 1 + n\beta\Delta + O(\Delta^2), \qquad n_\Delta = n - \tfrac{1}{2}n(1-n)\beta\Delta + O(\Delta^2).$$

The proof is in Appendix C.8. The bias of autoregressive contagion estimates is of order $\beta\Delta$, and it does not vanish for a fixed bin width, however long the series. Offspring born in their parent's bin are invisible to lagged regressors; they show up only as excess innovation dispersion $\varpi_\Delta > 1$, which (a) converts into attenuation. The leading relative bias is $-(1-n)\beta\Delta/2$, not $-\beta\Delta/2$ as the Riemann-sum heuristic suggests. With daily bins and a kernel half-life of a day, the exact formula should be used rather than the expansion. Two further consequences follow.

(i) *Link to Theorem 4.7.* Formula (a) reads $n_\Delta = 1 - (\varpi_\Delta/F_\infty)^{1/2}$. This is the dispersion functional $n_F = 1 - F_\infty^{-1/2}$ of Theorem 4.7(c), with ϖ_Δ in place of 1. The proof of (a) uses only stationarity and conditional overdispersion, so the identity also holds for a Cox process without contagion. There it gives $n_\Delta = 1 - \{\varpi_\Delta/(1+\kappa)\}^{1/2}$. This is positive when the covariance density of the Cox process is nonnegative, because then $\sigma_\Delta^2 \leq \operatorname{Var} X_t \leq (1+\kappa)\bar{\Lambda}\Delta$, with strict inequality unless the counts are uncorrelated. Binned autoregressions are therefore exposed to spurious contagion in the same way as continuous-time fits.

(ii) *Multivariate streams.* Within-bin cross-excitation, for example same-day online-to-offline spillover, never enters the lag coefficients. It appears as contemporaneous correlation of the innovations, which has no direction, so a lagged model on daily counts can miss a same-day effect entirely. Remedies are finer bins for the online stream; a hybrid likelihood in which the offline point process has lagged online counts as a piecewise-constant covariate; or the continuous-time likelihood with within-bin randomisation of time stamps (Appendix H.5). Spectral estimation from binned counts handles aliasing correctly (Cheysson and Lang, 2022), but it is a second-order method and so subject to Theorem 4.2.

C.5 The boundary likelihood-ratio test

We use the notation of Section 6.5: $\vartheta = (\alpha, \vartheta_2)$ with $\alpha = \alpha_{kj}$ and $\vartheta_2 \in \Theta_2 \subset \mathbb{R}^d$. The parameter space is $\Theta = [0,\infty) \times \Theta_2$, intersected with constraints that are inactive at ϑ_0, and the null set is $\Theta_0 = \{0\} \times \Theta_2$. Let $\hat{\vartheta}$ and $\tilde{\vartheta}$ maximise ℓ_T over Θ and Θ_0, and let $\mathrm{LR}_T = 2\{\ell_T(\hat{\vartheta}) - \ell_T(\tilde{\vartheta})\}$. Let $U_T = T^{-1/2}\nabla\ell_T(\vartheta_0)$, where the derivative in α is one-sided, and $J_T(\vartheta) = -T^{-1}\nabla^2\ell_T(\vartheta)$. The dimension d is fixed as $T \to \infty$. We assume the following:

(R1) *Consistency.* $\vartheta_0 = (0, \vartheta_{2,0})$ with $\vartheta_{2,0}$ interior to Θ_2, and $\hat{\vartheta} \to \vartheta_0$ and $\tilde{\vartheta} \to \vartheta_0$ in probability.

(R2) *Score.* $U_T \Rightarrow N(0, \boldsymbol{I})$ with $\boldsymbol{I}$ positive definite.

(R3) *Information.* For every $\delta_T \downarrow 0$, $\sup\{\|J_T(\vartheta) - \boldsymbol{I}\| : \vartheta \in \Theta,\ |\vartheta - \vartheta_0| \leq \delta_T\} \to 0$ in probability.

Partition $\boldsymbol{I}$ conformably with (α, ϑ_2), and let $I_{\alpha\alpha\cdot 2} = I_{\alpha\alpha} - I_{\alpha 2} I_{22}^{-1} I_{2\alpha}$ be the efficient information for α.

Proposition C.3 (Boundary likelihood-ratio test; Chernoff, 1954; Self and Liang, 1987). *Under (R1)–(R3):*

(a) $\mathrm{LR}_T = (Z_T^+)^2 + o_p(1)$, *where* $Z_T = (U_{T,\alpha} - I_{\alpha 2} I_{22}^{-1} U_{T,2}) / I_{\alpha\alpha\cdot 2}^{1/2}$ *is the standardised efficient score. Hence*

$$\mathrm{LR}_T \Rightarrow \tfrac{1}{2}\chi_0^2 + \tfrac{1}{2}\chi_1^2.$$

The asymptotic p-value of an observed value $\mathrm{LR}_T = x > 0$ *is* $\frac{1}{2}\mathbb{P}(\chi_1^2 \geq x)$, *and it is* 1 *if* $x = 0$. *The* 5% *critical value is* 2.71, *not* 3.84.

(b) $\sqrt{T}\,\hat{\alpha} \Rightarrow \max(0, W)$ *with* $W \sim N(0, I_{\alpha\alpha\cdot 2}^{-1})$. *Wald intervals for* α_{kj} *are therefore invalid near the boundary.*

Proof. Step 1: rate. Since $\vartheta_0 \in \Theta$, we have $\ell_T(\hat{\vartheta}) \geq \ell_T(\vartheta_0)$. Near ϑ_0 the set Θ is convex, so the segment from ϑ_0 to $\hat{\vartheta}$ lies in Θ. Taylor's theorem with integral remainder, applied along the segment, gives with $h = \sqrt{T}(\hat{\vartheta} - \vartheta_0)$

$$0 \leq \ell_T(\hat{\vartheta}) - \ell_T(\vartheta_0) = h^\top U_T - \tfrac{1}{2} h^\top \bar{J}_T h, \qquad \bar{J}_T = 2\int_0^1 (1-s)\, J_T\big(\vartheta_0 + s(\hat{\vartheta} - \vartheta_0)\big)\, \mathrm{d}s.$$

By (R1) there are $\delta_T \downarrow 0$ with $\mathbb{P}(|\hat{\vartheta} - \vartheta_0| \leq \delta_T) \to 1$, and then (R3) gives $\bar{J}_T = \boldsymbol{I} + o_p(1)$. Hence $\frac{1}{2}\{\lambda_{\min}(\boldsymbol{I}) - o_p(1)\}|h|^2 \leq |h|\,|U_T|$, and $|h| = O_p(1)$ by (R2). The same argument applies to $\tilde{\vartheta}$.

Step 2: local quadratic approximation. The same expansion shows that, uniformly in $h \in \sqrt{T}(\Theta - \vartheta_0)$ with $|h| \leq c$,

$$\ell_T(\vartheta_0 + h/\sqrt{T}) - \ell_T(\vartheta_0) = Q_T(h) + o_p(1), \qquad Q_T(h) = h^\top U_T - \tfrac{1}{2} h^\top \boldsymbol{I} h.$$

Because $\vartheta_{2,0}$ is interior, the only constraint active near ϑ_0 is $\alpha \geq 0$. For each fixed c and large T, the local parameter set inside $\{|h| \leq c\}$ is therefore $C \cap \{|h| \leq c\}$, where $C = [0, \infty) \times \mathbb{R}^d$. The local null set is likewise $\mathcal{L} \cap \{|h| \leq c\}$, where $\mathcal{L} = \{0\} \times \mathbb{R}^d$. The maximisers of ℓ_T (Step 1) and of Q_T over C and over $\mathcal{L}$ are all $O_p(1)$. Hence $2\{\ell_T(\hat{\vartheta}) - \ell_T(\vartheta_0)\} = 2\sup_C Q_T + o_p(1)$, and likewise with $\tilde{\vartheta}$ and $\mathcal{L}$. Let $Y_T = \boldsymbol{I}^{-1} U_T$ and $\|v\|^2 = v^\top \boldsymbol{I} v$. Then $2Q_T(h) = \|Y_T\|^2 - \|Y_T - h\|^2$, so

$$\mathrm{LR}_T = \inf_{h \in \mathcal{L}} \|Y_T - h\|^2 - \inf_{h \in C} \|Y_T - h\|^2 + o_p(1).$$

Step 3: geometry. Let $v = \boldsymbol{I}^{-1}\boldsymbol{e}_1$, where $\boldsymbol{e}_1$ is the α-coordinate. For $h \in \mathcal{L}$, $\langle h, v\rangle = h^\top \boldsymbol{I}\boldsymbol{I}^{-1}\boldsymbol{e}_1 = h_1 = 0$, so v is orthogonal to $\mathcal{L}$ in the inner product of $\boldsymbol{I}$. Every h decomposes as $h = h_{\mathcal{L}} + t\, v/\|v\|$, with $t = h_1/\|v\|$ and $\|v\|^2 = (\boldsymbol{I}^{-1})_{11} > 0$. Hence $C = \mathcal{L} \oplus \{t\, v/\|v\| : t \geq 0\}$. Write $Y_T = Y_{\mathcal{L}} + s_T\, v/\|v\|$. Then $\inf_{\mathcal{L}} \|Y_T - h\|^2 = s_T^2$ and $\inf_C \|Y_T - h\|^2 = \min(s_T, 0)^2$, so $\mathrm{LR}_T = (s_T^+)^2 + o_p(1)$. By the partitioned inverse, $(\boldsymbol{I}^{-1})_{11} = I_{\alpha\alpha\cdot 2}^{-1}$ and $(\boldsymbol{I}^{-1}U_T)_1 = I_{\alpha\alpha\cdot 2}^{-1}(U_{T,\alpha} - I_{\alpha 2} I_{22}^{-1} U_{T,2})$, so $s_T = (\boldsymbol{I}^{-1}U_T)_1 / \|v\| = Z_T$. By (R2), $Z_T \Rightarrow N(0,1)$. Hence $(Z_T^+)^2$ is 0 with probability tending to $\frac{1}{2}$ and is otherwise asymptotically χ_1^2.

(b) The maximiser of Q_T over C is the $\boldsymbol{I}$-projection $Y_{\mathcal{L}} + s_T^+ v/\|v\|$. Its α-coordinate is $s_T^+\|v\| = \max\{0, (\boldsymbol{I}^{-1}U_T)_1\}$. The limit $Q(h) = h^\top U - \frac{1}{2} h^\top \boldsymbol{I} h$ has a unique maximiser over the closed convex cone C, and the approximation of Step 2 is uniform on compacts. The argmax continuous mapping theorem (van der Vaart and Wellner, 1996, Thm 3.2.2) therefore gives the claim, with $W = (\boldsymbol{I}^{-1}U)_1 \sim N(0, (\boldsymbol{I}^{-1})_{11})$. □

Remark C.4 (When the conditions hold). (i) *The decay must be identified under* H_0. The decay of φ_{kj} must be fixed in advance, or shared with kernels whose branching ratios are positive under H_0, as

with a common decay within a block (Appendix H.4). Otherwise $\partial \ell_T / \partial \beta_{kj} \equiv 0$ at $\alpha_{kj} = 0$, the information is singular, and the score condition fails. Fixing β_{kj} at its unrestricted estimate does not remove the problem, and the test is then liberal. The sup-statistic (6.10) of Section 6.5 treats this case.

(ii) *Other conditions.* If another branching ratio or a loading is also zero, the limit is a chi-bar-square mixture with weights that depend on $\boldsymbol{I}$ (Self and Liang, 1987). For a stationary process with constant background conditions (R1)–(R3) can be verified along the lines of Ogata (1978); under the pre-registered fixed knot spacing h_b, the number K_b of basis functions grows with T and they do not apply as stated, so the size of the test is checked by parametric bootstrap and in the synthetic dry run (Appendix H.6).

(iii) *Only stream k needs refitting.* If no parameter is shared across rows, LR_T involves only $\ell_{T,k}$ (Section 6.1).

C.6 Decays identified only under the alternative

Suppose φ_{kj} has its own decay rate. Restrict it to a pre-registered interval $\mathcal{B} = [\beta_L, \beta_U]$ of half-lives, and let $\mathrm{LR}_T(\beta)$ be the statistic with $\beta_{kj} = \beta$ fixed. For each β, Proposition C.3 gives $\mathrm{LR}_T(\beta) = (Z_T(\beta)^+)^2 + o_p(1)$. Under uniform versions of (R1)–(R3), $Z_T(\cdot)$ converges to a centred, unit-variance Gaussian process Z, whose correlation function is that of the efficient scores. Then $\sup_\beta \mathrm{LR}_T(\beta) \Rightarrow \sup_\beta (Z(\beta)^+)^2$ (Davies, 1977, 1987). This limit is stochastically larger than $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$, because it dominates $(Z(\beta_0)^+)^2$ for every $\beta_0 \in \mathcal{B}$. Referring the sup-statistic to the chi-bar-square therefore inflates the size. We use two valid references.

- *An upcrossing bound.* If Z has continuously differentiable paths, Rice's formula for the expected number of upcrossings gives, for $c > 0$,

$$\mathbb{P}\Big(\sup_{\beta \in \mathcal{B}} Z(\beta) > c \Big) \le \Phi(-c) + \frac{\mathcal{V}}{2\pi} e^{-c^2/2}, \qquad \mathcal{V} = \int_{\mathcal{B}} \{ \mathrm{Var}\, Z'(\beta) \}^{1/2} \, \mathrm{d}\beta.$$

  Since $\mathbb{E}|Z'(\beta)| = (2/\pi)^{1/2} \{\mathrm{Var}\, Z'(\beta)\}^{1/2}$, we can estimate $\mathcal{V}$ by $(\pi/2)^{1/2}$ times the total variation of the observed path on a grid $\beta_1 < \cdots < \beta_G$. With $u = \sup_\beta \mathrm{LR}_T(\beta)$ and $\widehat{\mathrm{TV}} = \sum_g |\hat{Z}(\beta_g) - \hat{Z}(\beta_{g-1})|$, this gives

$$p_{\mathrm{D}} = \tfrac{1}{2}\mathbb{P}(\chi^2_1 > u) + \frac{\widehat{\mathrm{TV}}}{2\sqrt{2\pi}} \, e^{-u/2}.$$

  This is the one-sided form of Davies' bound, half of his two-sided bound. It is an upper bound only when $\mathcal{V}$ is known. The grid estimate of $\mathcal{V}$ is random and biased downward, because a coarse grid misses oscillations of the path. So p_{D} is an approximation, not a guaranteed upper bound, and a fine grid reduces the bias. For $\hat{Z}(\beta)$ we recommend the standardised efficient score at the null fit. It needs a single null fit and one recursion (6.3) per grid point. It also follows the whole Gaussian path, whereas the signed root of $\mathrm{LR}_T(\beta)$ is flattened at zero wherever $\hat{\alpha}_{kj} = 0$, which understates the total variation.

- *Parametric bootstrap.* Simulate the full model at the null fit $\tilde{\vartheta}$ (Appendix E). Recompute the sup-statistic with refits, and report $p_{\mathrm{B}} = (1 + \#\{\mathrm{LR}^{(b)}_{\sup} \ge \mathrm{LR}_{\sup}\})/(B+1)$ for B replicates, where $\mathrm{LR}_{\sup} = \sup_\beta \mathrm{LR}_T(\beta)$.

C.7 Time rescaling

Proposition C.5 (Time rescaling; Brown et al., 2002; Papangelou, 1972). *Let N be a simple point process on $[0,\infty)$ with intensity $\lambda > 0$ with respect to a right-continuous filtration $(\mathcal{F}_t)$. Suppose $\Lambda(t) = \int_0^t \lambda(s)\,\mathrm{d}s$ is finite for every t and $\Lambda(t) \to \infty$ almost surely. Then the points $\Lambda(t_1) < \Lambda(t_2) < \cdots$ form a homogeneous Poisson process with unit rate. In particular, the increments $z_i = \Lambda(t_i) - \Lambda(t_{i-1})$, with $t_0 = 0$, are i.i.d.* Exp(1)*, and the $u_i = 1 - e^{-z_i}$ are i.i.d. uniform on* $(0,1)$*. If the multivariate model is correct, this holds for each stream N_k with its compensator Λ_k relative to the joint history. Here $\lambda_k \geq \mu_k > 0$, so the positivity condition holds.*

Proof. Since $\lambda > 0$, Λ is continuous and strictly increasing from 0 to ∞, so it has a continuous inverse Λ^{-1}. Each $\Lambda^{-1}(u)$ is a stopping time, because $\{\Lambda^{-1}(u) \leq t\} = \{\Lambda(t) \geq u\} \in \mathcal{F}_t$. Let $\tilde{N}(u) = N(\Lambda^{-1}(u))$ and $\tilde{\mathcal{F}}_u = \mathcal{F}_{\Lambda^{-1}(u)}$. Then $\tilde{N}(u) = \#\{i : \Lambda(t_i) \leq u\}$, and $\tilde{N}$ is simple because Λ is injective.

We show that $\tilde{N}(u) - u$ is a $\tilde{\mathcal{F}}$-martingale. Fix v. The stopped process $N(t \wedge \Lambda^{-1}(v)) - \Lambda(t \wedge \Lambda^{-1}(v))$ is a local martingale whose compensator part is bounded by v. By localisation and monotone convergence, $\mathbb{E}N(\Lambda^{-1}(v)) = v$. The stopped process is therefore dominated by the integrable $N(\Lambda^{-1}(v)) + v$, so it is a uniformly integrable martingale. Optional sampling at $\Lambda^{-1}(u) \leq \Lambda^{-1}(v)$ gives $\mathbb{E}\{\tilde{N}(v) - \tilde{N}(u) \mid \tilde{\mathcal{F}}_u\} = v - u$.

A simple counting process whose compensator is the identity is a unit-rate Poisson process. This is Watanabe's characterisation (Brémaud, 1981, Ch. II). For the multivariate model, apply the argument to each N_k with the filtration $\mathcal{F}$. The positivity assumption only simplifies the proof: the result holds for any continuous compensator (Daley and Vere-Jones, 2003, Sec. 7.4). □

On a finite window, the rescaled points of stream k are those of a unit Poisson process on $[0, \Lambda_k(T)]$, where $\Lambda_k(T)$ is a stopping time for the time-changed filtration. The n_k increments are therefore not exactly a fixed-size i.i.d. sample, and the final censored gap is ignored. The discrepancy vanishes as $\Lambda_k(T) \to \infty$.

Estimated parameters. In the i.i.d. setting with maximum likelihood estimates, the direction of the error from plugging in $\hat{\vartheta}$ is known. The estimated empirical process converges to a Gaussian process whose covariance is that of the Brownian bridge minus a positive-semidefinite kernel (Durbin, 1973). By Anderson's inequality (Anderson, 1955), a centred Gaussian law with smaller covariance puts more mass on the symmetric convex set $\{\sup|z| \leq c\}$. The plug-in statistic is therefore asymptotically stochastically smaller, and the test is conservative, as in Lilliefors (1967). We know of no proof for rescaled point processes, but the same mechanism operates when $\hat{\vartheta}$ is the maximum likelihood estimator. The argument can fail for penalised, least-squares or posterior-mean plug-ins, whose covariance correction need not be positive semidefinite. Hence the parametric bootstrap of Section 6.6.

C.8 Proofs for Section 6

Proof of Proposition 6.1. Q separates into a background term for each k, a shock term for each k, and a term for each pair (k, j).

(a) The background term is $\sum_i p_i^{k,0} \log \mu_k(t_i^k) - \int_0^T \mu_k$. For constant μ_k it equals $N_k^0 \log \mu_k - T\mu_k$. For the log-linear form, the first part is linear in θ_k. The integral $\int e^{B^\top \theta}$ is strictly convex when the basis is linearly independent, and the penalty is convex.

(b) The pair term is $N_{kj} \log \alpha_{kj} - \alpha_{kj}\mathcal{G}_{kj}$, plus terms free of α_{kj}. It is maximised at $N_{kj}/\mathcal{G}_{kj}$. The shock term in υ_k is $(\sum_e N_{ke}) \log \upsilon_k - \upsilon_k \sum_e S_e \Psi_k^T(e)$.

(c) For $g_{kj}(u) = \beta e^{-\beta u}$ the pair term is $N \log \alpha + N \log \beta - \beta W - \alpha\beta\Upsilon(\beta)$, with $N = N_{kj}$, $W = W_{kj}$ and $\Upsilon(\beta) = \sum_l \int_0^{r_l} e^{-\beta u}\,\mathrm{d}u$; note that $\mathcal{G}_{kj} = \beta\Upsilon(\beta)$. Profiling $\alpha = N/\{\beta\Upsilon(\beta)\}$ gives $-\beta W - N \log \Upsilon(\beta)$

plus a constant. Now Υ is the Laplace transform of the measure $\nu(\mathrm{d}u) = \sum_l \mathbf{1}\{0 \le u \le r_l\}\,\mathrm{d}u$. Laplace transforms are log-convex, and here strictly so, because ν is not a point mass: $(\log \Upsilon)''(\beta) = \mathrm{Var}_\beta(u) > 0$, where Var_β is the variance under $e^{-\beta u}\nu(\mathrm{d}u)/\Upsilon(\beta)$. Hence the profiled objective is strictly concave, and its derivative is $-W + N\,m_j(\beta)$, since $m_j = -\Upsilon'/\Upsilon$ is the mean of u under the tilted measure. Moreover $m_j'(\beta) = -\mathrm{Var}_\beta(u) < 0$. As $\beta \to \infty$ the tilted measure concentrates at 0. As $\beta \downarrow 0$ it tends to $\nu/\nu(\mathbb{R})$, whose mean is $\sum_l r_l^2/(2\sum_l r_l)$. A shared decay sums strictly concave terms. The response rate is the same calculation with $\nu = \sum_e S_e \mathbf{1}\{0 \le u \le T - \tau_e\}\,\mathrm{d}u$ for $\tau_e \ge 0$.

(d) This is the usual EM argument (Dempster et al., 1977). Write $\ell_T(\vartheta) = Q(\vartheta) - H(\vartheta)$, where $H(\vartheta) = \mathbb{E}\{\log p_\vartheta(Z \mid \text{data}) \mid \text{data}, \vartheta^{(r)}\}$. By Gibbs' inequality, $H(\vartheta) \le H(\vartheta^{(r)})$. The penalty is a deterministic function of ϑ, so it passes through unchanged. □

Proof of Lemma 6.2. Write $\boldsymbol{A}_x = \boldsymbol{A}^{(-kj)} + x\boldsymbol{e}_k\boldsymbol{e}_j^\top$ and $\rho(x) = \varrho(\boldsymbol{A}_x)$, so $\rho(0) = \varrho(\boldsymbol{A}^{(-kj)}) < 1$. By Perron–Frobenius monotonicity, ρ is nondecreasing and continuous on $[0,\infty)$. Let $x^* = \sup\{x : \rho(x) < 1\}$. If $\rho(x) < 1$, the eigenvalues λ_i of $\boldsymbol{A}_x$ satisfy $\det(\boldsymbol{I} - \boldsymbol{A}_x) = \prod_i(1-\lambda_i) > 0$: complex eigenvalues come in conjugate pairs, and real ones are below 1. By the matrix determinant lemma,

$$\det(\boldsymbol{I} - \boldsymbol{A}_x) = \det(\boldsymbol{I} - \boldsymbol{A}^{(-kj)})\,\big(1 - x\,\boldsymbol{e}_j^\top \boldsymbol{M}^{(-kj)}\boldsymbol{e}_k\big),$$

which is affine in x and positive at $x = 0$. Suppose first that $x^* < \infty$. Then $\rho(x^*) = 1$ by continuity, and 1 is an eigenvalue of the nonnegative matrix $\boldsymbol{A}_{x^*}$ by Perron–Frobenius. So the determinant vanishes at x^*, and x^* is its unique root $1/\boldsymbol{M}_{jk}^{(-kj)}$. Conversely, $\boldsymbol{M}^{(-kj)} = \sum_{m\ge 0}(\boldsymbol{A}^{(-kj)})^m \ge \boldsymbol{0}$. If $\boldsymbol{M}_{jk}^{(-kj)} > 0$, then $x^* < \infty$: otherwise the determinant would be positive for every x, yet it vanishes at $x = 1/\boldsymbol{M}_{jk}^{(-kj)}$. If $\boldsymbol{M}_{jk}^{(-kj)} = 0$, the determinant never vanishes, so $x^* = \infty$. □

Proof of Proposition C.2. (a) Write $\pi(z) = 1 - \sum_\ell \pi_\ell z^\ell$ and f_X for the spectral density of X. The Wold representation gives $f_X(\omega) = \sigma_\Delta^2/\{2\pi|\pi(e^{-i\omega})|^2\}$ (Brockwell and Davis, 1991, Ch. 5). At $\omega = 0$, $2\pi f_X(0) = \sum_h \mathrm{Cov}(X_0, X_h) = \lim_m m^{-1}\,\mathrm{Var}\,N(0, m\Delta] = F_\infty\bar\Lambda\Delta$, and $F_\infty = (1-n)^{-2}$ by Proposition 3.2. The function π is continuous on the closed unit disc. It has no zeros there: inside, $\pi\psi = 1$ for the Wold transfer function ψ, and on the circle $|\pi|^2 = \sigma_\Delta^2/(2\pi f_X) > 0$ because f_X is bounded. It is real on $[0,1]$ with $\pi(0) = 1$, so $\pi(1) > 0$. Hence

$$1 - n_\Delta = \pi(1) = \frac{\sigma_\Delta}{\{2\pi f_X(0)\}^{1/2}} = (1-n)\,\varpi_\Delta^{1/2}.$$

For $\varpi_\Delta \ge 1$, note that the predictor is a function of $\mathcal{F}_{(t-1)\Delta}$, so $\sigma_\Delta^2 \ge \mathbb{E}\,\mathrm{Var}(X_t \mid \mathcal{F}_{(t-1)\Delta})$. Given the past, the future of a Hawkes process is again a Poisson cluster process. Its cluster seeds form a Poisson process with the deterministic intensity $\mu + \int_{(-\infty,(t-1)\Delta]} \varphi(\cdot - s)\,\mathrm{d}N(s)$, and each seed carries an independent cluster (Hawkes and Oakes, 1974). So X_t is conditionally a Poisson sum $\sum_c S_c$ of within-bin cluster sizes $S_c \ge 1$. Its conditional variance is $\mathbb{E}[\sum_c S_c^2 \mid \mathcal{F}] \ge \mathbb{E}[\sum_c S_c \mid \mathcal{F}] = \mathbb{E}[X_t \mid \mathcal{F}]$. Taking expectations gives $\sigma_\Delta^2 \ge \bar\Lambda\Delta$.

(b) The covariance density is $c(u) = Ke^{-\gamma|u|}$ by (3.5). Integrating it over pairs of bins gives V and $\mathrm{Cov}(X_t, X_{t+\ell}) = K(1-e^{-\gamma\Delta})^2 e^{-\gamma\Delta(\ell-1)}/\gamma^2 = C\rho^\ell$. Hence $Y_t = X_t - \rho X_{t-1}$ has $\mathrm{Var}\,Y_t = v_0$, $\mathrm{Cov}(Y_t, Y_{t-1}) = v_1$, and zero autocovariance at lags ≥ 2. So Y is an MA(1) process, $Y_t = \varepsilon_t - \zeta\varepsilon_{t-1}$. Binning and aliasing map spectra to spectra monotonically, and the Bartlett spectrum (3.5) dominates the Poisson floor $\bar\Lambda/(2\pi)$. Hence $f_X \ge \bar\Lambda\Delta/(2\pi) > 0$, the binned spectrum of a Poisson process. So $f_Y(\omega) = |1 - \rho e^{-i\omega}|^2 f_X(\omega) = v_0 + 2v_1\cos\omega > 0$, and the invertible solution has $|\zeta| < 1$. Matching $\sigma^2(1+\zeta^2) = v_0$ and $-\zeta\sigma^2 = v_1$ shows that σ^2 is the larger root of $s^2 - v_0 s + v_1^2 = 0$. Inverting

$(1-\rho z)/(1-\zeta z) = \pi(z)$ gives π_ℓ and n_Δ. For the expansion, put $u = V - C = \bar{\Lambda}\Delta + O(\Delta^3)$ and note $C = K\Delta^2 + O(\Delta^4)$. Then $v_0 = u(1+\rho^2) + C(1-\rho^2) = 2\bar{\Lambda}\Delta - 2\gamma\bar{\Lambda}\Delta^2 + O(\Delta^3)$, and

$$v_0^2 - 4v_1^2 = (1-\rho^2)\{u^2(1-\rho^2) + 2uC(1+\rho^2) + C^2(1-\rho^2)\} = 4\gamma\bar{\Lambda}(\gamma\bar{\Lambda} + 2K)\Delta^4 + O(\Delta^5).$$

Since $\gamma\bar{\Lambda} + 2K = \bar{\Lambda}\beta/(1-n)$, the square root is $2\bar{\Lambda}\beta\Delta^2 + O(\Delta^3)$. So $\sigma_\Delta^2 = \bar{\Lambda}\Delta\{1 + (\beta-\gamma)\Delta\} + O(\Delta^3)$, with $\beta - \gamma = n\beta$, and (a) gives the expansion of n_Δ. □

Proof of Lemma 6.3. Given the conditioning variables, $\Gamma^{(b)} = \Gamma(C_b; \text{data})$ applies one fixed function to exchangeable arguments, so $(\Gamma^{(0)}, \dots, \Gamma^{(B)})$ is exchangeable. Let $R_b = \#\{b' \in \{0,\dots,B\} : \Gamma^{(b')} \geq \Gamma^{(b)}\}$, so that $p = R_0/(B+1)$. Order the $B+1$ values decreasingly, breaking ties arbitrarily. The position of $\Gamma^{(b)}$ is at most R_b, and distinct indices have distinct positions, so at most m indices satisfy $R_b \leq m$. By exchangeability, $\mathbb{P}(R_0 \leq m) = (B+1)^{-1}\sum_b \mathbb{P}(R_b \leq m) = (B+1)^{-1}\mathbb{E}\#\{b : R_b \leq m\} \leq m/(B+1)$. Take $m = \lfloor a(B+1) \rfloor$, and integrate over the conditioning variables. □

Appendix D Supplement to Section 7

This appendix collects material behind Section 7: exact predictive distributions, an imputation algorithm for alerting from delayed data, the link between cost-weighted losses and the Brier score, and a monitor for regime shifts. Proofs are in Appendix B. None of this material is evaluated in the simulation study.

D.1 Exact predictive distributions

Remark D.1 (Exact predictive distributions). The predictive law can also be computed without simulation. Fix a target stream k and the window $W = (t, t+\Delta_a]$. For a cluster rooted at a stream-j incident at time s, let $G_j(z;s)$ be the probability generating function of the number of its members in stream k and in W, root included. The branching structure gives

$$G_j(z;s) = z^{\mathbf{1}\{j=k,\ s\in W\}} \exp\Big\{\sum_{i=1}^{M}\int_s^{t+\Delta_a} \varphi_{ij}(r-s)\big(G_i(z;r)-1\big)\,\mathrm{d}r\Big\}, \tag{D.1}$$

and the Poisson immigrants of Lemma 7.1 give

$$\mathbb{E}\big[z^{N_k(W)} \mid \mathcal{F}_t\big] = \exp\Big\{\sum_{j=1}^{M}\int_t^{t+\Delta_a} \nu_{t,j}(s)\big(G_j(z;s)-1\big)\,\mathrm{d}s\Big\}. \tag{D.2}$$

For exponential kernels, the inner integrals in (D.1) satisfy M^2 ordinary differential equations, solved backwards in s. In the univariate case, $I(s) = \int_s^{t+\Delta_a}\varphi(r-s)\{G(z;r)-1\}\,\mathrm{d}r$ solves

$$I' = \beta I - n\beta\,(ze^I - 1), \qquad I(t+\Delta_a) = 0,$$

and $G = ze^I$ on W. Evaluating (D.2) at the Lth roots of unity and applying the discrete Fourier transform gives the predictive probability mass function. The aliasing error in each probability is at most the predictive probability of L or more incidents.

Because ν_t is affine in $\chi(t)$, the logarithm of (D.2) is affine in $\chi(t)$. The predictive law is therefore the law at $\chi = \mathbf{0}$ convolved with independent compound Poisson laws whose rates are proportional to the coordinates of $\chi(t)$. This is the analytic form of the coupling in the proof of Theorem 7.2.

Algorithm 2 Real-time alerting with imputation of unreported incidents

Require: incidents available at t; parameter draws $\vartheta^{(1)}, \dots, \vartheta^{(R)}$; recording probabilities p_k and delay distributions F_k; imputation window L with $F_k(L) \approx 1$; window Δ_a; surge set $\mathcal{U}$; threshold $\pi^* = c_{\mathrm{FA}}/(c_{\mathrm{FA}} + c_{\mathrm{miss}})$; particles per draw J

Ensure: alert a_t and estimate $\hat{\pi}_t$ of $\mathbb{P}(Y_t = 1 \mid \mathcal{G}_t)$

for $r = 1, \dots, R$ **do**
  $\hat{\boldsymbol{h}}^{(r)} \leftarrow$ (7.2) with $\delta = L$, under $\vartheta^{(r)}$ ▷ state at $t - L$
  for $\ell = 1, \dots, J$ **do**
    impute the unavailable incidents in $(t - L, t]$ by thinning with intensity $\{1 - p_k F_k(t - s)\}\lambda_k(s)$, inserting the available ones
    $w_{r\ell} \leftarrow$ (D.3) for the completed path ▷ Lemma D.2
    $\pi_{r\ell} \leftarrow G_t(\boldsymbol{\chi}_{r\ell}(t))$ for the completed state ▷ simulation or Remark D.1
  $\hat{\pi}^{(r)} \leftarrow \sum_\ell w_{r\ell}\pi_{r\ell} \big/ \sum_\ell w_{r\ell}$
$\hat{\pi}_t \leftarrow R^{-1} \sum_r \hat{\pi}^{(r)}$
return $a_t = \mathbf{1}\{\hat{\pi}_t \geq \pi^*\}$ and $\hat{\pi}_t$

D.2 Alerting from delayed data by imputation

We use the notation of Section 7.3. The plug-in rule that compares a nowcast state with the threshold curve of Theorem 7.2(d) is not the Bayes rule for delayed data, because G_t is nonlinear. The following lemma and Algorithm 2 approximate the predictive probability given the available data instead. The algorithm has not been run in this paper; the simulation study evaluates only the plug-in rule (Section 8.6).

Lemma D.2 (Importance weights for imputation). *Assume Assumptions A1 and A2 with known parameters, and condition on the history up to $t - L$. Let O be the incidents in $(t - L, t]$ that are available by t. Simulate the remaining incidents U in $(t - L, t]$ by thinning with intensity $\{1 - p_k F_k(t - s)\}\lambda_k(s)$, where λ is computed from the history up to $t - L$, from O and from the points of U simulated so far. Then the conditional law of U given O has density, with respect to this proposal, proportional to*

$$w(U) = \prod_{i \in O} \lambda_{k_i}(s_i) \, \exp\Big\{ - \sum_{k=1}^{M} \int_{t-L}^{t} p_k F_k(t - s) \, \lambda_k(s) \, \mathrm{d}s \Big\}. \tag{D.3}$$

Consequently, if $U_1, \dots, U_J$ are independent proposals and $\boldsymbol{\chi}_\ell(t)$ is the state of the completed history, then $\sum_\ell w(U_\ell) G_t(\boldsymbol{\chi}_\ell(t)) / \sum_\ell w(U_\ell)$ converges almost surely, as $J \to \infty$, to $\mathbb{P}(Y_t = 1 \mid O,$ history up to $t - L)$.

In practice the history up to $t - L$ is itself incomplete. Algorithm 2 replaces it by the weighted carried-over excitation (7.2) with $\delta = L$. The window L is chosen so that $F_k(L) \approx 1$ and the excitation carried over from before $t - L$ has largely decayed by t. The only approximations, beyond Monte Carlo error, are this substitution and the parameter draws. When $p_k < 1$, unrecorded incidents never become available. A fraction $1 - p_k$ of the intensity is then imputed throughout the window, not only near t.

Several points about Algorithm 2 follow.

- *Normalising the weights.* Weights are normalised within each parameter draw. The draws already condition on all the data, so weighting across draws would count the recent data twice.
- *Effective sample size.* The effective sample size $(\sum_\ell w_{r\ell})^2 / \sum_\ell w_{r\ell}^2$ should be monitored. If it is small, J should be increased or L shortened.

- *Monte Carlo error.* The Monte Carlo standard error of $\hat{\pi}_t$ is estimated from the spread of the R per-draw estimates, or by the delta method for ratio estimators when $R = 1$. If $|\hat{\pi}_t - \pi^*|$ is within two standard errors, the simulation size should be increased before deciding.
- *Parameter uncertainty.* With posterior draws of ϑ, p_k and F_k, $\hat{\pi}_t$ estimates the posterior predictive probability. This is π_t in Theorem 7.2(a) when the information includes parameter uncertainty.
- *Calibration in practice.* The algorithm is calibrated only if the model is. Before operational use, the PIT (Lemma 7.4) and a reliability diagram of $\hat{\pi}_t$ should be checked on held-out data. If they show miscalibration, $\hat{\pi}_t$ should be recalibrated, for instance by isotonic regression (Remark D.3), before the threshold π^* is applied.
- *Plug-in shortcuts.* Comparing a nowcast state with the threshold curve of Theorem 7.2(d) is fast but is not the Bayes rule for delayed data. If it is used, it should be recalibrated in the same way.
- *Reuse.* With G_t replaced by simulated future counts, the same weighted particles give the predictive distributions used for scoring in Section 7.4.

D.3 Cost-weighted losses and the Brier score

Remark D.3 (Cost-weighted losses and the Brier score). Normalise the costs so that $c_{\mathrm{FA}} = c$ and $c_{\mathrm{miss}} = 1-c$. Then $\pi^* = c$, and the loss of the rule $\mathbf{1}\{\pi_t \geq c\}$ is $S_c(\pi_t, Y_t) = c\,(1 - Y_t)\mathbf{1}\{\pi_t \geq c\} + (1 - c)\,Y_t\mathbf{1}\{\pi_t < c\}$. For all π_t and Y_t,

$$\int_0^1 S_c(\pi_t, Y_t)\,\mathrm{d}c = \tfrac{1}{2}(\pi_t - Y_t)^2.$$

The Brier score is therefore twice the average realised cost when the normalised cost $c = \pi^*$ is drawn uniformly from $(0, 1)$. Every proper scoring rule for binary events is a mixture of such elementary losses (Schervish, 1989). When partners cannot agree on one cost ratio, the Brier score summarises performance across normalised costs. When they can, the realised cost at π^* is the relevant measure.

Theorem 7.2(a) presumes that π_t is the true conditional probability. A forecast that is not calibrated should therefore be recalibrated before the threshold π^* is applied, for instance by isotonic regression of Y_t on π_t over the training period.

D.4 Detecting regime shifts

Alerts concern the next Δ_{a} days with the parameters held fixed. A different question is whether the parameters themselves have changed, through a sustained rise either in the background or in contagion.

Remark D.4 (A CUSUM monitor). For a pre-change model ϑ_0 and an alternative ϑ_1, let

$$\ell(t) = \sum_{k=1}^{M}\Big[\sum_{i:\,t_i^k \leq t} \log\frac{\lambda_k^{\vartheta_1}(t_i^k)}{\lambda_k^{\vartheta_0}(t_i^k)} - \int_0^t \big\{\lambda_k^{\vartheta_1}(s) - \lambda_k^{\vartheta_0}(s)\big\}\,\mathrm{d}s\Big]$$

be the log-likelihood ratio of the data on $[0, t]$, with both intensities computed from the same observed history. Page's CUSUM statistic $\mathcal{W}(t) = \ell(t) - \min_{s \leq t} \ell(s)$ signals at the first t with $\mathcal{W}(t) \geq h$ (Page, 1954). Two detectors can run in parallel:

- one whose alternative is a shift in the background, $\boldsymbol{\mu} \mapsto (1 + \epsilon_\mu)\boldsymbol{\mu}$ with $\boldsymbol{A}$ fixed;
- one whose alternative is a shift in contagion, $\boldsymbol{A} \mapsto (1 + \epsilon_A)\boldsymbol{A}$ with $\boldsymbol{\mu}$ fixed.

Comparing them shows which kind of change better explains a sustained departure. The threshold h is calibrated by simulation under ϑ_0 to a target average run length to false alarm. The optimality theory of the CUSUM is for independent observations; here it serves as a monitoring statistic.

Two cautions apply. First, the detectors inherit the identification limits of Section 4. Over short stretches, a shift in the background and a shift in contagion can produce similar data, so the comparison is a diagnostic, not a test that separates the two. Second, the statistics must be computed on settled data or with the nowcast of Proposition 7.3, because reporting delays mimic a recent decline.

Appendix E Computational details

This appendix collects the recursions, quadrature rules, simulation algorithms and error calculations behind Section 6. The notation is that of Sections 6.1 and 6.2. Exponential kernels $g_{kj}(u) = \beta_{kj} e^{-\beta_{kj} u}$ and exponential shock responses $\psi_k(u) = \eta_k e^{-\eta_k u}$ are assumed unless stated otherwise. The methods are implemented in the accompanying code (`code/hawkes`).

E.1 Recursions for exponential kernels

Fix a pair (k, j). Let $s_1 < s_2 < \cdots$ be the merged event times of streams j and k only, d_m the stream of s_m, and $\Delta_m = s_m - s_{m-1}$. Define the decay sums

$$R_{kj}(t) = \sum_{l:\, t_l^j < t} e^{-\beta_{kj}(t - t_l^j)}, \qquad R_{kj}^{(1)}(t) = \sum_{l:\, t_l^j < t} (t - t_l^j)\, e^{-\beta_{kj}(t - t_l^j)}.$$

Write $t - t_l^j = \Delta_m + (s_{m-1} - t_l^j)$ for $t = s_m$, and note that the incident at s_{m-1} contributes lag Δ_m. This gives

$$R_{kj}(s_m) = e^{-\beta_{kj}\Delta_m}\big\{R_{kj}(s_{m-1}) + \mathbf{1}\{d_{m-1} = j\}\big\}, \qquad R_{kj}^{(1)}(s_m) = e^{-\beta_{kj}\Delta_m} R_{kj}^{(1)}(s_{m-1}) + \Delta_m\, R_{kj}(s_m), \tag{E.1}$$

with $R_{kj}(s_1) = R_{kj}^{(1)}(s_1) = 0$. These two recursions give every quantity that Section 6 needs at the incidents t_i^k.

- *Likelihood.* The intensity $\lambda_k(t_i^k)$ is given by (6.1) and the compensator by (6.2).
- *Compensator at the incidents* (for time rescaling). Since $\sum_{t_l^j < t}\{1 - e^{-\beta_{kj}(t - t_l^j)}\} = N_j(t-) - R_{kj}(t)$,

$$\Lambda_k(t) = \int_0^t \mu_k + \upsilon_k \sum_e S_e\{\Psi_k(t - \tau_e) - \Psi_k(-\tau_e)\} + \sum_j \alpha_{kj}\{N_j(t-) - R_{kj}(t)\}.$$

- *Gradient.* With $\lambda = \lambda_k(t_i^k)$ and sums over the incidents of stream k,

$$\frac{\partial \ell_T}{\partial \alpha_{kj}} = \sum_i \frac{\beta_{kj} R_{kj}(t_i^k)}{\lambda} - G_{kj},$$

$$\frac{\partial \ell_T}{\partial \beta_{kj}} = \alpha_{kj} \sum_i \frac{R_{kj}(t_i^k) - \beta_{kj} R_{kj}^{(1)}(t_i^k)}{\lambda} - \alpha_{kj} \sum_l (T - t_l^j) e^{-\beta_{kj}(T - t_l^j)},$$

$$\frac{\partial \ell_T}{\partial \theta_k} = \sum_i \frac{\mu_k(t_i^k) B(t_i^k)}{\lambda} - \int_0^T \mu_k(t) B(t)\, \mathrm{d}t,$$

$$\frac{\partial \ell_T}{\partial \upsilon_k} = \sum_i \frac{\sum_e S_e \psi_k(t_i^k - \tau_e)}{\lambda} - \sum_e S_e \Psi_k^T(e).$$

The derivative in η_k uses the shock analogues of R and $R^{(1)}$ below.

- *EM statistics.* N_{kj} and W_{kj} of (6.7) are the sums over i of $\alpha_{kj}\beta_{kj}R_{kj}(t_i^k)/\lambda$ and $\alpha_{kj}\beta_{kj}R_{kj}^{(1)}(t_i^k)/\lambda$.
- *Shocks.* The shock terms use the same recursions, with the calendar merged into the event sequence. The decay sum is $R_k^{\mathrm{sh}}(t) = \sum_{\tau_e<t} S_e e^{-\eta_k(t-\tau_e)}$, which jumps by S_e after τ_e. For short calendars, direct summation over e is simpler.

Numerical stability and vectorisation. The sequential recursion multiplies by factors $e^{-\beta\Delta_m} \le 1$ and is stable. Our implementation vectorises it. Source events are cut into chunks that span at most $200/\beta_{kj}$ time units. Within a chunk, R_{kj} is a cumulative sum of $e^{\beta_{kj}(t_l^j - a)}$ relative to an anchor a at the start of the chunk. A short loop over chunks carries the state forward. The result is exact up to rounding, with relative error of order 10^{-13}; there is no truncation window.

Ties. All sums use the strict past. If two incidents share a time stamp, the recursion with $\Delta_m = 0$ would count one as the parent of the other. Ties are therefore broken first, by randomisation within the recorded resolution (Appendix H.5).

Cost. The recursion for pair (k, j) costs $O(n_j + n_k)$, so all pairs together cost $O(Mn_{\mathrm{ev}})$. Running it over all n_{ev} merged events for every pair, as one state vector updated at every incident, would cost $O(M^2 n_{\mathrm{ev}})$.

Parallelism. If no parameter is shared across rows of the model, the likelihood separates by target stream (Section 6.1). Each row can then be fitted on its own processor.

E.2 Background integrals

The log-linear background needs $\int_0^T e^{B(t)^\top\theta}\,\mathrm{d}t$, together with its gradient $\int Be^{B^\top\theta}$ and Hessian $\int BB^\top e^{B^\top\theta}$. We use a composite 8-point Gauss–Legendre rule and reuse the same nodes for all three integrals. The cells must be chosen so that the integrand is smooth on each of them. For cubic B-splines alone, the exponent is a polynomial on each knot interval, and our implementation splits each knot interval into 16 cells. Day-of-week and holiday indicators make the integrand jump at day boundaries. With such indicators, the cells must also be split at every day boundary, giving at least one cell per day; our implementation currently supports smooth bases only. Seasonal (Fourier) bases use uniform cells. The basis evaluated at the incidents costs $O(n_{\mathrm{ev}}K_b)$ per gradient evaluation. The M-step for θ_k (Proposition 6.1(a)) takes Newton steps with gradient $\sum_i p_i^{k,0}B(t_i^k) - \int B\mu_k - \lambda_k^{\mathrm{pen}}P\theta_k$ and negative Hessian $\int BB^\top\mu_k + \lambda_k^{\mathrm{pen}}P$. Step halving ensures that the M-step objective increases, so the iteration remains a generalised EM and Proposition 6.1(d) applies.

E.3 Simulation

Synthetic data for the simulation study, and parametric bootstrap replicates, are drawn by one of two exact algorithms.

Ogata's thinning. Between incidents and away from calendar times, the excitation and shock terms of λ_k are nonincreasing, because they decay exponentially. A time-varying background need not be monotone, but it is bounded by $\bar\mu_k \ge \sup_{[0,T]}\mu_k$. So the sum of $\bar\mu_k$ and the current excitation and shock terms bounds the future intensity until the next incident or calendar time. Algorithm 3 exploits this. Candidates come from a Poisson process at the current bound and are accepted with probability $\lambda/\bar\lambda$. By the thinning theorem (Lewis and Shedler, 1979; Ogata, 1981), the accepted points have intensity λ. After a rejection the bound

Algorithm 3 Ogata's thinning for the shock-augmented model with exponential kernels.

Require: ϑ; calendar $\{(\tau_e, S_e)\}$; horizon T; bounds $\bar{\mu}_k \geq \sup_{[0,T]} \mu_k$.
Ensure: Events $\mathcal{S}$ on $[0, T]$ with intensity (6.1).

$t \leftarrow 0$; $R_{kj} \leftarrow 0$ for all (k, j); $\mathcal{S} \leftarrow \emptyset$.
loop
  $h_k \leftarrow \upsilon_k \sum_{\tau_e \leq t} S_e \eta_k e^{-\eta_k (t - \tau_e)}$ for all k; $\bar{\lambda} \leftarrow \sum_k \{\bar{\mu}_k + h_k + \sum_j \alpha_{kj} \beta_{kj} R_{kj}\}$.
  Draw $E \sim \mathrm{Exp}(\bar{\lambda})$; $s \leftarrow t + E$; let τ^+ be the first calendar time in $(t, T]$ (or $+\infty$).
  if $\tau^+ \leq s$ **then**
    $R_{kj} \leftarrow R_{kj} e^{-\beta_{kj} (\tau^+ - t)}$ for all (k, j); $t \leftarrow \tau^+$; **next iteration**.
  if $s > T$ **then**
    return $\mathcal{S}$
  $R_{kj} \leftarrow R_{kj} e^{-\beta_{kj} (s - t)}$ for all (k, j); $t \leftarrow s$.
  $\lambda_k \leftarrow \mu_k(t) + \upsilon_k \sum_{\tau_e < t} S_e \eta_k e^{-\eta_k (t - \tau_e)} + \sum_j \alpha_{kj} \beta_{kj} R_{kj}$ for all k; draw $U \sim U(0, 1)$.
  if $U \bar{\lambda} \leq \sum_k \lambda_k$ **then**
    Draw k with probability $\lambda_k / \sum_{k'} \lambda_{k'}$; add (t, k) to $\mathcal{S}$; $R_{k'k} \leftarrow R_{k'k} + 1$ for all k'.

is recomputed at the new time, which is valid because exponential waiting times are memoryless. At a calendar time the shock term jumps up, so the algorithm restarts there.

Exact cluster simulation. The cluster representation (Section 3.3) gives a second exact algorithm. It is our default, because it is faster and records the true branching labels.

(1) *Background immigrants.* Draw a Poisson process with intensity μ_k on $[0, T]$ for each stream. A piecewise-constant μ_k is sampled directly; any other μ_k by thinning against $\bar{\mu}_k$.

(2) *Shock immigrants.* For each (e, k), draw Poisson$\{\upsilon_k S_e \Psi_k^T(e)\}$ points at τ_e plus delays from ψ_k, truncated to the window. The delays are drawn by inverting the truncated distribution function.

(3) *Offspring.* Proceed generation by generation. Each point of stream j at s has Poisson(α_{kj}) children in each stream k, placed at s plus independent g_{kj} delays. Children after T are discarded; all their descendants would also lie after T.

When $\varrho(\boldsymbol{A}) < 1$, the generations form a subcritical multitype Galton–Watson process. The algorithm then terminates almost surely, after an expected $\boldsymbol{M}\boldsymbol{m}$ points, where $\boldsymbol{m}$ is the vector of expected immigrant counts. Each generation is a vectorised operation, so the cost is $O(M n_{\mathrm{ev}})$. For a stationary start, either simulate on $[-T_0, T]$ and discard $[-T_0, 0)$, or use perfect simulation (Møller and Rasmussen, 2005). The burn-in error decays with the cascade memory $1/\{\beta(1 - n)\}$ (Proposition 3.2). Comparing the two algorithms on the same parameters is a useful unit test, since they share no code beyond the model specification.

E.4 Standard errors

Standard errors come from the observed information $\hat{\boldsymbol{I}}_{\mathrm{obs}} = -\nabla^2 \ell_T(\hat{\vartheta})$. Our implementation computes it by central differences of the exact log-likelihood, with relative step 10^{-4}. Central differences of the analytic gradient of Appendix E.1 are more accurate. They need $2p$ gradient evaluations for p free parameters and have $O(h^2)$ error in each entry; one-sided differences need only p evaluations but have $O(h)$ error. Two further details matter.

- *Boundary parameters.* Free parameters estimated on or near the boundary ($\hat{\alpha}_{kj} \approx 0$, $\hat{\upsilon}_k \approx 0$) are excluded from the inversion and flagged. Their sampling distribution is not normal (Proposition C.3(b)). Intervals for them come from the profile likelihood, the parametric bootstrap, or the posterior.
- *Alternatives.* The information can also be assembled from complete-data quantities via Louis's identity (Louis, 1982). Two further consistent forms are positive semidefinite by construction: the optional (quadratic) variation of the score, $\sum_k \sum_i \lambda_k^{-2} \nabla\lambda_k \nabla\lambda_k^\top(t_i^k)$, a sum over incidents; and its predictable variation, $\sum_k \int_0^T \lambda_k^{-1} \nabla\lambda_k \nabla\lambda_k^\top \, \mathrm{d}t$.

Derived quantities use the delta method. For the multiplier, $\partial \boldsymbol{M}/\partial\alpha_{kj} = \boldsymbol{M}\boldsymbol{e}_k\boldsymbol{e}_j^\top\boldsymbol{M}$ (Equation (B.1)). If $\varrho(\boldsymbol{A})$ is a simple eigenvalue (for example, if $\boldsymbol{A}$ is irreducible) with right and left Perron vectors $\boldsymbol{v}$ and $\boldsymbol{w}$, then $\partial\varrho(\boldsymbol{A})/\partial\alpha_{kj} = w_k v_j/(\boldsymbol{w}^\top\boldsymbol{v})$. Near criticality and near the boundary the delta method is unreliable (Section 5.5), and we use bootstrap or posterior draws instead.

E.5 Tests and calibration

Likelihood-ratio tests. The alternative is fitted first, with the default multistart. The null fit fixes $\alpha_{kj} = 0$ exactly. It is started both from the alternative optimum, with α_{kj} set to 0, and from the default multistart, and the better null optimum is kept. Because the alternative nests the null, a null log-likelihood above the alternative's signals a local optimum, and the alternative is then refitted from the null optimum. The statistic is $\max\{0, 2(\hat{\ell}_1 - \hat{\ell}_0)\}$. For the sup-statistic, each alternative fit fixes β_{kj} at a grid value and starts from the null fit with a small positive α_{kj}.

Score process for the Davies problem. At the null fit $\tilde{\vartheta}$, the score in the nuisance parameters vanishes when they are interior. The standardised efficient score at decay β is therefore

$$\hat{Z}(\beta) = \frac{U_\alpha(\beta)}{\{I_{\alpha\alpha}(\beta) - I_{\alpha 2}(\beta) I_{22}^{-1} I_{2\alpha}(\beta)\}^{1/2}}, \qquad U_\alpha(\beta) = \sum_i \frac{\beta R_{kj}(t_i^k;\beta)}{\tilde{\lambda}_k(t_i^k)} - G_{kj}(\beta).$$

The information blocks are those of the optional-variation form in Appendix E.4, with $\partial\lambda_k/\partial\alpha_{kj} = \beta R_{kj}(\cdot;\beta)$. Each grid point costs one recursion (E.1). The whole path, and with it the sup-score statistic and the bound p_{D} of Section 6.5, needs a single null fit. The same holds in every bootstrap replicate, which makes the score-based bootstrap cheaper than the likelihood-ratio version by a factor of about the grid size.

Placebo calendars. Each placebo calendar requires one refit of the selected model with an extra shock class. The refits are independent and run in parallel. Starting each refit from the fit to the real calendar, with the placebo loading set to a small positive value, reduces the number of EM iterations substantially.

E.6 Complexity and memory

Table 5 summarises the costs. Here n_{ev} is the number of incidents, M the number of streams, E the number of calendar events, K_b the background basis size, Q the number of quadrature nodes, p the number of free parameters, G the size of a decay grid, and B the number of bootstrap or placebo replicates. No step stores an $n_{\mathrm{ev}} \times n_{\mathrm{ev}}$ array. Direct summation over pairs of incidents, by contrast, needs $O(n_{\mathrm{ev}}^2)$ time, and $O(n_{\mathrm{ev}}^2)$ memory if the parent probabilities are stored.

Table 5. Computational cost of the main steps (exponential kernels unless stated).

Step	Time	Memory
Log-likelihood or gradient	$O(Mn_{\text{ev}} + n_{\text{ev}}E + n_{\text{ev}}K_b + QK_bM)$	$O(n_{\text{ev}} + M^2 + QK_b)$
EM iteration (E-step, M-step root solves)	$O(Mn_{\text{ev}} + n_{\text{ev}}E + n_{\text{ev}}K_b + r\,Mn_{\text{ev}})$	$O(n_{\text{ev}} + M^2)$
Background Newton step (stream k)	$O\{(n_k + Q)K_b^2\}$	$O\{(n_k + Q)K_b\}$
Gibbs sweep (labels, conjugate updates)	$O(Mn_{\text{ev}} + n_{\text{ev}} \log n_{\text{ev}})$	$O(n_{\text{ev}} + M^2)$
Histogram or spline kernels (any shape)	$O(n_{\text{ev}}w)$ per EM iteration	$O(n_{\text{ev}})$
Ogata thinning (Algorithm 3)	$O\{(M^2 + E)\,n_{\text{cand}}\}$	$O(n_{\text{ev}} + M^2)$
Cluster simulation	$O(Mn_{\text{ev}})$ expected	$O(n_{\text{ev}})$
Observed information by differences	$O(p^2)$ likelihood evaluations	$O(p^2)$
Sup-LR test, bootstrap	$O(BG)$ fits	as a fit
Sup-score test, bootstrap	$O\{B(\text{null fit} + GMn_{\text{ev}})\}$	as a fit
INAR(p_ℓ) by CLS, N_{bin} bins	$O(p_\ell N_{\text{bin}}M^2 + p_\ell^3 M^3)$	$O(p_\ell^2 M^2)$

w: mean number of incidents within the kernel support. r: root-search steps per M-step. n_{cand}: number of thinning candidates, $n_{\text{ev}} \int \bar{\lambda} / \int \lambda$ in expectation. p_ℓ: number of INAR lags.

Appendix F Simulation details

This appendix documents the simulation study of Section 8: the software and the runs (Appendix F.1), the configurations (Appendix F.2), the estimator settings (Appendix F.3), further results of E1, E4 and E5 (Appendix F.4), the Monte Carlo standard errors (Appendix F.5) and how to reproduce the results (Appendix F.6). All data are synthetic. Every configuration value below is stored with the results in `results/sim_results.json`, and the per-replicate values are in `results/raw/E1.json`, . . . , `results/raw/E6.json`.

F.1 Software, seeds and runs

Software. The experiments are run by `code/run_experiments.py`. It calls the lab's Hawkes library in `code/hawkes/` (version 1.0.0) for simulation, EM fitting, standard errors, the likelihood-ratio test, the surge decomposition and predictive simulation. The figures and tables of Section 8 and of this appendix are written by `code/make_figures.py` from the results file. The recorded environment is Python 3.13.18, NumPy 2.5.3, SciPy 1.18.1 and Matplotlib 3.11.2 on Linux x86_64 (glibc 2.39). No other packages are used.

Seeds. The base seed is 2026110302. Every replicate has its own NumPy generator, created by `np.random.default_rng` from the seed sequence

$$\texttt{SeedSequence}([\,2026110302,\ i,\ c,\ r\,]),$$

where i is the experiment index (E1 to E6 are 1 to 6), c indexes the configuration and r the replicate. The configuration index is the index of T in E1; the position in Table 9 in E2 (0–11 for the main configurations, 12 and 13 for the long windows); the index of n for the homogeneous part of E3 and 100 plus the index of the recording case for its heterogeneous part; and the scenario index in E4 (size, three power values, common shock). E5 draws its label paths in 20 chunks of 1000, with $(i, c, r) = (5, 0, \text{chunk})$. E6 uses $c = 0$ for its replicates, $c = 1$ for the draws of the integrating forecasters, and $(i, c, r) = (6, 999, 0)$ for the threshold pilot. Because each replicate has its own stream, results do not depend on the number of worker

Table 6. Run times of the simulation study, in seconds. Main run: E5 is the rerun with 20,000 paths. Revision: the E2 long windows and check, and the full reruns of E3 and E6, which replace the main-run E3 and E6 entries in the results file. Task CPU time is summed over tasks.

		E1	E5	E3	E4	E2	E6	Total
Main run	wall	503.0	34.6	103.0	760.4	795.1	209.9	2406.0
	CPU	577.5	25.5	184.2	752.9	568.9	128.2	2237.2
Revision	wall	—	—	98.8	—	54.1	65.3	218.2
	CPU	—	—	191.3	—	102.9	119.6	413.8

processes or on the order of tasks, and adding configurations or methods with new seeds leaves the existing ones unchanged.

Runs. The main run was on 3 November 2026, from 21:08:49 to 21:48:24 UTC, with two worker processes, `OPENBLAS_NUM_THREADS=1` and `nice`, on a shared four-core machine. The experiments ran in the order E1, E5, E3, E4, E2, E6. Later steps changed the results file.

(1) *E6 post-processing* (21:49:52 UTC), which added the split of the PIT by whether a new external event arrives in the forecast window, from calendars regenerated from the seeds. It is superseded by step (5).

(2) *E5 rerun* (21:49:53 to 21:50:28 UTC) with 20,000 branching-label paths, replacing a first run with 4000 paths. The plug-in estimates of E5 reuse the stored E1 fits.

(3) *Merge* (21:55:25 UTC), which rewrote `sim_results.json` from the raw files without running any experiment.

(4) *Revision, E2* (22:53:27 to 22:54:22 UTC, option `--augment-e2`). The two long-window configurations were run; the 240 stored replicates of the 12 main configurations were kept, and the summary was recomputed from the replicate values (ridge counts, medians, shocks per path). Four stored main tasks, among them three ridge fits, were rerun as a check and reproduced every stored value exactly.

(5) *Revision, E3 and E6* (22:54:23 to 22:57:07 UTC). Both were rerun in full, E3 with the relative errors 30% and 50% in $\hat{p}$ added and E6 with the integrating forecasters added. The new quantities use no extra draws (E3) or their own generator (E6), and every value stored by the main run, including the seven original E6 forecasters and the post-processed split, was reproduced exactly.

Table 6 gives the run times. The main run took 2406.0 seconds of wall time, about five minutes over its planned budget of 35 minutes; the revision took 218.2 seconds. Task CPU time is summed over tasks; it exceeds wall time when two workers ran in parallel, and it falls short of it when the machine was busy with other jobs.

Code hashes. Each experiment records the first 16 hexadecimal digits of the SHA-256 hash of every source file (Table 7). The driver `run_experiments.py` has had four versions.

- `4214ba239bd036f8` (repository commit `10d25ad`) ran E1, E2, E3, E4 and E6 in the main run. The stored E1, E4 and main E2 results come from it.
- `97cd66769a8bd7a9` ran steps (1) and (2). It is not in the repository. The diff between the two committed versions, `4214ba23` and `caf14514`, contains only the default number of E5 paths, the E6

Table 7. Source files of the simulation study and the first 16 hexadecimal digits of their SHA-256 hashes. Library files are in `code/hawkes/`; for `inar.py` and `attribution.py` the second hash is the revision, which changed docstrings only.

File	Hash	File	Hash
`__init__.py`	`f8498f730e100620`	`likelihood.py`	`ae276c6c27e89823`
`alert.py`	`22a64c948e3a9b61`	`model.py`	`b66101b97399627c`
`attribution.py`	`e7f5e764f1adf374`, `3544a33f11cc23c2`	`rescaling.py`	`9ab2c9b32e3003fc`
		`simulate.py`	`aaedac31380652f5`
`em.py`	`2f6aebebccfc91e7`	`tests_alpha.py`	`9a70835c25addd10`
`inar.py`	`fd6f659749824310`, `fc3110841faaf12b`	`thinning.py`	`0374a889ddb6ddb6`
`run_experiments.py`: `4214ba239bd036f8` (main run), `97cd66769a8bd7a9` (steps 1–2), `caf14514836f2036` (merge), `7bb84df8f8f3c4ee` (revision)			
`make_figures.py`	`eec5d91235c0017f`		

split with its post-processing option, and metadata written by the merge, so this version lies between them. Its outputs no longer depend on it: rerunning E5 with the current driver reproduces the stored E5 results exactly, and step (5) superseded the post-processing.

- `caf14514836f2036` (commit `65f86dd`) performed the merge (3).
- `7bb84df8f8f3c4ee`, the current version, ran steps (4) and (5).

The library files were identical in all runs except `inar.py` and `attribution.py`, whose docstrings were corrected before the revision: the binning limit of the INAR approximation, $n_\Delta \approx n - \frac{1}{2}n(1-n)\beta\Delta$ (Proposition C.2), and the fact that passing a stream index j to the spillover function zeroes the whole column $\boldsymbol{A}_{\cdot j}$, which is not Definition 5.3; the experiments pass the online-to-offline mask of that definition. Their code is unchanged: the syntax trees without docstrings are identical. The INAR code is not used in E1–E6.

F.2 Configurations

Table 8 summarises the six experiments.

Time is in arbitrary units, called days in the text. Unless stated otherwise, EM stops when the gain in log-likelihood is at most $10^{-8}(1 + |\ell_T|)$, or after 2000 iterations.

E1 (recovery). $M = 3$ streams: offline region A, offline region B and online. $\boldsymbol{\mu} = (0.25, 0.20, 0.50)$; $\boldsymbol{A}$ as in Appendix F.4, with $\varrho(\boldsymbol{A}) = 0.4848$; one decay $\beta = 1.5$ for all kernels. Events at days $60, 180, \ldots, 1980$ (every 120 days), with sizes $1, 2, 3, 1, 2, 1, 3, 2$ repeated: 8 events before $T = 1000$ and 17 before $T = 2000$. Loadings $\boldsymbol{\upsilon} = (3, 2, 6)$; response kernel $\eta e^{-\eta t}$ with $\eta = 0.4$ for all streams. $T \in \{1000, 2000\}$; $R = 100$ per T. The mean numbers of incidents per stream are 621.0, 511.2 and 1076.8 at $T = 1000$, and 1252.8, 1026.1 and 2173.0 at $T = 2000$. Fits estimate 17 parameters with the calendar known; the mean fitting time was 2.6 seconds at $T = 1000$ and 6.9 at $T = 2000$.

E5 (attribution). The E1 truth on $[0, 315]$, which contains the first three events. The attributed event is the third, at $\tau_e = 300$ with $S_e = 3$; horizons $H \in \{7, 14\}$, and $H = \infty$ for the plug-in totals. 20,000 paths are simulated with their branching labels. An incident in $(\tau_e, \tau_e + H]$ is counted in D_k if it is an immigrant of

Table 8. Design of the simulation study (synthetic data only). R is the number of Monte Carlo replicates. Configurations are given below.

Exp.	Claim tested	Truth and estimators	R
E1	EM recovery and standard errors (Algorithm 1)	$M = 3$ (two offline regions, one online stream), $\varrho(\boldsymbol{A}) = 0.485$, a known calendar of events; $T \in \{1000, 2000\}$	100 per T
E5	Surge decomposition (Theorem 5.1 and Propositions 5.2 and 5.4)	E1 truth; exact formulas against branching labels; plug-in estimates from the E1 fits	20,000 paths; E1 fits
E2	Spurious contagion (Theorem 4.7) and its remedies (Corollary 4.9(i))	No contagion: shot-noise Cox processes (rare large or frequent small shocks) and slow regime switching; Hawkes fits with a constant or spline background or the shock times known; a dispersion estimator; long windows for rare shocks	20 per configuration (12 at $T = 1000$, 2 at $T = 10{,}000$)
E3	Thinning and the dark-figure correction (Theorem 4.10, Corollary 4.11, and Proposition A.6)	Univariate Hawkes, $n \in \{0.3, 0.5, 0.7\}$, thinned at $p \in \{1, 0.8, 0.6, 0.4, 0.2\}$; correction with p known or estimated with 10, 30 or 50% error; two marks with heterogeneous recording	40 per n; 30 per mark case (4)
E4	Boundary test of $\alpha_{\text{off,on}} = 0$ (Proposition C.3) and common shocks (Remark 4.13(i))	$M = 2$ (offline, online); size, power, and a calendar of common shocks omitted from or included in the model	160 (size); 60 per other scenario (4)
E6	Alerting and nowcasting (Theorem 7.2, Proposition 7.3, and Lemma 7.4)	$M = 2$ with random external events; weekly Hawkes forecasts (true or fitted parameters; no future events or integrating over them; complete or delayed data) and two baselines	12 replicates × 150 windows

event e, in C_k if it is a later-generation descendant of such an immigrant, in C_k^{on} if, in addition, its ancestral line contains a step from the online stream to an offline stream, and in B_k otherwise. The exact values are those of Theorem 5.1 for the process started empty at time 0, with B_k including the clusters rooted in the first two events. They are computed with matrix exponentials of the linear system of Proposition 5.2, as in (5.13), and the online channel uses the online-to-offline mask of Definition 5.3; the plug-in estimates apply the same computation to the E1 fits.

E2 (spurious contagion). Mean rate $\Lambda = 2$; lead-in of 40 days; $R = 20$ per configuration; $T = 1000$ for the 12 main configurations and $T = 10{,}000$ for the two long windows (rare family, $\kappa \in \{2, 4\}$, constant background and dispersion estimator only). Shot-noise families: shock decay $\gamma = 0.5$, m incidents per shock on average, shock share $s = \kappa/m$, shock rate $r = s\Lambda/m$, amplitude $a = m\gamma$ and constant part $\nu = \Lambda(1 - s)$. The process is simulated as a Hawkes process with $\boldsymbol{A} = \mathbf{0}$ whose calendar is the shocks in $[-40, T]$, each with size 1, loading m and response rate γ; this is the law of $C(\nu, r, a, \gamma)$, and it lets the shocks before time 0 act on the window. Slow family: a two-state rate $\Lambda \pm \delta$ with switching rate 0.01 in each direction ($q = 0.02$ in total), $\delta = (\kappa\Lambda q/2)^{1/2}$, and a random initial state. Table 9 lists the configurations and the theoretical values; the shocks per path are summarised in Table 2(a). The spline background is a clamped cubic B-spline basis with 20 functions and equally spaced interior knots on $[0, T]$, without penalty. The dispersion estimator uses bins of 25 days.

Table 9. E2 configurations, in the order of their seed index. s: share of shock-driven incidents; r: shock rate; ν: constant part of the rate; shocks: mean number per path, including the 40-day lead-in; n^*, n^*_w and β^*: second-order pseudo-true branching ratio, its 25-day-bin version and the pseudo-true decay (Theorem 4.7). Fits: c = constant background, s = spline background, k = shock times known.

Family	κ	s	r	ν	Shocks	n^*	n^*_w	β^*	Fits
Rare ($m = 16$)	0.5	0.0312	0.00391	1.9375	4.2	0.184	0.172	0.612	c, s, k
	1	0.0625	0.00781	1.8750	9.2	0.293	0.278	0.707	c, s, k
	2	0.1250	0.01562	1.7500	16.1	0.423	0.407	0.866	c, s, k
	4	0.2500	0.03125	1.5000	34.1	0.553	0.538	1.118	c, s, k
	8	0.5000	0.06250	1.0000	65.7	0.667	0.654	1.500	c, s, k
Frequent ($m = 4$)	0.5	0.1250	0.06250	1.7500	66.8	0.184	0.172	0.612	c, s, k
	1	0.2500	0.12500	1.5000	131.7	0.293	0.278	0.707	c, s, k
	2	0.5000	0.25000	1.0000	262.1	0.423	0.407	0.866	c, s
	3	0.7500	0.37500	0.5000	388.9	0.500	0.484	1.000	c, s
Slow ($q = 0.02$)	0.5	$\delta = 0.1$; rates 1.9 and 2.1				0.184	—	0.0245	c, s
	2	$\delta = 0.2$; rates 1.8 and 2.2				0.423	—	0.0346	c, s
	8	$\delta = 0.4$; rates 1.6 and 2.4				0.667	—	0.0600	c, s
Rare, $T = 10^4$	2	0.1250	0.01562	1.7500	158.4	0.423	0.407	0.866	c
	4	0.2500	0.03125	1.5000	320.8	0.553	0.538	1.118	c

The shocks-known fit was restricted to configurations with at most about 130 shocks per path on average. The library's E-step stores an array of size (incidents × shocks), so its cost grows with the number of shocks: for the frequent family at $\kappa = 1$, with 131.7 shocks per path, a shocks-known fit took 10.6 seconds on average, against 1.3 seconds for the constant-background fit. This is a limitation of the implementation, not of Algorithm 1: the M-step needs only sums over shocks, which admit the same recursion as (6.3). For the same reason the long windows were fitted with the constant background only.

E3 (thinning). Univariate Hawkes process with $\beta = 1$, $\mu = 2(1 - n)$ (so $\Lambda = 2$), $T = 1500$, $n \in \{0.3, 0.5, 0.7\}$ and $R = 40$. Recording probabilities $p \in \{1, 0.8, 0.6, 0.4, 0.2\}$, applied with one uniform variable per incident. The theoretical apparent branching ratios $n' = h_p(n)$ are 0.3000, 0.2613, 0.2154, 0.1597 and 0.0902 for $n = 0.3$; 0.5000, 0.4577, 0.4024, 0.3258 and 0.2094 for $n = 0.5$; and 0.7000, 0.6683, 0.6238, 0.5548 and 0.4248 for $n = 0.7$. Estimated p: $\hat{p} = \min\{1, \max(0.02,\ p + rpZ)\}$ with $Z \sim N(0, 1)$ drawn once for each replicate and p, and $\mathrm{SE}(\hat{p}) = r\hat{p}$, for $r \in \{0.1, 0.3, 0.5\}$; all three use the same Z, so the cells with $r = 0.1$ are those of the main run. The floor of Table 2(b) is the second term of (4.6) with $\mathrm{Var}(\hat{p}) = r^2p^2$, whose square root is $r\,n(1 - n)(2 - n)/2$. Heterogeneous part: marks $x \in \{1, 2\}$ with probabilities $(0.5, 0.5)$ and offspring means $n_x = (0.1, 0.9)$, simulated as a two-type Hawkes process with $\alpha_{yx} = \pi_y n_x$ (the type of a child is drawn independently of its parent), $\beta = 1$, $\Lambda = 2$, $T = 1500$ and $R = 30$. The recording cases and the values of Proposition A.6 are: $(0.2, 0.8)$ with $\mathrm{Cov}(p_X, n_X) = 0.12$, $n'' = 0.4495$ and $n_{\text{naive}} = 0.5774$; $(0.5, 0.5)$ with 0.4045 and 0.5358; $(0.8, 0.2)$ with $\mathrm{Cov}(p_X, n_X) = -0.12$, 0.3463 and 0.4787; and $(1, 1)$ with 0.5358 and 0.5358. In all cases $\bar{n} = 0.5$ and $\mathrm{Var}(n_X) = 0.16$.

E4 (testing). $M = 2$ (offline, online); $\boldsymbol{\mu} = (0.4, 0.8)$; $\boldsymbol{A} = \left(\begin{smallmatrix} 0.3 & \alpha_{\text{off,on}} \\ 0.2 & 0.4 \end{smallmatrix}\right)$; $\beta = 1$; $T = 600$. Size: $\alpha_{\text{off,on}} = 0$, $R = 160$. Power: $\alpha_{\text{off,on}} \in \{0.05, 0.1, 0.2\}$, $R = 60$ each. Common shocks: $\alpha_{\text{off,on}} = 0$; events uniform

on $[0, T]$ at rate 0.05 (29.4 per path on average); sizes $e^{0.5Z}$; loadings $(4, 12)$; response rates $(0.25, 2.0)$; $R = 60$. EM tolerance 10^{-9}.

E6 (alerting). $M = 2$ (offline target, online); $\boldsymbol{\mu} = (0.3, 0.8)$; $\boldsymbol{A} = \left(\begin{smallmatrix} 0.35 & 0.10 \\ 0.25 & 0.45 \end{smallmatrix}\right)$; $\beta = 1$. Events uniform at rate $1/30$; sizes $e^{0.75Z}$ (mean 1.325); loadings $(6, 15)$; common response rate 0.3. Training period $[0, 1000]$, followed by 150 windows of $\Delta_a = 7$ days; $R = 12$. The surge threshold $c = 21$ was set from a pilot path of 30,000 days, using the 4271 complete weekly windows after a burn-in of 100 days; in the pilot the exceedance rate is 0.0974, and weekly counts have mean 9.59 and variance 95.96. Predictive distributions use 2000 simulated continuations per forecast. The moving-average forecast uses the last four weekly counts. Cost ratios $c_{\text{miss}}/c_{\text{FA}} \in \{2, 5, 10\}$ with $c_{\text{FA}} = 1$. Delays on the offline stream only: zero with probability 0.3, otherwise exponential with mean 3 days, so $F(d) = 0.3 + 0.7(1 - e^{-d/3})$ for $d \geq 0$. PIT histograms have 10 bins. In the training periods there were 35.3 events per replicate on average (1.8), so the estimated event rate averages 0.0353, and the mean training size is 1.334.

F.3 Estimator settings

EM. All fits use Algorithm 1 as implemented in `hawkes.em.fit_em`, with the exact window terms, one decay rate shared by all excitation kernels, and one response rate shared by all streams (stream-specific response rates in the E4 fit with the calendar included). Starting values come from the profile log-likelihood in the decay rate, evaluated by 40 EM iterations at each point of an 8-point logarithmic grid on $[\min(0.05\, n_{\text{ev}}/T,\ 20/T),\ 200\, n_{\text{ev}}/T]$; full EM is then run from up to three local maxima of the grid profile, best first, and the best optimum is kept. The decay and response rates are bounded to $[0.01/T,\ 10^4 n_{\text{ev}}/T]$. SQUAREM steps are accepted only when they raise the log-likelihood. The spline background is fitted with three Newton steps per M-step and no penalty.

Standard errors and intervals. The observed information is the numerical Hessian of the exact log-likelihood, by central differences with relative step 10^{-4}, over the free parameters that exceed 10^{-6}; parameters at or below this value are treated as on the boundary and flagged. Parameter intervals are $\hat{x} \pm 1.96\,\text{SE}$. Derived quantities (multiplier, spectral radius, surge components) use the delta method with a central-difference Jacobian, and shares use the logit scale. A replicate without a standard error counts as a miss; no such replicate occurred in E1. In E3, the corrected intervals use (4.6).

E2 summaries. The in-window branching ratio is $\hat{n}_{\text{eff}} = \hat{\alpha}\, N(T)^{-1} \sum_l \{1 - e^{-\hat{\beta}(T - t_l)}\}$. A fit is flagged as on the ridge if $\hat{\beta}T < 1$. The main run flagged only decays at the lower bound, $\hat{\beta} \leq 1.01 \times 0.01/T$, which caught two of the nine ridge fits; the revision recomputed the flag from the stored estimates. The nine fits (configuration, replicate, $\hat{\alpha}$, $\hat{\beta}T$, $\hat{n}_{\text{eff}}$) are: shocks known, rare $\kappa = 1$, 17, 16.3, 0.010, 0.078; rare $\kappa = 2$, 17 and 19, 6.77 and 5.72, 0.012 and 0.010, 0.041 and 0.028; frequent $\kappa = 0.5$, 13, 2.06, 0.077, 0.076; frequent $\kappa = 1$, 16 and 19, 1.73 and 3.90, 0.040 and 0.014, 0.034 and 0.027; constant background, slow $\kappa = 2$, 5 and 12, 0.28 and 2.22, 0.88 and 0.038, 0.092 and 0.041; spline, slow $\kappa = 8$, 7, 29.1, 0.015, 0.231. Their $\hat{n}_{\text{eff}}$ is a valid summary, so they are kept in the means and medians. For the constant-background fits to the shot-noise families, $|\hat{n}_{\text{eff}} - \hat{\alpha}|$ is at most 0.0023 (0.0001 in the long windows).

E4 test. The alternative is fitted first. The null fit, with $\alpha_{\text{off,on}} = 0$ exactly, is started both from the alternative optimum and from the default multistart, and the better optimum is kept; if the null log-likelihood exceeds the alternative's, the alternative is refitted from the null optimum (Appendix E.5). The statistic is $\max\{0, 2(\hat{\ell}_1 - \hat{\ell}_0)\}$, and the p-value is $\frac{1}{2}\mathbb{P}(\chi^2_1 \geq x)$ for $x > 0$ and 1 for $x = 0$. The decay is shared and re-estimated under each hypothesis.

E6 forecasts. The predictive distribution at origin t is simulated from the state at t (Lemma 7.1), including the remaining response to events up to t. The *no-future-event* forecasters stop there. The *integrating* forecasters add, to each continuation, an independent draw of the events arriving in $(t, t+7]$: their number is Poisson with mean 7× the event rate, their times are uniform, each sends a Poisson number of immigrants with mean $\upsilon_k S_e$ to stream k at $\mathrm{Exp}(\eta)$ delays (those inside the window are kept), and the immigrants cascade with the model's $\boldsymbol{A}$ and β. Given the past these clusters are independent of the rest of the continuation, so the sum is a draw from the conditional law given the past. The true version uses rate $1/30$, log-normal sizes and the true loadings and response rate; the fitted version uses the rate of the training calendar (events per day), sizes resampled from it, and the fitted loadings and response rate. A unit test checks the mean of these draws against the exact expected response integrated over the arrival time. The surge probability is the fraction of simulated counts at or above c. The randomised PIT uses one uniform variable per window and method. The log score uses a floor of $0.5/2000$ for simulated distributions and 0.5 divided by the number of training windows for climatology. The moving-average mean is floored at 0.1. Climatology is the empirical distribution of the weekly counts in the training period from day 100. The naive state uses the incidents available at t without weights. The Horvitz–Thompson state weights each available offline incident at s by $1/F(t-s)$, which is finite because $F(0) = 0.3$. Both states are plugged into the same continuation as the complete-data forecast, so state uncertainty is not integrated over and Algorithm 2 is not run. The fitted forecasts use the EM estimates from the training period as fixed values. The state ratios of Section 8.6 are computed with the true parameters, from the excitation carried by the offline stream.

F.4 Further results: E1, E4 and E5

E1 design. There are three streams: two offline regions, A and B, and one online stream. The truth is $\boldsymbol{\mu} = (0.25, 0.20, 0.50)$, a common decay $\beta = 1.5$ and

$$\boldsymbol{A} = \begin{pmatrix} 0.25 & 0.08 & 0.12 \\ 0.08 & 0.20 & 0.12 \\ 0.10 & 0.10 & 0.35 \end{pmatrix}, \qquad \varrho(\boldsymbol{A}) = 0.485,$$

with rows indexing the receiving stream. Events occur every 120 days from day 60, with sizes cycling through $(1, 2, 3, 1, 2, 1, 3, 2)$, loadings $\boldsymbol{\upsilon} = (3, 2, 6)$ and response rate $\eta = 0.4$. With $R = 100$ replicates for each T, the mean number of incidents per path is 2209.0 ($T = 1000$) and 4451.8 ($T = 2000$). The fit estimates all 17 parameters $(\mu_k, \alpha_{kj}, \beta, \upsilon_k, \eta)$ with the calendar known.

E1 results. All 200 fits converged, all information matrices were positive definite and no estimate was on the boundary (Table 10, Figure 7). At $T = 2000$ the nine α_{kj} have mean |bias| 0.002, mean RMSE 0.026 (maximum 0.038) and coverage between 0.92 and 0.99. The background rates have mean RMSE 0.022 and coverage 0.92–0.99. The decay has mean 1.501, RMSE 0.117 and coverage 0.95; the response rate has mean 0.406, RMSE 0.055 and coverage 0.97; the loadings have mean RMSE 0.602 and coverage 0.93–0.95. At $T = 1000$ the α_{kj} have mean |bias| 0.003, mean RMSE 0.036 (maximum 0.048) and coverage 0.92–0.98, and the decay has RMSE 0.173 and coverage 0.92. For every α_{kj} the mean standard error lies between 0.91 and 1.21 times the Monte Carlo standard deviation of the estimates. Delta-method intervals for the nine entries of $\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{A})^{-1}$ cover in 0.93–0.99 of replicates at $T = 1000$ and 0.92–0.99 at $T = 2000$. The spectral radius is estimated at 0.476 and 0.481 on average, against 0.485, with coverage 0.96 and 0.97; its small downward bias (−0.008, MC SE 0.003, at $T = 1000$) halves when T doubles. The online loading $\upsilon_3 = 6$ is overestimated by 0.199 (MC SE 0.109) at $T = 1000$ and 0.137 (0.079) at $T = 2000$. Each is

Table 10. E1, recovery by EM (synthetic data, $R = 100$ per T). Cov.: coverage of nominal 95% intervals. Group rows: mean |bias|, mean RMSE and range of coverage over the entries. Stream 1 is offline region A, stream 3 the online stream.

Quantity	Setting	Target	Mean	Bias	RMSE	Cov. (%)
μ_1	$T = 1000$	0.250	0.253	+0.003	0.032	92
α_{11}	$T = 1000$	0.250	0.248	−0.002	0.039	95
α_{13} (on→off)	$T = 1000$	0.120	0.117	−0.003	0.025	96
α_{31} (off→on)	$T = 1000$	0.100	0.099	−0.001	0.046	94
β	$T = 1000$	1.500	1.516	+0.016	0.173	92
υ_3 (online loading)	$T = 1000$	6.000	6.199	+0.199	1.105	94
η	$T = 1000$	0.400	0.405	+0.005	0.077	96
M_{13}	$T = 1000$	0.289	0.280	−0.009	0.059	93
$\varrho(\boldsymbol{A})$	$T = 1000$	0.485	0.476	−0.008	0.035	96
all α_{kj} (9)	$T = 1000$	—	—	0.003	0.036	92–98
all M_{kj} (9)	$T = 1000$	—	—	0.008	0.076	93–99
all υ_k (3)	$T = 1000$	—	—	0.094	0.837	92–95
μ_1	$T = 2000$	0.250	0.252	+0.002	0.022	95
α_{11}	$T = 2000$	0.250	0.246	−0.004	0.028	93
α_{13} (on→off)	$T = 2000$	0.120	0.121	+0.001	0.020	93
α_{31} (off→on)	$T = 2000$	0.100	0.094	−0.006	0.033	92
β	$T = 2000$	1.500	1.501	+0.001	0.117	95
υ_3 (online loading)	$T = 2000$	6.000	6.137	+0.137	0.801	95
η	$T = 2000$	0.400	0.406	+0.006	0.055	97
M_{13}	$T = 2000$	0.289	0.292	+0.003	0.046	92
$\varrho(\boldsymbol{A})$	$T = 2000$	0.485	0.481	−0.004	0.025	97
all α_{kj} (9)	$T = 2000$	—	—	0.002	0.026	92–99
all M_{kj} (9)	$T = 2000$	—	—	0.005	0.055	92–99
all υ_k (3)	$T = 2000$	—	—	0.056	0.602	93–95

within two MC SEs of zero, but the sign is the same at both lengths, and the bias carries over to the direct online component in E5. Near the boundary the Wald intervals are invalid (Proposition C.3(b)); E3 shows what happens there.

E5 design and results. For each simulated path, stream and horizon $H \in \{7, 14\}$, the incidents in $(\tau_e, \tau_e + H]$ were counted in the four classes of Appendix F.2, and the Monte Carlo means were compared with the exact values by $z = (\text{MC mean} - \text{exact})/\text{MC SE}$, for 11 quantities per horizon. The largest $|z|$ is 1.48 at $H = 7$ and 1.50 at $H = 14$. For offline region A at $H = 14$, the contagion component is 9.694 exactly and 9.710 (MC SE 0.034) by simulation, and its online-channel part is 5.779 exactly and 5.800 (0.026). The contagion share of region A at $H = 14$ is 0.519, about twice $\alpha_{11} = 0.25$: through cross-excitation, a stream's share can exceed its own self-excitation (Section 5). The plug-in estimates are in Table 11(b) and Figure 8. Over all 45 attribution quantities (B_k, D_k, C_k and shares for three streams, and C_k^{on} and σ_k for the two offline streams, at $H = 7$, 14 and ∞), coverage ranges from 0.90 to 0.99 at $T = 1000$ and from 0.91 to 0.96 at $T = 2000$. The overestimated online loading of E1 appears as a direct online component of 18.33 against 17.93 at $H = 14$ (bias +0.392, MC SE 0.234).

Table 11. Results of E4 and E5 (synthetic data). (a) E4, test of $\alpha_{\text{off,on}} = 0$: mean estimate, rejection rates at the 5% level with the chi-bar-square and the naive χ^2_1 reference, and the share of statistics equal to zero. (b) E5, plug-in surge decomposition from the E1 fits ($T = 2000$, $R = 100$) for the event with $S_e = 3$, horizon $H = 14$ unless marked ∞; stream 1 is offline region A, stream 3 the online stream. Cov.: coverage of nominal 95% intervals.

(a) E4: the online-to-offline test

Scenario	R	True $\alpha_{\text{off,on}}$	Mean $\hat{\alpha}_{\text{off,on}}$	Reject., $\bar{\chi}^2$ (%)	Reject., χ^2_1 (%)	$\mathbb{P}(\text{LR} = 0)$ (%)
size	160	0	0.010	6.3	2.5	56.3
power	60	0.05	0.056	61.7	53.3	1.7
power	60	0.1	0.105	98.3	95.0	0.0
power	60	0.2	0.200	100.0	100.0	0.0
common shocks, calendar omitted	60	0	0.072	100.0	100.0	0.0
common shocks, calendar in model	60	0	0.010	8.3	6.7	55.0

(b) E5: surge attribution

Quantity	Truth	Mean	Bias	RMSE	Cov. (%)
B_1	7.381	7.406	+0.025	0.368	92
D_1	8.967	8.976	+0.009	1.598	94
C_1	9.694	9.779	+0.085	1.153	92
$C_1/(D_1 + C_1)$	0.519	0.523	+0.004	0.049	92
σ_1 (online channel)	0.596	0.602	+0.005	0.065	94
D_3 (online)	17.933	18.326	+0.392	2.357	95
C_3 (online)	14.667	14.693	+0.026	1.464	96
$C_1(\infty)$	9.769	9.866	+0.096	1.174	91

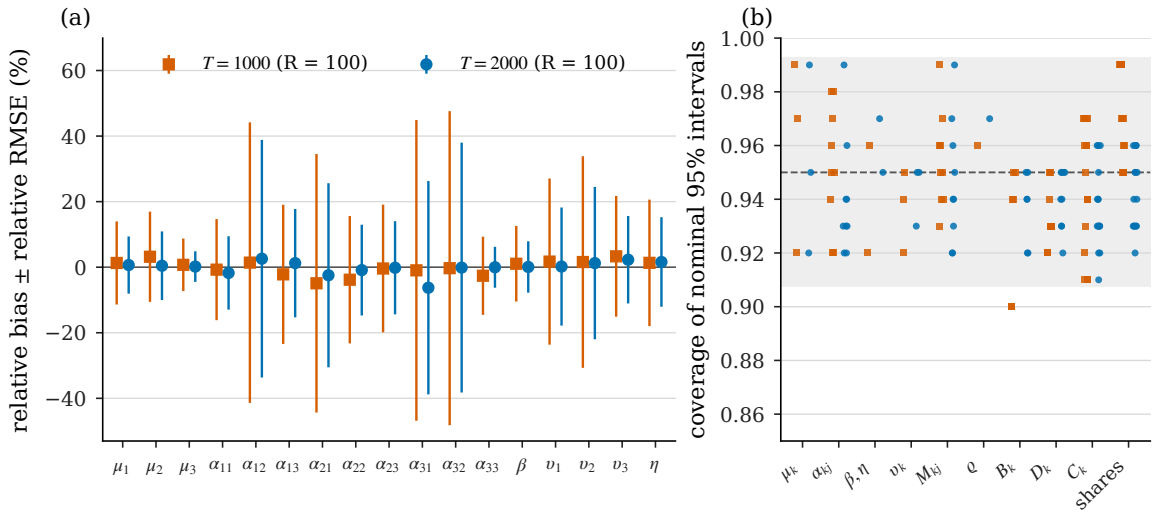


Figure 7. E1, recovery by EM (synthetic data, $R = 100$ per T). (a) Relative bias (markers) ± relative RMSE (bars), in per cent, for the 17 parameters. (b) Coverage of nominal 95% intervals for the parameters and the derived quantities: multiplier entries M_{kj}, spectral radius ϱ and the surge components of E5. The grey band is the 95% Monte Carlo range around 0.95 for $R = 100$.

E4 details. Under H_0 a path has 1253.4 incidents on average. The naive χ^2_1 reference has power 0.533, 0.95 and 1.0 against 0.617, 0.983 and 1.0 for the chi-bar-square reference (Figure 9). In the common-shock scenario there are 29.4 events per path on average; with the calendar in the model the loadings (3.98, 12.29) and response rates (0.257, 1.988) are recovered.

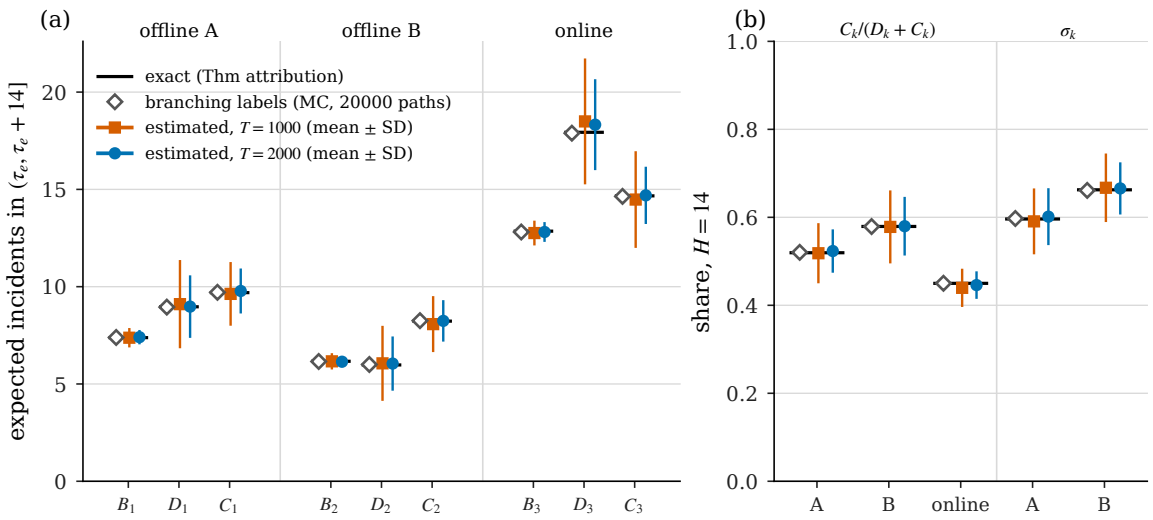


Figure 8. E5, surge attribution after an event of size $S_e = 3$ (synthetic data). (a) Expected counts in $(\tau_e, \tau_e + 14]$ by stream: exact values (Theorem 5.1; lines), branching-label Monte Carlo means (20,000 paths; diamonds), and estimates from the E1 fits (mean ± SD over 100 replicates). (b) Contagion shares $C_k/(D_k + C_k)$ and online-channel shares σ_k at $H = 14$, by stream.

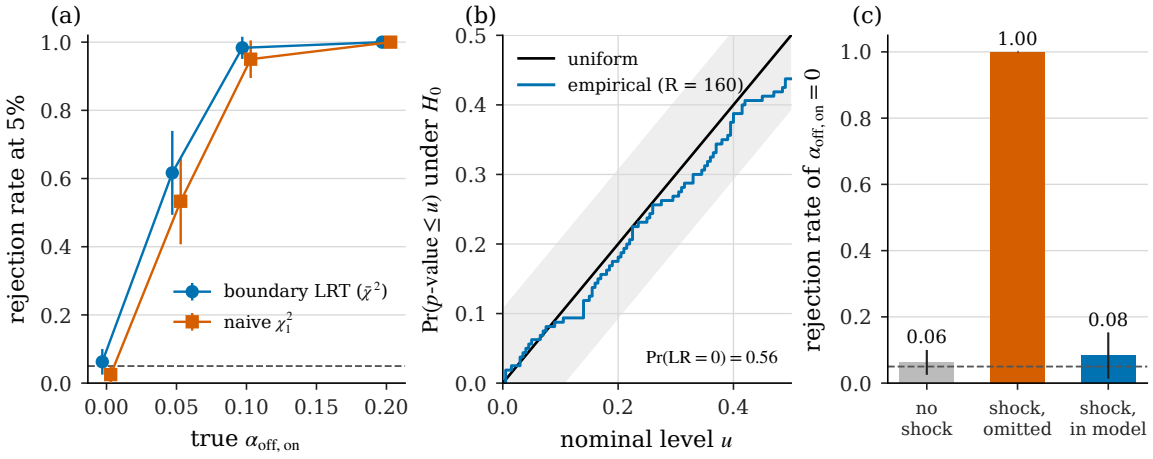


Figure 9. E4, the test of $\alpha_{\text{off,on}} = 0$ (synthetic data; bars ±1.96 MC SE). (a) Rejection rate at the 5% level against the true $\alpha_{\text{off,on}}$, for the chi-bar-square and the naive χ^2_1 reference. (b) Empirical distribution function of the chi-bar-square p-values under H_0 ($R = 160$), with a 95% Kolmogorov–Smirnov band; the p-value is 1 whenever the statistic is 0. (c) Rejection rates without shocks, and with common shocks omitted from or included in the model ($R = 60$).

F.5 Monte Carlo standard errors

Let $x_1, \dots, x_R$ be the estimates of a quantity with true value x_0 over R replicates, with mean $\bar{x}$, standard deviation s_x and errors $\varepsilon_r = x_r - x_0$.

- *Mean and bias.* The bias is $\bar{x} - x_0$. The MC SE of the mean and of the bias is $s_x/\sqrt{R}$. Medians are reported without MC SE.
- *RMSE.* RMSE $= (R^{-1}\sum_r \varepsilon_r^2)^{1/2}$, with MC SE $s_{\varepsilon^2}/(2\,\mathrm{RMSE}\sqrt{R})$ by the delta method, where s_{ε^2} is the standard deviation of the squared errors.
- *Coverage and rejection rates.* For an estimated proportion $\hat{\pi}$, the MC SE is $\{\hat{\pi}(1-\hat{\pi})/R\}^{1/2}$. The band of Figure 7(b) is $0.95 \pm 1.96\,(0.95 \times 0.05/R)^{1/2}$.
- *E5.* For each quantity, the mean over the $N = 20{,}000$ paths has MC SE $(\hat{\sigma}^2/N)^{1/2}$, with $\hat{\sigma}^2$ the sample variance over paths, and $z = (\text{mean} - \text{exact})/\text{MC SE}$.
- *E6.* Costs per window, Brier, CRPS and log scores and the mean forecast bias are averaged within each replicate; the MC SE is the standard deviation of the 12 replicate averages divided by $\sqrt{12}$. Differences between two forecasters are paired: the MC SE is that of the 12 replicate differences. Alert rates, precision and recall are pooled over all 1800 windows and have no reported MC SE. The PIT statistic is $\chi^2 = \sum_{b=1}^{10}(h_b - e)^2/e$, with h_b the bin counts and e the number of windows divided by 10, referred to χ^2_9. The windows do not overlap and the outcomes are settled, so by Lemma 7.4(b) the PIT values are independent under a correct forecast; the reference ignores the Monte Carlo error of the predictive distributions. The state ratios are ratios of means over all windows.

In the figures, error bars are ±1.96 MC SE, except in Figure 7(a) (± relative RMSE), Figure 8 (±1 SD over replicates) and Figure 5(c) (no bars).

F.6 Reproduction

From `papers/contagion/code/`, the commands

```
OPENBLAS_NUM_THREADS=1 nice python run_experiments.py --workers 2
python make_figures.py
```

run all six experiments, write `results/raw/E*.json` and `results/sim_results.json`, and then the figures and the tables `tables/tab-sim-e1.tex`, ..., `tab-sim-e6.tex`. With the current driver a single run replaces the steps of Appendix F.1: E5 uses 20,000 paths, E2 includes the long windows, E3 the three error levels and E6 the integrating forecasters, and the seeds of every value stored earlier are unchanged. One experiment can be rerun with `--only E5` (E5 reads the stored E1 results when E1 is not run in the same invocation); `--augment-e2` repeats revision step (4); and `--merge-only` rebuilds the results file from the raw files. The option `--quick` runs tiny configurations in seconds; the tests in `code/tests/test_experiments.py` use it to check the output schema, the absence of failed replicates, that two runs give identical results and that the integrating forecasters leave the other E6 forecasters unchanged. With the recorded software versions, the seeds determine every replicate. We have not checked whether other versions of NumPy or SciPy reproduce the results to the last digit.

Appendix G Candidate data sources

This appendix gives the source-by-source detail summarised in Section 2.2. For every source we record what can be checked in the provider's own documentation: the timestamp the record carries, its temporal

and geographic resolution, the anti-Jewish bias category, and how the data can be obtained. Properties that we could not confirm from primary documentation at the time of writing (November 2026) are marked as such and are not used in any argument. Table 12 summarises the candidate streams.

G.1 Incident streams

City-level open data: NYPD hate crimes. The New York City Police Department publishes a file of confirmed hate-crime incidents on the NYC Open Data portal (dataset `bqiq-cu78`) (New York City Police Department, 2026), beginning in 2019. According to the portal's data dictionary, *Complaint Year Number* and *Month Number* give the year and month in which the incident occurred, and *Record Create Date* gives the date on which the report was filed. A *Bias Motive Description* field holds the NYPD's bias category, whose values include `ANTI-JEWISH`. Geography is given by the complaint precinct, the patrol borough and the county; each of the city's five counties is coterminous with a borough. For point-process work the file thus has a two-level time structure: the occurrence time is interval-censored to a calendar month, while the report date is known to the day. We treat the record-creation date as a report date and model the delay between occurrence and record creation explicitly (Section 2.3), with the occurrence month as a constraint. Whether day-level occurrence dates can be obtained from the NYPD, for example under a data-use agreement, is an open feasibility question for the pilot (Section 9); we do not assume it. Changes to reporting systems are structural breaks for time-series use. The New York State Comptroller reported that the state's Division of Criminal Justice Services did not publish 2023 hate-crime data for New York City because of the NYPD's transition in data-reporting systems as part of its move to NIBRS (Office of the New York State Comptroller, 2024). Rosenbaum (2025) analyses NYPD administrative hate-crime records for 2019–2024 at monthly resolution. Adjusting for the Jewish population of each borough, the study reports that monthly anti-Jewish hate crimes were on average about twice as common in the first year of the Israel–Hamas war as in the preceding years of the study period (prevalence ratio 1.97, 95% CI 1.64–2.35). Monthly resolution suffices for such before–after contrasts but not for kernels on the scale of days.

FBI NIBRS.. The FBI's Uniform Crime Reporting Program retired its Summary Reporting System on 1 January 2021 and moved to collection through the National Incident-Based Reporting System (NIBRS) (Federal Bureau of Investigation, 2026). For hate crimes, however, the FBI reopened the 2021 collection to summary-format submissions (Federal Bureau of Investigation, 2023). According to the Congressional Research Service, it has accepted both formats since 2022 (Congressional Research Service, 2025). Each NIBRS incident carries the mandatory Data Element 3, *Incident Date/Hour*. The date is recorded as `YYYYMMDD` and the hour on a 24-hour clock. When the incident date is unknown, the report date is entered instead and flagged with a report-date indicator (Federal Bureau of Investigation, 2025a,b). Anti-Jewish bias is code 21 of Data Element 8A, *Bias Motivation* (Federal Bureau of Investigation, 2025a). The geographic unit is the reporting agency, identified by its ORI number. NIBRS submissions thus supply occurrence dates, and hours where recorded, for individual anti-Jewish incidents. For summary-format submissions we have not confirmed comparable date fields, so the stream is restricted to NIBRS records. The data are also slow. National statistics for data year 2024 were released on 5 August 2025 (Federal Bureau of Investigation, 2025c), so for that year the lag from occurrence to national availability was roughly 7 to 19 months, depending on when in the year an incident occurred. Agency participation varies by agency and year; the companion paper models it.

ADL Audit and H.E.A.T. Map. ADL describes its annual Audit as covering criminal and non-criminal incidents of harassment, vandalism and assault, as reported to ADL by victims, law enforcement, the media and partner organisations (Anti-Defamation League, 2025). An ADL stream therefore measures a different quantity from a police stream, and the two reach the record through different funnels (Section 2.6). ADL states that since 7 October 2023 the Audit has included certain expressions of opposition to Zionism and of support for violent resistance against Israel or Zionists, while continuing to exclude legitimate political protest (Anti-Defamation League, 2025). For a point-process model this matters for a structural reason, separate from any substantive judgement. A change in recording scope at a date that is also a candidate shock time τ_e is not identified separately from the shock term unless the change is modelled explicitly. An ADL stream therefore needs a scope-break indicator in $\mu_k(t)$, or records coded under a constant scope rule. ADL's H.E.A.T. Map (Hate, Extremism, Antisemitism, Terrorism) publishes incident-level listings that can be downloaded as a CSV file, with some details removed to protect victims' privacy (Anti-Defamation League, n.d.). We have not confirmed whether the listed date is the date of occurrence or of report to ADL, nor how redaction affects dates. These checks are part of the pilot's feasibility stage.

CST (United Kingdom). The Community Security Trust records antisemitic incidents in the United Kingdom and publishes annual reports, most recently for 2025 (Community Security Trust, 2026). CST data would serve as an international comparison stream. That report illustrates a further threat to time-series analysis. CST states that since March 2024 a technical problem has disrupted the flow of reports to it from the Metropolitan Police (Community Security Trust, 2026). In the notation of Section 2.6, an interruption in a referral feed is a step change in the recording probability $p_k(t)$, which a model with constant p_k would read as a change in the incident process itself. Incident-level records with dates would require a data-sharing agreement, and a CST stream is therefore conditional.

Table 12. Candidate data streams for the pre-registered analysis. "TBC": not confirmed from the provider's documentation at the time of writing; to be confirmed before registration. Delays are from occurrence to availability to analysts.

Stream	Time stamp (resolution)	Geography	Delay to availability	Access
NYPD hate crimes (New York City Police Department, 2026)	Occurrence: month; record creation: day. From 2019.	Precinct, patrol borough, county	Report delay bounded by occurrence month and record date; refresh schedule TBC	Public download (NYC Open Data)
FBI NIBRS (Congressional Research Service, 2025; Federal Bureau of Investigation, 2025a, 2026)	Incident date and hour; report date substituted (flagged) if incident date unknown. NIBRS from 2021; hate-crime collection also accepts summary format since 2022.	Reporting agency (ORI)	Annual national release; data year 2024 released 5 Aug. 2025	Public (FBI Crime Data Explorer)
ADL Audit / H.E.A.T. Map (Anti-Defamation League, 2025, n.d.)	Incident date listed; occurrence vs. report date TBC. Scope change from 7 Oct. 2023.	Place listed; resolution TBC	TBC	Public map; CSV download with redactions
CST (UK) (Community Security Trust, 2026)	TBC	TBC	Annual report; 2025 report published Feb. 2026	Incident-level records by agreement (TBC)
ACLED (Raleigh et al., 2010)	TBC	Event location	TBC	Provider terms (TBC)
GDELT (Leetaru and Schrodt, 2013)	TBC	Event location	TBC	Provider terms (TBC)
Platform data (European Parliament and Council of the European Union, 2022)	Post time stamps (binned for analysis)	Varies	Depends on access route	DSA Art. 40 procedure; status TBC
Classifier volume (Kornstein Lab for Antisemitism Research, 2026b,c)	Bin chosen by design (e.g. day)	None or coarse	Construction lag only	Lab-constructed from a defined corpus

G.2 Platform data access

Article 40 of the EU Digital Services Act provides a procedure by which vetted researchers may obtain data from very large online platforms and very large online search engines for research that contributes to the detection, identification and understanding of systemic risks in the Union. It also contains a separate provision on access to publicly accessible data, and it leaves technical conditions to delegated acts (European Parliament and Council of the European Union, 2022). Whether a given project qualifies, and on what terms, is decided through that procedure. We have not confirmed its operational status for the platforms and periods of interest, and the plan in Section 9 treats online streams as optional, so that the offline analysis does not depend on them. The terms of any research interface used will be recorded at registration.

G.3 Prior empirical literature

Mohler et al. (2011) developed self-exciting point-process models of crime; Reinhart (2018) reviews spatio-temporal versions and their applications. Related applications include a spatiotemporal test of whether gun violence is contagious (Loeffler and Flaxman, 2018) and self-exciting models of civilian deaths in Iraq (Lewis et al., 2012). For hate crime specifically, King and Sutton (2013) study the temporal clustering of hate-motivated offending. Two studies examine links between online content and offline hate crime: Müller and Schwarz (2021) study social media and hate crime, and Williams et al. (2020) examine anti-Black and anti-Muslim social-media posts as predictors of offline racially and religiously aggravated crime. For anti-Jewish incidents, Rosenbaum (2025) compares monthly NYPD counts before and after a salient external event (see above). We use this literature to motivate the questions, not as prior estimates. None of these studies estimates a multivariate branching matrix for antisemitic incidents, and our framework does not take its parameter values from them. Separating exogenous shocks from contagion is the identification problem of Section 4.

Appendix H The pre-registered analysis plan in full

This appendix sets out in full the analysis plan summarised in Section 9, which we intend to register before any outcome data are examined. It fixes the candidate data, the estimands, the procedure for listing external events, the model hierarchy and its tests, the sensitivity analyses, the validation checks and the decision rules. The numerical settings below (horizons, thresholds, grids, replicate numbers) are proposals. They will be frozen at registration; any change made before registration will be documented, and any change made after it will be reported as a deviation. The simulation study (Section 8) evaluates, on synthetic data with known ground truth, some of the tools these rules rely on: EM estimation, surge attribution, the boundary test, the thinning correction and a plug-in alert. It does not evaluate the block tests of Appendix H.4, the placebo calibration, the time-rescaling check, the Bayesian fit or the selection and decision rules themselves; they are checked in the synthetic dry run (V4). No part of the plan has yet been carried out on real data.

H.1 Candidate pilot

Setting. The candidate pilot is New York City, using the NYPD hate-crime file described in Section 2.2 (New York City Police Department, 2026). We restrict it to records whose bias motive is `ANTI-JEWISH` and use the period from 2019 onwards. The offline streams k are the five boroughs. Sparse streams are handled by shrinkage on $\boldsymbol{A}$ (Section 6), not by merging streams on the basis of observed counts, which would let outcome data shape the model. The offence category serves as a mark in a secondary analysis.

The final 12 months of data available at the freeze date form the hold-out test period, and all earlier data form the training period.

Two designs, chosen before outcome data are seen. The NYPD file records the month of occurrence and the day of record creation (Section 2.2). The design is chosen on data availability alone:

(A) *Design A* (conditional): if day-level occurrence dates can be obtained under a data-use agreement, the model is fitted to occurrence days, with within-day ties handled as in Section 2.1.

(B) *Design B* (default): the model is fitted to record-creation dates, treated as a report-time process. The occurrence month enters as an interval constraint on each delay, and the delay distribution is modelled explicitly (Section 2.3). Contagion parameters then describe the report-time process, and sensitivity to the delay model is analysis S4.

Additional streams. Three further sources are conditional on access being confirmed during the feasibility stage: (i) a classifier-based online volume series (Section 2.5); (ii) an ADL stream, if its export provides day-level dates; and (iii) a CST stream as an international replication. A national NIBRS analysis of anti-Jewish incidents is retrospective only, because of the release lag (Section 2.2). If a conditional stream is unavailable it is dropped, and the primary analysis is unaffected.

Feasibility gates. Before registration we will: confirm the data dictionaries and archive the data vintages (F1); choose between Designs A and B (F2); record the licence terms of the event sources (F3); confirm or abandon the online streams (F4); and obtain ethics approval (F5; Section 10).

H.2 Primary estimands

E1. *Branching matrix.* The matrix $\boldsymbol{A} = [\alpha_{kj}]$ for the recorded streams has three parts: the diagonal α_{kk} (within-borough self-excitation); the off-diagonal offline entries (cross-borough excitation); and, when online streams are included, the online→offline and offline→online blocks. We report each entry with its interval, together with $\varrho(\boldsymbol{A})$ and $\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{A})^{-1}$. Estimates are reported on the recorded scale first. Thinning-corrected values (Theorem 4.10) are reported alongside and labelled as model-based (Section 2.6).

E2. *Surge decomposition.* For each pre-registered event e and stream k we report the decomposition $\mathbb{E}[N_k(\tau_e, \tau_e + H]] = B_k + D_k + C_k$ of Theorem 5.1 and, where online streams are included, the online→offline spillover share. The primary horizon is $H = 14$ days, with $H \in \{7, 28\}$ as secondary horizons. Results are summarised by event class. Event-level results appear in an appendix under neutral identifiers (Appendix H.3).

E3. *Alert performance.* We evaluate the rule of Theorem 7.2 and Algorithm 2, with window $\Delta_a = 7$ days. For each stream the surge threshold c is the 95th percentile of the training-period distribution of 7-day counts. The cost ratio $c_{\text{miss}}/c_{\text{FA}}$ is agreed with community-security partners before registration, with $\{2, 5, 10\}$ as the default grid. Metrics are the hit rate, false-alarm rate, realised cost and the Brier and log scores for the exceedance event. Two baselines serve as comparators: the background-only model M0 and a persistence rule that raises an alert when the previous window exceeded c.

Secondary estimands are the kernel time scales, the class-level shock loadings υ_k and the seasonal components of $\mu_k(t)$.

H.3 Pre-specified external events

The event list is fixed by the following procedure. Every step is completed before the analysis team loads any incident data for the study period.

1. *Classes.* Event classes are defined by type: (a) onset or major escalation of armed conflict; (b) terrorist or other mass-casualty attacks; (c) high-profile public statements by public officials or other prominent figures. The operational definitions are applied identically whatever the location, the parties, the perpetrators or the victims. We do not select events by their presumed relevance to antisemitism. Relevance is estimated through the loadings υ_k, and events that are irrelevant to the outcome should receive loadings near zero. A secondary list restricts the pool to events whose coverage matches a keyword list that is fixed and deposited in advance.

2. *Salience threshold.* For each candidate event, a salience index is computed from the event datasets alone (Section 2.4). It is the coverage attributed to the event over $[\tau_e, \tau_e + 2 \text{ days}]$, divided by the median daily coverage over the preceding 365 days. An event enters the list if its index exceeds a threshold $s^\star$. The threshold is set, using event data only, so that the list contains between 10 and 30 events over the study period. The size S_e is the logarithm of the index.

3. *Blind coding.* Two coders independently review the candidates above the threshold. They merge duplicates, assign each event to a class and set τ_e to the time of first public report in the pilot city's time zone, to the day (or the hour under Design A). The coders have no access to incident data, which a separate data custodian holds until the freeze. Agreement is reported as Cohen's κ for class and as the share of τ_e that agree to within one day. A third coder resolves disagreements. Events less than 48 hours apart are merged into a single shock at the earlier time.

4. *Freeze.* The list (τ_e, class, S_e, source identifiers) is written to a file. The file's SHA-256 hash is posted with the registration and the file is deposited with the registry. Events after the freeze, which are needed for prospective evaluation, are coded under the same rules within seven days and appended with time-stamped hashes.

5. *Neutral identifiers.* Each event is reported by class and date. Descriptions follow the conventions of Section 10.

H.4 Model hierarchy, block tests and selection rules

The models are nested:

M0 Background only: $\mu_k(t) = \exp(B(t)^\top \theta_k)$. The basis B is a cubic B-spline in calendar time with knot spacing $h_b = 60$ days, so that the number K_b of basis functions grows with the length of the study period, plus day-of-week effects and indicators for a holiday calendar fixed in advance.

M1 M0 plus shocks $\sum_e \upsilon_k S_e \psi_k(t - \tau_e)$, with an exponential response kernel ψ_k; the loading υ_k and the kernel are specific to each stream and shared by all events of a class.

M2 M1 plus self-excitation (diagonal $\boldsymbol{A}$).

M3 M2 plus cross-stream offline excitation (the off-diagonal entries of the offline block of $\boldsymbol{A}$).

M4 M3 plus the online→offline and offline→online blocks, when online streams are available.

The kernels are exponential, with one common decay rate within each block: the diagonal (M2), the offline off-diagonal entries (M3), and each of the two online blocks (M4). We estimate by maximum likelihood via

EM (Algorithm 1), with parametric bootstrap intervals from 500 replicates. A Bayesian fit with shrinkage on $\boldsymbol{A}$ (Section 6.3) is co-primary for M3 and M4.

Block tests. Each step of the hierarchy adds a block of parameters that share one rate: the loadings with their response rate (M0→M1), or a block of entries of $\boldsymbol{A}$ with its common decay (M1→M2, M2→M3, and for M4 first the online→offline block, then the offline→online block). Under the null that every entry of the block is zero, that rate is not identified, so the chi-bar-square reference (6.9) does not apply (Remark C.4(i)). Each step is therefore decided by the following registered test.

(i) *Statistic.* The sup-LR statistic (6.10), $\mathrm{LR}_{\sup} = \max_g 2\{\ell_T(\hat{\vartheta}_{j+1}(\beta_g)) - \ell_T(\tilde{\vartheta}_j)\}$, where $\tilde{\vartheta}_j$ is the fit of the smaller model and $\hat{\vartheta}_{j+1}(\beta)$ the fit of the larger model with the new block's rate fixed at β and every other parameter free.

(ii) *Grid.* $G = 9$ rates whose half-lives $\log 2/\beta_g$ are equally spaced on the log scale from 0.5 to 21 days (0.5, 0.8, 1.3, 2.0, 3.2, 5.2, 8.3, 13.2, 21 days). The longest half-life is about a third of the knot spacing h_b, which keeps the cascade time scale apart from the background (Section 6.1).

(iii) *Reference.* The parametric bootstrap (6.11) from the fit of the smaller model, with the observed calendar and $B = 499$ replicates. In each replicate both models are refitted, the larger one at every grid value, starting from the null fit with small positive block entries (Appendix E.5). The step is taken if $p_{\mathrm{B}} \le 0.05$. The upcrossing approximation p_{D} (Appendix C.6) is reported as a secondary quantity only.

(iv) *Order.* The block tests are applied in the fixed order of the hierarchy, each at level 0.05 and only if the previous step was taken. This fixed-sequence procedure controls the family-wise error rate of the block decisions at 0.05.

(v) *Entries and the Holm family.* Within the blocks that were accepted, each entry α_{kj} is tested by the boundary test (6.9), whose decay is identified through the other entries of its block. If every other entry of the block is estimated as zero under the entry's null, that entry is tested with the sup-LR procedure of (i)–(iii) instead. All entry-level tests in the accepted blocks form one family, and Holm's procedure (Holm, 1979) controls its family-wise error rate at 0.05. Decision rules D1 and D2 refer to these Holm-adjusted tests.

(vi) *Fallbacks.* The synthetic dry run (V4, Appendix H.6) checks the size and the cost of the block tests before any real data are loaded. If only the cost check fails, the score-based sup-statistic of Appendix C.6 replaces the sup-LR statistic, with the same grid and $B = 999$ replicates. If the size check fails, the block's rate is fixed in advance at the median of the grid (half-life 3.2 days), and the block is tested by the likelihood-ratio statistic at that rate, with the parametric bootstrap reference of (iii). If that test also fails the size check, the step is decided by the out-of-sample log score alone, its excitation estimates are reported as exploratory, and no contagion claim under D1 or D2 is made for that block. The fallback used is reported for every step.

Selection rules. Moving from Mj to M(j+1) requires both of the following:

(a) the block test of the step rejects at level 0.05;

(b) the out-of-sample log score improves.

The reported model is the largest model that meets both criteria. Estimates from every model in the hierarchy are reported as well.

H.5 Sensitivity analyses

S1. *Background flexibility.* We vary the knot spacing $h_b \in \{30, 60, 120, 365\}$ days and also fit a random-walk log-background. A misspecified, too rigid background produces spurious contagion (Theorem 4.7), and second-order structure cannot settle the question (Theorem 4.2). Self-excitation that vanishes under a more flexible background while goodness of fit remains acceptable is reported as not robustly distinguishable from background variation. An overly flexible background can absorb genuine contagion. In the simulation study a spline background removed the spurious contagion of slow regime switches but not that of fast unrecorded shocks (Section 8.3); how much genuine contagion it absorbs was not studied and is checked in the synthetic dry run (V4).

S2. *Thinning.* We use (a) posterior draws of p_k from the companion paper, mapped to streams as in Section 2.6; (b) a grid $p_k \in \{0.1, 0.25, 0.5, 0.75, 1\}$; and (c) salience-dependent recording, $p_k(t) = p_k\{1 + \varepsilon \sum_e \psi(t - \tau_e)\}$ with $\varepsilon \in \{0, 0.25, 0.5, 1\}$ (capped at one), which violates the independence condition of Theorem 4.10. In addition, (d) known recording breaks, such as reporting-system transitions, scope changes and interrupted referral feeds (Section 2.2), enter $\mu_k(t)$ as step indicators, and results are reported with and without the affected periods.

S3. *Kernel family.* We compare exponential, gamma, power-law and nonparametric (histogram) kernels (Marsan and Lengliné, 2008).

S4. *Delay distribution.* Under Design B we fit the delay distribution in three versions: estimated from the month and record-creation bounds; with its scale shifted by ±50%; and with a heavy-tailed (log-normal) form. We compare the report-time analysis with an occurrence-month aggregate analysis. Nowcast corrections follow Proposition 7.3.

S5. *Event list.* We halve and double $s^\star$, drop each event in turn, compare class-specific and pooled loadings, and vary the shock kernel ψ.

S6. *Spatial resolution.* We compare boroughs with patrol boroughs.

S7. *Online measurement error.* We compare uncorrected and corrected series (Equation (2.2)), with period-specific error rates and propagated uncertainty.

S8. *Ties.* We compare at least 20 within-day randomisations with the discrete-time INAR fit (Equation (6.8)).

H.6 Validation

V1. *Goodness of fit.* Following Brown et al. (2002), we apply the time-rescaling theorem (Proposition C.5) stream by stream: a Kolmogorov–Smirnov test on rescaled inter-event times, with times randomised within the recorded resolution. Because the parameters are estimated, the reference distribution is obtained by parametric bootstrap.

V2. *Out-of-sample performance.* We compute log scores for daily counts (Gneiting and Raftery, 2007) and randomised PIT histograms for calibration (Dawid, 1984; Gneiting et al., 2007), using rolling-origin one-step and Δ_a-step forecasts over the test period.

V3. *Placebo shocks.* Placebo calibration is done for one event class at a time (Section 6.7). For each class we draw 1,000 placebo calendars with the same number of events as the real class, and permute the real sizes over them. Their dates are drawn uniformly from *all* days of the training period, holidays

included, subject only to the 48-hour merging rule of Appendix H.3 within each calendar, and without reference to the real dates of that class. Excluding holidays, or any other set of days on which a real event could fall, would break the exchangeability on which the rank test rests. The base model is the selected model with that class removed. The real class and each placebo calendar are added to the base model in turn, as a shock class, and the same statistics are computed for all of them. The rejection frequency of the placebo loading tests at nominal level 0.05 is the empirical false-positive rate of the shock test. The rank of each real-event direct component D_k among its placebo counterparts is reported. The rank p-value is exact under the randomisation hypothesis of Lemma 6.3, that the real dates are exchangeable with draws from this design; otherwise it is reported as a calibration, not as a test. If the empirical false-positive rate exceeds 0.10, inference on shocks uses the placebo distribution in place of the asymptotic reference.

V4. *Synthetic dry run.* Before any real data are loaded, the full pipeline is run on synthetic data generated as in Section 8, at the pilot's number of streams, length of study period and calendar size. It uses $R = 200$ datasets from each of two configurations: a null configuration without excitation (M1), and an alternative with self-excitation 0.3 and cross-stream entries 0.05, both with a spline background. It checks (a) the size of each block test of Appendix H.4 at a configuration that satisfies its null (the null configuration for the excitation blocks, and the same configuration without shocks for M0→M1), which passes if the rejection rate is at most 0.075 (Monte Carlo standard error about 0.015); (b) its cost, which passes if one block test takes at most seven days on the project's computing allocation; (c) the coverage of the intervals for α_{kj} and for the surge decomposition at the alternative; and (d) the selection and decision rules, including whether a spline background with spacing h_b absorbs genuine contagion. Failures of (a) or (b) trigger the fallbacks of Appendix H.4.

H.7 Decision rules

D1. *Self-excitation* in stream k is claimed only if: the test of $\alpha_{kk} = 0$ rejects after the Holm correction; the 95% interval for α_{kk} excludes zero under the most flexible background in S1; goodness of fit (V1) is not rejected at level 0.01; and the placebo false-positive rate (V3) is at most 0.10. Otherwise we report that self-excitation is not distinguishable from background variation at the available resolution.

D2. *Cross-stream and online→offline excitation* are claimed under the same criteria applied to α_{kj}. They are described as Granger-type predictive dependence (Proposition 4.12) and never as causation.

D3. *Surge decompositions* are reported for every pre-registered event, whatever its size or significance. A surge is described as predominantly due to contagion only if the 95% interval for $C_k/(D_k + C_k)$ lies above 0.5 under the primary model and under the S1 and S2 variants.

D4. *Thinning-corrected* branching ratios are never reported without the p_k used and the S2 range.

D5. *Alerts* are recommended for operational use with partners only if, in the test period, the rule's realised cost is lower than that of both baselines, with a bootstrap 95% interval for the difference that excludes zero, and the PIT shows no marked miscalibration. Otherwise the alert analysis is reported as a research result only.

D6. *Null results and deviations* are reported in full.

H.8 Replication package

The package will contain:

(i) the estimation and simulation code with a pinned environment;

(ii) the frozen event list and its hash;

(iii) acquisition scripts with dataset identifiers, vintages, download dates and SHA-256 checksums;

(iv) derived counts by stream, at the resolutions and subject to the small-cell rules of Appendix I.6;

(v) for restricted streams (platform or partner data), the code together with a synthetic dataset of the same structure;

(vi) all results as machine-readable files, with scripts that regenerate every table and figure; and

(vii) the deviations log.

Raw records are redistributed only where the provider's terms permit.

H.9 Timeline

Table 13 gives the timeline in months from the start of the feasibility stage.

Table 13. Timeline of the pre-registered analysis, in months from the start of the feasibility stage. Each gate must be passed before the next phase begins.

Phase	Months	Activities	Gate / output
P0 Feasibility	0–2	Confirm data dictionaries and access terms (F1–F4); choose Design A or B; submit for ethics review	Ethics approval; design fixed
P1 Events and registration	2–3	Blind coding; freeze event list; register plan	Time-stamped registration; hash deposited
P2 Dry run	3	Full pipeline on synthetic data (V4)	Coverage and selection rules checked
P3 Training fit	4–6	Fit M0–M4; apply selection rules	Selected model
P4 Test evaluation	7	V1–V3; back-test of alerts	Out-of-sample report
P5 Sensitivity	7–9	S1–S8	Robustness tables
P6 Reporting	9–11	Paper; replication package; partner debrief under Section 10	Public release
P7 Prospective	12–24	Prospective alert evaluation under embargo	Embargoed evaluation report

Appendix I The ethics protocol in full

This appendix gives in full the protocol summarised in Section 10. It accompanies the plan of Appendix H and complements the companion paper's protocol for victim privacy and record linkage (Kornstein Lab for Antisemitism Research, 2026a).

I.1 Risks of forecasting

Stigmatisation. A map of elevated predicted intensity can be read as a statement about the people who live in an area. That reading applies to neighbourhoods where Jewish residents are targeted, which can come to be seen as "dangerous". It applies equally to any community that observers might wrongly treat as a source of incidents. The model contains no information about perpetrators, and its spatial units are recording areas, not communities. Alerts and published maps are therefore at the borough level or coarser,

and never street- or venue-level heat maps. Finer units used in sensitivity analysis S6 are reported only as summary statistics. No output ranks neighbourhoods or communities.

Self-fulfilling and self-defeating alerts. An alert can change what it predicts. Publicity can raise reporting, which inflates recorded counts (the time-varying p_k of analysis S2). Protective deployment can prevent incidents, which makes a correct alert look like a false alarm. Either effect contaminates prospective evaluation. Alerts during the prospective period (phase P7) are therefore shared only under the protocol below, and evaluation reports state which alerts were acted on.

Information hazards. Forecasts that name "hot" dates or locations could help someone planning an attack, and could encourage imitation. We therefore publish methods, code and retrospective results, but no live or near-live forecasts and no forecast tied to a specific site. Retrospective results are released only after an embargo (Appendix I.2).

I.2 Responsible disclosure of alerts

Alerts produced during the prospective evaluation are handled as follows.

1. *Recipients.* Alerts go only to vetted community-security partners, and to a police liaison where the partner agreement provides for this. Each partner signs an agreement that limits use to protective purposes, such as security staffing and victim support, and excludes enforcement targeting of any community.
2. *Content.* An alert gives the borough or coarser area, the window Δ_a, the predicted probability of exceeding the threshold with its uncertainty, and a fixed statement of the model's limitations. It contains no individual-level information and no attribution to persons, groups or communities.
3. *Human review.* A named member of the research team reviews every alert before release, and may withhold an alert that is plainly driven by a data artefact.
4. *Channel and logging.* Alerts are sent over an encrypted channel. Each alert, the decision to release or withhold it, and the partner's acknowledgement are logged for audit.
5. *Embargo.* Alerts are not published. Aggregate evaluation results (hit and false-alarm rates by quarter and borough) are published no earlier than 90 days after the end of the period they cover.
6. *Suspension.* Alerts are suspended if rolling calibration deteriorates beyond a pre-specified tolerance, or if a partner reports misuse.

I.3 Language and attribution

The model attributes *timing*, not blame. A positive shock loading υ_k, or a large direct component D_k after event e, means that recorded incidents in stream k rose after τ_e by more than the background and the contagion terms predict. It does not mean that the parties to the event, the communities associated with them, or the people who discussed it caused any incident. Incidents are committed by individual perpetrators, about whom the model contains no information. We therefore:

(i) describe events as *salient external events* of a stated class and date, and do not characterise them politically;

(ii) never describe a community, nationality or religious group as a source, driver or cause of incidents;

(iii) phrase every result in terms of recorded incidents and statistical association; this includes the online→offline coefficients, which are Granger-type (Proposition 4.12); and

(iv) report event-level decompositions in an appendix under neutral identifiers, with class-level summaries in the main text.

I.4 Political neutrality

The paper takes no position on any political conflict, government, movement or public figure. Neutrality is built into the procedure, not just asserted: event classes are defined by type and apply to all parties alike (Appendix H.3); the event list is frozen before outcome data are seen, so events cannot be selected after the fact; every pre-registered event is reported whatever the result (rule D3); and the event-coding rules and frozen list are public. Before registration, the event-coding protocol will be reviewed by at least two external reviewers with differing perspectives on the events likely to be coded. Their comments and our responses will be deposited with the registration.

I.5 Online data

Online data are used only under the terms of the access route through which they are obtained, including any conditions on data protection and security attached to access under Article 40 of the Digital Services Act (European Parliament and Council of the European Union, 2022). The analysis uses aggregate counts per time bin. We do not build account-level scores or profiles, do not link online accounts to offline incidents, victims or suspects, and do not publish post text, handles or quotations that would allow re-identification. Annotation samples for estimating classifier error (Equation (2.2)) are held under access control and deleted when the retention period agreed with the data provider and the ethics board ends. Classifier outputs are used only as aggregate series. They are never used to flag individuals.

I.6 Ethics review and data security

The protocol, including this section, will be submitted to an institutional review board or equivalent research ethics committee before any data are accessed (gate F5). This applies even to publicly downloadable data, because the plan combines several streams and may involve partner and platform data. Restricted data are kept on encrypted storage with role-based access, and use is logged. The custodian who holds the incident data during event coding (Appendix H.3) is separate from the coders. Outputs follow a minimum-cell rule: counts below 10 are not shown below the borough level, and any table that could reveal a single incident's date and location is suppressed or coarsened. Derived counts are released at a resolution no finer than the source itself publishes; for restricted streams, they are released weekly or more coarsely. Data are retained no longer than the agreements allow, and the replication package contains only data whose release is permitted (Appendix H.8).